Strings in AdS$_3$ - $SL(2,\mathbb{R})$ WZW Model

Departmental Project 1

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1 Introduction

We consider critical bosonic string theory propagating on a spacetime of the form $\mathrm{AdS}_3 \times \mathcal{M}$, where AdS_3 is three–dimensional anti–de Sitter space and $\mathcal{M}$ is a compact internal manifold. Here *critical* means that the total central charge of the worldsheet conformal field theory satisfies $c_{\text{total}} = 26$, which is required for quantum consistency of the bosonic string.

The background also contains an antisymmetric tensor field $B_{\mu\nu}$ from the Neveu–Schwartz sector. Its field strength $H = dB$ or $H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + \partial_\nu B_{\rho\mu} + \partial_\rho B_{\mu\nu}$ is non-zero on AdS_3. Our goal is to determine the *spectrum*, i.e., the complete set of physical string states and their energies.

AdS_3 is special because, for generic curved backgrounds, the string worldsheet theory is only approximately conformal. However, when AdS_3 is supported by NS–NS flux, the worldsheet theory becomes exactly equivalent to a Wess–Zumino–Witten (WZW) model with target space $\mathrm{SL}(2,\mathbb{R})$. Thus, string propagation on AdS_3 is described exactly by the $\mathrm{SL}(2,\mathbb{R})$ WZW model. This allows one to study full string theory, not merely its low-energy gravity approximation.

In flat spacetime, $g_{00} = -1$, which is constant. But in global coordinates on AdS_3, the metric takes the form

$$ds^2 = -\cosh^2(\rho)\, dt^2 + d\rho^2 + \sinh^2(\rho)\, d\phi^2 \tag{1}$$

$$\implies g_{00} = -\cosh^2(\rho) \tag{2}$$

which depends on position. This means the time coordinate interacts with spatial coordinates. Such interactions make quantization difficult and typically introduce negative-norm states. Therefore, one must carefully identify which quantum states are physical.

The group manifold of $\mathrm{SL}(2,\mathbb{R})$ contains a periodic time-like direction. Periodic time implies closed time-like curves, which are unphysical. To avoid this, we use the *universal cover* of $\mathrm{SL}(2,\mathbb{R})$, denoted by $\widetilde{\mathrm{SL}(2,\mathbb{R})}$, where the time coordinate is non-compact, $t \in (-\infty, \infty)$. Locally, the geometry is unchanged, but global identifications are removed.

The WZW model possesses left– and right–moving affine current algebras

$$\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R} \tag{3}$$

All worldsheet states transform in the representations of these algebras. Physical string states are obtained by imposing the Virasoro constraints

$$\begin{aligned} (L_0 - 1)\,|\psi\rangle &= 0, \\ L_n|\psi\rangle &= 0 \qquad (n > 0), \end{aligned} \tag{4}$$

Thus, determining the string spectrum reduces to determining which representations of $\widehat{\mathrm{SL}(2,\mathbb{R})}$ must be included. For compact groups, unitarity strongly restricts the allowed representations. But $\mathrm{SL}(2,\mathbb{R})$ being non-compact, almost all representations contain negative-norm states. Hence, unitarity alone does not specify which representations are allowed. Only the final physical states, after Virasoro constraints, must have a non-negative norm.

Earlier analyses considered only representations with L_0 bounded below. It was found that the discrete spectrum is ghost-free if $0 \leq j \leq \frac{k}{2}$, where $j \sim$ representation label and k is the WZW level. The quadratic Casimir behaves as $C_2 = -j(j-1)$. So, j is roughly related to spacetime mass. This bound implies an upper limit on string masses. There are two big Puzzles,

- If the target space-time manifold has a non-trivial one-cycle (a closed loop that cannot be shrunk), strings may wrap it. A winding string satisfies

$$\phi(\sigma + 2\pi) = \phi(\sigma) + 2\pi w \tag{5}$$

  where w is the winding number. Winding contributes to the energy. An upper bound on mass, therefore, implies an upper bound on w. Such a bound is non-physical, since winding should be arbitrary.

- Semi-classical analysis predicts the existence of *long strings*, where the strings are stretched near the boundary of AdS_3 with finite energy. These states do not appear in representations with L_0 bounded below.

The resolution is to include new representations generated by *spectral flow*, an automorphism of the affine algebra. These representations have L_0 not bounded below, but after imposing Virasoro constraints, they yield physical, ghost-free states. Including spectral–flowed representations, removes the mass upper bound, allows arbitrary winding, produces long string states, and preserves unitarity of physical states.

Throughout, we use global coordinates on AdS_3, given by (t, ρ, ϕ) where $t \in (-\infty, \infty)$ is the AdS_3 time, $\rho \in [0, \infty)$ is the radial distance and $\phi \in (0, 2\pi], \phi + 2\pi \sim \phi$ is the angular coordinate. Thus (t, ρ, ϕ) covers the entire spacetime and contains no horizons.

In AdS/CFT, energies in global AdS_3 correspond directly to conformal dimensions of operators in the boundary CFT. A consistent string spectrum on AdS_3 requires enlarging the Hilbert space of the $\mathrm{SL}(2,\mathbb{R})$ WZW model to include spectral–flowed representations. This produces a consistent, ghost-free theory with the correct semi-classical limit.

2 Classical Structure of Strings on AdS_3

2.1 Group Elements and Group Manifold

Choose the generators of $\mathfrak{sl}(2,\mathbb{R})$ as follows,

$$T^1 = \frac{1}{2}\sigma_3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T^2 = \frac{1}{2}\sigma_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{and } T^3 = -\frac{i}{2}\sigma_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{6}$$

whose commutation relations are

$$[T^1, T^2] = -T^3, \qquad [T^2, T^3] = T^1, \qquad [T^3, T^1] = T^2 \tag{7}$$

In the Cartan-Weyl Basis, we can choose the generators as

$$H \equiv -iT^3 \quad \text{and} \quad T^{\pm} = T^1 \pm iT^2 \qquad \Longrightarrow \quad [H, T^{\pm}] = \pm T^{\pm}, \quad [T^+, T^-] = -2H \tag{8}$$

This forms the $\mathfrak{sl}(2,\mathbb{R})$ algebra. The trace relations are given by,

$$\mathrm{Tr}(T^a T^b) = \frac{1}{2}\kappa^{ab} \qquad \text{where} \quad \kappa^{ab} = \mathrm{diag}(1,1,-1) = \kappa_{ab}. \tag{9}$$

We can also choose another set of basis denoted by T^{a^*} which are given by,

$$T^{\pm^*} = T^{\mp} \quad \text{and} \quad T^{3^*} = T^3 \qquad \Longrightarrow \qquad \mathrm{Tr}(T^{a^*} T^{b^*}) = \frac{\kappa^{ab}}{2} \tag{10}$$

κ^{ab} is the invariant symmetric bilinear form of $\mathfrak{sl}(2,\mathbb{R})$ which is used to raise and lower indices. Any element of $\mathrm{SL}(2,\mathbb{R})$ group can be written as,

$$g = \begin{pmatrix} X_{-1} + X_1 & X_0 - X_2 \\ -X_0 - X_2 & X_{-1} - X_1 \end{pmatrix} \tag{11}$$

and since the determinant has to be 1, we have a constraint,

$$X_{-1}^2 + X_0^2 - X_1^2 - X_2^2 = 1 \quad \text{or} \quad -X_{-1}^2 - X_0^2 + X_1^2 + X_2^2 = -1 \tag{12}$$

Thus we can think of X_{-1}, X_0, X_1, X_2 as the embedding coordinates that form a three–dimensional hyperboloid in ambient $\mathbb{R}^{2,2}$ space with the metric

$$ds^2_{\mathbb{R}^{2,2}} = -dX_{-1}^2 - dX_0^2 + dX_1^2 + dX_2^2 \tag{13}$$

We can parametrize the embedded hyperboloid using the intrinsic global coordinates (t, ρ, ϕ).

We can relate the ambient space coordinates with the intrinsic coordinates as follows,

$$\begin{aligned} X_{-1} &= \cosh(\rho)\cos(t), \\ X_0 &= \cosh(\rho)\sin(t), \\ X_1 &= \sinh(\rho)\cos(\phi), \\ X_2 &= \sinh(\rho)\sin(\phi). \end{aligned} \tag{14}$$

The induced metric becomes

$$ds^2 = -\cosh^2(\rho)\,dt^2 + d\rho^2 + \sinh^2(\rho)\,d\phi^2 \tag{15}$$

This is the metric of the AdS_3 space-time with (t, ρ, ϕ) as the intrinsic global coordinates. Thus, the $\mathrm{SL}(2, \mathbb{R})$ group manifold is locally AdS_3.

In terms of the global coordinates, we can write the group element as,

$$g = e^{iu\sigma_2} e^{\rho\sigma_3} e^{iv\sigma_2} = \begin{pmatrix} \cosh(\rho)\ \cos(t) + \sinh(\rho)\ \cos(\phi) & \cosh(\rho)\ \sin(t) - \sinh(\rho)\ \sin(\phi) \\ -\cosh(\rho)\ \sin(t) - \sinh(\rho)\ \sin(\phi) & \cosh(\rho)\ \cos(t) - \sinh(\rho)\ \cos(\phi) \end{pmatrix} \tag{16}$$

where

$$u = \frac{t+\phi}{2}, \qquad v = \frac{t-\phi}{2}.$$

Also, it can be shown that for any group manifold, the induced metric can be written as,

$$ds^2 = \frac{1}{2}\mathrm{Tr}\big(g^{-1}dg\ g^{-1}dg\big) \tag{17}$$

where the above ds^2 is called the natural bi-invariant metric on the group. We can check this explicitly for $\mathrm{SL}(2, \mathbb{R})$ group manifold, where for $g = e^{iu\sigma_2} e^{\rho\sigma_3} e^{iv\sigma_2}$, we get the induced metric exactly as shown before.

Let, $g = ABC$ where $A = e^{iu\sigma_2}$, $B = e^{\rho\sigma_3}$, $C = e^{iv\sigma_2}$, the Maurer–Cartan one-form becomes

$$g^{-1}dg = C^{-1}B^{-1}(A^{-1}dA)BC + C^{-1}(B^{-1}dB)C + C^{-1}dC. \tag{18}$$

$$\text{where} \qquad A^{-1}dA = i\sigma_2\,du, \quad B^{-1}dB = \sigma_3\,d\rho, \quad C^{-1}dC = i\sigma_2\,dv, \tag{19}$$

$$\begin{aligned} \text{together with} \quad B^{-1}\sigma_2 B &= \cosh(2\rho)\sigma_2 - i\sinh(2\rho)\sigma_1, \\ C^{-1}\sigma_1 C &= \cos(2v)\sigma_1 + \sin(2v)\sigma_3, \\ C^{-1}\sigma_3 C &= -\sin(2v)\sigma_1 + \cos(2v)\sigma_3, \end{aligned} \tag{20}$$

one finds $g^{-1}dg = \omega_1\sigma_1 + i\omega_2\sigma_2 + \omega_3\sigma_3$, where

$$\begin{aligned}
\omega_1 &= \sinh(2\rho)\cos(2v)\,du - \sin(2v)\,d\rho, \\
\omega_2 &= \cosh(2\rho)\,du + dv, \\
\omega_3 &= \sinh(2\rho)\sin(2v)\,du + \cos(2v)\,d\rho.
\end{aligned} \tag{21}$$

Since $\mathrm{Tr}(\sigma_1^2) = \mathrm{Tr}(\sigma_3^2) = 2$, and $\mathrm{Tr}((i\sigma_2)^2) = -2$ and all mixed traces vanish, the ds^2 gives

$$ds^2 = \omega_1^2 + \omega_3^2 - \omega_2^2. \tag{22}$$

Substituting the explicit expressions for the one-forms yields

$$ds^2 = d\rho^2 + \sinh^2(2\rho)\,du^2 - \left(\cosh(2\rho)\,du + dv\right)^2. \tag{23}$$

Finally, using

$$du = \frac{dt + d\phi}{2}, \qquad dv = \frac{dt - d\phi}{2}, \tag{24}$$

and simplifying, one obtains

$$\boxed{ds^2 = d\rho^2 - \cosh^2(\rho)\,dt^2 + \sinh^2(\rho)\,d\phi^2}. \tag{25}$$

This is precisely the metric of global AdS_3. Therefore, the $\mathrm{SL}(2,\mathbb{R})$ group manifold, equipped with the bi-invariant metric, is locally identical to AdS_3.

We can also show this directly using the ambient coordinates as follows. We have,

$$g = \begin{pmatrix} X_{-1} + X_1 & X_0 - X_2 \\ -X_0 - X_2 & X_{-1} - X_1 \end{pmatrix} \quad \Longrightarrow \quad dg = \begin{pmatrix} dX_{-1} + dX_1 & dX_0 + dX_2 \\ -dX_0 + dX_2 & dX_{-1} - dX_1 \end{pmatrix}$$

$$\text{and} \quad g^{-1} = \begin{pmatrix} X_{-1} - X_1 & -X_0 - X_2 \\ X_0 - X_2 & X_{-1} + X_1 \end{pmatrix}$$

Multiplying these matrices, one finds

$$\frac{1}{2}\mathrm{Tr}\big(g^{-1}dg\,g^{-1}dg\big) = -dX_{-1}^2 - dX_0^2 + dX_1^2 + dX_2^2 \tag{26}$$

Thus, the group manifold inherits precisely the flat metric of the embedding space $\mathbb{R}^{2,2}$, given by $ds^2 = -dX_{-1}^2 - dX_0^2 + dX_1^2 + dX_2^2$, restricted to the hyperboloid $X_{-1}^2 + X_0^2 - X_1^2 - X_2^2 = 1$. By substituting the global coordinates parametrization, we will recover the AdS_3 metric.

2.2 $\mathrm{SL}(2,\mathbb{R})$ WZW Action, equations of motions and currents

Let $g(\tau,\sigma) : \Sigma \to G$ be a map from a two-dimensional worldsheet Σ into the group manifold of a Lie group G whose Lie algebra is $\mathfrak{g}$. So, $g(\sigma,\tau) \in G$. Then, the WZW action is given by,

$$S[g] = \underbrace{-\frac{k}{8\pi\alpha'}\int_\Sigma d^2\sigma\,\sqrt{-h}\,h^{\alpha\beta}\,\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)\,g^{-1}(\partial_\beta g)\big)}_{S_{\text{kin}}} + \underbrace{\frac{k}{12\pi\alpha'}\int_B \mathrm{Tr}\big(g^{-1}dg\wedge g^{-1}dg\wedge g^{-1}dg\big)}_{\Gamma_{WZ}} \tag{27}$$

where B is a three–dimensional manifold such that $\partial B = \Sigma$ and $h_{\alpha\beta}$ is the world-sheet metric. The first term in the WZW action is known as the *sigma-model kinetic term.* To understand this terminology, recall that a non-linear sigma model describes maps $X : \Sigma \to$ Target space.

The Sigma-Model action is

$$S_\sigma = \frac{1}{4\pi\alpha'}\int d^2\sigma\,\sqrt{-h}\,h^{\alpha\beta}G_{MN}(X)\partial_\alpha X^M\partial_\beta X^N \tag{28}$$

In the WZW model, the fundamental field is instead group-valued $g(\tau,\sigma)\in G$. Introducing local coordinates X^M on the group manifold and writing $g = g(X^M)$, the Maurer–Cartan one-form may be expanded as

$$g^{-1}dg = e^a{}_M(X)\,T_a\,dX^M \qquad \Longrightarrow \qquad g^{-1}\partial_\alpha g = e^a{}_M(X)\,T_a\,\partial_\alpha X^M \tag{29}$$

where $e^a{}_M$ are the vielbeins on the group manifold. Using $\mathrm{Tr}(T_aT_b) = \frac{1}{2}\kappa_{ab}$, one finds

$$\begin{aligned} \mathrm{Tr}\big(g^{-1}dg\,g^{-1}dg\big) &= \frac{1}{2}\kappa_{ab}e^a{}_M e^b{}_N dX^M dX^N \\ \Longrightarrow \quad \mathrm{Tr}\big(g^{-1}(\partial_\alpha g)\,g^{-1}(\partial_\beta g)\big) &= \frac{1}{2}\kappa_{ab}e^a{}_M e^b{}_N(\partial_\alpha X^M)(\partial_\beta X^N) \end{aligned} \tag{30}$$

Define the target-space metric as shown below,

$$G_{MN} = \frac{k}{2}\left(\frac{1}{2}\kappa_{ab}e^a{}_M e^b{}_N\right) \tag{31}$$

Then,

$$\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)\,g^{-1}(\partial_\beta g)\big) = \frac{2}{k}\,G_{MN}\,(\partial_\alpha X^M)(\partial_\beta X^N) \tag{32}$$

Substituting the Maurer–Cartan form into the kinetic term gives

$$\mathrm{Tr}\left(g^{-1}\partial_a g\,g^{-1}\partial^a g\right) = \frac{1}{2}\kappa_{bc}e^b{}_\mu e^c{}_\nu\partial_a X^\mu\partial^a X^\nu = \frac{2}{k}\,G_{\mu\nu}(X)\partial_a X^\mu\partial^a X^\nu. \tag{33}$$

Consequently,

$$S_{\text{kin}} = -\frac{1}{4\pi\alpha'}\int_\Sigma d^2\sigma\,\sqrt{-h}\,h^{\alpha\beta}G_{MN}(X)(\partial_\alpha X^M)(\partial_\beta X^N) \tag{34}$$

This is precisely the action of a non-linear sigma model with target-space metric G_{MN}. For this reason, the first term of the WZW action is referred to as the sigma-model kinetic term.

The second term, known as the Wess–Zumino term, describes the coupling of the string to the background antisymmetric B-field or, equivalently, to the NS–NS three-form flux supporting the AdS_3 background.

To derive equations of motion, choose conformal gauge $h_{\alpha\beta} = e^{\omega}\eta_{\alpha\beta}$ and light–cone coordinates

$$x^{\pm} = \tau \pm \sigma, \qquad \partial_{\pm} = \frac{1}{2}(\partial_\tau \pm \partial_\sigma),$$

$$\implies S[g] = -\frac{k}{4\pi\alpha'}\int_\Sigma d^2x\, \mathrm{Tr}(J_+ J_-) + \frac{k}{12\pi\alpha'}\int_B \epsilon^{ijk}\,\mathrm{Tr}\big(g^{-1}\partial_i g\, g^{-1}\partial_j g\, g^{-1}\partial_k g\big)\,. \tag{35}$$

where $J_+ = g^{-1}(\partial_+ g)$ and $J_- = g^{-1}(\partial_- g)$. Note that $J_\pm$ are Lie-algebra valued objects.

Consider the left variation given by, $g \to (1+\varepsilon)g$, where $\delta g = \varepsilon g$. Thus $\varepsilon \sim \varepsilon(\tau,\sigma) \in \mathfrak{g}$ Lie algebra. Then g^{-1} varies as $\delta(g^{-1}) = -g^{-1}(\delta g)g^{-1} = -g^{-1}\varepsilon$. The current $J_\alpha = g^{-1}(\partial_\alpha g)$ varies as follows:

$$\begin{aligned}
\delta J_\alpha &= \delta(g^{-1})\partial_\alpha g + g^{-1}\partial_\alpha(\delta g) \\
&= (-g^{-1}\varepsilon)\partial_\alpha g + g^{-1}\partial_\alpha(\varepsilon g) \\
&= -g^{-1}\varepsilon\partial_\alpha g + g^{-1}(\partial_\alpha\varepsilon)g + g^{-1}\varepsilon\partial_\alpha g \\
&= g^{-1}(\partial_\alpha\varepsilon)g
\end{aligned} \tag{36}$$

Consider $\varepsilon = g\eta g^{-1}$. Taking the derivative on both sides, we can show that,

$$\partial_\alpha\varepsilon = g\,(\partial_\alpha\eta + [J_\alpha,\eta])\,g^{-1} \tag{37}$$

$$\implies g^{-1}(\partial_\alpha\varepsilon)g = \partial_\alpha\eta + [J_\alpha,\eta] \tag{38}$$

$$\implies \delta J_\alpha = \partial_\alpha\eta + [J_\alpha,\eta] \tag{39}$$

The equations of motion are given by extremizing the total WZW action $\delta S = \delta S_{\mathrm{kin}} + \delta\Gamma_{\mathrm{WZ}} = 0$. Now let's vary the kinetic term,

$$\begin{aligned}
\delta S_{\mathrm{kin}} &= -\frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\left((\delta J_+)J_- + J_+(\delta J_-)\right) \\
&= -\frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big((\partial_+\eta + [J_+,\eta])J_- + J_+(\partial_-\eta + [J_-,\eta])\Big) \\
&= -\frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big((\partial_+\eta)J_- + J_+(\partial_-\eta)\Big)
\end{aligned} \tag{40}$$

The commutators vanish by cyclicity of trace. Then, on integrating by parts,

$$\delta S_{\mathrm{kin}} = \frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\big(\eta(\partial_+ J_- + \partial_- J_+)\big). \tag{41}$$

The variation of Wess–Zumino Term is given by

$$\begin{aligned}\delta\Gamma_{\mathrm{WZ}} &= \frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big(g^{-1}\delta g\,\epsilon^{\alpha\beta}(\partial_\alpha J_\beta)\Big) = \frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big((g^{-1}\varepsilon g)\,\epsilon^{\alpha\beta}(\partial_\alpha J_\beta)\Big)\\ &= \frac{k}{4\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big(\eta(\partial_+J_- - \partial_-J_+)\Big).\end{aligned} \tag{42}$$

Adding both variations, $$\delta S = -\frac{k}{2\pi\alpha'}\int d^2x\,\mathrm{Tr}\big(\eta\,(\partial_+J_-)\big) = 0 \tag{43}$$

Since ε is arbitrary, $\eta = g^{-1}\varepsilon g$ is also arbitrary. Thus, we get the equation of motion as,

$$\partial_+J_- = 0 \qquad \Longrightarrow \qquad \boxed{\partial_+\left(g^{-1}(\partial_-g)\right) = 0} \tag{44}$$

This variation was done by multiplying g with $(1+\varepsilon)$ from the left. If we repeat the variation with $g \to g(1+\varepsilon)$ (varying from right), then we get the other form of the equation of motion,

$$\partial_-J_+ = 0 \qquad \Longrightarrow \qquad \boxed{\partial_-\left((\partial_+g)g^{-1}\right) = 0} \tag{45}$$

2.3 Gauge Symmetry, Noether Theorem and chiral currents

The full WZW action is invariant under independent left and right multiplications by constant elements of G,

$$g(x^+,x^-) \;\longrightarrow\; \Omega_L\,g(x^+,x^-)\,\Omega_R^{-1}, \qquad \Omega_L,\Omega_R \in G. \tag{46}$$

This is a global symmetry since Ω_L and Ω_R are constants. Infinitesimally,

$$\Omega_L \sim 1+\omega_L, \qquad \Omega_R \sim 1+\omega_R, \qquad \text{where} \quad \omega_L,\omega_R \in \mathfrak{g} \qquad \Longrightarrow \qquad \delta g = \omega_L g - g\omega_R \tag{47}$$

Variation of the WZW action gives $\partial_+J_L = 0$ and $\partial_-J_R = 0$ where

$$J_L(x^-) = \frac{k}{\alpha'}\,g^{-1}\partial_-g\;, \qquad J_R(x^+) = \frac{k}{\alpha'}\,(\partial_+g)g^{-1}. \tag{48}$$

Thus, the on-Shell conditions directly make the currents J_L and J_R conserved. So, we can modify the global symmetry to local symmetry. Also, since the currents themselves are holomorphic and anti-holomorphic, we can promote the global variational parameters ω_L and ω_R to chiral functions,

$$\boxed{\omega_L \to \omega_L(x^-), \qquad \omega_R \to \omega_R(x^+).} \tag{49}$$

Thus, the global symmetry of the WZW action becomes a local symmetry under

$$g(x^+,x^-) \;\longrightarrow\; \Omega_L(x^-)\,g(x^+,x^-)\,\Omega_R^{-1}(x^+), \qquad \text{where}\;\; \Omega_L(x^-),\;\Omega_R(x^+) \in G. \tag{50}$$

The equations of motion, therefore, force the theory to split into left-moving and right-moving sectors where the left(right) multiplications become purely left(right)-moving.

Consider $\delta g = \omega_L(x^-)\, g$, $\omega_L \in G$. Define $\eta_L \equiv g^{-1}\omega_L g$. Then, following the procedure as before, we get,

$$\delta S = \frac{k}{2\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big(\eta_L\,\partial_+\big(\,g^{-1}(\partial_- g)\big)\Big). \tag{51}$$

We shall expand the quantity $J_L(x^-) = \frac{k}{\alpha'}g^{-1}(\partial_- g)$ in terms of basis T^{a^*} of $\mathfrak{g}$, whose coefficients of expansion is $J_L^a(x^-)$.

$$\Longrightarrow\ J_L(x^-) = \frac{k}{\alpha'}g^{-1}(\partial_- g) = \sum_{a,b} J_L^a(x^-)T^{b^*}\kappa_{ab} \tag{52}$$

Multiplying throughout by T^{c^*} and taking trace, gives,

$$\begin{aligned}\frac{k}{\alpha'}\mathrm{Tr}\big(T^{c^*}g^{-1}(\partial_- g)\big) &= \sum_{a,b}\kappa_{ab}J_L^a(x^-)\mathrm{Tr}\big(T^{b^*}T^{c^*}\big)\\ &= \sum_{a,b}\kappa_{ab}J_L^a(x^-)\frac{\kappa^{bc}}{2} = \frac{1}{2}\sum_a J_L^a(x^-)\delta_a^c = \frac{J_L^c(x^-)}{2}\end{aligned} \tag{53}$$

$$\Longrightarrow\ \boxed{J_L^a(x^-) = \frac{2k}{\alpha'}\,\mathrm{Tr}\big(T^{a^*}g^{-1}(\partial_- g)\big)} \tag{54}$$

Also, expand $\eta_L = \eta_{L\,c}T^{c^*}$. Then,

$$\begin{aligned}\delta S &= \sum_{a,b,c}\frac{1}{2\pi}\int d^2x\,\kappa_{ab}\mathrm{Tr}\Big(\eta_{L\,c}\,\partial_+\big(T^{c^*}\,J^a(x^-)T^{b^*}\big)\Big)\\ &= \sum_{a,b,c}\frac{1}{2\pi}\int d^2x\,\kappa_{ab}\,\eta_{L\,c}\,\big(\partial_+ J_L^a(x^-)\big)\,\mathrm{Tr}\big(T^{c^*}T^{b^*}\big)\\ &= \sum_{a,b,c}\frac{1}{2\pi}\int d^2x\,\kappa_{ab}\,\eta_{L\,c}\,\big(\partial_+ J_L^a(x^-)\big)\,\frac{\kappa^{bc}}{2} = \sum_{a,c}\frac{1}{4\pi}\int d^2x\,\eta_{L\,c}\,\big(\partial_+ J_L^a(x^-)\big)\,\delta_a^c\end{aligned} \tag{55}$$

$$\Longrightarrow\ \delta S = \sum_a\frac{1}{4\pi}\int d^2x\,\eta_{L\,a}\,\big(\partial_+ J_L^a(x^-)\big) \qquad\qquad \Longrightarrow\ \partial_+ J_L^a(x^-) = 0$$

Hence, we can write the left current conservation in terms of coefficients of the $\mathfrak{sl}(2,\mathbb{R})$ basis expansion of $J_L(x^-)$.

Similarly, for $\delta g = g\,\omega_R(x^+)$, $\omega_R \in G$. Define $\eta_R \equiv g^{-1}\omega_R g$. This gives,

$$\delta S = \frac{k}{2\pi\alpha'}\int d^2x\,\mathrm{Tr}\Big(\eta_R\,\partial_-\big((\partial_+ g)\,g^{-1}\big)\Big). \tag{56}$$

We shall expand the quantity $J_R(x^+) = \frac{k}{\alpha'}(\partial_+ g)g^{-1}$ in terms of basis T^a of $\mathfrak{g}$, whose coefficients of expansion is $J_R^a(x^+)$.

$$\Longrightarrow\ J_R(x^+) = \frac{k}{\alpha'}(\partial_+ g)g^{-1} = \sum_{a,b} J_R^a(x^+)T^b\kappa_{ab} \tag{57}$$

Multiplying throughout by T^c and taking trace, gives,

$$\boxed{J_R^a(x^+) = \frac{2k}{\alpha'}\,\mathrm{Tr}\big(T^a(\partial_+ g)\,g^{-1}\big)} \tag{58}$$

With this definition of right currents and by expanding $\eta_R = \eta_{Rc}T^c$, we get the current conservation for right multiplication as,

$$\delta S = \sum_a \frac{1}{4\pi}\int d^2x\,\eta_{Ra}\left(\partial_- J_R^a(x^+)\right) \tag{59}$$

Hence, the complete symmetry algebra is therefore $\widehat{\mathfrak{g}}_L \times \widehat{\mathfrak{g}}_R$.

Aside: The WZW action is invariant under independent left and right global multiplications, $g \to \Omega_L\, g$ and $g \to g\,\Omega_R^{-1}$. The natural objects appearing in the equations of motion are the left- and right-invariant Maurer-Cartan forms, $g^{-1}(\partial_- g)$ and $(\partial_+ g)g^{-1}$, respectively. They behave differently under these symmetries. Under a global left multiplication, $g^{-1}(\partial_- g) \to (g^{-1}\Omega_L^{-1})\,(\partial_-(\Omega_L g)) = g^{-1}(\partial_- g)$. So, $g^{-1}(\partial_- g)$ is invariant, while $(\partial_+ g)g^{-1} \to (\partial_+(\Omega_L g))(\Omega_L g)^{-1} = \Omega_L\big[(\partial_+ g)g^{-1}\big]\Omega_L^{-1}$. So, J_R transforms by conjugation. But conversely, under a global right multiplication, $g^{-1}(\partial_- g)$ transforms under a conjugation but $(\partial_+ g)g^{-1}$ is invariant. So, when defining current components $J^a_{L,R}$, one chooses a basis so that both left- and right-moving currents transform in the adjoint representation under the symmetry that acts non-trivially on them. This choice explains why a conjugated generator appears in the projection of the left current, while no such conjugation is needed for the right current.

2.4 Chiral Factorization

The classical field equations of the left and right sectors suggest that the solution for $g(x^+, x^-)$ has to factorize into a product of matrices as follows. Consider first the left-moving sector. The equation $g^{-1}\partial_- g = \dfrac{1}{k}J_L(x^-)$ is a first-order differential equation in x^-. Its general solution can be written as

$$g(x^+, x^-) = h_+(x^+)\,g_-(x^-), \tag{60}$$

where $g_-(x^-)$ satisfies

$$J_L(x^-) = k\,[g_-(x^-)]^{-1}\partial_-(g_-(x^-)), \tag{61}$$

and $h_+(x^+)$ is an integration constant with respect to x^-. Similarly, the right-moving equation $(\partial_+ g)g^{-1} = \dfrac{1}{k}J_R(x^+)$ is a first-order differential equation in x^+. Its general solution is

$$g(x^+, x^-) = g_+(x^+)\,h_-(x^-), \tag{62}$$

where $g_+(x^+)$ satisfies

$$J_R(x^+) = k\,[\partial_+(g_+(x^+))][g_+(x^+)]^{-1} \tag{63}$$

So here, $h_-(x^-)$ is independent of x^+, and is a constant of integration with respect to x^+. Since both results represent the same field $g(x^+, x^-)$, the arbitrary functions $h_\pm$ can be absorbed into redefinitions of $g_\pm$. Therefore, the most general classical solution factorizes as

$$\boxed{g(x^+, x^-) = g_+(x^+)\, g_-(x^-)}. \tag{64}$$

This factorization reflects the complete decoupling of left-moving and right-moving degrees of freedom at the classical level in the WZW model.

2.5 Monodromy in the WZW Model

For a closed string, the spatial worldsheet coordinate is periodic $\sigma \sim \sigma + 2\pi$. Therefore, the target-space embedding must satisfy $g(\tau, \sigma + 2\pi) = g(\tau, \sigma)$. In light cone coordinates, under a shift $\sigma \to \sigma + 2\pi$, we get,

$$x^+ \to x^+ + 2\pi \quad \text{and} \quad x^- \to x^- - 2\pi$$

The equations of motion of the WZW model imply $g(x^+, x^-) = g_+(x^+)\, g_-(x^-)$ where g_+ is purely right-moving and g_- is purely left-moving. Substitute the factorized form into the periodicity condition:

$$g_+(x^+ + 2\pi)\, g_-(x^- - 2\pi) = g_+(x^+)\, g_-(x^-) \tag{65}$$

Multiply from the left by $g_+(x^+)^{-1}$ and from the right by $g_-(x^-)^{-1}$:

$$g_+(x^+)^{-1} g_+(x^+ + 2\pi) = g_-(x^-) g_-(x^- - 2\pi)^{-1} \tag{66}$$

The left-hand side depends only on x^+, while the right-hand side depends only on x^-. Since the equality must hold for all (x^+, x^-), both sides must equal a constant group element $M \in \mathrm{SL}(2, \mathbb{R})$ called the Monodromy Matrix. Hence,

$$g_+(x^+ + 2\pi) = g_+(x^+)\, M \qquad \text{and} \qquad g_-(x^- - 2\pi) = M^{-1}\, g_-(x^-) \tag{67}$$

The full field $g = g_+ g_-$ is strictly periodic, but the chiral factors are only periodic up to multiplication by a Monodromy. Thus M measures the failure of the left- and right-moving parts to be individually periodic. It characterizes how the string is "twisted" as one goes once around the spatial circle. Also, the decomposition $g = g_+ g_-$ is not unique. One may redefine

$$g_+(x^+) \to g_+(x^+)\, f \qquad \text{and} \qquad g_-(x^-) \to f^{-1} g_-(x^-) \tag{68}$$

with a constant $f \in \mathrm{SL}(2, \mathbb{R})$. Under this transformation, the definition of M goes to

$$M \to f^{-1} M f.$$

Therefore, the Monodromy matrix is defined up to a conjugacy class.

Elements of $\mathrm{SL}(2,\mathbb{R})$ fall into three types.

- **Elliptic:** $|\mathrm{Tr}\, M| < 2$ (rotation-like)
- **Hyperbolic:** $|\mathrm{Tr}\, M| > 2$ (boost-like)
- **Parabolic:** $|\mathrm{Tr}\, M| = 2$ (null-like)

Classical string solutions are grouped according to which class their Monodromy belongs to. With respect to string theory, we will see that,

- Elliptic Monodromy $\rightarrow$ short strings oscillating near the center of AdS_3.
- Hyperbolic Monodromy $\rightarrow$ long strings reaching toward the boundary.
- Parabolic Monodromy $\rightarrow$ marginal or threshold configurations.

2.6 Left and Right Currents of $\mathrm{SL}(2,\mathbb{R})$ WZW Model

Using the global coordinates parameterization of the $\mathrm{SL}(2,\mathbb{R})$ group element,

$$g = e^{iu\sigma_2} e^{\rho\sigma_3} e^{iv\sigma_2} = \begin{pmatrix} \cosh(\rho)\,\cos(t) + \sinh(\rho)\,\cos(\phi) & \cosh(\rho)\,\sin(t) - \sinh(\rho)\,\sin(\phi) \\ \\ -\cosh(\rho)\,\sin(t) - \sinh(\rho)\,\sin(\phi) & \cosh(\rho)\,\cos(t) - \sinh(\rho)\,\cos(\phi) \end{pmatrix}$$

where

$$u = \frac{t+\phi}{2}, \qquad v = \frac{t-\phi}{2}.$$

We get the expression for the Left and Right Currents as:

$$\boxed{\begin{aligned} J_L^1 &= \frac{2k}{\alpha'}\big((\partial_- u)\sinh(2\rho)\sin(2v) + (\partial_-\rho)\cos(2v)\big) \\ J_L^2 &= \frac{2k}{\alpha'}\big((\partial_-\rho)\sin(2v) - (\partial_- u)\sinh(2\rho)\cos(2v)\big) \\ J_L^3 &= \frac{2k}{\alpha'}\big(\cosh(2\rho)(\partial_- u) + (\partial_- v)\big) \\ J_L^\pm &= \frac{2k}{\alpha'}\Big[(\partial_-\rho) \pm i(\partial_- u)\sinh(2\rho)\Big]e^{\mp 2iv} \end{aligned}} \tag{69}$$

$$\boxed{\begin{aligned} J_R^1 &= \frac{2k}{\alpha'}\big((\partial_+\rho)\cos(2u) + (\partial_+ v)\sinh(2\rho)\sin(2u)\big) \\ J_R^2 &= \frac{2k}{\alpha'}\big((\partial_+ v)\sinh(2\rho)\cos(2u) - (\partial_+\rho)\sin(2u)\big) \\ J_R^3 &= \frac{2k}{\alpha'}\big(\cosh(2\rho)(\partial_+ v) + (\partial_+ u)\big) \\ J_R^\pm &= \frac{2k}{\alpha'}\Big[(\partial_+\rho) \pm i(\partial_+ v)\sinh(2\rho)\Big]e^{\mp 2iu} \end{aligned}} \tag{70}$$

Derivation of J_L^a and J_R^a in Global Coordinates

The left-moving current is $J_L^a = k\,\mathrm{Tr}\big(T^{a^*}\, g^{-1}\partial_- g\big)$. Then, define

$$A = e^{iu\sigma_2}, \quad B = e^{\rho\sigma_3}, \quad C = e^{iv\sigma_2}, \quad \Longrightarrow \quad g = ABC \quad \text{and} \quad g^{-1} = C^{-1}B^{-1}A^{-1} \tag{71}$$

$$\partial_\alpha g = (\partial_\alpha A)BC + A(\partial_\alpha B)C + AB(\partial_\alpha C). \tag{72}$$

Multiplying by g^{-1} from the left,

$$g^{-1}\partial_\alpha g = C^{-1}B^{-1}A^{-1}(\partial_\alpha A)BC + C^{-1}B^{-1}(\partial_\alpha B)C + C^{-1}(\partial_\alpha C) \qquad \text{(3rd term)} \tag{73}$$

Also, we can find,

$$\begin{aligned} A^{-1}\partial_\alpha A &= i(\partial_\alpha u)\sigma_2, \\ B^{-1}\partial_\alpha B &= (\partial_\alpha \rho)\sigma_3, \\ C^{-1}\partial_\alpha C &= i(\partial_\alpha v)\sigma_2, \end{aligned} \tag{74}$$

We substitute these into the previous expression. Using Baker–Campbell–Hausdorff identities, we get other useful results,

$$\begin{aligned} B^{-1}\sigma_2 B &= \cosh(2\rho)\,\sigma_2 + i\sinh(2\rho)\,\sigma_1, \\ B^{-1}\sigma_1 B &= \cosh(2\rho)\,\sigma_1 - i\sinh(2\rho)\,\sigma_2, \\ C^{-1}\sigma_1 C &= \cos(2v)\,\sigma_1 - \sin(2v)\,\sigma_3, \\ C^{-1}\sigma_3 C &= \cos(2v)\,\sigma_3 + \sin(2v)\,\sigma_1. \end{aligned} \tag{75}$$

Consider the first term in $g^{-1}\partial_\alpha g$,

$$\begin{aligned} C^{-1}B^{-1}A^{-1}(\partial_\alpha A)BC &= C^{-1}B^{-1}\big(i(\partial_\alpha u)\,\sigma_2\big)BC \\ &= i(\partial_\alpha u)C^{-1}\left(\cosh(2\rho)\sigma_2 + i\sinh(2\rho)\sigma_1\right)C \\ &= i(\partial_\alpha u)\left(\cosh(2\rho)\sigma_2 + i\sinh(2\rho)C^{-1}\sigma_1 C\right) \\ &= i(\partial_\alpha u)\big(\cosh(2\rho)\sigma_2 + i\sinh(2\rho)[\cos(2v)\sigma_1 - \sin(2v)\sigma_3]\big) \end{aligned} \tag{76}$$

For the 2nd term,

$$C^{-1}B^{-1}(\partial_\alpha B)C = C^{-1}(\partial_\alpha\rho)\sigma_3 C = (\partial_\alpha\rho)\big(\cos(2v)\sigma_3 + \sin(2v)\sigma_1\big) \tag{77}$$

Then adding all three terms, and collecting $\sigma_1, \sigma_2, \sigma_3$ components, we get,

$$\begin{aligned} g^{-1}\partial_\alpha g = \Big[&-(\partial_\alpha u)\sinh(2\rho)\cos(2v) + (\partial_\alpha\rho)\sin(2v)\Big]\sigma_1 \\ &+ i\Big[\cosh(2\rho)(\partial_\alpha u) + (\partial_\alpha v)\Big]\sigma_2 + \Big[(\partial_\alpha u)\sinh(2\rho)\sin(2v) + (\partial_\alpha\rho)\cos(2v)\Big]\sigma_3. \end{aligned} \tag{78}$$

Now, $J_L^a = \frac{2k}{\alpha'}\,\mathrm{Tr}\big(T^{a^*}\,g^{-1}(\partial_- g)\big)$. $T^{3^*} = T^3$, $T^{\pm^*} = T^{\mp}$. Then,

$$\begin{aligned} J_L^3 &= \frac{2k}{\alpha'}\,\mathrm{Tr}\big(T^{3^*}\,g^{-1}(\partial_- g)\big) = -\frac{2k}{\alpha'}\,\mathrm{Tr}\left(\frac{i}{2}\sigma_2\,g^{-1}(\partial_- g)\right) \quad \text{only } \sigma_2 \text{ component survives} \\ &= \frac{2k}{\alpha'}\big(\cosh(2\rho)(\partial_- u) + (\partial_- v)\big)\frac{\mathrm{Tr}(\sigma_2^2)}{2} \end{aligned} \tag{79}$$

Similarly, we can find even J_L^1 and J_L^2, giving us,

$$\begin{aligned} J_L^1 &= \frac{2k}{\alpha'}\big((\partial_- u)\sinh(2\rho)\sin(2v) + (\partial_-\rho)\cos(2v)\big) \\ J_L^2 &= \frac{2k}{\alpha'}\big((\partial_-\rho)\sin(2v) - (\partial_- u)\sinh(2\rho)\cos(2v)\big) \\ J_L^3 &= \frac{2k}{\alpha'}\big(\cosh(2\rho)(\partial_- u) + (\partial_- v)\big) \end{aligned} \tag{80}$$

Then we also get $J_L^{\pm} = \frac{2k}{\alpha'}\,\mathrm{Tr}\big(T^{\pm^*}\,g^{-1}(\partial_- g)\big) = J_L^1 \mp iJ_L^2$ as,

$$\begin{aligned} J_L^{\pm} &= \frac{2k}{\alpha'}\Big[(\partial_-\rho)\cos(2v) + (\partial_- u)\sinh(2\rho)\sin(2v) \mp i\big((\partial_-\rho)\sin(2v) - (\partial_- u)\sinh(2\rho)\cos(2v)\big)\Big] \\ &= \frac{2k}{\alpha'}\Big[(\partial_-\rho)(\cos(2v) \mp i\sin(2v)) \pm i(\partial_- u)\sinh(2\rho)(\cos(2v) \mp i\sin(2v)) \end{aligned} \tag{81}$$

$$\implies \quad J_L^{\pm} = \frac{2k}{\alpha'}\Big[(\partial_-\rho) \pm i(\partial_- u)\sinh(2\rho)e\Big]e^{\mp 2iv} \tag{82}$$

Similarly, we can find the components of the Right current. Firstly, we have,

$$(\partial_\alpha g)g^{-1} = (\partial_\alpha A)A^{-1} + A(\partial_\alpha B)B^{-1}A^{-1} + AB(\partial_\alpha C)C^{-1}B^{-1}A^{-1} \tag{83}$$

Now, we can show that,

$$\begin{aligned} (\partial_\alpha A)A^{-1} &= i(\partial_\alpha u)\sigma_2, \\ (\partial_\alpha B)B^{-1} &= (\partial_\alpha \rho)\sigma_3, \\ (\partial_\alpha C)C^{-1} &= i(\partial_\alpha v)\sigma_2. \end{aligned} \tag{84}$$

$$\implies \quad (\partial_\alpha g)g^{-1} = i(\partial_\alpha u)\sigma_2 + A(\partial_\alpha\rho)\sigma_3 A^{-1} + AB\,i(\partial_\alpha v)\sigma_2\,B^{-1}A^{-1}. \tag{85}$$

We also use the conjugation identities involving A,

$$\begin{aligned} A\,\sigma_3 A^{-1} &= \cos(2u)\,\sigma_3 - \sin(2u)\,\sigma_1, \\ A\,\sigma_1 A^{-1} &= \cos(2u)\,\sigma_1 + \sin(2u)\,\sigma_3 \end{aligned} \tag{86}$$

Then, the 2nd term becomes $A((\partial_\alpha\rho)\,\sigma_3)A^{-1} = (\partial_\alpha\rho)\big[\cos(2u)\sigma_3 - \sin(2u)\sigma_1\big]$.

And the 3rd term:

$$
\begin{aligned}
AB\, i(\partial_\alpha v)\sigma_2\, B^{-1}A^{-1} &= i(\partial_\alpha v)A\big(\cosh(2\rho)\sigma_2 - i\sinh(2\rho)\sigma_1\big)A^{-1} \\
&= i(\partial_\alpha v)\cosh(2\rho)\sigma_2 + (\partial_\alpha v)\sinh(2\rho)\big[\cos(2u)\sigma_1 + \sin(2u)\sigma_3\big].
\end{aligned}
\tag{87}
$$

Then by collecting all terms,

$$
\begin{aligned}
(\partial_\alpha g)g^{-1} = &\Big[(\partial_\alpha v)\sinh(2\rho)\cos(2u) - (\partial_\alpha\rho)\sin(2u)\Big]\sigma_1 \\
&+ i\Big[(\partial_\alpha u) + \cosh(2\rho)(\partial_\alpha v)\Big]\sigma_2 + \Big[(\partial_\alpha\rho)\cos(2u) + (\partial_\alpha v)\sinh(2\rho)\sin(2u)\Big]\sigma_3.
\end{aligned}
\tag{88}
$$

The right currents are defined by $J_R^a = \frac{2k}{\alpha'}\,\mathrm{Tr}(T^a(\partial_+ g)g^{-1})$. Then,

$$
\begin{aligned}
J_R^3 &= \frac{2k}{\alpha'}\,\mathrm{Tr}\big(T^3\,(\partial_+ g)\,g^{-1}\big) = \frac{2k}{\alpha'}\,\mathrm{Tr}\left(-\frac{i}{2}\sigma_2\, g^{-1}(\partial_- g)\right) \quad \text{only } \sigma_2 \text{ component survives} \\
&= \frac{2k}{\alpha'}\big(\cosh(2\rho)(\partial_+ v) + (\partial_+ u)\big)\frac{\mathrm{Tr}(\sigma_2^2)}{2}
\end{aligned}
\tag{89}
$$

Similarly, we can find even J_R^1 and J_R^2, giving us,

$$
\begin{aligned}
J_R^1 &= \frac{2k}{\alpha'}\big((\partial_+\rho)\cos(2u) + (\partial_+ v)\sinh(2\rho)\sin(2u)\big) \\
J_R^2 &= \frac{2k}{\alpha'}\big((\partial_+ v)\sinh(2\rho)\cos(2u) - (\partial_+\rho)\sin(2u)\big) \\
J_R^3 &= \frac{2k}{\alpha'}\big(\cosh(2\rho)(\partial_+ v) + (\partial_+ u)\big)
\end{aligned}
\tag{90}
$$

Then we also get $J_R^\pm = J_R^1 \pm iJ_R^2$ as,

$$
\begin{aligned}
J_R^\pm &= \frac{2k}{\alpha'}\Big[(\partial_+\rho)\cos(2u) + (\partial_+ v)\sinh(2\rho)\sin(2u) \pm i\big((\partial_+ v)\sinh(2\rho)\cos(2u) - (\partial_+\rho)\sin(2u)\big)\Big] \\
&= \frac{2k}{\alpha'}\Big[(\partial_+\rho)(\cos(2u) \mp i\sin(2u)) \pm i(\partial_+ v)\sinh(2\rho)(\cos(2u) \mp i\sin(2u))
\end{aligned}
\tag{91}
$$

$$
\Longrightarrow \qquad J_R^\pm = \frac{2k}{\alpha'}\Big[(\partial_+\rho) \pm i(\partial_+ v)\sinh(2\rho)\Big]e^{\mp 2iu} \tag{92}
$$

2.7 Relation between $J^3_{L,R}$ Zero Modes and Spacetime Charges

The SL$(2,\mathbb{R})$ group element is written as $g = e^{iu\sigma_2}\, e^{\rho\sigma_3}\, e^{iv\sigma_2}$ with $u = \frac{1}{2}(t+\phi)$ and $v = \frac{1}{2}(t-\phi)$. Here t is the global AdS$_3$ time coordinate and ϕ is the angular coordinate. Consider an infinitesimal time translation

$$t \to t + \delta t \qquad \Longrightarrow \qquad u \to u + \frac{\delta t}{2} \;\; , \quad v \to v + \frac{\delta t}{2} \qquad \Longrightarrow \qquad g \to e^{i\frac{\delta t}{2}\sigma_2}\, g\, e^{i\frac{\delta t}{2}\sigma_2} \tag{93}$$

Thus, time translations correspond to simultaneous left and right multiplications by $e^{i\frac{\delta t}{2}\sigma_2}$. Similarly, for an infinitesimal angular rotation

$$\phi \to \phi + \delta\phi \qquad \Longrightarrow \qquad u \to u + \frac{\delta\phi}{2} \;\; , \quad v \to v - \frac{\delta\phi}{2} \qquad \Longrightarrow \qquad g \to e^{i\frac{\delta\phi}{2}\sigma_2}\, g\, e^{-i\frac{\delta\phi}{2}\sigma_2} \tag{94}$$

Therefore, angular rotations correspond to opposite left and right multiplications. Observe that in both time translation and angular rotation, the left and right multiplication factors have the σ_2 matrix in the exponential, which can be written in terms of T^3.

$$g \underbrace{\longrightarrow}_{t\to t+\delta t} e^{-T^3\delta t}\; g\; e^{-T^3\delta t} \quad \Longrightarrow \quad g \to (1 - T^3\delta t)g(1 - T^3\delta t) \quad \Longrightarrow \quad \delta_t g = -\delta t(T^3 g + gT^3) \tag{95}$$

$$g \underbrace{\longrightarrow}_{\phi\to\phi+\delta\phi} e^{-T^3\delta\phi}\; g\; e^{T^3\delta\phi} \quad \Longrightarrow \quad g \to (1 - T^3\delta\phi)g(1 + T^3\delta\phi) \quad \Longrightarrow \quad \delta_\phi g = -\delta\phi(T^3 g - gT^3) \tag{96}$$

If we define the left and right group actions as $L^a[g] \equiv T^a g$ and $R^a[g] \equiv gT^a$, then,

$$\delta_t g = -\delta t(L^3 + R^3)[g] \quad \text{and} \quad \delta_\phi g = -\delta\phi(L^3 - R^3)[g] \tag{97}$$

Note that the global left and right multiplications on g by any SL$(2,\mathbb{R})$ group element is an isometry symmetry transformation. Now, ∂_t and ∂_ϕ are killing vectors of AdS$_3$ space time.

$$\partial_t g = -(L^3 + R^3)[g] \;\; \Longrightarrow \;\; \partial_t \sim R^3 + L^3 \quad \text{and} \quad \partial_\phi g = (R^3 - L^3)[g] \;\; \Longrightarrow \;\; \partial_\phi \sim R^3 - L^3 \tag{98}$$

The corresponding conserved Noether currents associated with this isometry transformation are $J^3_R + J^3_L$ for time translation and $J^3_R - J^3_L$ for $\phi-$rotations. So, the corresponding conserved Noether spacetime charges are energy E, which is the generator of time-translation, and angular momentum ℓ, which is the generator of angular-rotation. They are given by,

$$E = J^3_{0,L} + J^3_{0,R} \quad \text{and} \quad \ell = J^3_{0,R} - J^3_{0,L} \tag{99}$$

where

$$J^3_{0,R} = \frac{1}{2\pi}\int_0^{2\pi} dx^+\; J^3_R(x^+) \quad \text{and} \quad J^3_{0,L} = \frac{1}{2\pi}\int_0^{2\pi} dx^-\; J^3_L(x^-) \tag{100}$$

Thus, the zero modes of the SL$(2,\mathbb{R})$ currents encode the AdS$_3$ energy and angular momentum.

2.8 Second Quadratic Casimir

Let $\mathfrak{g}$ be a Lie algebra with generators T^a satisfying $[T^a, T^b] = i f^{ab}{}_c T^c$. A Casimir operator C is an element of the universal enveloping algebra of $\mathfrak{g}$ such that

$$[C, T^a] = 0 \qquad \forall\, a \tag{101}$$

By Schur's lemma, in any irreducible representation, the object (here the second Casimir) which commutes with all the group elements has to be proportional to the identity, with its eigenvalue as the proportionality constant. Thus, its eigenvalue therefore labels the irr-reps.

In the WZW model, the physical symmetry algebra acting on string states is the current algebra generated by operators $J^a_L(x^-)$ and $J^a_R(x^+)$ whose modes satisfy the affine Kac-Moody Algebra,

$$[J^a_m, J^b_n] = i f^{ab}{}_c J^c_{m+n} + \frac{k}{2} m\, \kappa^{ab} \delta_{m+n,0} \qquad \text{(for both Left and Right currents)} \tag{102}$$

String states are built by acting with J^a_{-n} on a ground state. Thus, we are interested in the representations of $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$, which is the tensor product of left and right affine extensions of the $\mathfrak{sl}(2,\mathbb{R})$ algebra. Therefore, the invariant operators must be constructed from the current modes J^a_n. Now, the zero modes J^a_0 generate the global $\mathfrak{sl}(2,\mathbb{R})$ algebra:

$$[J^a_0, J^b_0] = i f^{ab}{}_c J^c_0 \tag{103}$$

In the Cartan-Weyl Basis,

$$[J^3_0, J^\pm_0] = \pm J^\pm_0 \quad \text{and} \quad [J^+_0, J^-_0] = 2J^3_0 \tag{104}$$

It can be shown that the second Casimir is given by,

$$C_2 = \sum_{a,b} \kappa_{ab} J^a_0 J^b_0 = \frac{1}{2}\big(J^+_0 J^-_0 + J^-_0 J^+_0\big) - \big(J^3_0\big)^2 \tag{105}$$

Since there are two copies of the current algebra, there can be two copies of the second Casimir $C_{2,L}$ and $C_{2,R}$ whose eigenvalues are given by $j_L(j_L - 1)$ and $j_R(j_R - 1)$, respectively, whose irreducible representations will be labeled by the quantum numbers (j_L, j_R). But for the physical closed-string states, we must have identical left- and right-moving Casimir eigenvalues $j_L = j_R$ because their eigenvalue is directly related to the spacetime mass of the string.

On group manifolds, the second Casimir plays the role of the AdS_3 Laplacian

$$-\nabla^2 \;\leftrightarrow\; J^a_0 J^a_0 .$$

Thus, the spacetime Klein-Gordon equation

$$(-\nabla^2 + m^2)|\Phi\rangle = 0$$

becomes

$$(J_0^a J_0^a + m^2)|\Phi\rangle = 0.$$

$$\implies m^2 \sim C_2 = -j(j-1).$$

Thus, the eigenvalue of the second Casimir determines spacetime mass in AdS_3, governing the mass spectrum of string states.

2.9 Classical Sugawara Energy Momentum Tensor

Varying the action with respect to the worldsheet metric $h_{\alpha\beta}$ (Lorentzian) defines the Energy-Momentum tensor as follows.

$$T_{\alpha\beta} = -\frac{4\pi}{\sqrt{-h}}\frac{\delta S}{\delta h^{\alpha\beta}}. \tag{106}$$

The full WZW action and its metric variation are given below.

$$S = -\frac{k}{8\pi\alpha'}\int d^2\sigma\,\sqrt{-h}\,h^{\alpha\beta}\,\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)g^{-1}(\partial_\beta g)\big) + k\,\Gamma_{\mathrm{WZ}}. \tag{107}$$

The Wess-Zumino term Γ_{WZ} is topological and does not depend on $h_{\alpha\beta}$. Hence, it does not contribute to the stress tensor. Varying the kinetic term with respect to $h^{\alpha\beta}$ gives,

$$\begin{aligned}
\delta_h S &= -\frac{k}{8\pi\alpha'}\int d^2\sigma\big[(\delta\sqrt{-h})\,h^{\alpha\beta} + \sqrt{-h}\,\delta h^{\alpha\beta}\big]\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)g^{-1}(\partial_\beta g)\big)\\
&= -\frac{k}{8\pi\alpha'}\int d^2\sigma\Big[-\frac{1}{2}\sqrt{-h}h_{\mu\nu}(\delta h^{\mu\nu})\,h^{\alpha\beta} + \sqrt{-h}\,(\delta h^{\alpha\beta})\Big]\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)g^{-1}(\partial_\beta g)\big)\\
&= \frac{k}{8\pi\alpha'}\int d^2\sigma\sqrt{-h}\Big[\frac{1}{2}h_{\mu\nu}\,h^{\alpha\beta}\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)g^{-1}(\partial_\beta g)\big) - \mathrm{Tr}\big(g^{-1}(\partial_\mu g)g^{-1}(\partial_\nu g)\big)\Big](\delta h^{\mu\nu})
\end{aligned} \tag{108}$$

$$\implies \frac{\delta S}{\delta h^{\mu\nu}} = -\frac{k}{8\pi\alpha'}\sqrt{-h}\Big[\mathrm{Tr}\big(g^{-1}(\partial_\mu g)g^{-1}(\partial_\nu g)\big) - \frac{1}{2}h_{\mu\nu}\,h^{\alpha\beta}\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)g^{-1}(\partial_\beta g)\big)\Big] \tag{109}$$

$$\implies \boxed{T_{\alpha\beta} = \frac{k}{2\alpha'}\left[\mathrm{Tr}\big(g^{-1}(\partial_\alpha g)g^{-1}(\partial_\beta g)\big) - \frac{1}{2}h_{\alpha\beta}\,h^{\mu\nu}\mathrm{Tr}\big(g^{-1}(\partial_\mu g)g^{-1}(\partial_\nu g)\big)\right]} \tag{110}$$

In conformal gauge and light-cone coordinates, $h_{+-} = h_{-+} = -1$, $h_{++} = h_{--} = 0$ and $h^{+-} = h^{-+} = -1$, $h^{++} = h^{--} = 0$. This gives,

$$T_{--} = \frac{k}{2\alpha'}\big[\mathrm{Tr}\big(g^{-1}(\partial_- g)g^{-1}(\partial_- g)\big)\big] = \frac{\alpha'}{2k}\left[\mathrm{Tr}\left(\Big[\frac{k}{\alpha'}\,g^{-1}(\partial_- g)\Big]\Big[\frac{k}{\alpha'}\,g^{-1}(\partial_- g)\Big]\right)\right] \tag{111}$$

Expand $\dfrac{k}{\alpha'}g^{-1}(\partial_- g) = \sum_{a,b}\kappa_{ab}J_L^a T^{b^*}$ and using $\mathrm{Tr}(T^{a^*}T^{b^*}) = \dfrac{\kappa^{ab}}{2}$, we get,

$$\begin{aligned} T_{--}(x^-) &= \frac{\alpha'}{2k}\mathrm{Tr}\Big(J_L(x^-)\,J_L(x^-)\Big) = \frac{\alpha'}{2k}\sum_{a,b}\sum_{c,d}\kappa_{ab}\kappa_{cd}\left[\mathrm{Tr}\Big(\big[J_L^a T^{b^*}\big]\big[J_L^c T^{d^*}\big]\Big)\right] \\ &= \frac{\alpha'}{2k}\sum_{a,b}\sum_{c,d}\kappa_{ab}\kappa_{cd}J_L^a J_L^c\frac{\kappa^{bd}}{2} = \frac{\alpha'}{4k}\sum_{a,c}\kappa_{ac}J_L^a J_L^c \end{aligned} \tag{112}$$

We shall choose a different normalization constant (irrelevant) for the energy-momentum tensor component and write,

$$\Longrightarrow \boxed{T_{--}(x^-) = \frac{\alpha'}{2k}\sum_{a,b}\kappa_{ab}J_L^a(x^-)J_L^b(x^-)} \tag{113}$$

Similarly for the T_{++} component, we get,

$$T_{++} = \frac{k}{2\alpha'}\big[\mathrm{Tr}\big(g^{-1}(\partial_+ g)g^{-1}(\partial_+ g)\big)\big] = \frac{\alpha'}{2k}\left[\mathrm{Tr}\left(\left[\frac{k}{\alpha'}(\partial_+ g)g^{-1}\right]\left[\frac{k}{\alpha'}(\partial_+ g)g^{-1}\right]\right)\right] \tag{114}$$

Expand $\dfrac{k}{\alpha'}(\partial_+ g)g^{-1} = \sum_{a,b}\kappa_{ab}J_R^a T^b$ and using $\mathrm{Tr}(T^a T^b) = \dfrac{\kappa_{ab}}{2}$, we get,

$$\begin{aligned} T_{++}(x^+) &= \frac{\alpha'}{2k}\mathrm{Tr}\Big(J_R(x^+)\,J_R(x^+)\Big) = \frac{\alpha'}{2k}\sum_{a,b}\sum_{c,d}\kappa_{ab}\kappa_{cd}\left[\mathrm{Tr}\Big(\big[J_R^a T^{b}\big]\big[J_R^c T^{d}\big]\Big)\right] \\ &= \frac{\alpha'}{2k}\sum_{a,b}\sum_{c,d}\kappa_{ab}\kappa_{cd}J_R^a J_R^c\frac{\kappa^{bd}}{2} = \frac{\alpha'}{4k}\sum_{a,c}\kappa_{ac}J_R^a J_R^c \end{aligned} \tag{115}$$

In terms of the new normalization constant, we get,

$$\Longrightarrow \boxed{T_{++}(x^+) = \frac{\alpha'}{2k}\sum_{a,b}\kappa_{ab}J_R^a(x^+)J_R^b(x^+)} \tag{116}$$

Let, $\dfrac{2k}{\alpha'} = \mathcal{K}$. Then, the chiral components of the energy-momentum tensor are given by,

$$T_{--}(x^-) = \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\,J_L^a(x^-)J_L^b(x^-) \;\text{ and }\; T_{++}(x^+) = \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\,J_R^a(x^+)J_R^b(x^+) \tag{117}$$

Aside: The normalization constant in front of the energy-momentum tensor components is irrelevant classically, as the classical equations of motion, the classical constraint algebra, and the physical classical solutions remain unchanged. But quantum mechanically it is significant, and we will see the reason later.

2.10 Virasoro Constraints

We consider string propagation in a background of the form $\mathrm{AdS}_3 \times \mathcal{M}$, where the AdS_3 part is described by an $\mathrm{SL}(2,\mathbb{R})$ WZW model and $\mathcal{M}$ is some internal conformal field theory. Then the total worldsheet action is a sum of two decoupled sigma models:

$$S_{\text{total}} = \underbrace{S_{\mathrm{AdS}_3}[g, h_{\alpha\beta}]}_{WZW} + S_{\mathcal{M}}[X^i, h_{\alpha\beta}],$$

where $g(\tau,\sigma) \in \mathrm{SL}(2,\mathbb{R})$ describes motion in AdS_3, $X^i(\tau,\sigma)$ are coordinates on $\mathcal{M}$, and $h_{\alpha\beta}$ is the worldsheet metric. Since the action splits into two independent pieces, the worldsheet stress-energy tensor also splits as follows,

$$T^{\text{total}}_{\alpha\beta} = T^{AdS}_{\alpha\beta} + T^{\mathcal{M}}_{\alpha\beta} \tag{118}$$

The Polyakov string action is invariant under Worldsheet diffeomorphisms and Weyl transformations. Because of this gauge symmetry, the worldsheet metric $h_{\alpha\beta}$ has no propagating degrees of freedom. So, the equation of motion obtained by varying the action with respect to $h_{\alpha\beta}$ is $\dfrac{\delta S}{\delta h^{\alpha\beta}} = 0$. Using the definition of the stress tensor, this implies

$$T^{\text{total}}_{\alpha\beta} = T^{AdS}_{\alpha\beta} + T^{\mathcal{M}}_{\alpha\beta} = 0 \tag{119}$$

Thus, the Virasoro constraints are simply the equations of motion of the worldsheet metric given in the above equation. In terms of the light cone coordinates, it becomes,

$$T^{\text{total}}_{++} = T^{AdS}_{++} + T^{\mathcal{M}}_{++} \qquad \text{and} \qquad T^{\text{total}}_{--} = T^{AdS}_{--} + T^{\mathcal{M}}_{--} \tag{120}$$

The AdS_3 Energy-Momentum tensor is the same one derived previously using the Sugawara construction, and $T^{\mathcal{M}}_{++}$ is left unspecified, corresponding to the internal CFT.

Quantum Correction

At the quantum level, operator products of currents modify the stress tensor. The Sugawara form becomes

$$T_{--}(x^-) = \frac{\alpha'}{2(k-2)} \sum_{a,b} \kappa_{ab} : J^a_L(x^-) J^b_L(x^-) : \quad \text{and} \quad T_{++}(x^+) = \frac{\alpha'}{2(k-2)} \sum_{a,b} \kappa_{ab} : J^a_R(x^+) J^b_R(x^+) :$$

Thus in the quantum theory, one replaces

$$k \to k - h^{\wedge}$$

where $h^{\wedge}$ is the dual Coxeter number of the linear algebra $\mathfrak{g}$ which is 2 for $\mathfrak{sl}(2,\mathbb{R})$.

3 Geodesics in AdS_3

We previously derived that the classical solutions of the WZW model obey $g(x^+, x^-) = g_+(x^+)\, g_-(x^-)$. Also, if the solution depends only on τ and not on σ, then the string has no spatial extension. It is a point particle moving in target space. The trajectory of this point must then satisfy the geodesic equation of AdS_3. Thus, to find geodesics, we look for solutions where the solution $g(x^+, x^-)$ depends only on τ.

3.1 Time-like geodesic

Consider an ansatz solution

$$g_+(x^+) = U e^{iv_+(x^+)\sigma_2} \;,\; g_-(x^-) = e^{iu_-(x^-)\sigma_2} V \implies g(x^+, x^-) = U e^{i(v_+(x^+)+u_-(x^-))\sigma_2} V \tag{121}$$

where U and V are constant $\mathrm{SL}(2,\mathbb{R})$ matrices. Let, $\theta(x^+, x^-) = v_+(x^+) + u_-(x^-)$ Then, let us derive the currents $J_L(x^-)$ and $J_R(x^+)$ for this ansatz.

$$\partial_+ g = U\Big[i(\partial_+ v_+)\sigma_2 e^{i\theta(x^+,x^-)\sigma_2}\Big]V \tag{122}$$

$$\implies (\partial_+ g)g^{-1} = U\Big[i(\partial_+ v_+)\sigma_2 e^{i\theta(x^+,x^-)\sigma_2}\Big]V\,\Big(V^{-1}e^{-i\theta(x^+,x^-)\sigma_2}U^{-1}\Big) = i(\partial_+ v_+)U\sigma_2 U^{-1} \tag{123}$$

Then the right current components are,

$$J^a_R = \mathcal{K}\, i(\partial_+ v_+)\,\mathrm{Tr}\Big(T^a\, U\sigma_2 U^{-1}\Big) = -2\mathcal{K}(\partial_+ v_+)\,\mathrm{Tr}\Big((U^{-1}T^a U)T^3\Big) \tag{124}$$

The conjugation of any basis element can be expressed as a linear combination as follows,

$$U^{-1}T^a U = \sum_b R(U)^a{}_b\, T^b \tag{125}$$

$$\begin{aligned} \implies \mathrm{Tr}\Big((U^{-1}T^a U)T^c\Big) &= \sum_b R(U)^a{}_b \mathrm{Tr}\Big(T^b T^c\Big) = \sum_b R(U)^a{}_b \frac{\kappa^{bc}}{2} = \frac{1}{2}R(U)^{ac} \\ \implies \quad R(U)^{ab} &= 2\,\mathrm{Tr}\Big((U^{-1}T^a U)T^b\Big) \end{aligned} \tag{126}$$

$$\text{Also,}\quad R(U)^{ab} = 2\,\mathrm{Tr}\Big(U^{-1}T^a U T^b\Big) = 2\,\mathrm{Tr}\Big(UT^b U^{-1}T^a\Big) = R(U^{-1})^{ba} \tag{127}$$

Then, the right current can be written as,

$$J^a_R = -\mathcal{K}(\partial_+ v_+)R(U)^{a3} \tag{128}$$

$$\begin{aligned} \implies \sum_{a,b} \kappa_{ab} J^a_R T^b &= -\mathcal{K}(\partial_+ v_+)\sum_{a,b} \kappa_{ab} R(U)^{a3} T^b = -\mathcal{K}(\partial_+ v_+)\sum_{a,b} \kappa_{ab} R(U^{-1})^{3a} T^b \\ &= -\mathcal{K}(\partial_+ v_+)\sum_b R(U^{-1})^3{}_b T^b = -\mathcal{K}(\partial_+ v_+)\big(UT^3U^{-1}\big) \end{aligned} \tag{129}$$

$$\implies \boxed{\sum_a J^a_R T_a = -\mathcal{K}(\partial_+ v_+)\big(UT^3U^{-1}\big)} \tag{130}$$

Similarly, we can find the left-moving current as follows,

$$g(x^+,x^-) = Ue^{i(v_+(x^+)+u_-(x^-))\sigma_2}V \qquad \implies \partial_- g = U\, i(\partial_- u_-)\sigma_2 e^{i\theta(x^+,x^-)\sigma_2}V \tag{131}$$

$$\implies g^{-1}(\partial_- g) = i(\partial_- u_-)V^{-1}\sigma_2 V = -2(\partial_- u_-)V^{-1}T^{3^*}V \tag{132}$$

$$\implies J_L^a = -2\mathcal{K}(\partial_- u_-)\mathrm{Tr}\Big(T^{a^*}V^{-1}T^{3^*}V\Big) = -2\mathcal{K}(\partial_- u_-)\mathrm{Tr}\Big(VT^{a^*}V^{-1}T^{3^*}\Big) \tag{133}$$

The conjugation of any left basis (T^{a*}) element can be expressed as a linear combination, as

$$V^{-1}T^{a^*}V = \sum_b L(V)^a{}_b\, T^{b^*} \tag{134}$$

$$\begin{aligned}\implies \mathrm{Tr}\Big((V^{-1}T^{a^*}V)T^{c^*}\Big) &= \sum_b L(V)^a{}_b \mathrm{Tr}\Big(T^{b^*}T^{c^*}\Big) = \sum_b L(V)^a{}_b \frac{\kappa^{bc}}{2} = \frac{1}{2}L(V)^{ac} \\ \implies \quad L(V)^{ab} &= 2\,\mathrm{Tr}\Big((V^{-1}T^{a^*}V)T^{b^*}\Big)\end{aligned} \tag{135}$$

$$\text{Also,}\quad L(V)^{ab} = 2\,\mathrm{Tr}\Big(V^{-1}T^{a^*}VT^{b^*}\Big) = 2\,\mathrm{Tr}\Big(VT^{b^*}V^{-1}T^{a^*}\Big) = L(V^{-1})^{ba} \tag{136}$$

$$\implies J_L^a = -\mathcal{K}(\partial_- u_-)L(V)^{3a} \tag{137}$$

$$\begin{aligned}\implies \sum_{a,b}\kappa_{ab}J_L^aT^{b^*} &= -\mathcal{K}(\partial_- u_-)\sum_{a,b}\kappa_{ab}L(V)^{3a}T^{b^*} \\ &= -\mathcal{K}(\partial_- u_-)\sum_b L(V)^3{}_b T^{b^*} = -\mathcal{K}(\partial_- u_-)V^{-1}T^{3^*}V\end{aligned} \tag{138}$$

$$\implies \boxed{\sum_a J_L^a T_a^* = -\mathcal{K}(\partial_- u_-)V^{-1}T^{3^*}V} \tag{139}$$

Now, with this expression for left and right currents, we can find the chiral components of the energy-momentum tensors as follows. For the right sector,

$$\begin{aligned}T_{++}(x^+) &= \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\,J_R^a(x^+)J_R^b(x^+) = \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\Big[-\mathcal{K}(\partial_+v_+)R(U)^{a3}\Big]\Big[-\mathcal{K}(\partial_+v_+)R(U)^{b3}\Big] \\ &= \mathcal{K}(\partial_+v_+)^2\sum_{a,b}\kappa_{ab}\,R(U)^{a3}R(U)^{b3} = \mathcal{K}(\partial_+v_+)^2\sum_{a,b}\kappa_{ab}\,R(U)^a{}_3R(U)^b{}_3\end{aligned} \tag{140}$$

Now, in the adjoint representation, the invariant bilinear form is preserved.

$$\text{i.e.,}\quad \sum_{a,b}\kappa_{cd}\,R(U)^c{}_aR(U)^d{}_b = \kappa_{ab} \tag{141}$$

$$\implies T_{++}(x^+) = \mathcal{K}(\partial_+v_+)^2\kappa_{33} \tag{142}$$

Using, $\kappa_{33} = -1$, we get,

$$\boxed{T_{++}(x^+) = -\mathcal{K}(\partial_+v_+)^2} \tag{143}$$

Similarly, we can find $T_{--}(x^-)$ as follows.

$$T_{--}(x^-) = \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\,J_L^a(x^-)J_L^b(x^-) = \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\Big[-\mathcal{K}(\partial_- u_-)L(V)^{3a}\Big]\Big[-\mathcal{K}(\partial_- u_-)L(V)^{3b}\Big]$$
$$= \mathcal{K}(\partial_- u_-)^2\sum_{a,b}\kappa_{ab}\,L(V)^{3a}L(V)^{3b} = \mathcal{K}(\partial_- u_-)^2\sum_{a,b}\kappa^{ab}\,L(V)^3{}_aL(V)^3{}_b \tag{144}$$

Since the invariant bilinear form is preserved, where $\sum_{a,b}\kappa^{ab}\,L(V)^c{}_bL(V)^d{}_a = \kappa^{cd}$, and using $\kappa^{33} = -1$, we get,

$$\boxed{T_{--}(x^-) = -\mathcal{K}(\partial_- u_-)^2} \tag{145}$$

Now, we have determined the contributions to the energy-momentum tensor arising from the AdS_3 sector. However, the action also contains additional contributions from the compact manifold $\mathcal{M}$. If the string is excited along $\mathcal{M}$, these excitations are characterized by a conformal weight (h_R, h_L) in the corresponding sigma model on $\mathcal{M}$. Assuming equal left- and right-moving conformal weights, $h_R = h_L = h$, we set,

$$T_{++}^{\mathcal{M}} = T_{--}^{\mathcal{M}} = h.$$

Including this contribution, the Virasoro constraints become

$$-\mathcal{K}(\partial_+ v_+)^2 + h = 0 \qquad \text{and} \qquad -\mathcal{K}(\partial_- u_-)^2 + h = 0 \tag{146}$$

$$\implies (\partial_+ v_+) = \pm\sqrt{\frac{h}{\mathcal{K}}} = (\partial_- u_-) \tag{147}$$

Thus, the functions v_+ and u_- are linear functions of x^+ and x^-, respectively. Let,

$$v_+ = \frac{\alpha}{2}x^+ \qquad \text{and} \qquad u_+ = \frac{\alpha}{2}x^- \quad \text{where} \quad \alpha = \pm\sqrt{\frac{4h}{\mathcal{K}}} \tag{148}$$

Then the classical solution becomes,

$$g(x^+, x^-) = Ue^{i(v_+(x^+)+u_-(x^-))\sigma_2}V = Ue^{i\alpha\tau\sigma_2}V = U\Big(\mathbb{I}_2\cos(\alpha\tau) + i\sigma_2\sin(\alpha\tau)\Big)V \tag{149}$$

$$\implies g(x^+, x^-) = U\begin{pmatrix}\cos(\alpha\tau) & \sin(\alpha\tau)\\ -\sin(\alpha\tau) & \cos(\alpha\tau)\end{pmatrix}V \tag{150}$$

Since the solution depends only on τ and not on σ, the string embedding is independent of σ. This means every point along the string has an identical spatial position. Therefore, the string is collapsed to a point. This is a point particle-like solution evolving along a trajectory in AdS_3 parameterized by τ. The currents and stress tensor components are given by,

$$J_R(x^+) = -\frac{\mathcal{K}\alpha}{2}\big(UT^3U^{-1}\big)\ , \quad J_L(x^-) = -\frac{\mathcal{K}\alpha}{2}\big(V^{-1}T^{3^*}V\big) \ \text{ and } \ T_{\pm\pm}(x^\pm) = -\frac{1}{4}\mathcal{K}\alpha^2 \tag{151}$$

In the special case $U = V = 1$, the solution is just,

$$g(\tau) = e^{i\alpha\tau\sigma_2} \tag{152}$$

In terms of global coordinates,

$$t = \alpha\tau \quad \text{and} \quad \rho = 0 \tag{153}$$

Thus, this configuration corresponds to a particle located at the center of AdS_3.

Now, AdS_3 can be realized as the group manifold $\mathrm{AdS}_3 \simeq \mathrm{SL}(2,\mathbb{R})$, with induced metric $ds^2 = \frac{1}{2}\mathrm{Tr}\,(g^{-1}dg\, g^{-1}dg)$. This metric is invariant under the transformation

$$g \to \Omega_L\, g\, \Omega_R^{-1} \qquad \text{where } \Omega_L, \Omega_R \in \mathrm{SL}(2,\mathbb{R}) \text{ and are constants.} \tag{154}$$

Because, under this change, $g^{-1}dg \to \Omega_R(g^{-1}dg)\Omega_R^{-1}$ and as the trace is cyclic, $\mathrm{Tr}(g^{-1}dg\; g^{-1}dg)$ is unchanged. Hence, the isometry group is $\mathrm{SL}(2,\mathbb{R})_L \times \mathrm{SL}(2,\mathbb{R})_R$.

For the classical solution $g(x^+, x^-) = e^{i\alpha\tau\sigma_2}$, where $t = \alpha\tau$ and $\rho = 0$, the global AdS_3 metric becomes,

$$ds^2 = -dt^2 = -\alpha^2 d\tau^2 < 0 \tag{155}$$

So, the trajectory is timelike. Then, a general solution of the form $g = Ue^{i\alpha\tau\sigma_2}V$ is just an isometry transformation with $\Omega_L = U$ and $\Omega_R = V^{-1}$. Thus, the induced metric, even for this general solution, should be the same as that of $g = e^{i\alpha\tau\sigma_2}$. Thus, any solution of the form,

$$g(\tau) = Ue^{i\alpha\tau\sigma_2}V \tag{156}$$

represents a time-like geodesic in AdS_3.

The Monodromy matrix is defined by $g_+(x^+ + 2\pi) = g_+(x^+)\, M$ and $g_-(x^- - 2\pi) = M^{-1}\, g_-(x^-)$,

$$\begin{aligned}
\implies \; Ue^{iv_+(x^+)\sigma_2}M &= Ue^{iv_+(x^+ + 2\pi)\sigma_2} \\
\implies \; e^{i\frac{\alpha x^+}{2}\sigma_2}M &= e^{i\frac{\alpha(x^+ + 2\pi)}{2}\sigma_2} \\
\implies \; e^{i\frac{\alpha x^+}{2}\sigma_2}M &= e^{i\frac{\alpha x^+}{2}\sigma_2}e^{i\alpha\pi\sigma_2} \\
\implies \; M &= e^{i\alpha\pi\sigma_2}
\end{aligned} \tag{157}$$

Thus,

$$M = \begin{pmatrix} \cos(\alpha\pi) & \sin(\alpha\pi) \\ -\sin(\alpha\pi) & \cos(\alpha\pi) \end{pmatrix} \tag{158}$$

Now, $\mathrm{Tr}(M) = 2\cos(\alpha\pi)$. Thus, $-2 < \mathrm{Tr}(M) < 2$ or $|\mathrm{Tr}(M)| < 2$. Hence, this solution corresponds to the Elliptic conjugacy class of $\mathrm{SL}(2,\mathbb{R})$ which describes short strings oscillating near the center of AdS_3.

3.2 Space-like geodesic

We know that the classical equations of motion of the WZW model factorize into two chiral parts as $g(x^+, x^-) = g_+(x^+)g_-(x^-)$. Also, a point-like string (geodesic in target space) satisfies $\partial_\sigma g = 0$. And since, $\partial_\sigma = \partial_+ - \partial_- \implies \partial_+ g = \partial_- g \implies g_+^{-1}(\partial_+ g_+) = (\partial_- g_-)g_-^{-1}$. Consider the following ansatz, which satisfies the above equations,

$$\begin{aligned} g_+(x^+) &= Ue^{v_+(x^+)\sigma_3}, \quad \text{and} \quad g_-(x^-) = e^{u_-(x^-)\sigma_3}V \\ \implies g(x^+, x^-) &= Ue^{(v_+(x^+)+u_-(x^-))\sigma_3}V \end{aligned} \tag{159}$$

Then, following the same procedure as before, we get,

$$J_R^a = 2\mathcal{K}(\partial_+ v_+)\,\mathrm{Tr}\Big((U^{-1}T^aU)T^1\Big) = \mathcal{K}(\partial_+ v_+)R(U)^{a1} \tag{160}$$

$$J_L^a = 2\mathcal{K}(\partial_- u_-)\,\mathrm{Tr}\Big((VT^{a^*}V^{-1})T^1\Big) = \mathcal{K}(\partial_- u_-)L(V)^{1a} \tag{161}$$

Then the chiral components of the energy-momentum tensor for this solution are proportional to $k_{11} = 1$, Thus,

$$T_{++}(x^+) = \mathcal{K}(\partial_+ v_+)^2 \tag{162}$$

$$T_{--}(x^-) = \mathcal{K}(\partial_- u_-)^2 \tag{163}$$

We are looking for a hyperbolic conjugacy class of solutions. So let's allow negative conformal weight in $\mathcal{M}$ sector. Then, the Virasoro constraints give,

$$\mathcal{K}(\partial_+ v_+)^2 + h = 0 \quad \text{and} \quad \mathcal{K}(\partial_- u_-)^2 + h = 0 \tag{164}$$

$$\implies (\partial_+ v_+) = \pm\sqrt{\frac{-h}{\mathcal{K}}} = (\partial_- u_-) \tag{165}$$

$$\implies v_+ = \frac{\alpha}{2}x^+ \qquad \text{and} \qquad u_+ = \frac{\alpha}{2}x^- \qquad \text{where} \quad \alpha = \pm\sqrt{\frac{-4h}{\mathcal{K}}} \tag{166}$$

$$\implies \boxed{T_{\pm\pm} = \frac{1}{4}\mathcal{K}\alpha^2} \qquad \text{(positive)} \tag{167}$$

And the currents are,

$$J_R(x^+) = \frac{\mathcal{K}\alpha}{2}(UT^1U^{-1}) \quad \text{and} \quad J_L(x^-) = \frac{\mathcal{K}\alpha}{2}(V^{-1}T^1V) \tag{168}$$

Then, we get the solution to be,

$$g(\tau) = Ue^{\alpha\tau\sigma_3}V = U\begin{pmatrix} e^{\alpha\tau} & 0 \\ 0 & e^{-\alpha\tau} \end{pmatrix}V \tag{169}$$

So, those string states which have a negative conformal weight in the compact $\mathcal{M}$ manifold will be able to produce such solutions in AdS_3.

For $U = V = 1$, the solution becomes $e^{\alpha\tau\sigma_3}$, where in global coordinates, it is,

$$g(x^+, x^-) = e^{\rho\sigma_3} \qquad \text{where } \rho = \alpha\tau, \quad t = 0 \quad \text{and} \quad \phi = 0 \tag{170}$$

Hence, the geodesic lies entirely in a fixed time slice $t = 0$ and at a fixed angular direction $\phi = 0$. The trajectory is therefore a straight radial line emanating from the origin $\rho = 0$ in the spatial section of AdS_3 in the (ρ, ϕ) plane. The spatial slice $t = 0$ of global AdS_3 is a hyperbolic disk. In this geometry, radial curves at fixed ϕ are spacelike geodesics. Therefore, this trajectory represents a straight spacelike geodesic.

For this geodesic, the metric is given by $ds^2 = d\rho^2 > 0$. So it is space-like. Then, by acting $\mathrm{SL}(2, \mathbb{R})_L \times \mathrm{SL}(2, \mathbb{R})_R$ isometry using constant matrices U and V, we can generate all spacelike geodesics at different time-slices and angles.

The Monodromy matrix M satisfies $g_+(x^+ + 2\pi) = g_+(x^+)\, M$ and $g_-(x^- - 2\pi) = M^{-1}\, g_-(x^-)$,

$$\implies U e^{v_+(x^+)\sigma_3} M = U e^{v_+(x^+ + 2\pi)\sigma_3} \qquad \implies \qquad e^{\frac{\alpha x^+}{2}\sigma_3} M = e^{\frac{\alpha(x^+ + 2\pi)}{2}\sigma_3} \tag{171}$$

$$\implies M = \begin{pmatrix} e^{\alpha\pi} & 0 \\ 0 & e^{-\alpha\pi} \end{pmatrix} \tag{172}$$

Now, $\mathrm{Tr}(M) = 2\cosh(\alpha\pi) > 2$, thus this class of solutions belongs to the Hyperbolic conjugacy class which describes long strings reaching toward the spatial boundary of AdS_3.

3.3 Null Geodesics

In the $\mathfrak{sl}(2, \mathbb{R})$ Cartan–Weyl basis we have $T^{\pm}$ as the nilpotent raising / lowering generators whose $(T^+)^2 = (T^-)^2 = 0$. A representative of the parabolic conjugacy class is the exponential of a nilpotent element. Thus, consider the ansatz,

$$g_{(\pm)}(x^+, x^-) = U e^{(v_+(x^+) + u_-(x^-))\,\sigma^{\pm}} V = U e^{2\theta(x^+, x^-)T^{\pm}} V \ , \quad U, V \in \mathrm{SL}(2, \mathbb{R}) \ \text{constant} \tag{173}$$

where the scalar function $\theta(x^+, x^-) = v_+(x^+) + u_-(x^-)$ will be chosen such that it satisfies the Virasoro constraints and produces a nontrivial parabolic monodromy. Since $(T^+)^2 = 0$, the exponential truncates $e^{2\theta T^{\pm}} = 1 + 2\theta\, T^{\pm}$. Thus,

$$g_{(\pm)}(x^+, x^-) = UV + 2\theta(x^+, x-) U T^{\pm} V \tag{174}$$

$$\implies \partial_+ g_{(\pm)} = 2(\partial_+ v_+) U T^{\pm} V \quad \text{and} \quad \partial_- g_{(\pm)} = 2(\partial_- u_-) U T^{\pm} V \tag{175}$$

$$\Longrightarrow\ (\partial_+ g_{(\pm)})\, g_{(\pm)}^{-1} = 2(\partial_+ v_+) U T^{\pm} V\ \ V^{-1}(1-2\theta T^{\pm})U^{-1} = 2(\partial_+ v_+) U T^{\pm} U^{-1}$$
$$\text{and}\ \ g_{(\pm)}^{-1}(\partial_- g_{(\pm)}) = 2(\partial_- u_-) V^{-1}(1-2\theta T^{\pm})U^{-1}\ U T^{\pm} V = 2(\partial_- u_-) V^{-1} T^{\pm} V \tag{176}$$

Then, the left and right currents are,

$$\begin{aligned} J^a_{R\pm}(x^+) &= 2\mathcal{K}(\partial_+ v_+)\mathrm{Tr}\big(T^a U T^{\pm} U^{-1}\big) = \mathcal{K}(\partial_+ v_+) R(U)^{a\pm} \\ J^a_{L\pm}(x^-) &= 2\mathcal{K}(\partial_- u_-)\mathrm{Tr}\big(T^{a^*} V^{-1} T^{\mp^*} V\big) = \mathcal{K}(\partial_- u_-) L(V)^{\mp a} \end{aligned} \tag{177}$$

Then the energy-momentum tensor components are given by,

$$\begin{aligned} T_{++\pm} &= \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\, J^a_{R\pm}(x^+) J^b_{R\pm}(x^+) = \frac{1}{\mathcal{K}}\sum_{a,b}\kappa_{ab}\Big[\mathcal{K}(\partial_+ v_+) R(U)^{a\pm}\Big]\Big[\mathcal{K}(\partial_+ v_+) R(U)^{b\pm}\Big] \\ &= \mathcal{K}(\partial_+ v_+)^2 \sum_{a,b}\kappa_{ab}\, R(U)^{a\pm} R(U)^{b\pm} = \mathcal{K}(\partial_+ v_+)^2 \kappa^{\pm\pm} \end{aligned} \tag{178}$$

But $\kappa^{\pm\pm} = 2\mathrm{Tr}(T^{\pm}T^{\pm}) = 0$. Thus,

$$T_{++\pm} = 0 \tag{179}$$

Similarly, we get,

$$T_{--\pm} = 0 \tag{180}$$

Thus, the AdS part of the stress tensor vanishes for the parabolic (nilpotent) ansatz: the world-sheet tangent vector in target space is null. The Virasoro constraints require the conformal weight of the $\mathcal{M}$ sector to be zero for this solution to exist.

We can also show that the induced metric in the AdS$_3$ part has zero space-time displacement.

$$ds^2 = \frac{1}{2}\,\mathrm{Tr}\big(g^{-1}dg\, g^{-1}dg\big). \tag{181}$$

For our ansatz $g_{(\pm)} = U e^{2\theta T^{\pm}} V$,

$$\Longrightarrow\ dg_{(\pm)} = 2(\partial_\alpha\theta)dx^\alpha U e^{2\theta T^{\pm}} T^{\pm} V = 2(\partial_\alpha\theta)dx^\alpha g_{(\pm)} V^{-1} T^{\pm} V \tag{182}$$

$$\Longrightarrow\ g_{(\pm)}^{-1} dg_{(\pm)} = 2(\partial_\alpha\theta)dx^\alpha\, V^{-1} T^{\pm} V \tag{183}$$

Then,

$$\begin{aligned} ds^2 &= \frac{1}{2}\mathrm{Tr}\big(g_{(\pm)}^{-1}dg_{(\pm)}\ g_{(\pm)}^{-1}dg_{(\pm)}\big) \\ &= 2(\partial_\alpha\theta)dx^\alpha\,(\partial_\beta\theta)dx^\beta\ \mathrm{Tr}\big(V^{-1}T^{\pm}V\ V^{-1}T^{\pm}V\big) \\ &= 2(\partial_\alpha\theta)dx^\alpha\,(\partial_\beta\theta)dx^\beta\ \mathrm{Tr}\big(V^{-1}(T^{\pm})^2 V\big) = 0 \end{aligned} \tag{184}$$

This solution represents a string moving along the null direction in the target AdS$_3$ space-time. Now, let us try to find the Monodromy matrix. For a closed string $g(\tau,\sigma+2\pi) = g(\tau,\sigma)$,

$$\implies \theta(\tau,\sigma+2\pi) = \theta(\tau,\sigma) + \Delta v_+ + \Delta u_- = \theta(\tau,\sigma) \tag{185}$$

where

$$\Delta v_+ = v_+(x^+ + 2\pi) - v_+(x^+) \qquad \text{and} \qquad \Delta u_- = u_-(x^- - 2\pi) - u_-(x^-) \tag{186}$$

To obtain point-like solutions, we require $g_+^{-1}(\partial_+ g_+) = (\partial_- g_-)g_-^{-1} \implies \partial_+ v_+ = \partial_- u_-$. (Here g_+ and g_- are the chiral factors). Then, the possible solutions for v_+ and u_- are such that,

$$\partial_+ v_+ = \partial_- u_- = \frac{\lambda}{2} \qquad (\lambda \text{ is some constant}) \tag{187}$$

$$\implies v_+ = \frac{\lambda}{2}x^+ \qquad \text{and} \qquad u_- = \frac{\lambda}{2}x^- \tag{188}$$

The Monodromy matrix is then given by,

$$Ue^{2v_+(x^+ + 2\pi)T^\pm} = Ue^{2v_+(x^+)T^\pm}\, M \quad \text{and} \quad e^{2u_-(x^- - 2\pi)T^\pm}V = M^{-1}\, e^{2u_-(x^-)T^\pm}V \tag{189}$$

$$\implies e^{\lambda x^+ T^\pm} e^{2\lambda\pi T^\pm} = e^{\lambda x^+ T^\pm}\, M \quad \text{and} \quad e^{\lambda x^- T^\pm} e^{-2\lambda\pi T^\pm} = M^{-1}\, e^{\lambda x^- T^\pm} \tag{190}$$

$$\implies M = e^{2\lambda\pi T^\pm} = 1 + 2\lambda\pi T^\pm = \begin{pmatrix} 1+\lambda\pi & \pm\lambda\pi \\ \mp\lambda\pi & 1-\lambda\pi \end{pmatrix} \tag{191}$$

Thus, we get $\mathrm{Tr}\big(M\big) = 2$, which corresponds to the Parabolic class of geodesics where the string configuration is at the margin between short and long.

4 Spectral Flow and Strings with Winding Numbers

Consider a classical solution for the string configuration of the form $\tilde{g}(x^+,x^-) = \tilde{g}_+(x^+)\tilde{g}_-(x^-)$. We can generate a new solution as follows,

$$\begin{gathered} g_+(x^+) = e^{\frac{i}{2}w_R x^+ \sigma_2}\, \tilde{g}_+(x^+) \qquad \text{and} \qquad g_-(x^-) = \tilde{g}_-(x^-)\, e^{\frac{i}{2}w_L x^- \sigma_2} \\ \implies g(x^+,x^-) = g_+(x^+)g_-(x^-) = e^{\frac{i}{2}w_R x^+ \sigma_2}\, \tilde{g}(x^+,x^-)\, e^{\frac{i}{2}w_L x^- \sigma_2} \end{gathered} \tag{192}$$

We can observe that the chosen left and right multiplication factors for generating new solutions are elements of $\mathrm{SL}(2,\mathbb{R})$. It depends linearly on $x^\pm$. They are chosen such that they preserve equations of motion, preserve periodicity for integer $w_{L,R}$. This specific choice of action on a pre-existing solution is called Spectral Flow.

For integer $w_{L,R}$, under $\sigma \to \sigma + 2\pi$, we see that,

$$e^{\frac{i}{2}w_R x^+ \sigma_2} \to e^{\frac{i}{2}w_R x^+ \sigma_2} e^{i w_R \pi \sigma_2} = e^{\frac{i}{2}w_R x^+ \sigma_2}\Big(\cos(w_R\pi)\mathbb{I} + i\sigma_2 \sin(w_R\pi)\Big) = (-1)^{w_R} e^{\frac{i}{2}w_R x^+ \sigma_2} \tag{193}$$

$$e^{\frac{i}{2}w_L x^- \sigma_2} \to e^{\frac{i}{2}w_L x^- \sigma_2} e^{-i w_L \pi \sigma_2} = e^{\frac{i}{2}w_L x^- \sigma_2}\Big(\cos(w_L\pi)\mathbb{I} - i\sigma_2 \sin(w_L\pi)\Big) = (-1)^{w_L} e^{\frac{i}{2}w_L x^- \sigma_2} \tag{194}$$

For integer $w_{L,R}$, the left/right factors are periodic up to the central element $(-1)^{w}_{L,R}$. Hence, spectral flow is a large loop-group transformation: it preserves the equations of motion, modifies the Monodromy of g by a central phase, and for the diagonal choice $w_L = w_R = w$, it produces winding strings while acting properly on the field space. This is an action of the loop group $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$. For the universal cover, the diagonal choice $w_R = w_L \equiv w \in \mathbb{Z}$ defines the spectral flow symmetry and ensures.

Now, let us see how the target space global coordinates change under spectral flow.

$$g(x^+, x^-) = e^{\frac{i}{2}w_R x^+ \sigma_2}\, e^{i\tilde{u}\sigma_2} e^{\tilde{\rho}\sigma_3} e^{i\tilde{v}\sigma_2}\, e^{\frac{i}{2}w_L x^- \sigma_2} = e^{iu\sigma_2} e^{\rho\sigma_3} e^{iv\sigma_2} \tag{195}$$

where,

$$u = \tilde{u} + \frac{1}{2}w_R x^+ \qquad \text{and} \qquad v = \tilde{v} + \frac{1}{2}w_L x^- \tag{196}$$

Then the coordinates $t = u + v,\ \rho,\ \phi = u - v$ change as,

$$\begin{aligned} t &= \tilde{t} + \frac{1}{2}\big(w_R x^+ + w_L x^-\big) = \tilde{t} + \frac{1}{2}\Big((w_R + w_L)\tau + (w_R - w_L)\sigma\Big) \\ \rho &= \tilde{\rho} \\ \phi &= \tilde{\phi} + \frac{1}{2}\big(w_R x^+ - w_L x^-\big) = \tilde{\phi} + \frac{1}{2}\Big((w_R - w_L)\tau + (w_R + w_L)\sigma\Big) \end{aligned} \tag{197}$$

For the diagonal spectral flow $w_L = w_R = w$,

$$t = \tilde{t} + w\tau \ \ , \quad \rho = \tilde{\rho} \quad \text{and} \quad \phi = \tilde{\phi} + w\sigma \tag{198}$$

If the initial solution described a point-like string, then $\tilde{t} = t_0(\tau)\ ,\ \tilde{\rho} = \rho_0(\tau)\ ,\ \tilde{\phi} = \phi_0(\tau)$. But under spectral flow, a point-like geodesic acquires explicit σ-dependence and becomes a circular string winding w times around the center of AdS_3 ($\rho = 0$). The term $w\tau$ stretches the solution in the time direction, while the term $w\,\sigma$ produces angular winding. Under $\sigma \to \sigma + 2\pi$, the angular coordinate ϕ transforms as $\phi \to \phi + 2\pi w$, showing that the resulting configuration is a circular string winding w times around the center $\rho = 0$ of AdS_3. Thus, spectral flow maps a geodesic (point-like solution) into a winding string configuration.

4.0.1 Transformation of currents

The right-moving currents are $J^a_R(x^+) = \mathcal{K}\mathrm{Tr}(T^a\,(\partial_+ g)\,g^{-1})$. Now, under spectral flow, the solution changes as $g(x^+, x^-) = e^{\frac{i}{2}w_R x^+ \sigma_2}\,\tilde{g}(x^+, x^-)\,e^{\frac{i}{2}w_L x^- \sigma_2}$.

$$\partial_+ g = \frac{i}{2}w_R \sigma_2\, g + e^{\frac{i}{2}w_R x^+ \sigma_2}\,(\partial_+ \tilde{g})\,e^{\frac{i}{2}w_L x^- \sigma_2} \tag{199}$$

$$\begin{aligned} (\partial_+ g)g^{-1} &= \frac{i}{2}w_R\sigma_2 \ + e^{\frac{i}{2}w_R x^+ \sigma_2}\,(\partial_+\tilde{g})\,e^{\frac{i}{2}w_L x^- \sigma_2}\Big(e^{-\frac{i}{2}w_L x^- \sigma_2}\ \tilde{g}^{-1} e^{-\frac{i}{2}w_R x^+ \sigma_2}\Big) \\ &= \frac{i}{2}w_R\sigma_2 \ + e^{\frac{i}{2}w_R x^+ \sigma_2}\Big((\partial_+\tilde{g})\tilde{g}^{-1}\Big)e^{-\frac{i}{2}w_R x^+ \sigma_2} \\ \Longrightarrow\ (\partial_+ g)g^{-1} &= -w_R T^3 + e^{-w_R x^+ T^3}\Big(\,(\partial_+\tilde{g})\,\tilde{g}^{-1}\Big)e^{w_R x^+ T^3} \end{aligned} \tag{200}$$

$$\begin{aligned}\implies J_R^a &= \mathcal{K}\,\mathrm{Tr}\big(T^a(\partial_+ g)g^{-1}\big) = \mathcal{K}\,\mathrm{Tr}\Big(-w_R T^a T^3 + T^a e^{-w_R x^+ T^3}\,(\partial_+\tilde{g})\,\tilde{g}^{-1}\,e^{w_R x^+ T^3}\Big)\\ \implies J_R^a &= -\frac{\mathcal{K}}{2} w_R \kappa^{a3} + \mathcal{K}\,\mathrm{Tr}\Big(e^{w_R x^+ T^3} T^a e^{-w_R x^+ T^3}\,(\partial_+\tilde{g})\,\tilde{g}^{-1}\Big)\end{aligned} \tag{201}$$

Then, we can read off how the individual components for different 'a'. So,

$$J_R^3 = -\frac{\mathcal{K}}{2} w_R \kappa^{33} + \mathcal{K}\,\mathrm{Tr}\Big(e^{w_R x^+ T^3} T^3 e^{-w_R x^+ T^3}\,(\partial_+\tilde{g})\,\tilde{g}^{-1}\Big) \tag{202}$$

Using $\kappa^{33} = -1$ and $\tilde{J}_R^3(x^+) = \mathcal{K}\,\mathrm{Tr}\big(T^3\,(\partial_+\tilde{g})\,\tilde{g}^{-1}\big)$, we can say that under spectral flow,

$$\boxed{J_R^3(x^+) = \tilde{J}_R^3(x^+) + \frac{\mathcal{K}}{2} w_R} \tag{203}$$

Then, for the $J_R^\pm$ we get,

$$J_R^\pm = -\frac{\mathcal{K}}{2} w_R \kappa^{3\pm} + \mathcal{K}\,\mathrm{Tr}\Big(e^{w_R x^+ T^3} T^\pm e^{-w_R x^+ T^3}\,(\partial_+\tilde{g})\,\tilde{g}^{-1}\Big) \tag{204}$$

Using $\kappa^{3\pm} = \frac{1}{2}\mathrm{Tr}\big(T^3 T^\pm\big) = \frac{1}{2}\mathrm{Tr}\big(T^3(T^1 \pm iT^2)\big) = \kappa^{31} \pm i\kappa^{32} = 0$ (as κ^{ab} is diagonal), the standard result $e^{w_R x^+ T^3} T^\pm e^{-w_R x^+ T^3} = e^{\mp i w_R x^+} T^\pm$, and $\tilde{J}_R^\pm(x^+) = \mathcal{K}\,\mathrm{Tr}(T^\pm\,(\partial_+\tilde{g})\,\tilde{g}^{-1})$, we get,

$$\boxed{J_R^\pm = e^{\mp i w_R x^+}\,\tilde{J}_R^\pm(x^+)} \tag{205}$$

For the left-moving currents $J_L^a(x^-) = \mathcal{K}\,\mathrm{Tr}\big(T^{a^*}\,g^{-1}(\partial_- g)\big)$, under spectral flow, the solution changes as $g(x^+, x^-) = e^{\frac{i}{2} w_R x^+ \sigma_2}\,\tilde{g}(x^+, x^-)\,e^{\frac{i}{2} w_L x^- \sigma_2}$.

$$\partial_- g = g\Big(\frac{i}{2} w_L \sigma_2\Big) + e^{\frac{i}{2} w_R x^+ \sigma_2}\,(\partial_-\tilde{g})\,e^{\frac{i}{2} w_L x^- \sigma_2} \tag{206}$$

$$\begin{aligned}\implies g^{-1}(\partial_- g) &= \frac{i}{2} w_L \sigma_2 + \Big(e^{-\frac{i}{2} w_L x^- \sigma_2}\,\tilde{g}^{-1} e^{-\frac{i}{2} w_R x^+ \sigma_2}\Big) e^{\frac{i}{2} w_R x^+ \sigma_2}\,(\partial_-\tilde{g})\,e^{\frac{i}{2} w_L x^- \sigma_2}\\ &= \frac{i}{2} w_L \sigma_2 + e^{-\frac{i}{2} w_L x^- \sigma_2}\Big(\tilde{g}^{-1}(\partial_-\tilde{g})\Big) e^{\frac{i}{2} w_L x^- \sigma_2}\\ \implies g^{-1}(\partial_- g) &= -w_L T^{3^*} + e^{w_L x^- T^{3^*}}\Big(\tilde{g}^{-1}(\partial_-\tilde{g})\Big) e^{-w_L x^- T^{3^*}}\end{aligned} \tag{207}$$

$$\begin{aligned}\implies J_L^a &= \mathcal{K}\,\mathrm{Tr}\big(T^{a^*} g^{-1}(\partial_- g)\big) = -\mathcal{K}\,\mathrm{Tr}\Big(w_L T^{a^*} T^{3^*} + T^{a^*} e^{w_L x^- T^{3^*}}\Big(\tilde{g}^{-1}(\partial_-\tilde{g})\Big) e^{-w_L x^- T^{3^*}}\Big)\\ \implies J_L^a &= -\frac{\mathcal{K}}{2} w_L \kappa^{a3} + \mathcal{K}\,\mathrm{Tr}\Big(e^{-w_L x^- T^{3^*}} T^{a^*} e^{w_L x^- T^{3^*}}\,\big(\tilde{g}^{-1}(\partial_-\tilde{g})\big)\Big)\end{aligned} \tag{208}$$

Then, the individual left current components can be found.

$$J_L^3 = -\frac{\mathcal{K}}{2} w_L \kappa^{33} + \mathcal{K}\,\mathrm{Tr}\Big(e^{w_L x^- T^{3^*}} T^{3^*} e^{-w_L x^- T^{3^*}}\,\big(\tilde{g}^{-1}(\partial_-\tilde{g})\big)\Big) \implies \boxed{J_L^3 = \tilde{J}_L^3 + \frac{\mathcal{K}}{2} w_L} \tag{209}$$

$$J_L^\pm = \frac{\mathcal{K}}{2} w_L \kappa^{3\pm} + \mathcal{K}\,\mathrm{Tr}\Big(e^{-w_L x^- T^{3^*}} T^{\pm^*} e^{w_L x^- T^{3^*}}\,\big(\tilde{g}^{-1}(\partial_-\tilde{g})\big)\Big) \tag{210}$$

Use $\kappa^{3\pm} = 0$ and $e^{-w_L x^- T^{3^*}} T^{\pm^*} e^{w_L x^- T^{3^*}} = e^{-w_L x^- T^3} T^\mp e^{w_L x^- T^3} = e^{\mp i w_L x^-} T^\mp = e^{\mp i w_L x^-} T^{\pm^*}$

$$\implies \boxed{J_L^\pm = e^{\mp i w_L x^-}\,\tilde{J}_L^\pm} \tag{211}$$

4.0.2 Transformation of Sugawara Energy Momentum tensor

With the invariant metric $\kappa_{ab} = \text{diag}(1, 1, -1)$, the AdS_3 stress tensor is given by

$$T_{++} = \frac{1}{\mathcal{K}} \sum_{a,b} \kappa_{ab}\, J_R^a J_R^b = \frac{1}{\mathcal{K}}\Big((J_R^1)^2 + (J_R^2)^2 - (J_R^3)^2\Big) = \frac{1}{\mathcal{K}}\left(\frac{1}{2}\big(J_R^+ J_R^- + J_R^- J_R^+\big) - (J_R^3)^2\right) \tag{212}$$

Then, on substituting the spectral transformation of the right current components, we get,

$$T_{++} = \frac{1}{\mathcal{K}}\left(\frac{1}{2}\big(\tilde{J}_R^+ \tilde{J}_R^- + \tilde{J}_R^- \tilde{J}_R^+\big) - \Big(\tilde{J}_R^3 + \frac{\mathcal{K}}{2} w_R\Big)^2\right) = \frac{1}{\mathcal{K}}\left(\frac{1}{2}\big(\tilde{J}_R^+ \tilde{J}_R^- + \tilde{J}_R^- \tilde{J}_R^+\big) - (\tilde{J}_R^3)^2 - \mathcal{K} w_R \tilde{J}_R^3 - \frac{\mathcal{K}^2}{4} w_R^2\right)$$

$$\implies \boxed{T_{++} = \tilde{T}_{++} - w_R \tilde{J}_R^3 - \frac{\mathcal{K}}{4} w_R^2} \tag{213}$$

Similarly, for the T_{--} component,

$$T_{--} = \frac{1}{\mathcal{K}}\left(\frac{1}{2}\big(J_L^+ J_L^- + J_L^- J_L^+\big) - (J_L^3)^2\right) \tag{214}$$

Then, on substituting the spectral transformation of the right current components, we get,

$$T_{--} = \frac{1}{\mathcal{K}}\left(\frac{1}{2}\big(\tilde{J}_L^+ \tilde{J}_L^- + \tilde{J}_L^- \tilde{J}_L^+\big) - \Big(\tilde{J}_L^3 + \frac{\mathcal{K}}{2} w_L\Big)^2\right) = \frac{1}{\mathcal{K}}\left(\frac{1}{2}\big(\tilde{J}_L^+ \tilde{J}_L^- + \tilde{J}_L^- \tilde{J}_L^+\big) - (\tilde{J}_L^3)^2 - \mathcal{K} w_L \tilde{J}_L^3 - \frac{\mathcal{K}^2}{4} w_L^2\right)$$

$$\implies \boxed{T_{--} = \tilde{T}_{--} - w_L \tilde{J}_L^3 - \frac{\mathcal{K}}{4} w_L^2} \tag{215}$$

4.0.3 Spectral flow in the Fourier modes of $J_{L,R}^a$ and $T_{\pm\pm}$

The Fourier modes of the currents are defined by the equation,

$$J_{L,R}^a(x^\pm) = \sum_{n\in\mathbb{Z}} J_{n_{L,R}}^a\, e^{-inx^\pm}, \qquad \tilde{J}_{L,R}^a(x^\pm) = \sum_{n\in\mathbb{Z}} \tilde{J}_{n_{L,R}}^a\, e^{-inx^\pm} \tag{216}$$

For $J_{L,R}^3$ we get,

$$\sum_n J_{n_L}^3\, e^{-inx^-} = \sum_n \tilde{J}_{n_L}^3\, e^{-inx^-} + \frac{\mathcal{K}}{2} w_L = \sum_n \Big(\tilde{J}_{n_L}^3 + \frac{\mathcal{K}}{2} w_L \delta_{n,0}\Big)\, e^{-inx^-} \tag{217}$$

$$\implies \boxed{J_{n_L}^3 = \tilde{J}_{n_L}^3 + \frac{\mathcal{K}}{2} w_L \delta_{n,0}} \tag{218}$$

Similarly,

$$\sum_n J_{n_R}^3\, e^{-inx^+} = \sum_n \tilde{J}_{n_R}^3\, e^{-inx^+} + \frac{\mathcal{K}}{2} w_R = \sum_n \Big(\tilde{J}_{n_R}^3 + \frac{\mathcal{K}}{2} w_R \delta_{n,0}\Big)\, e^{-inx^+} \tag{219}$$

$$\implies \boxed{J_{n_R}^3 = \tilde{J}_{n_R}^3 + \frac{\mathcal{K}}{2} w_R \delta_{n,0}} \tag{220}$$

Mode relation for $J^{\pm}_{L,R}$ can be found as follows.

$$J^{\pm}_{L}(x^{-}) = \sum_{n} J^{\pm}_{n_L} e^{-inx^{-}} = e^{\mp i w_L x^{-}} \sum_{n} \tilde{J}^{\pm}_{n_L} e^{-inx^{-}} = \sum_{n} \tilde{J}^{\pm}_{n_L} e^{-i(n \pm w_L)x^{-}} \tag{221}$$

Put $n \pm w_L = m$. Then,

$$\sum_{n} J^{\pm}_{n_L} e^{-inx^{-}} = \sum_{m} \tilde{J}^{\pm}_{(m\mp w_L)_L} e^{-imx^{-}} \tag{222}$$

$$\Longrightarrow \boxed{J^{\pm}_{n_L} = \tilde{J}^{\pm}_{(m\mp w_L)_L}} \tag{223}$$

Similarly,

$$J^{\pm}_{R}(x^{+}) = \sum_{n} J^{\pm}_{n_R} e^{-inx^{+}} = e^{\mp i w_R x^{+}} \sum_{n} \tilde{J}^{\pm}_{n_R} e^{-inx^{+}} = \sum_{n} \tilde{J}^{\pm}_{n_R} e^{-i(n \pm w_R)x^{+}} \tag{224}$$

Put $n \pm w_R = m$. Then,

$$\sum_{n} J^{\pm}_{n_R} e^{-inx^{+}} = \sum_{m} \tilde{J}^{\pm}_{(m\mp w_R)_R} e^{-imx^{+}} \tag{225}$$

$$\Longrightarrow \boxed{J^{\pm}_{n_R} = \tilde{J}^{\pm}_{(m\mp w_R)_R}} \tag{226}$$

Now let's see how the Virasoro generators or the modes of $T_{\pm\pm}$ change under spectral flow. The Virasoro generators $L^{(\pm)}_n$ are defined as,

$$T_{\pm\pm}(x^{\pm}) = \sum_{n\in\mathbb{Z}} L^{(\pm)}_n e^{-inx^{\pm}} \tag{227}$$

The relation for the spectral flow of Virasoro generators is very simple and can be clearly seen to follow the relations below.

$$\boxed{L^{(+)}_n = \tilde{L}^{(+)}_n - w_R \tilde{J}^3_{n_R} - \frac{\mathcal{K}}{4} w_R^2\, \delta_{n,0}} \tag{228}$$

$$\text{and} \quad \boxed{L^{(-)}_n = \tilde{L}^{(-)}_n - w_L \tilde{J}^3_{n_L} - \frac{\mathcal{K}}{4} w_L^2\, \delta_{n,0}} \tag{229}$$

Spectral flow can map a point-particle solution with no σ dependence to an extended string configuration that winds w times around the center of AdS_3. Since the Sugawara tensor transforms as shown before, the Virasoro constraints must be imposed using the spectral-flowed energy-momentum tensor, leading to the physical-state conditions after spectral-flow.

4.1 Short strings as the spectral flow of timelike geodesics

For time-like geodesics, we have the classical solution of the form, $\tilde{g}(x^+, x^-) = U e^{i(\tilde{v}_+(x^+) + \tilde{u}_-(x^-))\sigma_2} V$ where, $\tilde{v}_+ = \frac{\alpha}{2} x^+$ and $\tilde{u}_+ = \frac{\alpha}{2} x^-$, with $\alpha = \pm\sqrt{\frac{4h}{\mathcal{K}}}$. The energy-momentum tensor components are, $\tilde{T}_{++}(x^+) = -\mathcal{K}(\partial_+ \tilde{v}_+)^2 = -\frac{\mathcal{K}}{4}\alpha^2$ and $\tilde{T}_{--}(x^-) = -\mathcal{K}(\partial_- \tilde{u}_-)^2 = -\frac{\mathcal{K}}{4}\alpha^2$.

The spectral flowed energy-momentum tensor is given by $T_{++} = \tilde{T}_{++} - w_R \tilde{J}^3_R - \frac{\mathcal{K}}{4} w_R^2$ and $T_{--} = \tilde{T}_{--} - w_L \tilde{J}^3_L - \frac{\mathcal{K}}{4} w_L^2$,

$$\implies T_{++} = -\frac{\mathcal{K}}{4}\alpha^2 - w_R \tilde{J}^3_R - \frac{\mathcal{K}}{4} w_R^2 \quad \text{and} \quad T_{--} = -\frac{\mathcal{K}}{4}\alpha^2 - w_L \tilde{J}^3_L - \frac{\mathcal{K}}{4} w_L^2 \tag{230}$$

If the spectral flow doesn't change anything in the compact manifold $\mathcal{M}$ sector, then we can consider $T^{\mathcal{M}}_{++} = T^{\mathcal{M}}_{--} = h$ as before.

After spectral flow, we shall impose the Virasoro constraint equation for the spectrally flowed energy-momentum tensor.

$$\begin{aligned} T_{++} + h = 0 \quad &\implies \quad -\frac{\mathcal{K}}{4}(\alpha^2 + w_R^2) - w_R \tilde{J}^3_R + h = 0 \\ T_{--} + h = 0 \quad &\implies \quad -\frac{\mathcal{K}}{4}(\alpha^2 + w_L^2) - w_L \tilde{J}^3_L + h = 0 \end{aligned} \tag{231}$$

$$\implies \quad \tilde{J}^3_R = \frac{1}{w_R}\Big(h - \frac{\mathcal{K}}{4}(\alpha^2 + w_R^2)\Big) \quad \text{and} \quad \tilde{J}^3_L = \frac{1}{w_L}\Big(h - \frac{\mathcal{K}}{4}(\alpha^2 + w_L^2)\Big) \tag{232}$$

Then the new spectral flowed currents $J^3_{R,L}$ are given by,

$$\begin{aligned} J^3_R = \tilde{J}^3_R + \frac{\mathcal{K}}{2} w_R = \frac{1}{w_R}\Big(h - \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4} w_R \\ J^3_L = \tilde{J}^3_L + \frac{\mathcal{K}}{2} w_L = \frac{1}{w_L}\Big(h - \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4} w_L \end{aligned} \tag{233}$$

Since we get J^3_R and J^3_L to be independent of $x^\pm$, they have only the zeroth mode, which is equal to themselves. Under the diagonal choice of winding numbers $w_R = w_L = w$, we get the total energy of the string, which is bounded above depending on α as,

$$E = J^3_{0_R} + J^3_{0_L} = 2J^3_{0_R} = \frac{1}{w}\Big(2h - \frac{\mathcal{K}}{2}\alpha^2\Big) + \frac{\mathcal{K}}{2} w \tag{234}$$

$$\text{Thus,} \quad E < \frac{2h}{w} + \frac{\mathcal{K} w}{2} \tag{235}$$

4.1.1 Explicit solution when $U = V^{-1}$

Without loss of generality, we can set,

$$U = V^{-1} = e^{\frac{1}{2}\rho_0\sigma_3} \tag{236}$$

Then, the unflowed short string solution will be,

$$\tilde{g}(x^+, x^-) = e^{\frac{1}{2}\rho_0\sigma_3} e^{i\alpha\tau\sigma_2} e^{-\frac{1}{2}\rho_0\sigma_3} \tag{237}$$

If a different choice of $U = V^{-1}$ was chosen, then we could consider a general constant $\mathrm{SL}(2, \mathbb{R})$ matrix, $U = V^{-1} = e^{iu_0\sigma_2} e^{\frac{1}{2}\rho_0\sigma_3} e^{v_0\sigma_2}$, then it can be shown that the spectrally flowed solution would have just a constant shift of ϕ which is irrelevant. Hence, let us choose $u_0 = v_0 = 0$.

Then, after spectral flow,

$$g(x^+, x^-) = e^{\frac{i}{2}wx^+\sigma_2} \tilde{g}(x^+, x^-) e^{\frac{i}{2}wx^-\sigma_2} = e^{\frac{i}{2}wx^+\sigma_2} e^{\frac{1}{2}\rho_0\sigma_3} e^{i\alpha\tau\sigma_2} e^{-\frac{1}{2}\rho_0\sigma_3} e^{\frac{i}{2}wx^-\sigma_2} \tag{238}$$

For a diagonal spectral flow, we can find the flowed solution and compare it to get the flowed global coordinates as follows. The exponentials of the required Pauli matrices are,

$$e^{i\theta\sigma_2} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \qquad \text{and} \qquad e^{\frac{1}{2}\rho_0\sigma_3} = \begin{pmatrix} e^{\frac{\rho_0}{2}} & 0 \\ 0 & e^{-\frac{\rho_0}{2}} \end{pmatrix}. \tag{239}$$

Then the unflowed solution evaluates to,

$$\begin{aligned} \tilde{g}(x^+, x^-) &= \begin{pmatrix} e^{\frac{\rho_0}{2}} & 0 \\ 0 & e^{-\frac{\rho_0}{2}} \end{pmatrix} \begin{pmatrix} \cos(\alpha\tau) & \sin(\alpha\tau) \\ -\sin(\alpha\tau) & \cos(\alpha\tau) \end{pmatrix} \begin{pmatrix} e^{-\frac{\rho_0}{2}} & 0 \\ 0 & e^{\frac{\rho_0}{2}} \end{pmatrix} \\ \implies \tilde{g}(x^+, x^-) &= \begin{pmatrix} \cos(\alpha\tau) & e^{\rho_0}\sin(\alpha\tau) \\ -e^{-\rho_0}\sin(\alpha\tau) & \cos(\alpha\tau) \end{pmatrix} \end{aligned} \tag{240}$$

Then the spectrally flowed solution is,

$$g(x^+, x^-) = \begin{pmatrix} \cos\left(\frac{wx^+}{2}\right) & \sin\left(\frac{wx^+}{2}\right) \\ -\sin\left(\frac{wx^+}{2}\right) & \cos\left(\frac{wx^+}{2}\right) \end{pmatrix} \begin{pmatrix} \cos(\alpha\tau) & e^{\rho_0}\sin(\alpha\tau) \\ -e^{-\rho_0}\sin(\alpha\tau) & \cos(\alpha\tau) \end{pmatrix} \begin{pmatrix} \cos\left(\frac{wx^-}{2}\right) & \sin\left(\frac{wx^-}{2}\right) \\ -\sin\left(\frac{wx^-}{2}\right) & \cos\left(\frac{wx^-}{2}\right) \end{pmatrix}$$

After multiplying these matrices, we get the entries of the flowed solution $g(x^+, x^-)$ as,

$$
\begin{aligned}
g_{11} &= \cosh(\rho_0)\, \cos(w\tau)\cos(\alpha\tau) - \sin(w\tau)\sin(\alpha\tau) - \sinh(\rho_0)\, \sin(\alpha\tau)\sin(w\sigma),\\
g_{12} &= \cosh(\rho_0)\, \sin(w\tau)\cos(\alpha\tau) + \cos(w\tau)\sin(\alpha\tau) - \sinh(\rho_0)\, \sin(\alpha\tau)\cos(w\sigma),\\
g_{21} &= -\cosh(\rho_0)\, \sin(w\tau)\cos(\alpha\tau) - \cos(w\tau)\sin(\alpha\tau) - \sinh(\rho_0)\, \sin(\alpha\tau)\cos(w\sigma),\\
g_{22} &= \cosh(\rho_0)\, \cos(w\tau)\cos(\alpha\tau) - \sin(w\tau)\sin(\alpha\tau) + \sinh(\rho_0)\, \sin(\alpha\tau)\sin(w\sigma)
\end{aligned}
\tag{241}
$$

But we can compare this with the global coordinates parameterization, which is written as,

$$
g(x^+, x^-) = \begin{pmatrix} \cosh(\rho)\, \cos(t) + \sinh(\rho)\, \cos(\phi) & \cosh(\rho)\, \sin(t) - \sinh(\rho)\, \sin(\phi) \\ -\cosh(\rho)\, \sin(t) - \sinh(\rho)\, \sin(\phi) & \cosh(\rho)\, \cos(t) - \sinh(\rho)\, \cos(\phi) \end{pmatrix}
$$

Then, by comparing both forms, we get,

$$
\begin{aligned}
\frac{g_{11}+g_{22}}{2} &= \cosh(\rho_0)\cos(w\tau)\cos(\alpha\tau) - \sin(w\tau)\sin(\alpha\tau) = \cosh(\rho)\, \cos(t)\\
\frac{g_{12}-g_{21}}{2} &= \cosh(\rho_0)\sin(w\tau)\cos(\alpha\tau) + \cos(w\tau)\sin(\alpha\tau) = \cosh(\rho)\, \sin(t)\\
-\frac{g_{12}+g_{21}}{2} &= \sinh(\rho_0)\, \sin(\alpha\tau)\cos(w\sigma) = \sinh(\rho)\, \sin(\phi)\\
\frac{g_{11}-g_{22}}{2} &= -\sinh(\rho_0)\, \sin(\alpha\tau)\sin(w\sigma) = \sinh(\rho)\, \cos(\phi)
\end{aligned}
\tag{242}
$$

Then, taking the ratio of the first and the second relation, we get,

$$
\tan(t) = \frac{\cosh(\rho_0)\sin(w\tau)\cos(\alpha\tau) + \cos(w\tau)\sin(\alpha\tau)}{\cosh(\rho_0)\cos(w\tau)\cos(\alpha\tau) - \sin(w\tau)\sin(\alpha\tau)} \tag{243}
$$

Dividing throughout by $\cosh(\rho_0)\cos(w\tau)\cos(\alpha\tau)$ we get,

$$
\boxed{\tan(t) = \frac{\tan(w\tau) + \tan(\alpha\tau)/\cosh(\rho_0)}{1 - \tan(w\tau)\tan(\alpha\tau)/\cosh(\rho_0)}} \tag{244}
$$

Also, finding $\frac{g_{11}-g_{22}}{2} + i\left(\frac{g_{12}+g_{21}}{2}\right)$ we get,

$$
\boxed{\sinh(\rho)\, e^{i\phi} = i\sinh(\rho_0)\, \sin(\alpha\tau)e^{iw\sigma}} \tag{245}
$$

The modulus of the above equation is $\sinh(\rho) = \sinh(\rho_0)\,|\sin(\alpha\tau)|$. Since $|\sin(\alpha\tau)| \leq 1$, it follows that $0 \leq \sinh(\rho) \leq \sinh(\rho_0)$. Thus, the radial coordinate is bounded for all worldsheet time. This makes it a short string solution. Moreover as τ increases, the radial coordinates oscillate ρ reaching the center of AdS_3 periodically.

Let us find the currents of this solution. Before spectral flow,

$$\tilde{J}^a_R = -2\mathcal{K}(\partial_+\tilde{v}_+)\,\mathrm{Tr}\Big((U^{-1}T^aU)T^3\Big) = \frac{i\mathcal{K}\alpha}{2}\,\mathrm{Tr}\Big(T^a(e^{\frac{1}{2}\rho_0\sigma_3}\sigma_2 e^{-\frac{1}{2}\rho_0\sigma_3})\Big) \tag{246}$$

Use, $e^{\frac{1}{2}\rho_0\sigma_3}\sigma_2 e^{-\frac{1}{2}\rho_0\sigma_3} = \cosh(\rho_0)\sigma_2 - i\sinh(\rho_0)\sigma_1$. Thus,

$$\begin{aligned}
\tilde{J}^a_R &= \frac{i\mathcal{K}\alpha}{2}\,\mathrm{Tr}\Big(T^a\big(\cosh(\rho_0)\sigma_2 - i\sinh(\rho_0)\sigma_1\big)\Big)\\
&= -\mathcal{K}\alpha\,\mathrm{Tr}\Big(T^a\big(\cosh(\rho_0)T^3 - \sinh(\rho_0)T^2\big)\Big)\\
&= -\mathcal{K}\alpha\Big(\cosh(\rho_0)\,\mathrm{Tr}\big(T^aT^3\big) - \sinh(\rho_0)\mathrm{Tr}\big(T^aT^2\big)\Big)\\
\Longrightarrow\ \tilde{J}^a_R &= -\frac{\mathcal{K}\alpha}{2}\Big(\cosh(\rho_0)\,\kappa^{a3} - \sinh(\rho_0)\kappa^{a2}\Big)
\end{aligned} \tag{247}$$

Then,

$$\tilde{J}^1_R = 0 \qquad \text{and} \qquad \tilde{J}^2_R = \frac{\mathcal{K}\alpha}{2}\sinh(\rho_0) \tag{248}$$

$$\tilde{J}^3_R = \frac{\mathcal{K}\alpha}{2}\cosh(\rho_0) \qquad \text{and} \qquad J^\pm_R = \pm i\frac{\mathcal{K}\alpha}{2}\sinh(\rho_0) \tag{249}$$

Then, after spectral flow, we get the currents to be,

$$\boxed{J^3_R(x^+) = \tilde{J}^3_R(x^+) + \frac{\mathcal{K}}{2}w_R = \frac{\mathcal{K}}{2}\Big(\alpha\cosh(\rho_0) + w_R\Big)} \tag{250}$$

$$\boxed{J^\pm_R(x^+) = \tilde{J}^\pm_R(x^+)e^{\mp i w_R x^+} = \pm i\frac{\mathcal{K}\alpha}{2}\sinh(\rho_0)e^{\mp i w_R x^+}} \tag{251}$$

Similarly, the left current before spectral flow is,

$$\Longrightarrow\ \tilde{J}^a_L = -2\mathcal{K}(\partial_-\tilde{u}_-)\mathrm{Tr}\Big(T^{a^*}V^{-1}T^{3^*}V\Big) = -\frac{i\mathcal{K}\alpha}{2}\mathrm{Tr}\Big(T^{a^*}e^{\frac{1}{2}\rho_0\sigma_3}\sigma_2 e^{-\frac{1}{2}\rho_0\sigma_3}\Big) \tag{252}$$

Use, $e^{\frac{1}{2}\rho_0\sigma_3}\sigma_2 e^{-\frac{1}{2}\rho_0\sigma_3} = \cosh(\rho_0)\sigma_2 - i\sinh(\rho_0)\sigma_1$. Thus,

$$\begin{aligned}
\tilde{J}^a_L &= -\frac{i\mathcal{K}\alpha}{2}\,\mathrm{Tr}\Big(T^{a^*}\big(\cosh(\rho_0)\sigma_2 - i\sinh(\rho_0)\sigma_1\big)\Big)\\
&= -\mathcal{K}\alpha\,\mathrm{Tr}\Big(T^{a^*}\big(\cosh(\rho_0)T^{3^*} + \sinh(\rho_0)T^{2^*}\big)\Big)\\
&= -\mathcal{K}\alpha\Big(\cosh(\rho_0)\,\mathrm{Tr}\big(T^{a^*}T^{3^*}\big) + \sinh(\rho_0)\mathrm{Tr}\big(T^{a^*}T^{2^*}\big)\Big)\\
\Longrightarrow\ \tilde{J}^a_L &= -\frac{\mathcal{K}\alpha}{2}\Big(\cosh(\rho_0)\,\kappa^{a3} + \sinh(\rho_0)\kappa^{a2}\Big)
\end{aligned} \tag{253}$$

Then,

$$\tilde{J}^1_L = 0 \qquad \text{and} \qquad \tilde{J}^2_L = \frac{\mathcal{K}\alpha}{2}\sinh(\rho_0) \tag{254}$$

$$\tilde{J}^3_L = \frac{\mathcal{K}\alpha}{2}\cosh(\rho_0) \qquad \text{and} \qquad \tilde{J}^\pm_L = \tilde{J}^1_L \mp i\tilde{J}^2_L = \mp i\frac{\mathcal{K}\alpha}{2}\sinh(\rho_0) \tag{255}$$

Then, after spectral flow, we get the currents to be,

$$\boxed{J^3_L(x^-) = \tilde{J}^3_L(x^-) + \frac{\mathcal{K}}{2}w_L = \frac{\mathcal{K}}{2}\Big(\alpha\cosh(\rho_0) + w_L\Big)} \tag{256}$$

$$\boxed{J^\pm_L(x^-) = \tilde{J}^\pm_L(x^-)e^{\mp iw_Lx^-} = \mp i\frac{\mathcal{K}\alpha}{2}\sinh(\rho_0)e^{\mp iw_Lx^-}} \tag{257}$$

But earlier, we got that under spectral flow, the third component of the current changes as $J^3_R = \frac{1}{w_R}\Big(h - \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4}w_R$ and $J^3_L = \frac{1}{w_L}\Big(h - \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4}w_L$. Then, for diagonal flow, we get,

$$\implies \frac{\mathcal{K}}{2}\Big(\alpha\cosh(\rho_0) + w\Big) = \frac{1}{w}\Big(h - \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4}w \tag{258}$$

$$\implies \alpha^2 + 2w\alpha\cosh(\rho_0) - \frac{4h}{\mathcal{K}} + w^2 = 0 \implies \alpha_\pm = -w\cosh(\rho_0) \pm \sqrt{w^2\cosh^2(\rho_0) + \frac{4h}{\mathcal{K}} - w^2}$$

$$\implies \boxed{\alpha_\pm = -w\cosh(\rho_0) \pm \sqrt{w^2\sinh^2(\rho_0) + \frac{4h}{\mathcal{K}}}} \tag{259}$$

Using these values of $\alpha_\pm$, we can find the 3$^{\text{rd}}$ components of the right and left currents in the diagonal flow as,

$$\begin{aligned} J^3_R &= \frac{\mathcal{K}}{2}\Bigg(\cosh(\rho_0)\Big(-w\cosh(\rho_0) \pm \sqrt{w^2\sinh^2(\rho_0) + \frac{4h}{\mathcal{K}}}\Big) + w\Bigg) \\ \implies J^3_R(\alpha_\pm) &= \frac{\mathcal{K}}{2}\Bigg(-w\sinh^2(\rho_0) \pm \cosh(\rho_0)\sqrt{w^2\sinh^2(\rho_0) + \frac{4h}{\mathcal{K}}}\Bigg) \end{aligned} \tag{260}$$

Similarly, we get the left current as,

$$J^3_L(\alpha_\pm) = \frac{\mathcal{K}}{2}\Bigg(-w\sinh^2(\rho_0) \pm \cosh(\rho_0)\sqrt{w^2\sinh^2(\rho_0) + \frac{4h}{\mathcal{K}}}\Bigg) \tag{261}$$

Then, the total energy of the string is given by $E = J^3_{0_R} + J^3_{0_L}$,

$$\boxed{E(\alpha_\pm) = \mathcal{K}\Bigg(\pm\cosh(\rho_0)\sqrt{w^2\sinh^2(\rho_0) + \frac{4h}{\mathcal{K}}} - w\sinh^2(\rho_0)\Bigg)} \tag{262}$$

Now, for the α_+ branch, the total space-time energy of the string is always positive. And for the α_- branch, the total space-time energy is negative always, which is not physical.

When $h = \dfrac{\mathcal{K}w^2}{4}$, we get, $\alpha^2 + 2\alpha\cosh(\rho_0)w = 0$,

$$\implies \alpha(\alpha + 2w\cosh(\rho_0)) = 0 \qquad \implies \alpha_+ = 0 \ \text{ and } \ \alpha_- = -2w\cosh(\rho_0) \tag{263}$$

Thus, at this value of the internal conformal weight, the positive Energy branch becomes,

$$E(\alpha_+ = 0) = \mathcal{K}w \tag{264}$$

where the energy is constant with respect to the size ρ_0 of the string. The spectral flowed solution is,

$$g(x^+, x^-) = e^{\frac{i}{2}wx^+\sigma_2} e^{\frac{1}{2}\rho_0\sigma_3} e^{i\alpha_+\tau\sigma_2} e^{-\frac{1}{2}\rho_0\sigma_3} e^{\frac{i}{2}wx^-\sigma_2} = e^{\frac{i}{2}wx^+\sigma_2} e^{\frac{i}{2}wx^-\sigma_2} \tag{265}$$

$$\implies u = \frac{wx^+}{2} \quad \text{and} \quad v = \frac{wx^-}{2} \quad \implies \quad t = u + v = w\tau \quad \text{and} \quad \phi = u - v = w\sigma \tag{266}$$

Also by using the explicit form of the solution where $\sinh(\rho)\, e^{i\phi} = i \sinh(\rho_0)\, \sin(\alpha\tau) e^{iw\sigma}$, and substituting $\phi = w\sigma$, and $\alpha = 0$, we get, $\rho = \rho_0$. Thus, for $h = \dfrac{\mathcal{K}w^2}{4}$,

$$t = w\tau \ , \quad \phi = w\sigma \ , \quad \text{and} \ \ \rho = \rho_0 \tag{267}$$

This solution represents a string staying at a fixed radius ρ_0, neither contracting nor expanding. The existence of this type of solution at any radius ρ_0 means that the string becomes marginally unstable in AdS_3. For other h values, the energy is a monotonously increasing function of ρ_0.

$$E(\rho_0 = 0) = 2\sqrt{h\mathcal{K}} \qquad \text{and} \qquad E(\rho_0 \to \infty) = \frac{\mathcal{K}w}{2} + \frac{2h}{w} \tag{268}$$

Thus, we can see that this solution is like a bound state of the string in AdS_3. The solution repeats expansion and contraction where the maximum size of the string is ρ_0.

4.1.2 Explicit solution when $U \neq V^{-1}$ or when $[UV,\, T^3] \neq 0$

If $[UV, T^3] \neq 0$, then U and V conjugate T^3 into different directions:

$$[UV, T^3] \neq 0 \implies UVT^3 \neq T^3UV \implies VT^3V^{-1} \neq U^{-1}T^3U.$$

Consequently, the geodesic is displaced from the center (it does not reach $\rho = 0$). Under spectral flow, this yields a circular string at finite radius; the string cannot collapse to a point because the base geodesic never reaches the center. Consider a time-like geodesic solution,

$$\tilde{g}(\tau) = Ue^{-\alpha(x^+ + x^-)T^3}V \qquad \implies \tilde{g}^{-1}\partial_-\tilde{g} = -\alpha V^{-1}T^3V \quad \text{and} \quad (\partial_+\tilde{g})\tilde{g}^{-1} = -\alpha UT^3U^{-1} \tag{269}$$

Then, by the current definitions,

$$\sum_a \tilde{J}^a_R\, T_a = -\frac{\mathcal{K}\alpha}{2}\, UT^3U^{-1} \neq -\frac{\mathcal{K}\alpha}{2}\, V^{-1}T^{3^*}V = \sum_a \tilde{J}^a_L\, T^*_a \tag{270}$$

This leads to different $J^3_{0_R}$ and $J^3_{0_L}$ modes, which implies the space-time angular momentum, $\ell = J^3_{0_R} - J^3_{0_L} \neq 0$. Physically, a nonzero ℓ corresponds to rotation of the string and produces a centrifugal effect that prevents collapse of the string to $\rho = 0$.

The spectral-flow action modifies each chiral Sugawara stress tensor component as,

$$T_{++}^{\text{AdS}} = \tilde{T}_{++}^{\text{AdS}} - w\tilde{J}_R^3 - \frac{\mathcal{K}}{4}w^2 \qquad \text{and} \qquad T_{--}^{\text{AdS}} = \tilde{T}_{--}^{\text{AdS}} - w\tilde{J}_L^3 - \frac{\mathcal{K}}{4}w^2 \tag{271}$$

For the time-like geodesic solutions, $\tilde{T}_{++}^{\text{AdS}} = \tilde{T}_{--}^{\text{AdS}} = -\dfrac{\mathcal{K}\alpha^2}{4}$. Then, on subtracting the two,

$$T_{++}^{\text{AdS}} - T_{--}^{\text{AdS}} = -\,w\big(\tilde{J}_R^3 - \tilde{J}_L^3\big) = -w\ell \tag{272}$$

If we include the internal $\mathcal{M}$ CFT chiral weights (h_R, h_L) and impose the Virasoro constraints,

$$\begin{aligned} T_{++}^{\text{AdS}} + T_{++}^{\mathcal{M}} = 0 \quad &\implies T_{++}^{\text{AdS}} = -h_R \\ T_{--}^{\text{AdS}} + T_{--}^{\mathcal{M}} = 0 \quad &\implies T_{++}^{\text{AdS}} = -h_L \end{aligned} \tag{273}$$

Then, combining the two results, we get,

$$\boxed{h_R - h_L = w\ell} \tag{274}$$

Thus, for the case where $[UV, T^3] \neq 0$, there will be different left and right currents, which implies there is non-zero angular momentum that preserves the string from collapsing to the center of AdS_3. The Virasoro constraints result in unequal right and left conformal weights (h_R, h_L) in the compact M sector. And their difference is precisely the winding number times the angular momentum.

4.2 Long strings as the spectral flow of space-like geodesics

For space-like geodesics, we have the classical solution of the form, $\tilde{g}(x^+, x^-) = Ue^{(v_+(x^+)+u_-(x^-))\sigma_3}V$ where, $\tilde{v}_+ = \frac{\alpha}{2}x^+$ and $\tilde{u}_+ = \frac{\alpha}{2}x^-$, with $\alpha = \pm\sqrt{\frac{-4h}{\mathcal{K}}}$. The energy-momentum tensor components are, $\tilde{T}_{++}(x^+) = \mathcal{K}(\partial_+\tilde{v}_+)^2 = \frac{\mathcal{K}}{4}\alpha^2$ and $\tilde{T}_{--}(x^-) = \mathcal{K}(\partial_-\tilde{u}_-)^2 = \frac{\mathcal{K}}{4}\alpha^2$. The spectral flowed energy-momentum tensor is given by $T_{++} = \tilde{T}_{++} - w_R\tilde{J}_R^3 - \frac{\mathcal{K}}{4}w_R^2$ and $T_{--} = \tilde{T}_{--} - w_L\tilde{J}_L^3 - \frac{\mathcal{K}}{4}w_L^2$,

$$\implies T_{++} = \frac{\mathcal{K}}{4}\alpha^2 - w_R\tilde{J}_R^3 - \frac{\mathcal{K}}{4}w_R^2 \quad \text{and} \quad T_{--} = \frac{\mathcal{K}}{4}\alpha^2 - w_L\tilde{J}_L^3 - \frac{\mathcal{K}}{4}w_L^2 \tag{275}$$

If the spectral flow doesn't change anything in the compact manifold $\mathcal{M}$ sector, then we can consider $T_{\pm\pm}^{\mathcal{M}} = h$ (negative) as before. Then the Virasoro constraint gives,

$$\begin{aligned} T_{++} + h = 0 \quad &\implies \quad \frac{\mathcal{K}}{4}(\alpha^2 - w_R^2) - w_R\tilde{J}_R^3 + h = 0 \\ T_{--} + h = 0 \quad &\implies \quad \frac{\mathcal{K}}{4}(\alpha^2 - w_L^2) - w_L\tilde{J}_L^3 + h = 0 \end{aligned} \tag{276}$$

$$\implies \quad \tilde{J}^3_R = \frac{1}{w_R}\Big(h + \frac{\mathcal{K}}{4}(\alpha^2 - w_R^2)\Big) \quad \text{and} \quad \tilde{J}^3_L = \frac{1}{w_L}\Big(h + \frac{\mathcal{K}}{4}(\alpha^2 - w_L^2)\Big) \tag{277}$$

Then the new spectral flowed currents $J^3_{R,L}$ are given by,

$$\begin{aligned} J^3_R &= \tilde{J}^3_R + \frac{\mathcal{K}}{2}w_R = \frac{1}{w_R}\Big(h + \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4}w_R \\ J^3_L &= \tilde{J}^3_L + \frac{\mathcal{K}}{2}w_L = \frac{1}{w_L}\Big(h + \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4}w_L \end{aligned} \tag{278}$$

Since we get J^3_R and J^3_L to be independent of $x^\pm$, they have only the zeroth mode, which is equal to themselves. Under the diagonal choice of winding numbers $w_R = w_L = w$, we get the total energy of the string, which is bounded below depending on α as,

$$E = J^3_{0_R} + J^3_{0_L} = 2J^3_{0_R} = \frac{1}{w}\Big(2h + \frac{\mathcal{K}}{2}\alpha^2\Big) + \frac{\mathcal{K}}{2}w \tag{279}$$

$$\text{Thus,} \quad E > \frac{2h}{w} + \frac{\mathcal{K}w}{2} \tag{280}$$

4.2.1 Spectral flow of space-like solution with $U = V = 1$

Here, we have $\tilde{g}(\tau) = e^{\alpha\tau\sigma_3}$, where $\tilde{\rho} = \alpha\tau$, $\tilde{t} = 0$, $\tilde{\phi} = 0$. Under spectral flow we get the flowed solution as, $g(x^+, x^-) = e^{\frac{i}{2}w_R x^+\sigma_2} e^{\alpha\tau\sigma_3} e^{\frac{i}{2}w_L x^-\sigma_2}$. And if the flow is diagonal ($w_L = w_R = w$) then, the new solutions are, $\rho = \alpha\tau$, $t = w\tau$, $\phi = w\sigma$.

$$\implies \rho e^{i\phi} = \alpha\tau e^{iw\sigma} \quad \text{and} \quad t = w\tau \quad \implies \boxed{\rho = \frac{\alpha|t|}{w}} \tag{281}$$

Thus, at the infinite past $t \to -\infty$, we see that the string is circular and has infinite radius located at the spatial boundary of AdS$_3$. Then, the string collapses to a point at $t = 0$, then again expand toward the boundary as $t \to \infty$. The currents before spectral flow are given by

$$\tilde{J}^a_R = \frac{\mathcal{K}\alpha\kappa^{a1}}{2} = \tilde{J}^a_L \quad \text{and} \quad \tilde{T}_{\pm\pm} = \frac{1}{4}\mathcal{K}\alpha^2 \tag{282}$$

After the spectral flow,

$$J^a_R = \tilde{J}^a_R + \frac{\mathcal{K}w}{2} = \frac{\mathcal{K}}{2}(\alpha\kappa^{a1} + w) = J^a_L \tag{283}$$

$$J^1_R = J^1_L = \frac{\mathcal{K}}{2}(\alpha + w) \quad \text{and} \quad J^2_L = J^2_R = J^3_L = J^3_R = \frac{\mathcal{K}w}{2} \tag{284}$$

$$\implies T_{\pm\pm} = \frac{\mathcal{K}}{4}(\alpha^2 - w^2) \tag{285}$$

But Virasoro constraint gives,

$$\alpha = \pm\sqrt{w^2 - \frac{4h}{\mathcal{K}}} \tag{286}$$

Here the energy is constant $E = \mathcal{K}w$ for any value of α, even for $\alpha = 0$, when $h = \dfrac{\mathcal{K}w^2}{4}$.

4.2.2 Spectral flow of space-like solution when $U = V^{-1} = e^{-\frac{1}{2}\rho_0\sigma_1}$

Here, the solution before spectral flow is,

$$\tilde{g}(x^+, x^-) = e^{-\frac{1}{2}\rho_0\sigma_1} e^{\alpha\tau\sigma_3} e^{\frac{1}{2}\rho_0\sigma_1} \tag{287}$$

Then, after a diagonal spectral flow,

$$g(x^+, x^-) = e^{\frac{i}{2}wx^+\sigma_2} e^{-\frac{1}{2}\rho_0\sigma_1} e^{\alpha\tau\sigma_3} e^{\frac{1}{2}\rho_0\sigma_1} e^{\frac{i}{2}wx^-\sigma_2} \tag{288}$$

To get the flowed global coordinate relations, we shall find the full matrix product and compare with the global coordinate parameterization. The exponentials of required Pauli matrices are,

$$e^{-\frac{1}{2}\rho_0\sigma_1} = \begin{pmatrix} \cosh\left(\frac{\rho_0}{2}\right) & -\sinh\left(\frac{\rho_0}{2}\right) \\ -\sinh\left(\frac{\rho_0}{2}\right) & \cosh\left(\frac{\rho_0}{2}\right) \end{pmatrix}, \quad e^{i\theta\sigma_2} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \quad \text{and} \quad e^{\alpha\tau\sigma_3} = \begin{pmatrix} e^{\alpha\tau} & 0 \\ 0 & e^{-\alpha\tau} \end{pmatrix}$$

Then we get the final matrix with the elements as follows,

$$\begin{aligned} g_{11} &= \cosh(\alpha\tau)\cos(w\tau) + \cosh(\rho_0)\sinh(\alpha\tau)\cos(w\sigma) - \sinh(\rho_0)\sinh(\alpha\tau)\sin(w\tau) \\ g_{12} &= \cosh(\alpha\tau)\sin(w\tau) - \cosh(\rho_0)\sinh(\alpha\tau)\sin(w\sigma) + \sinh(\rho_0)\sinh(\alpha\tau)\cos(w\tau) \\ -g_{21} &= \cosh(\alpha\tau)\sin(w\tau) + \cosh(\rho_0)\sinh(\alpha\tau)\sin(w\sigma) + \sinh(\rho_0)\sinh(\alpha\tau)\cos(w\tau) \\ g_{22} &= \cosh(\alpha\tau)\cos(w\tau) - \cosh(\rho_0)\sinh(\alpha\tau)\cos(w\sigma) - \sinh(\rho_0)\ \sinh(\alpha\tau)\sin(w\tau) \end{aligned} \tag{289}$$

Now we can compare with the usual parameterization and get the spectrally flowed global coordinates. Any solution should be of the form,

$$g(x^+, x^-) = \begin{pmatrix} \cosh(\rho)\ \cos(t) + \sinh(\rho)\ \cos(\phi) & \cosh(\rho)\ \sin(t) - \sinh(\rho)\ \sin(\phi) \\ -\cosh(\rho)\ \sin(t) - \sinh(\rho)\ \sin(\phi) & \cosh(\rho)\ \cos(t) - \sinh(\rho)\ \cos(\phi) \end{pmatrix}$$

Then, on comparing both the matrices we get,

$$\begin{aligned} \frac{g_{11} + g_{22}}{2} &= \cosh(\alpha\tau)\cos(w\tau) - \sinh(\rho_0)\ \sinh(\alpha\tau)\sin(w\tau) = \cosh(\rho)\ \cos(t) \\ \frac{g_{12} - g_{21}}{2} &= \cosh(\alpha\tau)\sin(w\tau) + \sinh(\rho_0)\sinh(\alpha\tau)\cos(w\tau) = \cosh(\rho)\ \sin(t) \\ -\frac{g_{12} + g_{21}}{2} &= \cosh(\rho_0)\sinh(\alpha\tau)\sin(w\sigma) = \sinh(\rho)\ \sin(\phi) \\ \frac{g_{11} - g_{22}}{2} &= \cosh(\rho_0)\ \sinh(\alpha\tau)\cos(w\sigma) = \sinh(\rho)\ \cos(\phi) \end{aligned} \tag{290}$$

Taking the ratio of the first two relations, we get,

$$\tan(t) = \frac{\cosh(\alpha\tau)\sin(w\tau) + \sinh(\rho_0)\sinh(\alpha\tau)\cos(w\tau)}{\cosh(\alpha\tau)\cos(w\tau) - \sinh(\rho_0)\,\sinh(\alpha\tau)\sin(w\tau)} \tag{291}$$

Dividing throughout by $\cosh(\alpha\tau)\cos(w\tau)$, we get,

$$\boxed{\tan(t) = \frac{\tan(w\tau) + \sinh(\rho_0)\tanh(\alpha\tau)}{1 - \sinh(\rho_0)\,\tanh(\alpha\tau)\tan(w\tau)}} \tag{292}$$

Then if we evaluate $\dfrac{g_{11} - g_{22}}{2} - i\Big(\dfrac{g_{12} + g_{21}}{2}\Big)$, we get,

$$\boxed{\sinh(\rho)e^{i\phi} = e^{iw\sigma}\cosh(\rho_0)\sinh(\alpha\tau)} \tag{293}$$

This represents a long string solution. Taking the absolute value of the above equation we get, $\sinh(\rho) = \cosh(\rho_0)\big|\sinh(\alpha\tau)\big|$. So ρ is unbounded here, which can reach the boundary of AdS_3 for large worldsheet time $\tau \to \pm\infty$. This is like a scattering state of the string.

Now, we can find the exact values of currents as follows.

$$\begin{aligned}
\tilde{J}^a_R &= \mathcal{K}\alpha\,\mathrm{Tr}\Big(T^a(UT^1U^{-1})\Big) = \frac{\mathcal{K}\alpha}{2}\mathrm{Tr}\Big(T^a(e^{-\frac{1}{2}\rho_0\sigma_1}\sigma_3 e^{\frac{1}{2}\rho_0\sigma_1})\Big)\\
&= \frac{\mathcal{K}\alpha}{2}\,\mathrm{Tr}\Big(T^a\big(\cosh(\rho_0)\sigma_3 + i\sigma_2\sinh(\rho_0)\big)\Big)\\
&= \mathcal{K}\alpha\,\Big(\cosh(\rho_0)\mathrm{Tr}\big(T^aT^1\big) - \sinh(\rho_0)\mathrm{Tr}\big(T^aT^3\big)\Big)\\
\implies \tilde{J}^a_R &= \frac{\mathcal{K}\alpha}{2}\big(\cosh(\rho_0)\kappa^{a1} - \sinh(\rho_0)\kappa^{a3}\big)
\end{aligned} \tag{294}$$

Thus, the right current components are,

$$\boxed{\tilde{J}^1_R = \frac{\mathcal{K}\alpha}{2}\cosh(\rho_0)\;\;,\;\;\tilde{J}^2_R = 0\;\;\text{ and }\;\;\tilde{J}^3_R = \frac{\mathcal{K}\alpha}{2}\sinh(\rho_0)} \tag{295}$$

Similarly, for the left currents we get,

$$\begin{aligned}
J^a_L &= \mathcal{K}\alpha\,\mathrm{Tr}\Big((VT^{a^*}V^{-1})T^1\Big) = \frac{\mathcal{K}\alpha}{2}\mathrm{Tr}\Big(T^{a^*}(e^{-\frac{1}{2}\rho_0\sigma_1}\sigma_3 e^{\frac{1}{2}\rho_0\sigma_1})\Big)\\
&= \frac{\mathcal{K}\alpha}{2}\,\mathrm{Tr}\Big(T^{a^*}\big(\cosh(\rho_0)\sigma_3 + i\sigma_2\sinh(\rho_0)\big)\Big)\\
&= \mathcal{K}\alpha\,\Big(\cosh(\rho_0)\mathrm{Tr}\big(T^{a^*}T^{1^*}\big) - \sinh(\rho_0)\mathrm{Tr}\big(T^{a^*}T^{3^*}\big)\Big)\\
\implies \tilde{J}^a_L &= \frac{\mathcal{K}\alpha}{2}\big(\cosh(\rho_0)\kappa^{a1} - \sinh(\rho_0)\kappa^{a3}\big)
\end{aligned} \tag{296}$$

Thus, the left current components are the same as the right ones, given by,

$$\boxed{\tilde{J}^1_L = \frac{\mathcal{K}\alpha}{2}\cosh(\rho_0)\;\;,\;\;\tilde{J}^2_L = 0\;\;\text{ and }\;\;\tilde{J}^3_L = \frac{\mathcal{K}\alpha}{2}\sinh(\rho_0)} \tag{297}$$

After a diagonal spectral flow, the 3rd component of left and right currents change as follows,

$$\boxed{J_R^3 = J_L^3 = \frac{\mathcal{K}}{2}\Big(\alpha\sinh(\rho_0) + w\Big)} \tag{298}$$

But in the diagonal flow, the Virasoro constraints give

$$J_R^3 = J_L^3 = \frac{\mathcal{K}}{2}\Big(\alpha\sinh(\rho_0) + w\Big) = \frac{1}{w}\Big(h + \frac{\mathcal{K}}{4}\alpha^2\Big) + \frac{\mathcal{K}}{4}w \tag{299}$$

$$\begin{aligned} &\Longrightarrow \alpha^2 - 2w\alpha\sinh(\rho_0) + \frac{4h}{\mathcal{K}} - w^2 = 0 \\ &\Longrightarrow \alpha = w\sinh(\rho_0) \mp \sqrt{w^2\sinh^2(\rho_0) + w^2 - \frac{4h}{\mathcal{K}}} \end{aligned} \tag{300}$$

$$\Longrightarrow \boxed{\alpha_\pm = w\sinh(\rho_0) \mp \sqrt{w^2\cosh^2(\rho_0) - \frac{4h}{\mathcal{K}}}} \tag{301}$$

We have chosen $\alpha_\pm$ such that for a critical value of $h = \dfrac{\mathcal{K}w^2}{4}$, we get $\alpha_+ = 0$. For this value of h, the 3rd component of spectrally flowed current becomes $J_R^3 = J_L^3 = \dfrac{\mathcal{K}w}{2}$, which gives the total energy $E = \mathcal{K}w$. This solution now coincides with that of the short string solution. For any other value of h, the total energy becomes,

$$E = J_R^3 + J_L^3 = \mathcal{K}\left(\sinh(\rho_0)\Big(w\sinh(\rho_0) \mp \sqrt{w^2\cosh^2(\rho_0) - \frac{4h}{\mathcal{K}}}\Big) + w\right) \tag{302}$$

$$\Longrightarrow \boxed{E(\alpha_\pm) = \mathcal{K}\left(w\cosh^2(\rho_0) \mp \sinh(\rho_0)\sqrt{w^2\cosh^2(\rho_0) - \frac{4h}{\mathcal{K}}}\right)} \tag{303}$$

For arbitrary h values, the long string energy is a monotonously increasing function of ρ_0.

$$E(\rho_0 = 0) = \mathcal{K}w \qquad \text{and} \qquad E(\rho_0 \to \infty) = \infty \ \ \text{(unbounded above)} \tag{304}$$

Thus, when $h = \dfrac{\mathcal{K}w^2}{4}$, we could see the transition of a short string into a long string, and it could escape to infinity.

Notice that since $U = V^{-1}$ or since $[UV, T^1] = 0$, we get that the 3rd component of left and right currents is the same. This allows the string to reach the center $\rho = 0$ (collapse) at least once in its lifetime. But this is not the case when $[UV, T^1] \neq 0$. In this case, we will get $J_L^3 \neq J_R^3$, thus there is non-zero angular momentum ℓ. And this is possible only when the compact manifold has unequal left- and right-conformal weights (h_R, h_L), which implies $h_R - h_L = w\ell$. It is this centrifugal force, due to the non-zero angular momentum, that prevents the string from collapsing to a point.

5 Semiclassical Analysis of Strings in AdS_3

In the study of classical string solutions in AdS_3 the integer parameter w is introduced and is interpreted as a winding number associated with the angular coordinate ϕ, through the identification $\phi \sim \phi + 2\pi$. Classically, solutions frequently take the form $\phi(\tau, \sigma) = w\sigma + \cdots$, suggesting that w counts how many times the string wraps the ϕ-circle. This interpretation is natural by analogy with flat or cylindrical target spaces, where winding numbers are topological invariants. However, this identification is *naive* in AdS_3 and requires careful re-examination. The global AdS_3 metric is $ds^2 = -\cosh^2(\rho)\, dt^2 + d\rho^2 + \sinh^2(\rho)\, d\phi^2$. And for any fixed $\rho > 0$, the coordinate ϕ parametrizes a circle. However, as $\rho \to 0$, $\sinh(\rho) \to 0$, and the ϕ-circle shrinks smoothly to a point. Therefore, the cycle $\phi \to \phi + 2\pi$ is *contractible* in AdS_3. As a consequence, there is no topologically conserved winding number associated with the angular direction. A string configuration that appears to wrap the ϕ-circle can be continuously deformed, by moving toward $\rho = 0$, into a configuration with no winding at all. Thus, the integer w cannot be interpreted as a topological invariant in general.

The situation changes when the string is located close to the AdS boundary. For large ρ, $\sinh(\rho) \sim \frac{1}{2}e^{\rho}$, and the proper size of the ϕ-circle becomes very large. A string that remains at large ρ for all worldsheet time cannot smoothly contract the angular cycle without traveling deep into the bulk. Such configurations are known as *long strings*. They satisfy $\lim\limits_{|\tau|\to\infty} \rho(\tau) = \infty$, and remain close to the boundary for long times. For these strings, the integer w labeling angular dependence is effectively conserved and well-defined. In this regime, it is meaningful to speak of winding, even though it is not topological in a strict sense. Therefore, long strings near the boundary admit a stable characterization by an integer w.

In contrast, *short strings* or point-like strings remain at finite radial position and often pass through the center $\rho = 0$. In this region, the ϕ-circle degenerates, and the notion of wrapping becomes ill-defined. For such configurations, $\rho(\tau) \leq \rho_0 < \infty$, and there is no obstruction to continuously changing the angular dependence of the solution. Consequently, the integer w does not label a conserved or even well-defined quantity for short strings. This explains why, for collapsing or point-like strings, the winding number loses its physical meaning.

Upon quantization of the string, the absence of a topological winding number implies that processes in which the effective winding changes are allowed, at least for short strings. Since such strings can pass through $\rho = 0$, nothing forbids transitions of the form $w \to w \pm n$, $n \in \mathbb{Z}$. This does not contradict the consistency of the quantum theory, because w is not a topological invariant of the target space. Despite this, string states can be organized by an integer w. The correct interpretation is that w labels *spectral flow sectors* of the affine $\widehat{\mathfrak{sl}}(2)_k$ current algebra. Spectral flow is an automorphism of the current algebra that shifts the zero modes and changes the representation sector. The integer w thus characterizes how a given state is embedded into the full Hilbert space, rather than counting a topological winding around ϕ.

5.1 The Nambu-Goto string action

From now we shall fix $\alpha' = 2$. The Nambu type action for the string in AdS_3 is given by,

$$S_{NG} = \frac{1}{2\pi} \int dt \, d\sigma \left[\sqrt{-\det(g_{\text{ind}})} \; - \; B_{t\phi} \, \partial_\sigma \phi \right] \tag{305}$$

where g_{ind} is the induced metric on the worldsheet, obtained by pulling back the full target–space metric G_{MN} given by,

$$(g_{\text{ind}})_{\alpha\beta} = G_{MN} \, \partial_\alpha X^M \, \partial_\beta X^N, \qquad \alpha, \beta \in \{0, 1\} = (\tau, \sigma) \tag{306}$$

The action has been gauge-fixed using the static gauge (also called static time gauge) condition, where $t(\tau, \sigma) = \tau$, so the worldsheet time coordinate equals the global AdS_3 global time. This gauge choice fixes worldsheet diffeomorphisms that move the target–space time coordinate.

$B_{t\phi}$ is the component of the NS–NS two–form (the B–field) with one index along the time direction t and one along the angular direction ϕ. The term $B_{t\phi} \, \partial_\sigma \phi$ is the pullback of the Wess–Zumino (WZ) coupling that appears for strings in an NS–NS background. In the AdS_3 NS–NS background of the $\mathrm{SL}(2, \mathbb{R})$ WZW model, the only physical B-field component allowed by symmetry is $B_{t\phi}$, since the field strength $H = dB$ must be proportional to the AdS_3 volume form. Fixing the static gauge $t = \tau$ eliminates all other possible antisymmetric couplings.

Now, the coordinate ϕ is not well–defined near $\rho = 0$. The angular coordinate ϕ is degenerate (the circle shrinks to a point). To avoid coordinate singularities, it is convenient to introduce local Cartesian coordinates in the (ρ, ϕ) plane,

$$X^1 = \rho \cos(\phi), \quad X^2 = \rho \sin(\phi) \quad \Longrightarrow \quad \Phi = X^1 + iX^2 \; = \; \rho \, e^{i\phi} \tag{307}$$

These are regular and are the natural variables for a small oscillation expansion around $\rho = 0$. The target–space metric for the AdS_3 factor in the usual global parametrization reads

$$ds^2_{\mathrm{AdS}_3} = G_{tt} \, dt^2 + G_{\rho\rho} d\rho^2 + G_{\phi\phi} \, d\phi^2 = k \Big(- \cosh^2(\rho) \, dt^2 + d\rho^2 + \sinh^2(\rho) \, d\phi^2 \Big)$$

Here k is the overall radius–squared (WZW level) factor that multiplies the AdS_3 metric. It appears in the induced metric components through every contraction with the AdS_3 part of G_{MN}.

The compact sector of the target space will be considered to be $\mathcal{M} = S^3 \times T^4$. The coordinates of T^4 internal directions will be denoted Y^i with metric $G^{(T^4)}_{ij}$. In the background $\mathrm{AdS}_3 \times S^3 \times T^4$, the worldsheet theory factorizes into independent sigma models for AdS_3, S^3, and T^4, so only those states whose nontrivial classical dynamics lie entirely in AdS_3 are focused. The T^4 factor has non-contractible 1-cycles, allowing genuine winding modes that provide a classical contribution to the worldsheet energy, encoded in the conformal weight h. In contrast, S^3 has no topological winding and only supports massive excitations due to curvature, so its coordinates are set to their ground state and treated as part of the spectator internal CFT.

The induced metric on the world sheet is given by the Pull-back of G_{MN}, as follows,

$$
\begin{aligned}
(g_{\mathrm{ind}})_{\tau\tau} &= G_{tt}\,(\partial_\tau t)^2 + G_{\rho\rho}\,(\partial_\tau \rho)^2 + G_{\phi\phi}\,(\partial_\tau \phi)^2 + G^{(T^4)}_{ij}(\partial_\tau Y^i)(\partial_\tau Y^j)\\
(g_{\mathrm{ind}})_{\tau\sigma} &= G_{tt}(\partial_\tau t)(\partial_\sigma t) + G_{\rho\rho}\,(\partial_\tau \rho)\,(\partial_\sigma \rho) + G_{\phi\phi}\,(\partial_\tau \phi)\,(\partial_\sigma \phi) + G^{(T^4)}_{ij}(\partial_\tau Y^i)(\partial_\sigma Y^j)\\
(g_{\mathrm{ind}})_{\sigma\sigma} &= G_{tt}(\partial_\sigma t)^2 + G_{\rho\rho}\,(\partial_\sigma \rho)^2 + G_{\phi\phi}\,(\partial_\sigma \phi)^2 + G^{(T^4)}_{ij}(\partial_\sigma Y^i)(\partial_\sigma Y^j)
\end{aligned}
\tag{308}
$$

Using the static gauge $t(\tau,\sigma) = \tau$, $(\partial_0 t = 1\ ,\ \partial_1 t = 0)$ and substituting the AdS_3 components $G_{tt} = -k\cosh^2(\rho),\ G_{\rho\rho} = k,\ G_{\phi\phi} = k\sinh^2(\rho)$, we get,

$$
\begin{aligned}
(g_{\mathrm{ind}})_{\tau\tau} &= -k\cosh^2(\rho) + k\,(\partial_\tau \rho)^2 + k\sinh^2(\rho)\,(\partial_\tau \phi)^2 + G^{(T^4)}_{ij}(\partial_\tau Y^i)(\partial_\tau Y^j)\\
(g_{\mathrm{ind}})_{\tau\sigma} &= k\,(\partial_\tau \rho)\,(\partial_\sigma \rho) + k\sinh^2(\rho)\,(\partial_\tau \phi)\,(\partial_\sigma \phi) + G^{(T^4)}_{ij}(\partial_\tau Y^i)(\partial_\sigma Y^j)\\
(g_{\mathrm{ind}})_{\sigma\sigma} &= k\,(\partial_\sigma \rho)^2 + k\sinh^2(\rho)\,(\partial_\sigma \phi)^2 + G^{(T^4)}_{ij}(\partial_\sigma Y^i)(\partial_\sigma Y^j)
\end{aligned}
\tag{309}
$$

Let us analyze the small perturbations near $\rho = 0$, where we keep terms only up to quadratic order in ρ (small oscillations). Thus, $\cosh^2(\rho) \approx 1+\rho^2$, $\sinh^2(\rho) \approx \rho^2$. And also we shall write ρ and ϕ in terms of the cartesian fields X^μ , $\mu = (1,\, 2)$. Since, $X^1 = \rho\cos(\phi)$, $X^2 = \rho\sin(\phi)$, we can write $\rho^2 = \delta_{\mu\nu}X^\mu X^\nu$ and $(\partial_\alpha \rho)(\partial_\beta \rho) + \rho^2(\partial_\alpha \phi)(\partial_\beta \phi) = \delta_{\mu\nu}(\partial_\alpha X^\mu)\,(\partial_\beta X^\nu)$. Thus,

$$
\begin{aligned}
(g_{\mathrm{ind}})_{\tau\tau} &= k\big[-(1+\delta_{\mu\nu}X^\mu X^\nu) + \delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\tau X^\nu)\big] + G^{(T^4)}_{ij}(\partial_\tau Y^i)(\partial_\tau Y^j)\\
(g_{\mathrm{ind}})_{\tau\sigma} &= k\,\delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\sigma X^\nu) + G^{(T^4)}_{ij}(\partial_\tau Y^i)(\partial_\sigma Y^j)\\
(g_{\mathrm{ind}})_{\sigma\sigma} &= k\,\delta_{\mu\nu}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu) + G^{(T^4)}_{ij}(\partial_\sigma Y^i)(\partial_\sigma Y^j)
\end{aligned}
\tag{310}
$$

For simplicity, only pure winding modes on T^4 are considered, where $\partial_\sigma Y^i \neq 0$ and $\partial_\tau Y^i = 0$. Physically, this picks a string that wraps cycles of T^4. The energy of such a configuration is stored in the spatial gradients $\partial_\sigma Y^i$. Thus, the conformal weight from the compact T^4 part is,

$$
h = \frac{1}{4}\oint \frac{d\sigma}{2\pi}\; G^{(T^4)}_{ij}\,(\partial_\sigma Y^i)\,(\partial_\sigma Y^j)
\tag{311}
$$

The RHS of the above equation is nothing but the $L_0^{(T^4)}$ Virasoro mode from the compact T^4 $\sigma-$model (assuming $h_L = h_R = h$). The first factor of $\frac{1}{2}$ comes from the fact that the energy includes contributions from both left-moving and right-moving worldsheet modes. The second factor of $\frac{1}{2}$ comes from the Hamiltonian density $\mathcal{H}^{(T^4)} = \frac{1}{2}\big[(\partial_\tau Y)^2 + (\partial_\sigma Y)^2\big]$ which for a pure winding configuration reduces to $\frac{1}{2}(\partial_\sigma Y)^2$. Thus, the internal conformal weight has an overall factor of $\frac{1}{4}$. Also, for the purely winding configuration, we have $Y^i(\tau,\sigma) = y^i + w^i\sigma$, such that $(\partial_\sigma Y)^2 = G^{(T^4)}_{ij}w^i w^j = \text{constant}$. Thus, $G^{(T^4)}_{ij}w^i w^j \approx 4h$. Then, finally, the conformal weights from the S^3 sector do not contribute as discussed before.

$$\implies \boxed{\begin{aligned}(g_{\text{ind}})_{\tau\tau} &= k\big[-(1+\delta_{\mu\nu}X^\mu X^\nu)+\delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\tau X^\nu)\big]\\ (g_{\text{ind}})_{\tau\sigma} &= k\,\delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\sigma X^\nu)\\ (g_{\text{ind}})_{\sigma\sigma} &= k\,\delta_{\mu\nu}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu)+4h\end{aligned}} \tag{312}$$

Then, we shall substitute these components of g_{ind} and the conformal weight h in the definition of $NG-$action, and keep terms only up to quadratic order in ρ (or X^μ).

$$\begin{aligned}\implies \det(g_{\text{ind}}) &= (g_{\text{ind}})_{\tau\tau}(g_{\text{ind}})_{\sigma\sigma} - \underbrace{(g_{\text{ind}})^2_{\tau\sigma}}_{\approx 0}\\ &= k^2\Big(-(1+\delta_{\mu\nu}X^\mu X^\nu)+\delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\tau X^\nu)\Big)\Big(\delta_{\mu\nu}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu)+\frac{4h}{k}\Big)\\ &= -4hk - 4hk\delta_{\mu\nu}X^\mu X^\nu + 4hk\delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\tau X^\nu) - k^2\delta_{\mu\nu}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu)\\ \implies -\det(g_{\text{ind}}) &= 4hk\Big(1+\delta_{\mu\nu}X^\mu X^\nu - \delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\tau X^\nu)+\frac{k}{4h}\delta_{\mu\nu}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu)\Big)\end{aligned} \tag{313}$$

Thus upto 2^{nd} order in X^μ we get,

$$\boxed{\sqrt{-\det(g_{\text{ind}})} = \sqrt{4hk}\Big(1+\frac{1}{2}\delta_{\mu\nu}X^\mu X^\nu - \frac{1}{2}\delta_{\mu\nu}(\partial_\tau X^\mu)(\partial_\tau X^\nu)+\frac{k}{8h}\delta_{\mu\nu}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu)\Big)} \tag{314}$$

Now, we should consider the effect of the $B_{t\phi}$ field. The bosonic $\mathrm{SL}(2,\mathbb{R})$ WZW model describes strings on AdS_3 supported by NS–NS three-form flux H. In the sigma-model language, this flux is fixed by the WZW term and is proportional to the AdS_3 volume form:

$$H = dB = k\,\upsilon_{\mathrm{AdS}_3} \tag{315}$$

where k is the WZW level. Using the AdS_3 metric $ds^2_{\mathrm{AdS}_3} = k\big(-\cosh^2(\rho)dt^2+d\rho^2+\sinh^2(\rho)d\phi^2\big)$, the invariant volume form is,

$$\upsilon_{\mathrm{AdS}_3} = \cosh(\rho)\sinh(\rho)\,dt\wedge d\rho\wedge d\phi \tag{316}$$

Now, we need to choose a gauge for the 2-form B-field such that it gives us $H = dB$ and is consistent with AdS_3 isometries and worldsheet gauge transforms. So, we can choose,

$$B = B_{t\phi}(\rho)\,dt\wedge d\phi \quad\implies\quad H = (\partial_\rho B_{t\phi})\,d\rho\wedge dt\wedge d\phi \tag{317}$$

$$\implies \partial_\rho B_{t\phi} = k\cosh(\rho)\sinh(\rho) \quad\implies\quad B_{t\phi} = k\sinh^2(\rho)+\text{constant} \tag{318}$$

Now, the classical solution can be chosen such that the integration constant is zero. And up to quadratic order, we get,

$$B_{t\phi} = k\sinh^2(\rho)\approx k\rho^2 \tag{319}$$

This is the expression used in the semiclassical expansion of the $NG-$Action.

Since only $B_{t\phi}$ component of the anti-symmetric B-field is non-zero, the $B-$part of the action contains just the below term,

$$S_B = \frac{1}{2\pi}\int d\tau\, d\sigma\, \varepsilon^{\alpha\beta} B_{t\phi}(\partial_\alpha t)(\partial_\beta \phi) \tag{320}$$

where $\varepsilon^{01} = 1 = -\varepsilon^{10}$ and $\varepsilon^{00} = \varepsilon^{11} = 0$. In static gauge, $t(\tau,\sigma) = \tau$, $(\partial_0 t = 1\ ,\ \partial_1 t = 0)$.

$$\implies S_B = -\frac{k}{2\pi}\int dt\, d\sigma\, \rho^2(\partial_\sigma \phi) \tag{321}$$

Since, $X^1 = \rho\cos(\phi)$, $X^2 = \rho\sin(\phi)$, and using $\varepsilon_{12} = 1 = -\varepsilon_{21}$, we can write the identity $X^1\partial_\sigma X^2 - X^2\partial_\sigma X^1 = \varepsilon_{\mu\nu}X^\mu\partial_\sigma X^\nu = \rho^2(\partial_\sigma\phi) = \delta_{\mu\nu}X^\mu X^\nu(\partial_\sigma\phi)$.

$$\implies \boxed{S_B = -\frac{k}{2\pi}\int dt\, d\sigma\, \varepsilon_{\mu\nu}X^\mu(\partial_\sigma X^\nu)} \tag{322}$$

Then the final action can be written as,

$$S_{NG} = \frac{\sqrt{4hk}}{2\pi}\int d\tau\, d\sigma\left(1+\frac{1}{2}\delta_{\mu\nu}\big(X^\mu X^\nu-(\partial_\tau X^\mu)(\partial_\tau X^\nu)+\frac{k}{4h}(\partial_\sigma X^\mu)(\partial_\sigma X^\nu)\big)-\sqrt{\frac{k}{4h}}\,\varepsilon_{\mu\nu}X^\mu(\partial_\sigma X^\nu)\right)$$

The above action can be simplified in terms of complex field $\Phi = X^1+iX^2$ and $\Phi^* = X^1-iX^2$. Then, we can use the identities, $|\Phi|^2 = \delta_{\mu\nu}X^\mu X^\nu$, $|\partial_\alpha\Phi|^2 = \delta_{\mu\nu}(\partial_\alpha X^\mu)(\partial_\alpha X^\nu)$ and also notice that $\epsilon_{\mu\nu}X^\mu\partial_\sigma X^\nu = \mathrm{Im}\big(\Phi^*(\partial_\sigma\Phi)\big) = \frac{1}{2i}\big(\Phi^*(\partial_\sigma\Phi) - \Phi(\partial_\sigma\Phi^*)\big)$. Then, the above action becomes

$$\begin{aligned} S_{NG} &= \frac{\sqrt{4hk}}{2\pi}\int d\tau\, d\sigma\left(1 - \frac{1}{2}|\partial_\tau\Phi|^2 + \frac{1}{2}\Big(|\Phi|^2 + \frac{k}{4h}|\partial_\sigma\Phi|^2\Big) - \sqrt{\frac{k}{4h}}\frac{1}{2i}\big(\Phi^*(\partial_\sigma\Phi) - \Phi(\partial_\sigma\Phi^*)\big)\right) \\ &= \frac{\sqrt{4hk}}{2\pi}\int d\tau\, d\sigma\left(1 - \frac{1}{2}|\partial_\tau\Phi|^2 + \frac{1}{2}\frac{k}{4h}\Big(A^2|\Phi|^2 + |\partial_\sigma\Phi|^2 + iA\big(\Phi^*(\partial_\sigma\Phi) - \Phi(\partial_\sigma\Phi^*)\big)\Big)\right) \end{aligned} \tag{323}$$

Also, we can use the identity, $|\partial_\sigma\Phi - iA\Phi|^2 = |\partial_\sigma\Phi|^2 + A^2|\Phi|^2 + iA\left(\Phi^*\partial_\sigma\Phi - \Phi\partial_\sigma\Phi^*\right)$ where $A = \sqrt{\frac{4h}{k}}$. Then, finally, we get,

$$\boxed{S_{NG} = \frac{\sqrt{4hk}}{2\pi}\int d\tau\, d\sigma\left(1 - \frac{1}{2}|\partial_\tau\Phi|^2 + \frac{1}{2}\frac{k}{4h}|\partial_\sigma\Phi - iA\Phi|^2\right)} \tag{324}$$

The simplified action describes a massless charged scalar field Φ on $\mathbb{R}\times S^1$ (cylinder) coupled to a constant background gauge field A. The action contains no term proportional to $|\Phi|^2$ without derivatives. Therefore, the effective mass vanishes, and the transverse fluctuations of the string behave as a massless scalar field. The spatial derivative appears through the covariant derivative $D_\sigma = \partial_\sigma - iA$, which is like the minimal coupling of a complex scalar of unit charge to a gauge field. The 'charge' originates from the NS–NS $B-$field of AdS_3.

5.2 Canonical Quantization of the simplified $NG-$Action

The Lagrangian density is given by,

$$\mathcal{L} = -\frac{kA}{2\pi}\left(1 - \frac{1}{2}|\partial_\tau \Phi|^2 + \frac{1}{2A^2}|\partial_\sigma \Phi - iA\Phi|^2\right) \tag{325}$$

To find the equation of motions of the complex field Φ and Φ^*, we shall vary the simplified NG action with respect to Φ^* and Φ, respectively. The conjugate 4-momentum densities are,

$$\begin{aligned}
\Pi^\tau_{\Phi^*} &= \frac{\partial \mathcal{L}}{\partial(\partial_\tau \Phi^*)} = \frac{kA}{4\pi}(\partial_\tau \Phi) \ , \quad \Pi^\sigma_{\Phi^*} = \frac{\partial \mathcal{L}}{\partial(\partial_\sigma \Phi^*)} = -\frac{k}{4\pi A}(\partial_\sigma \Phi - iA\Phi) \\
\Pi^\tau_{\Phi} &= \frac{\partial \mathcal{L}}{\partial(\partial_\tau \Phi)} = \frac{kA}{4\pi}(\partial_\tau \Phi^*) \ , \quad \Pi^\sigma_{\Phi} = \frac{\partial \mathcal{L}}{\partial(\partial_\sigma \Phi)} = -\frac{k}{4\pi A}(\partial_\sigma \Phi^* + iA\Phi^*)
\end{aligned} \tag{326}$$

$$\text{also,} \quad \frac{\partial \mathcal{L}}{\partial \Phi^*} = -iA\,\Pi^\sigma_{\Phi^*} \qquad \text{and} \qquad \frac{\partial \mathcal{L}}{\partial \Phi} = iA\,\Pi^\sigma_{\Phi} \tag{327}$$

Now the Euler-Lagrange equations of motion are $\dfrac{\partial \mathcal{L}}{\partial \Phi^*} - \partial_\tau \Pi^\tau_{\Phi^*} - \partial_\sigma \Pi^\sigma_{\Phi^*} = 0$,

$$\begin{aligned}
\Longrightarrow \quad & -iA\,\Pi^\sigma_{\Phi^*} - \frac{kA}{4\pi}(\partial^2_\tau \Phi) + \frac{k}{4\pi A}(\partial^2_\sigma \Phi - iA\partial_\sigma \Phi) = 0 \\
\Longrightarrow \quad & \frac{ik}{4\pi}(\partial_\sigma \Phi - iA\Phi) + \frac{kA}{4\pi}(\partial^2_\tau \Phi) - \frac{k}{4\pi A}(\partial^2_\sigma \Phi - iA\partial_\sigma \Phi) = 0 \\
\Longrightarrow \quad & \partial^2_\tau \Phi - \frac{1}{A^2}\partial^2_\sigma \Phi + 2iA\partial_\sigma \Phi + A^2\Phi = 0 \\
\Longrightarrow \quad & \partial^2_\tau \Phi - \frac{1}{A^2}(\partial_\sigma - iA)^2 \Phi = 0
\end{aligned} \tag{328}$$

Upon rescaling the time coordinate to $\tau = A\hat{\tau}$, the equation in terms of $\hat{\tau}$ and σ becomes,

$$\boxed{\partial^2_{\hat{\tau}} \Phi - (\partial_\sigma - iA)^2 \Phi = 0} \tag{329}$$

Redefine the complex field as,

$$\begin{aligned}
\Phi(\hat{\tau},\sigma) &= e^{iA\sigma}\,\psi(\hat{\tau},\sigma) \implies \partial_\sigma \Phi(\hat{\tau},\sigma) = e^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma) + iAe^{iA\sigma}\psi(\hat{\tau},\sigma) \\
\implies (\partial_\sigma - iA)\Phi(\hat{\tau},\sigma) &= e^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma) \\
\implies (\partial_\sigma - iA)^2\Phi(\hat{\tau},\sigma) &= (\partial_\sigma - iA)e^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma) \\
&= \partial_\sigma[e^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma)] - iAe^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma) \\
&= iAe^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma) + e^{iA\sigma}\,\partial^2_\sigma \psi(\hat{\tau},\sigma) - iAe^{iA\sigma}\,\partial_\sigma \psi(\hat{\tau},\sigma) \\
\implies (\partial_\sigma - iA)^2\Phi(\hat{\tau},\sigma) &= e^{iA\sigma}\,\partial^2_\sigma \psi(\hat{\tau},\sigma) \\
\text{and} \quad \partial^2_{\hat{\tau}}\Phi(\hat{\tau},\sigma) &= e^{iA\sigma}\partial^2_{\hat{\tau}}\psi(\hat{\tau},\sigma)
\end{aligned}$$

Thus, in terms of the redefined complex field, the differential equation of motion becomes,

$$\boxed{\left(\partial^2_{\hat{\tau}} - \partial^2_\sigma\right)\psi(\hat{\tau},\sigma) = 0} \tag{330}$$

This is a $1D$ wave equation where the spatial coordinate is $\sigma \sim \sigma + 2\pi$ (periodic) , $\sigma \in [0, 2\pi)$ and $\hat{\tau} \in (-\infty, \infty)$. If we define two coordinates, $\hat{x}^+ = \hat{\tau} + \sigma$ and $\hat{x}^- = \hat{\tau} - \sigma$. In terms of these new variables, the above wave equation can be written as,

$$\partial_{\hat{x}^+}(\partial_{\hat{x}^-}\psi) = \partial_{\hat{x}^-}(\partial_{\hat{x}^+}\psi) = 0 \tag{331}$$

Thus $\partial_{\hat{x}^+}\psi$ depends only on $\hat{x}^+$ and independent of $\hat{x}^-$. Similarly, $\partial_{\hat{x}^-}\psi$ depends only on $\hat{x}^-$ and independent of $\hat{x}^+$. Thus, the most general solution (without assuming periodicity) is,

$$\psi(\hat{\tau}, \sigma) = f(\hat{\tau} + \sigma) + g(\hat{\tau} - \sigma) \qquad \Longrightarrow \qquad \psi(\hat{x}^+, \hat{x}^-) = f(\hat{x}^+) + g(\hat{x}^-) \tag{332}$$

Now, we get the full solution of $\Phi = e^{iA\sigma}\psi$ as,

$$\Phi(\hat{\tau}, \sigma) = e^{iA\sigma}\big(f(\hat{\tau} + \sigma) + g(\hat{\tau} - \sigma)\big) \tag{333}$$

Now we impose the periodic boundary condition of $\Phi(\hat{\tau}, \sigma + 2\pi) = \Phi(\hat{\tau}, \sigma)$. Thus,

$$\begin{aligned} &e^{iA(\sigma+2\pi)}\big(f(\hat{\tau} + \sigma + 2\pi) + g(\hat{\tau} - \sigma - 2\pi)\big) = e^{iA\sigma}\big(f(\hat{\tau} + \sigma) + g(\hat{\tau} - \sigma)\big) \\ &\Longrightarrow \quad e^{2i\pi A} f(\hat{x}^+ + 2\pi) + e^{2i\pi A} g(\hat{x}^- - 2\pi) = f(\hat{x}^+) + g(\hat{x}^-) \end{aligned} \tag{334}$$

Thus individually the functions $f(x^+)$ and $g(\hat{x}^-)$ should follow,

$$f(\hat{x}^+ + 2\pi) = e^{-2i\pi A} f(\hat{x}^+) \quad \text{and} \quad g(\hat{x}^- - 2\pi) = e^{-2i\pi A} g(\hat{x}^-) \tag{335}$$

Thus, the functions $f(\hat{x}^+)$ and $g(\hat{x}^-)$ are periodic up to a phase factor (quasi-periodicity). For this kind of periodicity, these functions can be expanded in terms of Fourier modes as,

$$f(\hat{x}^+) = \sum_{n\in\mathbb{Z}} \frac{a_n^\dagger}{n - A}\, e^{i(n-A)\hat{x}^+} \qquad \text{and} \qquad g(\hat{x}^-) = \sum_{n\in\mathbb{Z}} \frac{b_n}{n - A}\, e^{-i(n-A)\hat{x}^-} \tag{336}$$

Then the modes $a_n^\dagger$ and b_n can be extracted as follows,

$$\begin{aligned} f(\hat{x}^+) e^{-i(m-A)\sigma} &= \sum_{n\in\mathbb{Z}} \frac{a_n^\dagger}{n - A}\, e^{i(n-A)\hat{x}^+} e^{-i(m-A)\sigma} = \sum_{n\in\mathbb{Z}} \frac{a_n^\dagger}{n - A}\, e^{i(n-A)\hat{\tau}} e^{i(n-m)\sigma} \\ \Longrightarrow \quad \frac{1}{2\pi}\int_0^{2\pi} d\sigma\, f(\hat{x}^+) e^{-i(m-A)\sigma} &= \sum_{n\in\mathbb{Z}} \frac{a_n^\dagger}{n - A}\, e^{i(n-A)\hat{\tau}} \delta_{mn} = \frac{a_m^\dagger}{m - A}\, e^{i(m-A)\hat{\tau}} \end{aligned} \tag{337}$$

$$\Longrightarrow \boxed{a_n^\dagger = (n - A)\int_0^{2\pi} \frac{d\sigma}{2\pi}\, f(\hat{x}^+) e^{-i(n-A)\hat{x}^+}} \tag{338}$$

In the same way, we get b_n as, $\boxed{b_n = (n - A)\int_0^{2\pi} \frac{d\sigma}{2\pi}\, g(\hat{x}^-) e^{i(n-A)\hat{x}^-}}$ (339)

Thus, the full solution of Φ can be written as,

$$\boxed{\begin{aligned}\Phi(\tau,\sigma) &= e^{iA\sigma}\sum_{n\in\mathbb{Z}}\Big(\frac{a_n^\dagger\, e^{i(n-A)\hat{x}^+} + b_n\, e^{-i(n-A)\hat{x}^-}}{n-A}\Big)\\ \Phi^*(\tau,\sigma) &= e^{-iA\sigma}\sum_{n\in\mathbb{Z}}\Big(\frac{a_n\, e^{-i(n-A)\hat{x}^+} + b_n^\dagger\, e^{i(n-A)\hat{x}^-}}{n-A}\Big)\end{aligned}} \tag{340}$$

where $\hat{x}^+ = \hat{\tau}+\sigma \;\;,\quad \hat{x}^- = \hat{\tau}-\sigma \quad \text{with} \quad \hat{\tau} = \dfrac{\tau}{A}$. Then, the momentum density is given by,

$$\begin{aligned}\Pi_\Phi^\tau &= \frac{kA}{4\pi}(\partial_\tau\Phi^*) = \frac{kA}{4\pi}\left(\frac{\partial\hat{\tau}}{\partial\tau}\right)(\partial_{\hat{\tau}}\Phi^*)\\ &= \frac{k}{4\pi}e^{-iA\sigma}\sum_{n\in\mathbb{Z}}\Big(\frac{a_n\,(-i(n-A))e^{-i(n-A)\hat{x}^+} + b_n^\dagger(i(n-A))\, e^{i(n-A)\hat{x}^-}}{n-A}\Big)\end{aligned} \tag{341}$$

$$\boxed{\begin{aligned}\Longrightarrow\; \Pi_\Phi^\tau(\tau,\sigma) &= \frac{k}{4\pi i}e^{-iA\sigma}\sum_{n\in\mathbb{Z}}\Big(a_n\, e^{-i(n-A)\hat{x}^+} - b_n^\dagger\, e^{i(n-A)\hat{x}^-}\Big)\\ \Longrightarrow\; \Pi_{\Phi^*}^\tau(\tau,\sigma) &= -\frac{k}{4\pi i}e^{iA\sigma}\sum_{n\in\mathbb{Z}}\Big(a_n^\dagger\, e^{i(n-A)\hat{x}^+} - b_n\, e^{-i(n-A)\hat{x}^-}\Big)\end{aligned}} \tag{342}$$

Now we can find a useful relation to write the Fourier modes in terms of the fields. Consider the mode expansion of $\Phi(\tau,\sigma)$. It can be written as,

$$\begin{aligned}\Phi(\tau,\sigma) = e^{iA\sigma}\sum_{n\in\mathbb{Z}}\Big(\frac{a_n^\dagger\, e^{i(n-A)\hat{x}^+} + b_n\, e^{-i(n-A)\hat{x}^-}}{n-A}\Big) &= \sum_{n\in\mathbb{Z}}\frac{e^{in\sigma}}{n-A}\Big(a_n^\dagger\, e^{i(n-A)\hat{\tau}} + b_n\, e^{-i(n-A)\hat{\tau}}\Big)\\ \Longrightarrow \int_0^{2\pi}\frac{d\sigma}{2\pi}\,\Phi(\tau,\sigma)e^{-in\sigma} &= \frac{1}{n-A}\Big(a_n^\dagger\, e^{i(n-A)\hat{\tau}} + b_n\, e^{-i(n-A)\hat{\tau}}\Big)\\ \Longrightarrow \int_0^{2\pi}\frac{d\sigma}{2\pi}\,(n-A)\Phi(\tau,\sigma)e^{-in\sigma} &= a_n^\dagger\, e^{i(n-A)\hat{\tau}} + b_n\, e^{-i(n-A)\hat{\tau}}\\ \Longrightarrow \int_0^{2\pi}\frac{d\sigma}{2\pi}\,(n-A)\Phi^*(\tau,\sigma)e^{in\sigma} &= a_n\, e^{-i(n-A)\hat{\tau}} + b_n^\dagger\, e^{i(n-A)\hat{\tau}}\end{aligned} \tag{343}$$

Now, consider the mode expansion of the momentum density fields.

$$\begin{aligned}\Pi_\Phi^\tau(\tau,\sigma) &= \frac{k}{4\pi i}e^{-iA\sigma}\sum_{n\in\mathbb{Z}}\Big(a_n\, e^{-i(n-A)\hat{x}^+} - b_n^\dagger\, e^{i(n-A)\hat{x}^-}\Big)\\ \Longrightarrow\; \frac{4\pi i}{k}\,\Pi_\Phi^\tau(\tau,\sigma) &= \sum_{n\in\mathbb{Z}}\Big(a_n\, e^{-i(n-A)\hat{\tau}} - b_n^\dagger\, e^{i(n-A)\hat{\tau}}\Big)e^{-in\sigma}\\ \Longrightarrow\; \frac{4\pi i}{k}\int_0^{2\pi}\frac{d\sigma}{2\pi}\,\Pi_\Phi^\tau(\tau,\sigma)\, e^{in\sigma} &= a_n\, e^{-i(n-A)\hat{\tau}} - b_n^\dagger\, e^{i(n-A)\hat{\tau}}\\ \Longrightarrow\; -\frac{4\pi i}{k}\int_0^{2\pi}\frac{d\sigma}{2\pi}\,\Pi_{\Phi^*}^\tau(\tau,\sigma)\, e^{-in\sigma} &= a_n^\dagger\, e^{i(n-A)\hat{\tau}} - b_n\, e^{-i(n-A)\hat{\tau}}\end{aligned} \tag{344}$$

Hence, we can find the explicit formula for the Fourier modes as,

$$\boxed{\begin{aligned} a_n &= \int_0^{2\pi} \frac{d\sigma}{4\pi}\Big((n-A)\Phi^*(\tau,\sigma) + \frac{4\pi i}{k}\,\Pi^\tau_\Phi(\tau,\sigma)\Big)e^{i(n-A)\hat{x}^+}e^{iA\sigma} \\ a_n^\dagger &= \int_0^{2\pi} \frac{d\sigma}{4\pi}\Big((n-A)\Phi(\tau,\sigma) - \frac{4\pi i}{k}\,\Pi^\tau_{\Phi^*}(\tau,\sigma)\Big)e^{-i(n-A)\hat{x}^+}e^{-iA\sigma} \\ b_n &= \int_0^{2\pi} \frac{d\sigma}{4\pi}\Big((n-A)\Phi(\tau,\sigma) + \frac{4\pi i}{k}\,\Pi^\tau_{\Phi^*}(\tau,\sigma)\Big)e^{i(n-A)\hat{x}^-}e^{-iA\sigma} \\ b_n^\dagger &= \int_0^{2\pi} \frac{d\sigma}{4\pi}\Big((n-A)\Phi^*(\tau,\sigma) - \frac{4\pi i}{k}\,\Pi^\tau_\Phi(\tau,\sigma)\Big)e^{-i(n-A)\hat{x}^-}e^{iA\sigma} \end{aligned}} \tag{345}$$

Now, to achieve quantization, we shall promote the fields to operators and impose the equal-time commutation relations between the fields Φ and Φ^* and their canonical conjugate momentum density fields Π^τ_Φ and $\Pi^\tau_{\Phi^*}$ respectively.

$$\boxed{[\Phi(\tau,\sigma),\Pi^\tau_\Phi(\tau,\sigma')] = i\,\delta(\sigma-\sigma') \qquad \text{and} \qquad [\Phi^*(\tau,\sigma),\Pi^\tau_{\Phi^*}(\tau,\sigma')] = i\,\delta(\sigma-\sigma')} \tag{346}$$

And all other combinations of commutation relations are zero. Following the above canonical commutation relations, we can find the commutation relations between the modes.

$$\begin{aligned} [a_n, a_m^\dagger] &= \int_0^{2\pi}\frac{d\sigma}{4\pi}\int_0^{2\pi}\frac{d\sigma'}{4\pi} e^{i(n-A)\hat{x}^+}e^{iA\sigma}e^{-i(m-A)\hat{x}^{+\prime}}e^{-iA\sigma'} \\ &\qquad \times\left[(n-A)\Phi^*(\tau,\sigma) + \frac{4\pi i}{k}\,\Pi^\tau_\Phi(\tau,\sigma)\ ,\ (m-A)\Phi(\tau,\sigma') - \frac{4\pi i}{k}\,\Pi^\tau_{\Phi^*}(\tau,\sigma')\right] \\ &= \int_0^{2\pi}\frac{d\sigma}{4\pi}\int_0^{2\pi}\frac{d\sigma'}{4\pi} e^{i(n-m)\hat{\tau}}e^{i(n\sigma-m\sigma')}\Bigg(\Big[(n-A)\Phi^*(\tau,\sigma)\ ,\ -\frac{4\pi i}{k}\,\Pi^\tau_{\Phi^*}(\tau,\sigma')\Big] \\ &\qquad + \Big[\frac{4\pi i}{k}\,\Pi^\tau_\Phi(\tau,\sigma)\ ,\ (m-A)\Phi(\tau,\sigma')\Big]\Bigg) \\ &= \int_0^{2\pi}\frac{d\sigma}{4\pi}\int_0^{2\pi}\frac{d\sigma'}{4\pi} e^{i(n-m)\hat{\tau}}e^{i(n\sigma-m\sigma')}(n-A+m-A)\,\frac{4\pi i}{k}(-i)\delta(\sigma-\sigma') \\ &= \int_0^{2\pi}\frac{d\sigma}{4\pi}\,\frac{1}{4\pi}e^{i(n-m)\hat{\tau}}e^{i(n-m)\sigma}(n-A+m-A)\frac{4\pi}{k} \\ &= \frac{1}{2}\,\frac{1}{4\pi}e^{i(n-m)\hat{\tau}}\delta_{nm}(n-A+m-A)\frac{4\pi}{k} \end{aligned} \tag{347}$$

Thus, the above commutation is non-zero only when $m=n$. But then, the factors $e^{i(n-m)\hat{\tau}} \to 1$ and $(n-A+m-A) \to 2(n-A)$. Thus,

$$[a_n, a_m^\dagger] = \frac{1}{k}\delta_{nm}(n-A) \tag{348}$$

Now, had we redefined the Lagrangian by an overall factor and neglected the overall constant factor k, then we would similarly get all the commutation relations as,

$$\boxed{[a_n, a_m^\dagger] = (n-A)\delta_{nm} \qquad \text{and} \qquad [b_n, b_m^\dagger] = (n-A)\delta_{nm}} \tag{349}$$

All other commutation relations between the Fourier modes will be zero. Here, the coefficient $(n-A)$ plays the role of the norm of the one–particle state. Acting on a would-be vacuum $|0\rangle$,

$$\langle 0|\, a_n a_n^\dagger \,|0\rangle = (n-A)\langle 0|0\rangle \tag{350}$$

To ensure a positive-definite Hilbert space, operators whose commutator coefficient is positive must be treated as annihilation operators, while those with a negative coefficient must be treated as creation operators. Hence:

- For $n > A$, we have $n - A > 0$, so a_n and b_n annihilate the vacuum.
- For $n < A$, we have $n - A < 0$, so $a_n^\dagger$ and $b_n^\dagger$ annihilate the vacuum.

Therefore, the vacuum is defined by,

$$\boxed{a_n|0\rangle = b_n|0\rangle = 0 \quad (n > A), \qquad a_n^\dagger|0\rangle = b_n^\dagger|0\rangle = 0 \quad (n < A).} \tag{351}$$

This shifted notion of creation and annihilation operators directly gives rise to spectral flow, where the integer $w = \lfloor A \rfloor$ shifts the Fermi/Bose sea level. Hence, the vacuum is not the standard highest-weight state of the unflowed current algebra.

5.3 Currents in terms of Fourier Modes

Consider the exact classical right–moving current of the $\mathrm{SL}(2,\mathbb{R})$ WZW model (with $\alpha' = 2$).

$$J_R^{\pm} = k\Big[(\partial_+\rho) \pm i(\partial_+ v)\sinh(2\rho)\Big]\, e^{\mp 2iu}, \qquad \text{where} \quad u = \frac{t+\phi}{2}\,, \qquad v = \frac{t-\phi}{2}. \tag{352}$$

Near the center of AdS_3, $\rho \ll 1 \implies \sinh(2\rho) = 2\rho + \mathcal{O}(\rho^3)$. Also, $2(\partial_+ v) = (\partial_+ t) - (\partial_+\phi)$. Thus,

$$J_R^{\pm} = k\left[(\partial_+\rho) \pm i\rho\left((\partial_+ t) - (\partial_+\phi)\right)\right] e^{\mp i(t+\phi)} + \mathcal{O}(\rho^3) \approx k\left[(\partial_+\rho)e^{\mp i\phi} \pm i\rho e^{\mp i\phi}\left((\partial_+ t) - (\partial_+\phi)\right)\right] e^{\mp it}$$

$$\implies J_R^{\pm} = k\left[(\partial_+\rho)e^{\mp i\phi} + \rho(\partial_+ e^{\mp i\phi}) \pm i\rho(\partial_+ t)e^{\mp i\phi}\right] e^{\mp it} = k\left[\partial_+(\rho e^{\mp i\phi}) \pm i\rho(\partial_+ t)e^{\mp i\phi}\right] e^{\mp it}$$

In terms of $\Phi = \rho e^{i\phi}$ and $\Phi^* = \rho e^{-i\phi}$ we get,

$$\boxed{J_R^{+} = k\left[(\partial_+\Phi^*) + i\Phi^*(\partial_+ t)\right] e^{-it}} \qquad \text{and} \qquad \boxed{J_R^{-} = k\left[(\partial_+\Phi) - i\Phi(\partial_+ t)\right] e^{it}} \tag{353}$$

Now, the left currents are given by,

$$J_L^{\pm} = k\Big[(\partial_-\rho) \pm i(\partial_- u)\sinh(2\rho)\Big]e^{\mp 2iv} \tag{354}$$

Near the center of AdS_3, $\sinh(2\rho) = 2\rho + \mathcal{O}(\rho^3)$. Also, $2(\partial_- u) = (\partial_- t) + (\partial_-\phi)$. Thus,

$$J_L^{\pm} = k\left[(\partial_-\rho) \pm i\rho\left((\partial_- t) + (\partial_-\phi)\right)\right]e^{\mp i(t-\phi)} + \mathcal{O}(\rho^3) \approx k\left[(\partial_-\rho)e^{\pm i\phi} \pm i\rho e^{\pm i\phi}\left((\partial_- t) + (\partial_-\phi)\right)\right]e^{\mp it}$$

$$\implies J_L^{\pm} = k\left[(\partial_-\rho e^{\pm i\phi}) \pm i\rho e^{\pm i\phi}\left(\partial_- t\right)\right]e^{\mp it}$$

In terms of $\Phi = \rho e^{i\phi}$ and $\Phi^* = \rho e^{-i\phi}$ we get,

$$\boxed{J_L^{+} = k\left[(\partial_-\Phi) + i\Phi\left(\partial_- t\right)\right]e^{-it}} \qquad \text{and} \qquad \boxed{J_L^{-} = k\left[(\partial_-\Phi^*) - i\Phi^*\left(\partial_- t\right)\right]e^{it}} \tag{355}$$

Now we shall try to derive the mode expansion of $J_{R,L}^{\pm}$. Firstly, we know that $x^{\pm} = \tau \pm \sigma$ and $\hat{x}^{\pm} = \hat{\tau} \pm \sigma$ where $\tau = A\hat{\tau}$. So these two coordinates are related to each other by,

$$x^{\pm} = \left(\frac{A \pm 1}{2}\right)\hat{x}^{+} + \left(\frac{A \mp 1}{2}\right)\hat{x}^{-} \tag{356}$$

Define

$$\alpha = \frac{A+1}{2}, \qquad \beta = \frac{A-1}{2}, \qquad \alpha^2 - \beta^2 = A.$$

By chain rule: $$\partial_+ = \frac{1}{A}\left(\alpha\,\partial_{\hat{x}^+} - \beta\,\partial_{\hat{x}^-}\right), \qquad \partial_- = \frac{1}{A}\left(-\beta\,\partial_{\hat{x}^+} + \alpha\,\partial_{\hat{x}^-}\right).$$

Also, in static coordinates, $\partial_+ t = \partial_- t = \dfrac{1}{2}$. Now, consider the mode expansion of Φ and Φ^*.

$$\Phi(\tau,\sigma) = e^{iA\sigma}\sum_{n\in\mathbb{Z}}\left(\frac{a_n^\dagger\, e^{i(n-A)\hat{x}^+} + b_n\, e^{-i(n-A)\hat{x}^-}}{n-A}\right)$$
$$\Phi^*(\tau,\sigma) = e^{-iA\sigma}\sum_{n\in\mathbb{Z}}\left(\frac{a_n\, e^{-i(n-A)\hat{x}^+} + b_n^\dagger\, e^{i(n-A)\hat{x}^-}}{n-A}\right)$$

Define

$$S(\hat{x}^+,\hat{x}^-) = \sum_{n\in\mathbb{Z}}\left(\frac{a_n\, e^{-i(n-A)\hat{x}^+} + b_n^\dagger\, e^{i(n-A)\hat{x}^-}}{n-A}\right) \qquad \implies \qquad \Phi^* = e^{-iA\sigma}S.$$

Now, we can calculate,

$$\partial_{\hat{x}^+}e^{-iA\sigma} = -\frac{iA}{2}e^{-iA\sigma}, \qquad \partial_{\hat{x}^-}e^{-iA\sigma} = +\frac{iA}{2}e^{-iA\sigma}.$$

$$\partial_{\hat{x}^+}S = -i\sum_n a_n e^{-i(n-A)\hat{x}^+}, \qquad \partial_{\hat{x}^-}S = +i\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}.$$

$$\partial_{\hat{x}^+}\Phi^* = e^{-iA\sigma}\left(-\frac{iA}{2}S - i\sum_n a_n e^{-i(n-A)\hat{x}^+}\right) \quad \text{and} \quad \partial_{\hat{x}^-}\Phi^* = e^{-iA\sigma}\left(+\frac{iA}{2}S + i\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right)$$

$$\implies \quad \partial_+\Phi^* = -ie^{-iA\sigma}\left(\frac{A}{2}S + \frac{\alpha}{A}\sum_n a_n e^{-i(n-A)\hat{x}^+} + \frac{\beta}{A}\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right) \tag{357}$$

The right current component is $J_R^+ = k\left[(\partial_+\Phi^*) + i\Phi^*(\partial_+ t)\right]e^{-it} = k\left(\partial_+\Phi^* + \frac{i}{2}\Phi^*\right)e^{-it}$. Then, on substituting the mode expansions, we get,

$$J_R^+ = -ik\, e^{-it}e^{-iA\sigma}\left(\frac{A-1}{2}S + \frac{\alpha}{A}\sum_n a_n e^{-i(n-A)\hat{x}^+} + \frac{\beta}{A}\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right) \tag{358}$$

Now, we use, $e^{-it}e^{-iA\sigma} = e^{-iA\hat{x}^+} \implies e^{-it}e^{-iA\sigma}e^{-i(n-A)\hat{x}^+} = e^{-in\hat{x}^+}$ and $e^{-it}e^{-iA\sigma}e^{i(n-A)\hat{x}^-} = e^{-iA\hat{x}^+ + i(n-A)\hat{x}^-} = e^{i(n-A)\hat{\tau}}e^{-in\sigma} = e^{in\hat{x}^-}e^{-2it}$. Then, the final expression for the current is,

$$\boxed{J_R^+(\hat{x}^+,\hat{x}^-) = -ik\sum_{n\in\mathbb{Z}}\left[\left(\frac{(A+1)n-2A}{2A(n-A)}\right)a_n e^{-in\hat{x}^+} + \left(\frac{(A-1)n}{2A(n-A)}\right)b_n^\dagger e^{in\hat{x}^-}e^{-2it}\right]} \tag{359}$$

Had we defined the chiral currents as

$$J_R^+(\hat{x}^+) = k\left[(\partial_{\hat{x}^+}\Phi^*) + i\Phi^*(\partial_+ t)\right]e^{-it} \tag{360}$$

Then, to find its modes, all that we have to do is replace $A = 1$ in the above derivation (note that we are not fixing the gauge field A, we are just defining new currents in this way). Thus, we would get the mode expansion to be,

$$J_R^+(\hat{x}^+) = -ik\sum_{n\in\mathbb{Z}} a_n e^{-in\hat{x}^+} \qquad \implies \qquad \boxed{J_{n_R}^+ \sim -ik\ a_n} \tag{361}$$

Similarly, we can define $J_R^- = k\left[(\partial_{\hat{x}^+}\Phi) - i\Phi(\partial_+ t)\right]e^{+it}$ which is nothing but the hermitian conjugate of J_R^+. Then,

$$J_R^-(\hat{x}^+) = ik\sum_{n\in\mathbb{Z}} a_n^\dagger e^{in\hat{x}^+} = ik\sum_{n\in\mathbb{Z}} a_{-n}^\dagger e^{-in\hat{x}^+} \qquad \implies \qquad \boxed{J_{n_R}^- \sim ik\ a_{-n}^\dagger} \tag{362}$$

Now, let us calculate the left current modes. Using the previous results, we have,

$$\partial_{\hat{x}^+}\Phi^* = e^{-iA\sigma}\left(-\frac{iA}{2}S - i\sum_n a_n e^{-i(n-A)\hat{x}^+}\right) \quad \text{and} \quad \partial_{\hat{x}^-}\Phi^* = e^{-iA\sigma}\left(+\frac{iA}{2}S + i\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right)$$

$$\implies \partial_-\Phi^* = \frac{1}{A}e^{-iA\sigma}\left[-\beta\left(-\frac{iA}{2}S - i\sum_n a_n e^{-i(n-A)\hat{x}^+}\right) + \alpha\left(+\frac{iA}{2}S + i\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right)\right]$$

$$\implies \partial_-\Phi^* = i\, e^{-iA\sigma}\left[\frac{A}{2}S + \frac{\beta}{A}\sum_n a_n e^{-i(n-A)\hat{x}^+} + \frac{\alpha}{A}\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right] \tag{363}$$

Consider the following left moving current component given by $J_L^- = k\left[(\partial_-\Phi^*) - i\Phi^*(\partial_- t)\right] e^{+it}$, where $\partial_- t = \frac{1}{2}$. On substituting,

$$\begin{aligned}
J_L^- &= ke^{it}\left[i\, e^{-iA\sigma}\left(\frac{A}{2}S + \frac{\beta}{A}\sum_n a_n e^{-i(n-A)\hat{x}^+} + \frac{\alpha}{A}\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right) - i\frac{1}{2}e^{-iA\sigma}S\right] \\
&= ik\, e^{it}e^{-iA\sigma}\left[\frac{A-1}{2}S + \frac{\beta}{A}\sum_n a_n e^{-i(n-A)\hat{x}^+} + \frac{\alpha}{A}\sum_n b_n^\dagger e^{i(n-A)\hat{x}^-}\right]
\end{aligned}$$

Now, we use, $e^{it}e^{-iA\sigma} = e^{iA\hat{x}^-} \implies e^{it}e^{-iA\sigma}e^{-i(n-A)\hat{x}^+} = e^{iA\hat{x}^-}e^{iA\hat{x}^+}e^{-in\hat{x}^+} = e^{2it}e^{-in\hat{x}^+}$ and $e^{it}e^{-iA\sigma}e^{i(n-A)\hat{x}^-} = e^{in\hat{x}^-}$. Then, the final expression for the current is,

$$\boxed{J_L^-(\hat{x}^+,\hat{x}^-) = ik\sum_{n\in\mathbb{Z}}\left[\left(\frac{(A+1)n-2A}{2A(n-A)}\right) b_n^\dagger\, e^{in\hat{x}^-} + \left(\frac{(A-1)n}{2A(n-A)}\right) a_n\, e^{iA\hat{x}^- - i(n-A)\hat{x}^+}\right]} \tag{364}$$

Similarly, J_L^+ is given by the conjugation of J_L^-. Thus,

$$\boxed{J_L^+(\hat{x}^+,\hat{x}^-) = -ik\sum_{n\in\mathbb{Z}}\left[\left(\frac{(A+1)n-2A}{2A(n-A)}\right) b_n e^{-in\hat{x}^-} + \left(\frac{(A-1)n}{2A(n-A)}\right) a_n^\dagger e^{-iA\hat{x}^- + i(n-A)\hat{x}^+}\right]} \tag{365}$$

Had we defined the chiral left currents as,

$$J_L^- = k\left[(\partial_{\hat{x}^-}\Phi^*) - i\Phi^*(\partial_- t)\right] e^{+it} \quad \text{and} \quad J_L^+ = (J_L^-)^\dagger \tag{366}$$

Then, to get its modes, all that we have to do is replace $A = 1$ in the above derivation (note that we are not fixing the gauge field A, we are just defining new currents in this way). Thus, we would get the mode expansions to be,

$$\begin{aligned}
J_L^+(\hat{x}^-) &= -ik\sum_{n\in\mathbb{Z}} b_n e^{-in\hat{x}^-} & &\implies \boxed{J_{n_L}^+ = -ik\, b_n} \\
J_L^-(\hat{x}^-) = ik\sum_{n\in\mathbb{Z}} b_n^\dagger e^{in\hat{x}^-} &= ik\sum_{n\in\mathbb{Z}} b_{-n}^\dagger e^{-in\hat{x}^-} & &\implies \boxed{J_{n_L}^- = ik\, b_{-n}^\dagger}
\end{aligned} \tag{367}$$

5.4 Spectral flow to restore highest weight state

The oscillator algebra implies that $\mathrm{sgn}(n-A)$ determines whether a_n, b_n acts as an annihilation or creation operator. So, the vacuum $|0\rangle$ is defined by $a_n|0\rangle = b_n|0\rangle = 0 \quad (n > A)$ and $a_n^\dagger|0\rangle = b_n^\dagger|0\rangle = 0 \quad (n < A)$. Translating this into current modes gives

$$\begin{aligned} J^+_{n_R}|0\rangle = J^+_{n_L}|0\rangle = 0 \quad (\text{when } \ n > A) \\ \text{and} \quad J^-_{-n_R}|0\rangle = J^-_{-n_L}|0\rangle = 0 \quad (\text{when } \ n < A) \end{aligned} \tag{368}$$

This is not the standard highest-weight condition of an affine algebra, since the annihilation threshold is shifted from the usual $n > 0$. The standard highest-weight condition is $J^+_{n_{R,L}}|0\rangle = 0\ , \forall\, n \geq 1$ and $J^-_{n_{R,L}}|0\rangle = 0\ , \forall\, n \geq 0$. To restore the highest weight state form for the vacuum, let us introduce a new set of current modes $\tilde{J}^\pm_n$ by an integer shift as follows,

$$J^\pm_n = \tilde{J}^\pm_{n\mp w} \ \text{ or } \ \tilde{J}^\pm_n = J^\pm_{n\pm w} \quad \text{where} \quad w = \lfloor A \rfloor \qquad \Longrightarrow \ w \leq A < w+1 \ \ , w \in \mathbb{Z} \tag{369}$$

With this choice, the vacuum conditions become,

$$\begin{aligned} &\tilde{J}^+_n|0\rangle = J^+_{n+w}|0\rangle = 0 \qquad (\text{when } n+w > A \ \text{ or } \ n > A-w \geq 0 \ \text{ or } \ n \geq 1) \\ &\tilde{J}^-_n|0\rangle = J^-_{n-w}|0\rangle = 0 \qquad (\text{when } \ -n+w < A \ \text{ or } \ n > -(A-w) > -1 \ \text{ or } \ n \geq 0) \end{aligned} \tag{370}$$

Thus, the vacuum is the highest-weight state for these new currents, which are nothing but the resultant of *spectral flow*. This also shows that the original vacuum is not the highest-weight state in the unflowed basis, but becomes a regular highest-weight state after spectral flow. If we write the parameter $A = w + \epsilon\ , \quad w \in \mathbb{Z}, \quad 0 \leq \epsilon < 1$. Then, the integer w labels the spectral-flow sector, while ϵ determines the fractional moding of the oscillators. Different values of w correspond to inequivalent representations of the affine $\widehat{\mathfrak{sl}}(2,\mathbb{R})$ algebra.

Semi-classically, the parameter A is related to the internal conformal weight h of the string state by $A = \sqrt{\frac{4h}{k}}$. As h varies continuously, so does A. When A becomes an integer, one of the oscillator frequencies (the mode with $n = A$) vanishes. For $A \in \mathbb{Z}$, the quadratic potential for the mode $n = A$ becomes flat. Classically, this mode costs no energy to excite. Quantum mechanically, this leads to a continuum of states which are δ-function normalizable, rather than a discrete normalizable ground state. If h is increased further, normalizable states reappear, but they belong to the next spectral-flow sector labeled by $w + 1$. The appearance of the continuum corresponds to the emergence of long-string states in AdS_3. The integer w is the amount of spectral flow needed to bring the state into a standard highest-weight representation. It is not identical to geometric winding, but it reorganizes the Fock space in a way that makes the Hilbert space ghost-free and consistent with the no-ghost theorem.

6 Kac-Moody affine Lie algebra and Virasoro algebra

6.1 Deriving the $JJ-$OPE

In this section, we consider the WZW model on an arbitrary group G where we work on the complex worldsheet with coordinates $z = \tau + i\sigma$, $\bar{z} = \tau - i\sigma$. Group element is written as $g(z, \bar{z}) \in G$. In complex coordinates, the currents come in two flavors, holomorphic and anti-holomorphic. They correspond to the symmetry of the WZW action under the transformation $g(z, \bar{z}) \to \Omega(z)\, g(z, \bar{z})\, \bar{\Omega}(\bar{z})^{-1} \quad \Omega(z),\ \bar{\Omega}(\bar{z}) \in G$. The currents are given by,

$$J(z) = -\frac{k}{2}\,(\partial g)\, g^{-1} = \sum_{a,b} \kappa_{ab}\ J^a(z)\, T^a \quad \text{and} \quad \bar{J}(z) = \frac{k}{2}\, g^{-1}(\bar{\partial} g) = \sum_{a,b} \kappa_{ab}\ \bar{J}^a(\bar{z})\, T^a \tag{371}$$

where T^a are the generators of the lie algebra $\mathfrak{g}$, κ_{ab} is the invariant bilinear form defined by $\mathrm{Tr}(T^a T^b) = \frac{\kappa^{ab}}{2}$. And the structure constants are defined by $[T^a, T^b] = i f^{ab}{}_c T^c$.

Consider an infinitesimal left transformation parameterized by a Lie-algebra valued holomorphic test function $\omega(z) = \omega^a(z) T^a$. Thus $\delta_\omega g(z, \bar{z}) = \omega(z)\, g(z, \bar{z})$. Now, the variation of the WZW action under this local transformation (after integration by parts and by standard manipulations (Di Francisco, Sect. 15.1) can be written as a contour functional,

$$\delta_\omega S = -\frac{1}{2\pi i} \oint_C dz\ \sum_a \omega^a(z) J^a(z) \tag{372}$$

For any product of fields $X = \prod_i \Phi_i(z_i, \bar{z}_i)$, the standard path-integral Ward identity gives,

$$\delta_\omega \langle X \rangle = \langle (\delta_\omega S)\, X \rangle = -\frac{1}{2\pi i} \oint_C dz\ \sum_a \omega^a(z)\, \langle J^a(z)\, X \rangle \tag{373}$$

From the variation of the current under an infinitesimal left transformation, one finds,

$$\begin{aligned}
\delta_\omega J &= -\frac{k}{2}\big(\partial(\delta_\omega g)\ g^{-1} + (\partial g)(\delta_\omega g^{-1})\big) \\
&= -\frac{k}{2}\big(\partial(\omega g)\ g^{-1} + (\partial g)(-g^{-1}(\omega g) g^{-1})\big) \\
&= -\frac{k}{2}\omega(\partial g) g^{-1} - \frac{k}{2}(\partial \omega) + \frac{k}{2}(\partial g) g^{-1} \omega \\
&= \omega(z) J(z) - J(z)\omega(z) - \frac{k}{2}\partial\omega(z) \\
\implies \delta_\omega J(z) &= [\omega(z), J(z)] - \frac{k}{2}\partial\omega(z)
\end{aligned} \tag{374}$$

In terms of components,

$$[\omega(z), J(z)] = \kappa_{ab}\,\kappa_{cd}\,\omega^a J^c [T^b, T^d] = \kappa_{ab}\,\kappa_{cd}\, i f^{bd}{}_{e}\,\omega^a J^c T^e = i f_{abc}\,\omega^a J^b T^c \tag{375}$$

$$\begin{aligned}
&\Longrightarrow \delta_\omega J(z) = i f_{abc}\,\omega^a J^b T^c - \frac{k}{2}(\partial\omega)\\
&\Longrightarrow \delta_\omega\big(\kappa_{ac}\, J^a T^c\big) = i f_{abc}\,\omega^a J^b T^c - \frac{k}{2}\kappa_{ac}\big(\partial\omega^a\big)T^c\\
&\Longrightarrow \kappa_{ac}\,\delta_\omega J^a = i f_{abc}\,\omega^a J^b - \frac{k}{2}\kappa_{ac}(\partial\omega^a) \qquad (\text{use } \; f_{abc} = f_{cab})\\
&\Longrightarrow \boxed{\delta_\omega J^a = i f^a{}_{bc}\,\omega^b J^c - \frac{k}{2}(\partial\omega^a)}
\end{aligned} \tag{376}$$

This relation is the infinitesimal version of the current's gauge transformation. In the ward identity, take X to be a single current insertion $J^b(z')$ (the single-current case suffices to determine the JJ OPE). Insert the above transformation into the Ward identity. Concretely, choose ω with a simple pole at z': $\omega^a(z) \sim \dfrac{1}{z-z'}$ and shrink the contour to a small circle around z'. Equating the two sides yields the singular terms of $\langle J^a(z)J^b(z')\rangle$. Matching coefficients, one obtains the OPE relation,

$$\boxed{J^a(z)\,J^b(z') \;\sim\; \frac{\frac{k}{2}\,\kappa^{ab}}{(z-z')^2} \;+\; \frac{i f^{ab}{}_c\, J^c(z')}{z-z'}} \tag{377}$$

This single OPE encodes the quantum affine Kac-Moody algebra. Similarly, for the anti-holomorphic sector, we get,

$$\boxed{\bar{J}^a(\bar{z})\,\bar{J}^b(\bar{z}') \;\sim\; \frac{\frac{k}{2}\,\kappa^{ab}}{(\bar{z}-\bar{z}')^2} \;+\; \frac{i f^{ab}{}_c\, \bar{J}^c(\bar{z}')}{\bar{z}-\bar{z}'}} \tag{378}$$

The $\frac{k}{2}$ in the double pole is chosen so that the Kac-Moody affine Lie algebra matches with the authors of the SL$(2,\mathbb{R})$ WZW Model.

6.2 Sugawara construction of the Energy-Momentum Tensor

Classically, we found that the energy-momentum tensor for the WZW model, by the variation of the action with respect to the metric, is given by $T_{++} = \dfrac{\kappa_{ab}}{k}\, J^a_R J^b_R$. But due to the conformal normal ordering, this might not be true quantum mechanically. We should account for the quantum correction. We shall work in the complex coordinates. Let us assume

$$T(z) = \gamma\,\kappa_{ab} : J^a(z) J^b(z) : \tag{379}$$

We shall fix γ such that the OPE $T(z)\,J^a(z')$ makes the current $J^a(z)$ a tensor operator (primary field) of conformal weight 1.

So, let us first find the OPE $\langle J^a(z)\, T(z')\rangle$, which is easier to compute. Now, we shall represent the normal ordered product by a contour integral. By the standard contour representation of normal ordering,

$$:J^b(z')J^c(z'):= \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\; J^b(x)J^c(z') \tag{380}$$

where x is some complex number and $C_{z'}$ is a contour centered at z'. So,

$$\begin{aligned}
\kappa_{bc}\, J^a(z)\ :J^b(z')J^c(z'):&= \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\,\kappa_{bc}\left[\langle J^a(z)J^b(x)\rangle\, J^c(z') + J^b(x)\,\langle J^a(z)J^c(z')\rangle\right]\\
&= \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\,\kappa_{bc}\Bigg[\left(\frac{\frac{k}{2}\,\kappa^{ab}}{(z-x)^2} + \frac{i f^{ab}{}_{d}\, J^d(x)}{z-x}\right) J^c(z')\\
&\qquad\qquad + J^b(x)\left(\frac{\frac{k}{2}\,\kappa^{ac}}{(z-z')^2} + \frac{i f^{ac}{}_{d}\, J^d(z')}{z-z'}\right)\Bigg]\\
&= \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\Bigg[\left(\frac{\frac{k}{2}\, J^a(z')}{(z-x)^2} + \frac{i f^{a}{}_{cd}\,\langle J^d(x)J^c(z')\rangle}{z-x}\right)\\
&\qquad\qquad + \left(\frac{\frac{k}{2}\, J^a(x)}{(z-z')^2} + \frac{i f^{a}{}_{bd}\,\langle J^b(x)J^d(z')\rangle}{z-z'}\right)\Bigg]\\
&= \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\Bigg[\left(\frac{\frac{k}{2}\, J^a(z')}{(z-x)^2} + \frac{i f^{a}{}_{cd}}{z-x}\left(\frac{\frac{k}{2}\,\kappa^{dc}}{(x-z')^2} + \frac{i f^{dc}{}_{b}\, J^b(z')}{x-z'}\right)\right)\\
&\qquad\qquad + \left(\frac{\frac{k}{2}\, J^a(x)}{(z-z')^2} + \frac{i f^{a}{}_{bd}}{z-z'}\left(\frac{\frac{k}{2}\,\kappa^{bd}}{(x-z')^2} + \frac{i f^{bd}{}_{c}\, J^c(z')}{x-z'}\right)\right)\Bigg]
\end{aligned} \tag{381}$$

Now, $f^{a}{}_{cd}\,\kappa^{dc} = 0 = f^{a}{}_{bd}\kappa^{bd}$ as the structure constant is completely anti-symmetric and invariant form is symmetric. Thus, we get,

$$\begin{aligned}
\kappa_{bc}\, J^a(z)\ :J^b(z')J^c(z'):= \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\Bigg[&\left(\frac{\frac{k}{2}\, J^a(z')}{(z-x)^2} + \frac{i f^{a}{}_{cd}}{z-x}\left(\frac{i f^{dc}{}_{b}\, J^b(z')}{x-z'}\right)\right)\\
&+ \left(\frac{\frac{k}{2}\, J^a(x)}{(z-z')^2} + \frac{i f^{a}{}_{bd}}{z-z'}\left(\frac{i f^{bd}{}_{c}\, J^c(z')}{x-z'}\right)\right)\Bigg]
\end{aligned} \tag{382}$$

We use $-f^{a}{}_{cd}\, f^{dc}{}_{b} = f^{a}{}_{cd}\, f^{cd}{}_{b} = h^\vee \delta^a_b$ and $f^{a}{}_{bd}\, f^{bd}{}_{c} = h^\vee \delta^a_c$, where $h^\vee = -2$ for $\mathfrak{sl}(2,\mathbb{R})$ Lie algebra. Also, other normal-ordered terms vanish as the structure constants are anti-symmetric, whereas the normal-ordered products are symmetric (commutative). So, we have used,

$$f^{a}{}_{cd}(:J^d(z')J^c(z'):) + f^{a}{}_{bd}(:J^b(z')J^d(z'):) = 0 \tag{383}$$

Thus,

$$\begin{aligned}
\kappa_{bc}\, J^a(z)\ :J^b(z')J^c(z'):= &\ \frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\left[\frac{\frac{k}{2}\, J^a(z')}{(z-x)^2} - \frac{h^\vee J^a(z')}{(x-z)(x-z')}\right. \\
&\left. \qquad + \frac{\frac{k}{2}\, J^a(x)}{(z-z')^2} + \frac{h^\vee J^a(z')}{(z-z')(x-z')}\right] \\
= &\ \frac{k\, J^a(z')}{(z-z')^2} + \frac{1}{2\pi i}\oint_{C_{z'}} dx\left[\frac{h^\vee J^a(z')}{(z-z')(x-z')^2} - \frac{h^\vee J^a(z')}{(x-z)(x-z')^2}\right]
\end{aligned} \tag{384}$$

The second term vanishes as the coefficient of $(x-z')^{-2}$ is independent of x. But, in the third term, after the contour integral, we should replace $(x-z)^{-1} \to -(z'-z)^{-2} = -(z-z')^{-2}$

$$\implies \kappa_{bc}\, J^a(z)\ :J^b(z')J^c(z'):= \frac{k\, J^a(z')}{(z-z')^2} + \frac{h^\vee J^a(z')}{(z'-z)^2} = (k+h^\vee)\frac{J^a(z')}{(z-z')^2} \tag{385}$$

Due to the associativity of OPEs, $\kappa_{ab}\,\langle\, (:J^a(z)J^b(z)\ :)\ J^c(z')\,\rangle$ is got exactly same but interchanging $z' \leftrightarrow z$. Thus,

$$\begin{aligned}
\gamma\,\kappa_{ab}\,\langle\, (:J^a(z)J^b(z)\ :)\ J^c(z')\,\rangle = \langle T(z)J^a(z')\rangle &= \gamma(k+h^\vee)\frac{J^a(z)}{(z-z')^2} \\
&= \gamma(k+h^\vee)\ \frac{J^a(z') + (z-z')\partial J^a(z')}{(z-z')^2}
\end{aligned} \tag{386}$$

Thus, to have the current as a primary of weight 1, we shall fix $\gamma = \dfrac{1}{k+h^\vee}$. This gives,

$$\boxed{T(z) = \frac{1}{k+h^\vee}\ \kappa_{ab}\ :J^a(z)J^b(z):} \tag{387}$$

And, the conserved currents follow the OPE,

$$\boxed{\langle T(z)J^a(z')\rangle = \frac{J^a(z')}{(z-z')^2} + \frac{\partial J^a(z')}{(z-z')}} \tag{388}$$

This is the Sugawara construction of the WZW energy-momentum tensor.

6.3 Deriving the $TT-$OPE

Express the second T in the $TT-$OPE as a normal–ordered product of $J's$. Using the contour representation of normal ordering, we write,

$$\langle T(z)T(z')\rangle = \frac{1}{(k+h^\vee)}\frac{1}{2\pi i}\oint_{C_{z'}} \frac{dx}{x-z'}\,\kappa_{ab}\Big[\langle T(z)J^a(x)\rangle J^b(z') + J^a(x)\langle T(z)J^b(z')\rangle\Big] \tag{389}$$

Substituting the $TJ-$OPE, we get,

$$
\begin{aligned}
\langle T(z)T(z')\rangle &\sim \frac{1}{(k+h^\vee)}\frac{1}{2\pi i}\oint_{C_{z'}}\frac{dx}{x-z'}\,\kappa_{ab}\Bigg[\left(\frac{J^a(x)}{(z-x)^2}+\frac{\partial J^a(x)}{z-x}\right)J^b(z') \\
&\qquad\qquad + J^a(x)\left(\frac{J^b(z')}{(z-z')^2}+\frac{\partial J^b(z')}{z-z'}\right)\Bigg] \\
&\sim \frac{1}{(k+h^\vee)}\frac{1}{2\pi i}\oint_{C_{z'}}\frac{dx}{x-z'}\,\kappa_{ab}\Bigg[\frac{\langle J^a(x)J^b(z')\rangle}{(z-x)^2}+\frac{\partial_x\langle J^a(x)J^b(z')\rangle}{z-x} \\
&\qquad\qquad + \frac{\langle J^a(x)J^b(z')\rangle}{(z-z')^2}+\frac{\partial_{z'}\langle J^a(x)J^b(z')\rangle}{z-z'}\Bigg]
\end{aligned}
\tag{390}
$$

After substituting the $JJ-$OPE we get,

$$
\begin{aligned}
\langle T(z)T(z')\rangle \sim \frac{1}{(k+h^\vee)}\frac{1}{2\pi i}\oint_{C_{z'}}\frac{dx}{x-z'}\,\kappa_{ab}\Bigg[&\frac{\frac{k}{2}\kappa^{ab}}{(x-z')^2(z-x)^2} + \frac{if^{ab}{}_c\,J^c(z')}{(x-z')(z-x)^2} + \frac{:J^a(x)J^b(z'):}{(z-x)^2} \\
&+ \frac{k\,\kappa^{ab}}{(x-z)(x-z')^3} + \frac{if^{ab}{}_c\,J^c(z')}{(x-z)(x-z')^2} + \frac{:(\partial_x J^a(x))J^b(z'):}{z-x} \\
&+ \frac{\frac{k}{2}\kappa^{ab}}{(x-z')^2(z-z')^2} + \frac{if^{ab}{}_c\,J^c(z')}{(x-z')(z-z')^2} + \frac{:J^a(x)J^b(z'):}{(z-z')^2} \\
&+ \frac{k\,\kappa^{ab}}{(z-z')(x-z')^3} + \frac{if^{ab}{}_c\,J^c(z')}{(z-z')(x-z')^2} \\
&+ \frac{if^{ab}{}_c\,\partial_{z'}J^a(z')}{(z-z')(x-z')} + \frac{:J^a(x)\,(\partial_{z'}J^b(z')):}{z-z'}\Bigg]
\end{aligned}
\tag{391}
$$

use $\kappa_{ab}\kappa^{ab}=\dim(\mathfrak{g}),\;$ and $\;\kappa_{ab}f^{ab}{}_c=0$

$$
\begin{aligned}
\langle T(z)T(z')\rangle \sim \frac{1}{(k+h^\vee)}\frac{1}{2\pi i}\oint_{C_{z'}}dx\Bigg[&\frac{\frac{k}{2}\dim(\mathfrak{g})}{(x-z')^3(z-x)^2}+\frac{k\dim(\mathfrak{g})}{(x-z)(x-z')^4}+\frac{\frac{k}{2}\dim(\mathfrak{g})}{(x-z')^3(z-z')^2} \\
&+\frac{k\dim(\mathfrak{g})}{(z-z')(x-z')^4}+\frac{\kappa_{ab}:J^a(x)J^b(z'):}{(x-z')(z-x)^2}+\frac{\kappa_{ab}:J^a(x)J^b(z'):}{(x-z')(z-z')^2} \\
&+\frac{\kappa_{ab}:(\partial_x J^a(x))J^b(z'):}{z-x}+\frac{\kappa_{ab}:J^a(x)\,(\partial_{z'}J^b(z')):}{z-z'}\Bigg]
\end{aligned}
\tag{392}
$$

Then, on doing the x integral, the terms that are more than one order singular as $x\to z'$, and have no other x dependence, will vanish.

$$
\begin{aligned}
\implies \langle T(z)T(z')\rangle &\sim \frac{1}{(k+h^\vee)}\Bigg[\frac{(3-2)k\ \dim(\mathfrak{g})}{2(z-z')^4}+2\,\frac{:J^a(z')J^b(z'):}{(z-z')^2}\\
&\qquad\qquad+\frac{:(\partial_{z'}J^a(z'))J^b(z'):}{z-z'}+\frac{:J^a(z')\,(\partial_{z'}J^b(z')):}{z-z'}\Bigg]\\
&\sim\Bigg[\left(\frac{k\ \dim(\mathfrak{g})}{(k+h^\vee)}\right)\frac{1}{2(z-z')^4}+2\,\frac{\kappa_{ab}}{(k+h^\vee)}\frac{:J^a(z')J^b(z'):}{(z-z')^2}\\
&\qquad\qquad+\frac{\kappa_{ab}}{(k+h^\vee)}\frac{\partial_{z'}(:J^a(z')J^b(z'):)}{z-z'}\Bigg]
\end{aligned}
\tag{393}
$$

$$
\implies \boxed{\langle T(z)T(z')\rangle \sim \frac{c/2}{(z-z')^4}+\frac{2\,T(z')}{(z-z')^2}+\frac{\partial_{z'}T(z')}{z-z'} \qquad \text{where} \qquad c=\frac{k\ \dim(\mathfrak{g})}{(k+h^\vee)}}
\tag{394}
$$

For $\mathfrak{sl}(2,\mathbb{R})$, $\dim(\mathfrak{sl}(2,\mathbb{R}))=3$, and $h^\vee=-2$, thus we get,

$$
\boxed{c_{\mathfrak{sl}(2,\mathbb{R})}=\frac{3k}{k-2}}
\tag{395}
$$

Hence, Sugawara construction gave an appropriate stress tensor with conformal weight $h_T=2$.

6.4 Deriving the Kac-Moody current algebra

Since the current $J(z)$ is holomorphic with conformal weight $h_J=1$, it can be written as a Laurent expansion as,

$$
J^a(z)=\sum_{n\in\mathbb{Z}}\frac{J^a_n}{z^{n+1}} \qquad \text{where} \qquad J^a_n=\oint_{C_0}\frac{dz}{2\pi i}\,z^n\,J^a(z)
\tag{396}
$$

where C_0 is any contour encircling the origin. Since the current is holomorphic, the modes are contour independent (except that it has to encircle the origin). Now, we shall try to derive the commutation relation between the current modes. Since the modes are contour independent, we can use the usual contour deformation technique. Consider three concentric contours encircling the origin $c_0^{(1)}, c_0^{(2)}, c_0^{(3)}$ with decreasing radius. Then,

$$
\begin{aligned}
[J^a_m, J^b_n] &= \oint_{c_0^{(1)}}\frac{dz_1}{2\pi i}\oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}\;z_1^m z_2^n\,\langle J^a(z_1)J^b(z_2)\rangle-\oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}\oint_{c_0^{(3)}}\frac{dz_3}{2\pi i}\;z_2^n z_3^m\,\langle J^b(z_2)J^a(z_3)\rangle\\
&=\oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}z_2^n\left(\oint_{c_0^{(1)}}\frac{dz_1}{2\pi i}\;z_1^m\,\langle J^a(z_1)J^b(z_2)\rangle-\oint_{c_0^{(3)}}\frac{dz_3}{2\pi i}\;z_3^m\,\langle J^b(z_2)J^a(z_3)\rangle\right)\\
&=\oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}z_2^n\;\underset{z_1\to z_2}{\operatorname{Res}}\left(z_1^m\langle J^a(z_1)J^b(z_2)\rangle\right)
\end{aligned}
\tag{397}
$$

Use, $\langle J^a(z_1)J^b(z_2)\rangle \sim \dfrac{\frac{k}{2}\,\kappa^{ab}}{(z_1-z_2)^2} + \dfrac{if^{ab}{}_c\,J^c(z_2)}{z_1-z_2}$ and expand $z_1^m = z_2^m + (z_1-z_2)\,m\,z_2^{m-1} + \cdots$. Thus, the relevant substitution is,

$$\begin{aligned} z_1^m\langle J^a(z_1)J^b(z_2)\rangle &\to (z_2^m + (z_1-z_2)\,m\,z_2^{m-1} + \cdots)\Big(\frac{\frac{k}{2}\,\kappa^{ab}}{(z_1-z_2)^2} + \frac{if^{ab}{}_c\,J^c(z_2)}{z_1-z_2}\Big) \\ &\sim \frac{1}{(z_1-z_2)}\Big(m\,z_2^{m-1}\,\frac{k}{2}\,\kappa^{ab} + z_2^m\,if^{ab}{}_c\,J^c(z_2)\Big) \end{aligned} \tag{398}$$

$$\begin{aligned} \implies \qquad [J^a_m, J^b_n] &= \oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}\,z_2^n\Big(m\,z_2^{m-1}\,\frac{k}{2}\,\kappa^{ab} + z_2^m\,if^{ab}{}_c\,J^c(z_2)\Big) \\ &= \frac{k}{2}\,m\,\kappa^{ab}\oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}z_2^{m+n-1} + if^{ab}{}_c\oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}z_2^{(m+n)}\;J^c(z_2) \end{aligned} \tag{399}$$

$$\implies \boxed{[J^a_m, J^b_n] = \frac{k}{2}\,m\,\kappa^{ab}\,\delta_{m+n,0} + if^{ab}{}_c\,J^c_{n+m}}$$

This is the affine *Kac-Moody* current algebra denoted by $\widehat{\mathfrak{g}}_k$ (for the k^{th} level).

6.5 $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ Current algebra

For the $\mathrm{SL}(2,\mathbb{R})$ WZW model, we have two sets of chiral currents $J^a_R(x^+)$ and $J^a_L(x^-)$ which can be expanded in terms of their current modes as,

$$J^a_R(x^+) = \sum_{n\in\mathbb{Z}} J^a_n\,e^{-inx^+} \qquad \text{and} \qquad J^a_L(x^-) = \sum_{n\in\mathbb{Z}} \bar{J}^a_n\,e^{-inx^-} \tag{400}$$

These current modes also follow the above commutation relations. For the left currents, we have the generators $T^1 = \frac{1}{2}\sigma_3$, $T^2 = \frac{1}{2}\sigma_1$, $T^3 = -\frac{i}{2}\sigma_2$, and $T^{\pm} = T^1 \mp iT^2$. Then, the elements of the invariant bilinear form are given by,

$$\kappa^{ab} = 2\,\mathrm{Tr}(T^aT^b) = \mathrm{diag}(1,1,-1)$$

$$\kappa^{a,\pm} = 2\,\mathrm{Tr}(T^aT^{\pm}) = 2\,\mathrm{Tr}(T^a(T^1 \mp iT^2)) = \kappa^{a1} \mp i\kappa^{a2}$$

$$\kappa^{\pm,\pm} = 2\,\mathrm{Tr}(T^{\pm}T^{\pm}) = 2\,\mathrm{Tr}\big((T^1 \mp iT^2)(T^1 \mp iT^2)\big) = 2\,\mathrm{Tr}\big((T^1)^2 - (T^2)^2\big) = \kappa^{11} - \kappa^{22} = 0$$

$$\kappa^{\pm,\mp} = 2\,\mathrm{Tr}(T^{\pm}T^{\mp}) = 2\,\mathrm{Tr}\big((T^1 \mp iT^2)(T^1 \pm iT^2)\big) = 2\,\mathrm{Tr}\big((T^1)^2 + (T^2)^2\big) = \kappa^{11} + \kappa^{22} = 2$$

$$\boxed{\text{For } J_n: \quad \kappa^{ab} = \mathrm{diag}(1,1,-1)\,,\; \kappa^{1,\pm} = 1\,,\; \kappa^{2,\pm} = \mp i\,,\; \kappa^{3,\pm} = 0\,,\; \kappa^{\pm\pm} = 0\,,\; \kappa^{\pm\mp} = 2} \tag{401}$$

To find structure constants, use,

$$[T^1, T^2] = -T^3\;,\; [T^3, T^1] = T^2\;,\; [T^2, T^3] = +T^1$$

$$[T^3, T^{\pm}] = [T^3, T^1] \mp i[T^3, T^2] = T^2 \pm iT^1 = \pm i\big(T^1 \mp iT^2\big) = \pm i\,T^{\pm}$$

$$[T^{\pm}, T^{\mp}] = [T^1 \mp iT^2\;, T^1 \pm iT^2] = \pm 2i[T^1, T^2] = \mp 2iT^3$$

$$\boxed{\text{For } J_{n_L}: \quad f^{12}{}_3 = i \ , \ f^{23}{}_1 = -i \ , \ f^{31}{}_2 = -i \ , \ f^{3,\pm}{}_{\pm} = \pm 1 \ , \ f^{\pm,\mp}{}_3 = \mp 2} \tag{402}$$

Thus, the $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ affine lie algebra is given by,

$$\boxed{\begin{aligned} &[J^1_m, J^1_n] = [J^2_m, J^2_n] = \frac{k}{2} m \delta_{m+n,0} \\ &[J^3_m, J^3_n] = -\frac{k}{2} m \delta_{m+n,0} \\ &[J^1_m, J^2_n] = -J^3_{n+m} \ , \quad [J^2_m, J^3_n] = J^1_{n+m} \ , \quad [J^3_m, J^1_n] = J^2_{n+m} \\ &[J^3_m, J^{\pm}_n] = \pm J^{\pm}_{n+m} \qquad \text{and} \qquad [J^+_m, J^-_n] = k\, m\, \delta_{m+n,0} - 2J^3_{n+m} \end{aligned}} \tag{403}$$

We get the same commutation relations between the modes even for $\bar{J}^a_m$ (left current modes).

6.6 Deriving the Virasoro Algebra

The Sugawara stress tensor and the $TT-$OPE are given by,

$$T(z) = \frac{1}{k+h^\vee} \ \kappa_{ab} \ :J^a(z) J^b(z):$$

$$\langle T(z) T(z') \rangle \sim \frac{c/2}{(z-z')^4} + \frac{2\,T(z')}{(z-z')^2} + \frac{\partial_{z'} T(z')}{z-z'} \qquad \text{where} \qquad c = \frac{k \ \dim(\mathfrak{g})}{(k+h^\vee)}$$

Since $T(z)$ is holomorphic, we can define it in terms of the Laurent modes called the Virasoro Modes L_n defined by,

$$T(z) = \sum_{n\in\mathbb{Z}} \frac{L_n}{z^{n+2}} \qquad \text{where} \qquad L_n = \oint_{C_0} \frac{dz}{2\pi i}\, z^{n+1}\, T(z) \tag{404}$$

where C is any contour enclosing the origin. Consider,

$$\begin{aligned} [L_m, L_n] &= L_m L_n - L_n L_m \\ &= \oint_{C_0^{(1)}} \frac{dz_1}{2\pi i} \oint_{C_0^{(2)}} \frac{dz_2}{2\pi i}\, z_1^{m+1} z_2^{n+1}\, T(z_1)\, T(z_2) - \oint_{C_0^{(2)}} \frac{dz_2}{2\pi i} \oint_{C_0^{(3)}} \frac{dz_3}{2\pi i}\, z_2^{n+1} z_3^{m+1}\, T(z_2) T(z_3) \\ &= \oint_{C_0^{(2)}} \frac{dz_2}{2\pi i}\, z_2^{n+1} \left(\oint_{C_0^{(1)}} \frac{dz_1}{2\pi i}\, z_1^{m+1}\, T(z_1)\, T(z_2) - \oint_{C_0^{(3)}} \frac{dz_3}{2\pi i}\, z_3^{m+1}\, T(z_2) T(z_3) \right) \end{aligned} \tag{405}$$

where $C_0^{(1)}$ encloses $C_0^{(2)}$, and $C_0^{(2)}$ encloses $C_0^{(3)}$. Fix z_2 in the inner integral. To do this, let us first fix z_2 and do the other integrals. There are two poles in the first inner integral. One is at origin $z = 0$ and at other at $z_1 = z_2$ due to the $T(z_1)T(z_2)$ OPE. But in the second inner integral, only $z_3 = 0$ pole contributes as z_2 is outside the $C_0^{(3)}$ contour.

$$
\begin{aligned}
\implies [L_m, L_n] &= \oint_{C_0^{(2)}} \frac{dz_2}{2\pi i}\, z_2^{n+1} \left(L_m + \operatorname*{Res}_{z_1\to z_2} \left[z_1^{m+1} T(z_1) T(z_2) \right] - L_m \right) \\
&= \oint_{C_0^{(2)}} \frac{dz_2}{2\pi i}\, z_2^{n+1} \operatorname*{Res}_{z_1\to z_2} \left[z_1^{m+1} T(z_1) T(z_2) \right]
\end{aligned}
\tag{406}
$$

Expand z_1^{m+1} in terms of z_2 and insert the $T(z_1)T(z_2)$ OPE, and to find the residue, collect the terms proportional only to $(z_1 - z_2)^{-1}$. Thus,

$$
\begin{aligned}
[L_m, L_n] &= \oint_{C_0^{(2)}} \frac{dz_2}{2\pi i}\, z_2^{n+1} \left[\left(z_2^{m+1} (\partial_{z_2} T(z_2)) + 2(m+1)\, z_2^{m}\, T(z_2) + m\,(m^2-1) \frac{c}{12} z_2^{m-2} \right) \right] \\
&= \oint_{C_0} \frac{dz}{2\pi i} \left(z^{m+n+2} (\partial T(z)) + 2(m+1) z^{m+n+1}\, T(z) + m(m^2-1) \frac{c}{12} z^{m+n-1} \right)
\end{aligned}
\tag{407}
$$

Use, $T(z) = \sum_{k\in\mathbb{Z}} \frac{L_k}{z^{k+2}}$ and $\partial T(z) = -\sum_{k\in\mathbb{Z}} \frac{(k+2)L_k}{z^{k+3}}$. Thus,

$$
\begin{aligned}
[L_m, L_n] &= -\sum_{k\in\mathbb{Z}} (k+2) L_k \oint_{C_0} \frac{dz}{2\pi i} \frac{1}{z^{k-m-n+1}} + 2(m+1) \sum_{k\in\mathbb{Z}} L_k \oint_{C_0} \frac{dz}{2\pi i} \frac{1}{z^{k-m-n+1}} \\
&\qquad + \frac{c}{12} m(m^2-1) \oint_{C_0} \frac{dz}{2\pi i} z^{m+n-1} \\
&= -\sum_{k\in\mathbb{Z}} (k+2)\, L_k\, \delta_{k,m+n} + 2(m+1) \sum_{k\in\mathbb{Z}} L_k \delta_{k,m+n} + \frac{c}{12} m(m^2-1)\, \delta_{m,-n} \\
&= -(m+n+2)\, L_{m+n} + 2(m+1) L_{m+n} + m(m^2-1) \frac{c}{12} \delta_{m,-n}
\end{aligned}
\tag{408}
$$

$$
\implies \boxed{[L_m, L_n] = (m-n)\, L_{m+n} + \frac{c}{12} m(m^2-1) \delta_{m,-n}}
\tag{409}
$$

This set of commutation relations forms the Virasoro Algebra.

6.7 Virasoro modes in terms of current modes

The current has the Laurent expansion $J^a(z) = \sum_{n\in\mathbb{Z}} \frac{J_n^a}{z^{n+1}}$. The Sugawara stress tensor is $T(z) = \frac{1}{(k+h^\vee)} \kappa_{ab} : J^a(z) J^b(z) :$. Substitute the current expansion. Therefore,

$T(z) = \frac{1}{(k+h^\vee)} \sum_{m,n} \frac{\kappa_{ab} : J_m^a J_n^b :}{z^{m+n+2}}$ But, by definition, $T(z) = \sum_{k\in\mathbb{Z}} \frac{L_k}{z^{k+2}}$ Matching powers gives $k = m+n$. Thus, $L_k = \frac{1}{(k+h^\vee)} \sum_{m+n=k} \kappa_{ab} : J_m^a J_n^b :$ $\implies$ $\boxed{L_n = \frac{1}{(k+h^\vee)} \sum_{m\in\mathbb{Z}} \kappa_{ab} : J_{n-m}^a J_m^b :}$

Here, the Normal ordering of the modes means $\quad : J^a_m J^b_n := \begin{cases} J^a_m J^b_n, & m < 0 \\ J^b_n J^a_m, & m \geq 0 \end{cases}$ (410)

This ensures that annihilation operators are to the right.

$$\implies L_0 = \frac{1}{(k+h^\vee)} \sum_{m\in\mathbb{Z}} \kappa_{ab} : J^a_{-m} J^b_m : = \frac{1}{(k+h^\vee)} \left(\kappa_{ab}\, J^a_0 J^b_0 + 2 \sum_{m=1}^{\infty} \kappa_{ab}\, J^a_{-m} J^b_m \right) \tag{411}$$

And, for $n \neq 0$, we have,

$$L_n = \frac{\kappa_{ab}}{(k+h^\vee)} \left(: J^a_n J^b_0 : + \sum_{m=1}^{\infty} \left(J^a_{n-m} J^b_m + J^a_{-m} J^b_{n+m} \right) \right) \tag{412}$$

For $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ affine algebra, $h^\vee = -2$, $\kappa_{\pm,\mp} = \frac{1}{2}$, $\kappa_{\pm,\pm} = 0 = \kappa_{3,\pm}$ and $\kappa_{33} = -1$. Thus,

$$\boxed{L_0 = \frac{1}{(k-2)} \left(\frac{1}{2}(J^+_0 J^-_0 + J^-_0 J^+_0) - (J^3_0)^2 + \sum_{m=1}^{\infty} (J^+_{-m} J^-_m + J^-_{-m} J^+_m) - 2\, J^3_{-m} J^3_m \right)} \tag{413}$$

And for $n \neq 0$, we cannot collapse the $m > 0$ and $m < 0$ parts. So,

$$\boxed{\begin{aligned} L_{n\neq 0} = \frac{1}{(k-2)} \Bigg(& \frac{1}{2}(: J^+_n J^-_0 : + : J^-_n J^+_0 :) - : J^3_n J^3_0 : \\ & + \sum_{m=1}^{\infty} \left(\frac{1}{2}\left(J^+_{n-m} J^-_m + J^-_{n-m} J^+_m + J^+_{-m} J^-_{n+m} + J^-_{-m} J^+_{n+m}\right) - J^3_{n-m} J^3_m - J^3_{-m} J^3_{n+m} \right) \Bigg) \end{aligned}} \tag{414}$$

6.8 Commutation relations between Virasoro and Current modes

The Virasoro and current modes are given by,

$$L_m = \oint_{C_0} \frac{dz}{2\pi i}\, z^{m+1}\, T(z) \qquad \text{and} \qquad J^a_n = \oint_{C_0} \frac{dz}{2\pi i}\, z^n\, J^a(z)$$

Consider three concentric contours encircling the origin $c_0^{(1)}, c_0^{(2)}, c_0^{(3)}$ with decreasing radius.

$$\begin{aligned} [L_m, J^a_n] &= \oint_{c_0^{(1)}} \frac{dz_1}{2\pi i} \oint_{c_0^{(2)}} \frac{dz_2}{2\pi i}\, z_1^{m+1} z_2^n \langle T(z_1) J^a(z_2)\rangle - \oint_{c_0^{(2)}} \frac{dz_2}{2\pi i} \oint_{c_0^{(3)}} \frac{dz_3}{2\pi i}\, z_2^n z_3^{m+1} \langle J^a(z_2) T(z_3)\rangle \\ &= \oint_{c_0^{(2)}} \frac{dz_2}{2\pi i} z_2^n \left(\oint_{c_0^{(1)}} \frac{dz_1}{2\pi i}\, z_1^{m+1} \langle T(z_1) J^a(z_2)\rangle - \oint_{c_0^{(3)}} \frac{dz_3}{2\pi i}\, z_3^{m+1} \langle J^a(z_2) T(z_3)\rangle \right) \\ &= \oint_{c_0^{(2)}} \frac{dz_2}{2\pi i} z_2^n \operatorname*{Res}_{z_1\to z_2} \left(z_1^{m+1} \langle T(z_1) J^a(z_2)\rangle \right) \end{aligned} \tag{415}$$

Use, $\langle T(z_1)J^a(z_2)\rangle \sim \dfrac{J^a(z_2)}{(z_1-z_2)^2}+\dfrac{\partial_{z_2}J^a(z_2)}{z_1-z_2}$ and expand $z_1^{m+1}=z_2^{m+1}+(z_1-z_2)\,(m+1)\,z_2^m+\cdots$
Thus the relevant substitution is,

$$\begin{aligned}
z_1^{m+1}\langle T(z_1)J^a(z_2)\rangle &\to (z_2^{m+1}+(z_1-z_2)\,(m+1)\,z_2^m+\cdots)\left(\frac{J^a(z_2)}{(z_1-z_2)^2}+\frac{\partial_{z_2}J^a(z_2)}{z_1-z_2}\right)\\
&\sim \frac{1}{(z_1-z_2)}\Big((m+1)\,z_2^m\,J^a(z_2)+z_2^{m+1}\,\partial_{z_2}J^a(z_2)\Big)
\end{aligned} \tag{416}$$

$$\begin{aligned}
\implies \qquad [L_m,J_n^a] &= \oint_{c_0^{(2)}}\frac{dz_2}{2\pi i}\,z_2^n\Big((m+1)\,z_2^m\,J^a(z_2)+z_2^{m+1}\,\partial_{z_2}J^a(z_2)\Big)\\
&=(m+1)\oint_{c_0}\frac{dz}{2\pi i}z^{m+n}J^a(z)\;+\oint_{c_0}\frac{dz}{2\pi i}z^{m+n+1}\,\partial J^a(z)\\
&=(m+1)\sum_{k\in\mathbb{Z}}J_k^a\oint_{c_0}\frac{dz}{2\pi i}\frac{1}{z^{k-m-n+1}}-\sum_{k\in\mathbb{Z}}J_k^a(k+1)\oint_{c_0}\frac{dz}{2\pi i}\,\frac{1}{z^{k-m-n+1}}\\
&=(m+1)J_{m+n}^a-(m+n+1)J_{m+n}^a\\
\implies &\boxed{[L_m,J_n^a]=-nJ_{n+m}^a}
\end{aligned} \tag{417}$$

Thus, the 0^{th} current mode commutes with all the Virasoro modes. And $L_0J_n^a=J_n^aL_0-nJ_n^a$. So, the current modes $J_n^a\ ,\ n>0$ reduce the eigen value of L_0 by n and $J_{-n}^a\ ,\ n>0$ increase the eigen value of L_0 by n. Then, the complete spectrum generating (Kac-Moody and Virasoro) algebra is given by,

$$\boxed{\begin{aligned}
&[J_m^a,J_n^b]=\frac{k}{2}\,m\,\kappa^{ab}\,\delta_{m+n,0}\;+if^{ab}{}_c\,J_{n+m}^c\\
&[L_m,J_n^a]=-nJ_{n+m}^a\\
&[L_m,L_n]=(m-n)\,L_{m+n}+\frac{c}{12}m(m^2-1)\delta_{m,-n}
\end{aligned}} \tag{418}$$

6.9 Spectral Flow as an automorphism of Current+Virasoro Algebra

The spectral flow is defined as a transformation of the modes as follows,

$$\begin{aligned}
\tilde{J}_n^3 &= J_n^3-\frac{k}{2}w\,\delta_{n,0}\\
\tilde{J}_n^{\pm} &= J_{n\pm w}^{\pm}\\
\tilde{L}_n &= L_n+wJ_n^3-\frac{k}{4}w^2\,\delta_{n,0}
\end{aligned} \tag{419}$$

It can be shown that this transformation of the modes also has the exact same commutation relations as before the spectral flow, hence an automorphism.

$$[\tilde{J}^3_m, \tilde{J}^3_n] = \left[J^3_m - \frac{k}{2}w\,\delta_{m,0}, J^3_n - \frac{k}{2}w\,\delta_{n,0}\right] = [J^3_m, J^3_n] = -\frac{k}{2}m\delta_{m+n,0} \quad\Longrightarrow\quad [\tilde{J}^3_m, \tilde{J}^3_n] = -\frac{k}{2}m\delta_{m+n,0}$$

$$[\tilde{J}^3_m, \tilde{J}^\pm_n] = \left[J^3_m - \frac{k}{2}w\,\delta_{m,0}\;,\; J^\pm_{n\pm w}\right] = [J^3_m, J^\pm_{n\pm w}] = \pm J^\pm_{m+n\pm w} = \pm\tilde{J}^\pm_{m+n} \quad\Longrightarrow\quad [\tilde{J}^3_m, \tilde{J}^\pm_n] = \pm\tilde{J}^\pm_{m+n}$$

$$\begin{aligned}[\tilde{J}^+_m, \tilde{J}^-_n] &= [J^+_{m+w}, J^-_{n-w}] = k(m+w)\delta_{m+n,0} - 2J^3_{m+n} \\ &= km\delta_{m+n,0} + kw\delta_{m+n,0} - 2\left(\tilde{J}^3_{m+n} + \frac{k}{2}w\delta_{m+n,0}\right) \quad\Longrightarrow\quad [\tilde{J}^+_m, \tilde{J}^-_n] = km\delta_{m+n,0} - 2\tilde{J}^3_{n+m}\end{aligned}$$

$$\begin{aligned}[\tilde{L}_m, \tilde{J}^3_n] &= \left[L_m + wJ^3_m - \frac{k}{4}w^2\,\delta_{m,0}\;,\; J^3_n - \frac{k}{2}w\delta_{n,0}\;\right] \\ &= \left[L_m, J^3_n\right] + w\left[J^3_m, J^3_n\right] = -nJ^3_{n+m} + w\left(-\frac{k}{2}m\delta_{m+n,0}\right) \quad\Longrightarrow\quad [\tilde{L}_m, \tilde{J}^3_n] = -n\tilde{J}^3_{m+n} \\ &= -n\Big(J^3_{n+m} - \frac{k}{2}w\delta_{m+n,0}\Big) = -n\tilde{J}^3_{m+n}\end{aligned}$$

$$\begin{aligned}[\tilde{L}_m, \tilde{J}^\pm_n] &= [L_m + wJ^3_m,\ J^\pm_{n\pm w}] = [L_m, J^\pm_{n\pm w}] + w[J^3_m, J^\pm_{n\pm w}] \\ &= -(n\pm w)J^\pm_{m+n\pm w} \pm wJ^\pm_{m+n\pm w} = -(n\pm w)\tilde{J}^\pm_{m+n} \pm w\tilde{J}^\pm_{m+n} \quad\Longrightarrow\quad [\tilde{L}_m, \tilde{J}^\pm_n] = -n\tilde{J}^\pm_{m+n}\end{aligned}$$

For the Virasoro algebra,

$$\begin{aligned}[\tilde{L}_m, \tilde{L}_n] &= [L_m + wJ^3_m - \frac{k}{4}w^2\delta_{m,0}, L_n + wJ^3_n - \frac{k}{4}w^2\delta_{n,0}] \\ &= [L_m, L_n] + w[L_m, J^3_n] + w[J^3_m, L_n] + w^2[J^3_m, J^3_n]\end{aligned} \tag{420}$$

Use the known commutation relations,

$$\Longrightarrow\quad [\tilde{L}_m, \tilde{L}_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0} + w(m-n)J^3_{m+n} - \frac{k}{2}w^2m\delta_{m+n,0} \tag{421}$$

Use

$$L_{m+n} = \tilde{L}_{m+n} - wJ^3_{m+n} + \frac{k}{4}w^2\delta_{m+n,0}$$

$$\begin{aligned}\Longrightarrow\ [\tilde{L}_m, \tilde{L}_n] &= (m-n)\tilde{L}_{m+n} - w(m-n)J^3_{m+n} + (m-n)\frac{k}{4}w^2\delta_{m+n,0} \\ &\qquad + \frac{c}{12}m(m^2-1)\delta_{m+n,0} + w(m-n)J^3_{m+n} - \frac{k}{2}w^2m\delta_{m+n,0} \\ &= (m-n)\tilde{L}_{m+n} + 2m\,\frac{k}{4}w^2\delta_{m+n,0} + \frac{c}{12}m(m^2-1)\delta_{m+n,0} - \frac{k}{2}w^2m\delta_{m+n,0}\end{aligned} \tag{422}$$

$$\Longrightarrow\quad [\tilde{L}_m, \tilde{L}_n] = (m-n)\tilde{L}_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}. \qquad \text{`}c\text{' is unchanged} \tag{423}$$

Thus, spectral flow is an automorphism of the affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ current+Virasoro algebra.

7 Representation Theory of strings in $\mathrm{SL}(2,\mathbb{R})$ WZW Model

The Hilbert space of the WZW model is the space of all quantum states of the conformal field theory. In this theory the natural operators are the currents $J^a_L(x^-)$ and $J^a_R(x^+)$ whose modes are J^a_n and $\bar{J}^a_n$ which generate the affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ current algebra. Physical states are constructed by acting with these current modes on primary states. Therefore, the Hilbert space is organized in terms of representations of the affine current algebra.

A key feature of the $\mathrm{SL}(2,\mathbb{R})$ WZW model is that the Hilbert space is not a single representation of the affine algebra. Instead, it decomposes into distinct sectors labeled by an integer w, called the **spectral flow number**. Schematically, for a particular chiral sector (say right), $\mathcal{H}_{\text{full}} = \bigoplus_{w\in\mathbb{Z}} \mathcal{H}^{(w)}$. Each $\mathcal{H}^{(w)}$ represents a sector obtained by applying spectral flow w times. Moreover, spectral flow is an automorphism of the affine $\mathrm{SL}(2,\mathbb{R})$ algebra as shown previously. So, starting from any representation of the affine algebra, applying spectral flow produces a new representation. The proposal for the full Hilbert space of the WZW model is that it is obtained by taking certain basic representations of the current algebra and including all of their spectral flow images. Thus, every sector of the Hilbert space corresponds to a different amount of spectral flow. In the string theory interpretation on AdS_3, the integer w corresponds to the amount of spectral flow applied to a representation. Physically, w is interpreted as the winding number of strings in AdS_3, and different values of w describe different classes of string states.

Now, the Casimir operator C_2 is constructed only using the zero modes, $C_2 = \kappa_{ab}\, J^a_0\, J^b_0$ and it commutes with all the zero current modes $[C_2, J^a_0] = 0$. Also, the zero current modes commute all the Virasoro modes, $[L_n, J^a_0] = 0$. This also implies that C_2 commutes with all the Virasoro modes $[L_n, C_2] = 0$. Hence, the conformal transformations on the world-sheet generated by L_n modes preserve the global $\mathrm{SL}(2,\mathbb{R})$ representation label. So, we can choose one of the Virasoro modes and one of the zero-th current mode along with C_2 to form a complete set of commuting operators. By convention, it is chosen to be CSCO $= \{L_0\,,\, J^3_0\,,\, C_2\}$. Moreover,

$$L_0 = \frac{C_2}{(k-2)} + \frac{2}{(k-2)} \sum_{n=1}^{\infty} \kappa_{ab}\, J^a_{-n} J^b_n$$

L_0 measures the conformal weight (energy) of a state, whereas C_2 labels a representation.

For now, let us consider a representation in which the eigenvalue of L_0 is bounded below. These are called *Positive energy representations* or equivalently *unflowed representations*. Then there should exist states which give the minimum most eigen value of L_0. These states are called Primary states. Now, the commutator $[L_0, J_n] = -n\, J^a_n$ suggests that, in positive energy representation, the action of $J^a_n\,, n > 0$ on the primary states should result in zero. Hence, we define any state $|\psi\rangle$ to be 'Primary', if and only if $J^a_n|\psi\rangle = 0\,,\ n > 0$. Also, with respect to these primary states, $L_0 \sim \dfrac{C_2}{(k-2)}$. So, knowing C_2 will fix the L_0 eigenvalue immediately.

7.1 Representation of the Zero Modes

The zero modes of the SL$(2,\mathbb{R})$ affine currents follow the global $\mathfrak{sl}(2,\mathbb{R})$ algebra given by, $[J_0^3, J_0^{\pm}] = \pm J_0^{\pm}$ and $[J_0^+, J_0^-] = -2J_0^3$. The $\mathfrak{sl}(2,\mathbb{R})$ quadratic Casimir operator is given by, $C_2 = \kappa_{ab} J_0^a J_0^b = -(J_0^3)^2 + \frac{1}{2}\left(J_0^+ J_0^- + J_0^- J_0^+\right)$. Since $[C_2, J_0^a] = 0$, we shall choose to diagonalize the operators C_2 and J_0^3 simultaneously. We define the simultaneous eigen-states of C_2 and J_0^3 as $|j,m\rangle$, which obey the following eigenvalue equations:

$$\begin{aligned} C_2|j,m\rangle &= -j(j-1)|j,m\rangle \\ J_0^3|j,m\rangle &= m|j,m\rangle \end{aligned} \tag{424}$$

Here, j labels the representation and m labels the SL$(2,\mathbb{R})$ charge inside the representation. Now, consider the action of ladder operators $J_0^{\pm}$ on these states.

$$\begin{aligned} J_0^3(J_0^{\pm}|j,m\rangle) = (J_0^{\pm}J_0^3 + [J_0^3, J_0^{\pm}])|j,m\rangle = (J_0^{\pm}J_0^3 \pm J_0^{\pm})|j,m\rangle \\ \implies J_0^3(J_0^{\pm}|j,m\rangle) = (m \pm 1)(J_0^{\pm}|j,m\rangle) \end{aligned} \tag{425}$$

Thus, $J_0^{\pm}$ changes the m value by one unit $\implies J_0^{\pm}|j,m\rangle \propto |j,m\pm 1\rangle$.

Let $J_0^+|j,m\rangle = A_m|j,m+1\rangle \quad \implies \quad \langle j,m|J_0^- = A_m^*\langle j,m+1|$. Thus,

$$\langle j,m|J_0^- J_0^+|j,m\rangle = |A_m|^2\langle j,m+1|j,m+1\rangle \quad \implies \quad |A_m|^2 = \langle j,m|J_0^- J_0^+|j,m\rangle \tag{426}$$

Similarly, $J_0^-|j,m\rangle = B_m|j,m-1\rangle \quad \implies \quad \langle j,m|J_0^+ = B_m^*\langle j,m-1|$. Thus,

$$\langle j,m|J_0^+ J_0^-|j,m\rangle = |B_m|^2\langle j,m-1|j,m-1\rangle \quad \implies \quad |B_m|^2 = \langle j,m|J_0^+ J_0^-|j,m\rangle \tag{427}$$

Now, take the expectation value of this relation $[J_0^+, J_0^-] = -2J^3$ with respect to $|j,m\rangle$. Thus,

$$\begin{aligned} &\langle j,m|[J_0^+, J_0^-]|j,m\rangle = -2\langle j,m|J^3|j,m\rangle \\ &\implies \langle j,m|J_0^+J_0^-|j,m\rangle - \langle j,m|J_0^-J_0^+|j,m\rangle = -2m \\ &\implies |B_m|^2 - |A_m|^2 = -2m \end{aligned} \tag{428}$$

Now, take the expectation value of the Casimir $C_2 = -(J_0^3)^2 + \frac{1}{2}\left(J_0^+ J_0^- + J_0^- J_0^+\right)$, with respect to $|j,m\rangle$. This gives,

$$\begin{aligned} &\langle j,m|C_2|j,m\rangle = -\langle j,m|(J_0^3)^2|j,m\rangle + \frac{1}{2}\left(\langle j,m|J_0^+J_0^-|j,m\rangle + \langle j,m|J_0^-J_0^+|j,m\rangle\right) \\ &\implies -j(j-1) = -m^2 + \frac{1}{2}\left(|B_m|^2 + |A_m|^2\right) \end{aligned} \tag{429}$$

Thus, we can use the above two results to find $|A_m|^2$ and $|B_m|^2$.

First, substitute $|B_m|^2 = |A_m|^2 - 2m$ in $-j(j-1) = -m^2 + \frac{1}{2}\left(|B_m|^2 + |A_m|^2\right)$,

$$\begin{aligned}
&\implies -j(j-1) = -m^2 + \frac{1}{2}\left(|A_m|^2 - 2m + |A_m|^2\right) = -m^2 + |A_m|^2 - m \\
&\implies |A_m|^2 = m(m+1) - j(j-1) = m^2 - j^2 + m + j = (m+j)(m-j) + (m+j) \\
&\implies |A_m|^2 = (m-j+1)(m+j)
\end{aligned} \tag{430}$$

Similarly, to find $|B_m|^2$, substitute $|A_m|^2 = |B_m|^2 + 2m$ in $-j(j-1) = -m^2 + \frac{1}{2}\left(|B_m|^2 + |A_m|^2\right)$,

$$\begin{aligned}
&\implies -j(j-1) = -m^2 + \frac{1}{2}\left(|B_m|^2 + |B_m|^2 + 2m\right) = -m^2 + |B_m|^2 + m \\
&\implies |B_m|^2 = m(m-1) - j(j-1) = m^2 - m - j^2 + j = (m+j)(m-j) - (m-j) \\
&\implies |B_m|^2 = (m+j-1)(m-j)
\end{aligned} \tag{431}$$

We can choose the coefficients A_m and B_m to be real, and thus we get,

$$\begin{aligned}
J_0^+|j,m\rangle &= \sqrt{(m-j+1)(m+j)}\;|j,m+1\rangle, \\
J_0^-|j,m\rangle &= \sqrt{(m+j-1)(m-j)}\;|j,m-1\rangle.
\end{aligned} \tag{432}$$

These relations generate the full $\mathfrak{sl}(2,\mathbb{R})$ representation. Starting from one state $|j,m\rangle$, repeated action of the ladder operators generates

$$\ldots, |j,m-1\rangle, |j,m\rangle, |j,m+1\rangle, \ldots$$

Thus, the states form a ladder of $\mathrm{SL}(2,\mathbb{R})$ charge m within a representation labelled by j.

$$\boxed{\begin{aligned}
C_2|j,m\rangle &= -j(j-1)|j,m\rangle \\
J_0^3|j,m\rangle &= m|j,m\rangle \\
J_0^+|j,m\rangle &= \sqrt{(m-j+1)(m+j)}\;|j,m+1\rangle \\
J_0^-|j,m\rangle &= \sqrt{(m+j-1)(m-j)}\;|j,m-1\rangle
\end{aligned}} \tag{433}$$

Note that we also have to take account of the L_0 eigenvalues. For each L_0 eigenvalue, we can have all possible states of the form $|j,m\rangle$. So, let us fix the convention that these states, denoted by $|j,m\rangle$, are the lowest weight states. i.e.,

$$L_0|j,m\rangle = \Delta_{\text{min}}^{(j)}|j,m\rangle \quad \implies \quad J_n^3|j,m\rangle = J_n^{\pm}|j,m\rangle = 0 \;, \forall\, n > 0$$

But, with respect to these lowest weight states,

$$L_0 \sim \frac{C_2}{k-2} \qquad \implies \qquad \Delta_{\text{min}}^{(j)} = \frac{-j(j-1)}{k-2} \tag{434}$$

Thus, we have found the representation of the zero current modes in terms of the kets $|j, m\rangle$, which are also called *Affine Primaries*, and have the lowest weight or energy (L_0 eigenvalue) and hence they are the ground states of the world-sheet theory.

All non-primary states in the representation are obtained by acting with the negative modes of the currents J^3_{-n} and $J^{\pm}_{-n}$, $n > 0$ on the primary states. Thus, they are called the descendant states, which have the general form $(J^{a_1}_{-n_1} J^{a_2}_{-n_2} \cdots)|\text{primary}\rangle$.

The affine primaries themselves transform under the global $\mathfrak{sl}(2, \mathbb{R})$ algebra generated by the zero modes J^3_0, $J^{\pm}_0$. Therefore, they form a representation of $\mathrm{SL}(2, \mathbb{R})$, and the full affine representation (Affine Module) is built by acting with the negative current modes on these ground states. These representations form the basic building blocks of the Hilbert space of the $\mathrm{SL}(2, \mathbb{R})$ WZW model before including spectral flow sectors.

In string theory, the physical Hilbert space must consist of only those states with positive norm $\langle\psi|\psi\rangle > 0$. The states with negative norm correspond to ghosts that are not allowed in the physical spectrum. Therefore, the zero-mode representation of $\mathfrak{sl}(2, \mathbb{R})$ must be **unitary**. In a unitary representation, the generators satisfy appropriate Hermiticity conditions, and the inner product on the Hilbert space is positive definite. Physical states of the string must satisfy the Virasoro constraints $L^{(\text{total})}_n|\text{physical}\rangle = 0, \quad \forall\, n > 0$. The zero modes commute with all the Virasoro generators $[J^a_0, L_n] = 0$. Therefore, if a state is physical, acting with J^a_0 produces another physical state. Hence, the physical spectrum must organize into complete representations of the $\mathrm{SL}(2, \mathbb{R})$ algebra generated by the zero modes.

If the Cartan generator J^3_0 corresponding to time translation is compact, then the spectrum of the Cartan generator is bounded (m has both lower and upper limit), similar to compact groups like $SU(2)$. However, in the AdS_3 background, the time direction is non-compact. Therefore, the relevant representations are those of $\mathrm{SL}(2, \mathbb{R})$ with non-compact time, where the spectrum of the Cartan generator J^3_0 can be unbounded.

7.1.1 Five types of Unitary Representations of zero current modes

Unitarity requires all squared norms appearing above to be non-negative for every 'm' that occurs in the representation. Thus, for each 'm' in the spectrum, we must have

$$\begin{aligned} \|J^+_0|j, m\rangle\|^2 = |A_m|^2 \geq 0 \quad &\Longrightarrow \quad (m - j + 1)(m + j) \geq 0 \\ \|J^-_0|j, m\rangle\|^2 = |B_m|^2 \geq 0 \quad &\Longrightarrow \quad (m + j - 1)(m - j) \geq 0 \end{aligned} \tag{435}$$

From these inequalities, one can determine the allowed values for m for a given j value.

(A) Lowest-weight principal discrete representation $\mathcal{D}_j^+$

Suppose there exists a lowest weight $m_{\rm min}$ such that $J_0^-|j, m_{\rm min}\rangle = 0$. Then from the expression of $|B_m|^2$, we get,

$$
\begin{aligned}
|B_{m_{\rm min}}|^2 &= \|J_0^-|j, m_{\rm min}\rangle\|^2 = (m_{\rm min} - j)(m_{\rm min} + j - 1) = 0 \\
\implies m_{\rm min} &= j \quad \text{or} \quad m_{\rm min} = 1 - j
\end{aligned}
\tag{436}
$$

Now, the states higher in the ladder are given by $|j, m_{\rm min}{+}1\rangle\,,\ |j, m_{\rm min}{+}2\rangle\,, \cdots\ |j, m_{\rm min}{+}r\rangle\,, \cdots$. Consider the norm of some state $|j, m_{\rm min} + r\rangle$ for $r > 0$. Then,

$$
\begin{aligned}
\||j, m_{\rm min} + r\rangle\|^2 &= \|J_0^+|j, m_{\rm min} + r - 1\rangle\|^2 = |A_{m_{\rm min}+r-1}|^2 \\
&= (m_{\rm min} + r - 1 - j + 1)(m_{\rm min} + r - 1 - j) \\
\implies \||j, m_{\rm min} + r\rangle\|^2 &= (m_{\rm min} + r - j)(m_{\rm min} + r - j - 1)
\end{aligned}
\tag{437}
$$

Similarly, we can find

$$
\begin{aligned}
\||j, m_{\rm min} + r\rangle\|^2 &= \|J_0^-|j, m_{\rm min} + r + 1\rangle\|^2 = |B_{m_{\rm min}+r+1}|^2 \\
&= (m_{\rm min} + r + 1 + j - 1)(m_{\rm min} + r + 1 - j) \\
\implies \||j, m_{\rm min} + r\rangle\|^2 &= (m_{\rm min} + r + j)(m_{\rm min} + r - j + 1)
\end{aligned}
\tag{438}
$$

$$
\begin{aligned}
\text{For } m_{\rm min} = j \quad : \quad & \||j, j + r\rangle\|^2 = |A_{j+r-1}|^2 = r\,(r - 1) \geq 0 \ \ \forall\, r \geq 0 \\
& \||j, j + r\rangle\|^2 = |B_{j+r+1}|^2 = (2j + r)\,(r + 1) \geq 0 \ \ \forall\, r \geq 0 \ \text{ provided } \ j \geq 0
\end{aligned}
$$

$$
\begin{aligned}
\text{For } m_{\rm min} = 1 - j : \ & \||j, j + r\rangle\|^2 = |A_{r-j}|^2 = (r + 1 - 2j)\,r \geq 0 \ \ \forall\, r \ \text{ when } \ 2j \leq 1 \\
& \||j, j + r\rangle\|^2 = |B_{2-j+r}|^2 = (r + 1)\,(2 - 2j + r) \geq 0 \ \ \forall\, r \geq 0 \ \text{ provided } \ j \leq 1
\end{aligned}
$$

Thus, by convention, we can choose $m_{\rm min} = j$ with $j > 0$. Hence, the standard lowest-weight choice $m_{\rm min} = j$ gives the zero mode spectrum by repeatedly applying J^+ on $|j, j\rangle$ to get the ladder of states $|j, j\rangle\,,\ |j, j + 1\rangle\,,\ |j, j + 2\rangle\,, \cdots$, whose $m = j,\ j + 1,\ j + 2,\ \cdots$. This is the principal lowest-weight discrete series, denoted $\mathcal{D}_j^+$. Unitarity further requires $j > 0$. For the compact time $\mathrm{SL}(2, \mathbb{R})$ group (not the universal cover), j has to be a half-integer, but for the universal cover AdS_3, j may be any positive real number $j > 0$.

$$
\implies \mathcal{D}_j^+ = \Big\{ |j, m\rangle : m = j, j + 1, j + 2, \cdots \Big\} \tag{439}
$$

Note that the presence of a lower bound for 'm' doesn't necessarily imply an upper bound, as the presence of a lower bound was itself a choice.

(B) Highest-weight principal discrete representation $\mathcal{D}_j^-$

Analogously, if there exists a highest weight $m_{\max}$ with $J^+|j, m_{\max}\rangle = 0$, then,

$$\|J^+|j, m_{\max}\rangle\|^2 = |A_{\max}|^2 = (m_{\max} - j + 1)(m_{\max} + j) = 0 \tag{440}$$

Similarly, as before, we can choose $m_{\max} = -j$ where the spectrum is $\Big\{|j\,,\,-j-r\rangle \;\;\forall\, r \geq 0\Big\}$. For the universal cover AdS$_3$, unitarity requires 'j' to be any positive real number $j > 0$. This spectrum is the charge conjugate of $\mathcal{D}_j^+$ and is denoted by $\mathcal{D}_j^-$.

$$\implies \mathcal{D}_j^- = \Big\{|j, m\rangle : m = -j, -j-1, -j-2, \cdots\Big\} \tag{441}$$

(C) Principal continuous representation $\mathcal{C}_j^\alpha$

Here, there is no lower or upper bound on the spectrum. Thus, neither A_m nor B_m vanishes on the entire spectrum; the ladder extends infinitely in both directions. A general such spectrum is of the form,

$$\mathcal{C}_j^\alpha = \Big\{|j, \alpha, m\rangle : m = \alpha + n, \;\; n \in \mathbb{Z}\;,\; 0 \leq \alpha < 1\Big\} \tag{442}$$

Different choices of α correspond to different parities or sectors. For unitarity we must require $|A_m|^2 \geq 0$ and $|B_m|^2 \geq 0$ for all these m. Using,

$$\begin{aligned} |A_m|^2 &= m(m+1) - j(j-1) \geq 0 \quad \implies \quad m(m+1) \geq j(j-1) \\ |B_m|^2 &= m(m-1) - j(j-1) \geq 0 \quad \implies \quad m(m-1) \geq j(j-1) \end{aligned} \tag{443}$$

These conditions are simply satisfied if $-j(j-1) > 0$. Thus, if $j = \mathrm{Re}(j) + is \in \mathbb{C}$, then, we need $-j^* = j - 1 \implies -\mathrm{Re}(j) + is = -1 + \mathrm{Re}(j) + is$, This fixes $\mathrm{Re}(j) = \frac{1}{2}$. Thus, we need

$$j = \tfrac{1}{2} + is, \quad s \in \mathbb{R},$$

$$\implies -j(j-1) = -\big(\tfrac{1}{2} + is\big)\big(\tfrac{1}{2} + is - 1\big) = \big(\tfrac{1}{2} + is\big)\big(\tfrac{1}{2} - is\big) = \tfrac{1}{4} + s^2 > 0$$

Thus, $|A_m|^2 = m(m+1) + \frac{1}{4} + s^2 \geq 0$ and $|B_m|^2 = m(m-1) + \frac{1}{4} + s^2 \geq 0$ for all 'm' values.

These are *principal continuous* representations, written as C_j^α, $j = \frac{1}{2} + is$ and $m = \alpha + n$. For these, $|A_m|^2$ and $|B_m|^2$ are strictly positive for all m, so the representation is unitary.

For $j = 1/2$ and $\alpha = 1/2$, the spectrum becomes,

$$\mathcal{C}_{j=1/2}^{\alpha=1/2} = \Big\{|j = 1/2, \alpha = 1/2\,,\, m\rangle : m = 1/2\,,\, 1/2 \pm 1\,,\, 1/2 \pm 2\,, \cdots\Big\}$$

$$\implies \mathcal{C}_{j=1/2}^{\alpha=1/2} = \Big\{|j = 1/2,\, m\rangle : m = 1/2\,,\, 3/2\,,\, 5/2\,, \cdots\Big\} \bigcup \Big\{|j = 1/2,\, m\rangle : m = -1/2\,,\, -3/2\,,\, -5/2\,, \cdots\Big\}$$

Thus, $\mathcal{C}_{j=1/2}^{\alpha=1/2} = \mathcal{D}_{j=1/2}^+ \bigoplus \mathcal{D}_{j=1/2}^-$ is a reducible representation.

(D) Complementary Representations $\mathcal{E}_j^\alpha$

The unitarity requires $(m-j+1)(m+j) \geq 0$ and $(m+j-1)(m-j) \geq 0$. Thus, the allowed range for m to be unbounded and also to satisfy both inequalities is $m \leq -j$ or $m \geq j$. So, the requirement is $|m| \geq j$. And the forbidden region is $-j < m < j$.

For continuous-type representations the eigenvalues of J^3 take the form $m = \alpha + n, \quad n \in \mathbb{Z}$, where the fractional part satisfies $0 \leq \alpha < 1$. The inequalities must hold for every integer n.

Hence, we require,

$$\begin{aligned} &(\alpha+n-j+1)(\alpha+n+j) \geq 0 \quad \text{and} \quad (\alpha+n+j-1)(\alpha+n-j) \geq 0 \;\; \forall \;\; n \in \mathbb{Z} \\ \implies \; &\alpha+n \leq -j \quad \text{and} \quad \alpha+n \geq j \quad \implies \quad |\alpha+n| \geq j \;\text{ and }\; \alpha+n \notin (-j\,,j\,) \end{aligned} \tag{444}$$

So, the lattice of $\alpha + n$ must avoid the forbidden interval. But note that if $0 < j < 1/2$, then the length of forbidden region $2j$ lies in $0 < 2j < 1$. Thus, the conditions are satisfied trivially as the forbidden region is too small for the points in 'm' lattice to fall in $(-j\,,j\,)$. So, this merges with the usual continuous representations. Moreover, if $j \geq 1$ then the length of forbidden region $2j > 2$ is very high that the 'm' lattice points inevitably belong to $(-j\,,j\,)$.

But, to have a complementary representation, we shall choose $\frac{1}{2} < j < 1$. Then, we require $|m| = |\alpha+n| \geq j \;\; \forall \; n \in \mathbb{Z}$ or $\min(|m|) \geq j$. The lattice point closest to zero occurs when $n = 0, -1$. Thus, the points that can enter the forbidden region are α and $\alpha - 1$. Thus, we require α to satisfy $\alpha \notin (-j,j)$ and $\alpha - 1 \notin (-j,j) \quad \implies \quad \alpha > j$ and $\alpha - 1 < -j$.

$$\implies \alpha > j \quad \text{or} \quad \alpha < 1-j \quad \text{where} \quad \frac{1}{2} < j < 1$$

These two equations can be combined to,

$$\left|\alpha - \frac{1}{2}\right| \geq j - \frac{1}{2} \quad \text{where} \quad \frac{1}{2} < j < 1 \tag{445}$$

Thus, the complementary series of unitary representations of $\mathrm{SL}(2,\mathbb{R})$ is given by,

$$\mathcal{E}_j^\alpha = \left\{ |j,\alpha,m\rangle : m = \alpha+n\;,\; n \in \mathbb{Z},\; \left|\alpha - \frac{1}{2}\right| \geq j - \frac{1}{2} \;\text{ where }\; \frac{1}{2} < j < 1 \right\} \tag{446}$$

(E) Identity (trivial) representation.

The trivial representation is given by $j = 0$ with Casimir eigenvalue $C_2 = 0$, and it is unitary only for $m = 0$. Thus,

$$\mathbb{I} = |j = 0, m = 0\rangle \tag{447}$$

7.1.2 Quantum Mechanics of a particle in AdS_3

The representations discussed earlier were obtained purely from the algebraic properties of the $\mathfrak{sl}(2,\mathbb{R})$ algebra just by using the commutation relations and unitarity conditions. However, this classification is purely mathematical. In a given physical system, only a subset of these representations may actually appear. We shall study a simplifying limit of this theory to know which of these representations appear in the physical Hilbert space. The WZW model depends on the level k of the $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ algebra. To obtain intuition about the spectrum, one first considers the large k limit. In this limit, the curvature of the target space becomes small and quantum corrections are suppressed. The theory simplifies considerably.

In AdS_3 string theory, there exist classical solutions corresponding to strings moving along geodesics. Short strings correspond to strings that oscillate close to these geodesics. When we expand the WZW model around such solutions and take the limit $k \to \infty$, the dynamics reduces to that of a particle moving on AdS_3. Thus, the complicated string theory problem simplifies to the quantum mechanics of a particle on AdS_3.

For a particle moving on a manifold $\mathcal{M}$, the Hilbert space consists of square-integrable functions $\mathcal{H} = \mathcal{L}^2(\mathcal{M})$. Therefore, in the present case, $\mathcal{H} = \mathcal{L}^2(\mathrm{AdS}_3)$. Since AdS_3 is non-compact, the ordinary square integrability condition must be interpreted in a generalized sense (similar to plane-wave normalization) where we allow delta-function normalization of states.

The AdS_3 spacetime has the isometry group of $\mathrm{SL}(2,\mathbb{R}) \times \mathrm{SL}(2,\mathbb{R})$. Thus, the Hilbert space must carry representations of this symmetry group. In the WZW model, these correspond to the left-moving and right-moving current algebras. According to the Peter-Weyl theorem, functions on a compact group manifold can be decomposed into irreducible unitary representations of the group. But for the Hilbert space $\mathcal{L}^2(\mathrm{AdS}_3)$, harmonic analysis decomposition using infinite-dimensional irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R}) \times \mathrm{SL}(2,\mathbb{R})$, specifically the principal series, discrete series, and complementary series, can be performed. This decomposition is analogous to Fourier decomposition. So, square integrable functions on AdS_3 can be expanded in terms of matrix elements of $\mathrm{SL}(2,\mathbb{R}) \times \mathrm{SL}(2,\mathbb{R})$ representations.

A convenient basis of functions on AdS_3 is given by,

$$F^j_{m,\bar{m}}(g) = \langle j, m|g|j, \bar{m}\rangle \tag{448}$$

where $g \in \mathrm{AdS}_3$ is viewed as an element of the group $\mathrm{SL}(2,\mathbb{R})$. The indices m and $\bar{m}$ transform by the rules dictated by g where $|j,m\rangle$ and $|j,\bar{m}\rangle$ are eigen states of J_0^3 and $\bar{J}_0^3$ in a representation of $\mathrm{SL}(2,\mathbb{R}) \times \mathrm{SL}(2,\mathbb{R})$. These functions are the matrix elements of the representation. This is similar to the Wigner D-functions of $SU(2)$, where $D^j_{m,\bar{m}} = \langle j, m|g|j, \bar{m}\rangle$, $g \in SU(2)$.

The states satisfy $J_0^3|j,m\rangle = m|j,m\rangle$, and similarly for the right-moving sector, $\bar{J}_0^3|j,\bar{m}\rangle = \bar{m}|j,\bar{m}\rangle$. Thus the indices m and $\bar{m}$ label the eigenvalues of the Cartan generators of the $\mathrm{SL}(2,\mathbb{R}) \times \mathrm{SL}(2,\mathbb{R})$ where the Hilbert space decomposes as

$$\mathcal{H} = \bigoplus_{\mathcal{R}^{(j)}} \mathcal{R}^{(j)} \otimes \mathcal{R}^{(j)}, \tag{449}$$

$\mathcal{R}^{(j)}$ denotes a representation of $\mathrm{SL}(2,\mathbb{R})$. Thus the functions $F^j_{m,\bar{m}}(g)$ belong to the tensor product representation $\mathcal{R}^{(j)} \times \mathcal{R}^{(j)}$. Each function $F^j_{m,\bar{m}}(g)$ represents a quantum state of a particle in AdS_3. These states organize themselves into representations of the isometry group. So, the structure of the Hilbert space is determined by the representation theory of this group.

7.1.3 Wave functions in the discrete representation $\mathcal{D}_j^{\pm}$

In global coordinates the metric of AdS_3 is $ds^2 = -\cosh^2(\rho)\,dt^2 + d\rho^2 + \sinh^2(\rho)\,d\phi^2$. The Laplace-Beltrami operator associated with this metric is

$$\nabla^2 = \frac{1}{\sinh(2\rho)}\partial_\rho\big(\sinh(2\rho)\,\partial_\rho\big) - \frac{1}{\cosh^2(\rho)}\partial_t^2 + \frac{1}{\sinh^2(\rho)}\partial_\phi^2 \tag{450}$$

For functions on AdS_3, the above operator is related to the quadratic Casimir of $\mathrm{SL}(2,\mathbb{R})$ as $\nabla_\mu\nabla^\mu = -C_2$. For a representation labeled by j, the Casimir eigenvalue is $C_2 = -j(j-1)$. Thus, the eigen wave function $f^{(j)}(t,\rho,\phi)$ should satisfy the equation,

$$\nabla^2 f^{(j)}(t,\rho,\phi) = j(j-1)f^{(j)}(t,\rho,\phi) \tag{451}$$

The time t and the ϕ coordinates are decoupled into separate free harmonic equations that admit plane-wave-like solutions. So, we separate variables as $f^{(j)}(t,\rho,\phi) = e^{-i\omega t}e^{im\phi}R^{(j)}(\rho)$. Substituting this into the wave equation yields the radial equation,

$$\frac{1}{\sinh(2\rho)}\partial_\rho(\sinh(2\rho)\,\partial_\rho R^{(j)}) + \left(\frac{\omega^2}{\cosh^2(\rho)} - \frac{m^2}{\sinh^2(\rho)}\right)R^{(j)} = j(j-1)R^{(j)} \tag{452}$$

The solution of this equation has some hyperbolic functions multiplied by hypergeometric polynomials. But we want the large ρ approximation. For large ρ, $\sinh(\rho) \sim \cosh(\rho) \sim \frac{1}{2}e^\rho$. Thus $\dfrac{1}{\cosh^2(\rho)} \sim \dfrac{1}{\sinh^2(\rho)} \sim e^{-2\rho}$. These terms vanish exponentially at large ρ.

$$\implies \partial_\rho^2 R^{(j)} + 2\,\partial_\rho R^{(j)} = j(j-1)R^{(j)} \qquad \text{(at large } \rho) \tag{453}$$

Assume a solution of the form $R^{(j)}(\rho) = e^{\lambda_j\rho}$. Substituting this into the equation gives,

$$(\lambda_j^2 + 2\lambda_j - j(j-1))e^{\lambda_j\,\rho} = 0 \tag{454}$$

Thus, the solutions for λ_j are $\lambda_j = -j, \quad \lambda_j = j-1$. Therefore, the radial behavior is,

$$R^{(j)}(\rho) \sim e^{-j\rho} \;\; \text{(converges to 0 at large } \rho) \quad \text{or} \quad R^{(j)}(\rho) \sim e^{(j-1)\rho} \;\; \text{(diverges at large } \rho) \tag{455}$$

In the WZW model, the wavefunctions are matrix elements $F^j_{m,\bar{m}}(g) = \langle j, m|g|j, \bar{m}\rangle$. Since both the left and right sectors contribute equally to the radial dependence, the full asymptotic behavior for the non-diverging radial solution is,

$$R^{(j)}(\rho) \sim e^{-2j\rho} \qquad \text{(at large } \rho) \tag{456}$$

Now, for normalizable solutions, we require $\displaystyle\int_0^\rho |R^{(j)}(\rho)|^2\, d\mu$ to converge at large ρ, where $d\mu$ is the radial measure in the volume element. For AdS_3, volume element is $\frac{1}{2}\sinh(2\rho)\, d\rho\, dt\, d\phi$. Thus, $d\mu = \sinh(2\rho)\, d\rho$. Thus, at large ρ we require, $e^{-4j\rho}e^{2\rho} \to 0$. Thus, we need

$$\int e^{-4j\rho}e^{2\rho} d\rho = \int e^{-2(2j-1)\rho} d\rho < \infty \tag{457}$$

Hence, the normalization condition restricts $j > \frac{1}{2}$. Thus, $f^{(j)}(t,\rho,\phi) \in \mathcal{L}^2(AdS_3)$ only if $j > \frac{1}{2}$.

The quadratic Casimir operator eigenvalue is $C_2 = -j(j-1)$. For $0 < j < 1$, the two values j and $1-j$, give the same Casimir eigenvalue. Thus, there are two representations with the same Casimir. However, only the representation with $j > \frac{1}{2}$ is square-integrable in $\mathcal{L}^2(AdS_3)$. One could modify the inner product so that representations with $0 < j < \frac{1}{2}$ become normalizable. This would correspond to defining a modified inner product

$$\langle f\,,\, g\rangle = \int f^*(X)g(X)\, W_j(X)\, dX \tag{458}$$

where the weight function $W_j(X)$ depends on j. Such constructions correspond to nonstandard quantization of the theory. But the authors of the paper consider only the standard $\mathcal{L}^2$ normalizable states on AdS_3. Similarly, complementary series representations also require a j-dependent modification of the norm to become normalizable. Therefore, they would appear only in nonstandard quantization that is not included in the spectrum.

Note that there is a possibility of descendant states with smaller J^3_0 eigenvalues, even if the primary state satisfies $j > \frac{1}{2}$. For example, consider $J_{-1}|j,j\rangle$. Since, $[J^3_0, J^-_{-1}] = -J^-_{-1}$, this state has J_0's eigenvalue $m = j-1$. For $1 < j < \frac{3}{2}$, we get, $m = j - 1 < \frac{1}{2}$. Thus, descendants can have smaller values of m even though the representation label j satisfies $j > \frac{1}{2}$. But this will not cause any problem with the normalization condition, as it depends only on the j value. Since $J^\pm_n$ commutes with C_2, the representation label j remains the same with the same Casimir. Hence, the corresponding wave function is still normalizable if $j > \frac{1}{2}$.

For discrete representations $\mathcal{D}_j^{\pm}$, since we require $j > \frac{1}{2}$, we get, $C_2 = -j(j-1) \leq \frac{1}{4}$. Now, the Klein-Gordon equation is given by, $(-\nabla^2 + \mu^2)|\psi\rangle = 0$ ($\mu \sim$ space-time mass). But $C_2 = -\nabla^2$. Thus, $C_2 = -j(j-1) = -\mu^2 \implies m^2 = j(j-1)$. Now, since $j > \frac{1}{2}$, we get, $\mu^2 > -\dfrac{1}{4}$. Thus, the discrete representations satisfy the Breitenlohner–Freedman stability bound.

7.1.4 Wave functions in the continuous series representations $\mathcal{C}_j^{\alpha}$

For the principal continuous representation $\mathcal{C}_j^{\alpha}$ the parameter j takes the form $j = \frac{1}{2} + is$, $s \in \mathbb{R}$. Thus, at large ρ, the Casimir' eigenvalue equation becomes,

$$\implies \partial_\rho^2 R^{(s)} + 2\,\partial_\rho R^{(s)} = -\left(\frac{1}{4} + s^2\right) R^{(s)} \qquad \text{(at large } \rho) \tag{459}$$

where the full wave function is $f^{(s)}(t,\rho,\phi) = e^{-i\omega t} e^{im\phi} R^{(s)}(\rho)$. Assume a solution of the form $R^{(s)}(\rho) = e^{\lambda_s \rho}$. Substituting this into the equation gives,

$$\left(\lambda_s^2 + 2\lambda_s + \frac{1}{4} + s^2\right) e^{\lambda_s\,\rho} = 0 \tag{460}$$

Thus, the solutions for λ_s are $\lambda_s = -\frac{1}{2} \pm is$. But, including the contributions from both the left and right sectors, we get the full asymptotic behavior for the radial solution as,

$$R^{(s)}(\rho) \sim e^{-\rho} e^{\pm is\rho} \tag{461}$$

These are oscillatory plane-wave type solutions in the radial direction. The invariant measure on AdS_3 behaves for large ρ as $d\mu \sim e^{2\rho} d\rho$. Thus,

$$|R^{(s)}(\rho)|^2 d\mu \sim e^{-2\rho} e^{2\rho} d\rho \sim d\rho \tag{462}$$

Hence, the integral diverges. But this resembles the plane-wave conditions. Thus, we can use the delta-function normalization condition for such wave functions, where,

$$\int_0^\infty R^{(s')^*}(\rho) R^{(s)}(\rho) d\mu \sim \delta(s - s') \qquad \implies \qquad \langle f_s | f_{s'} \rangle \propto \delta(s - s') \tag{463}$$

Thus, discrete series wavefunctions behave as $f(\rho) \sim e^{-2j\rho}$, which correspond to normalizable bound states, whereas the continuous series wavefunctions behave as $f(\rho) \sim e^{-\rho} e^{\pm is\rho}$, which correspond to scattering states with plane-wave normalization. So, they belong to $\mathcal{L}^2(\text{AdS}_3)$ in the generalized sense, similar to plane waves in flat space. For these representations, the Casimir becomes $C_2 = \frac{1}{4} + s^2 > \frac{1}{4}$. This corresponds to $m^2 < -\frac{1}{4}$. Thus, it violates the Breitenlohner–Freedman bound for ordinary particle theories.

In bosonic string theory, such states correspond to Tachyon modes. The only physical state of this type in bosonic string theory is the tachyon. In a perturbatively stable string theory, such particle states would normally be excluded from the physical spectrum. Despite violating the Breitenlohner–Freedman bound for ordinary particles, the continuous representations appear naturally in the decomposition of $\mathcal{L}^2(\mathrm{AdS}_3)$ and correspond to scattering states or long-string sectors of the theory. The space of square-integrable functions on AdS_3 decomposes into irreducible representations of $\mathrm{SL}(2,\mathbb{R}) \times \mathrm{SL}(2,\mathbb{R})$. The basis consists of discrete series representations and also principal continuous series representations. Schematically,

$$\mathcal{L}^2(\mathrm{AdS}_3) = \bigoplus_{j>1/2} \mathcal{D}_j^\pm \otimes \mathcal{D}_j^\pm \oplus \int ds\, \mathcal{C}^\alpha_{1/2+is} \otimes \mathcal{C}^\alpha_{1/2+is} \tag{464}$$

Therefore, $\mathcal{C}_j^\alpha$ is expected to appear in the Hilbert space of the WZW model (these will determine which states correspond to physical string states).

7.2 Representations of the current algebra and no-ghost theorem

Suppose we start with a unitary representation $\mathcal{H}$ of the global group $\mathrm{SL}(2,\mathbb{R})$. This representation can be used to construct a representation of the full affine current algebra of the WZW model. The idea is to treat the states in $\mathcal{H}$ as affine **primary states** of the affine algebra, $|\mathrm{Primary}\rangle \in \mathcal{H}$. These primary states are annihilated by all positive modes of the currents:

$$J_n^{3,\pm}|\mathrm{Primary}\rangle = 0, \quad n \geq 1 \qquad \text{where} \quad |\mathrm{Primary}\rangle \sim |j,m\rangle$$

These serve as the ground-state representations of the affine modules. The complete affine representation is then obtained by acting with the negative current modes $J_{-n}^{3,\pm}|\mathrm{Primary}\rangle$. We consider only the representations where $\mathcal{H} = \mathcal{C}^\alpha_{j=1/2+is}$ and $\mathcal{D}_j^\pm, \quad j > \frac{1}{2}$ as they are the ones with standard normalization conditions in $\mathcal{L}^2(\mathrm{AdS}_3)$. The full affine current algebra representations constructed from these are denoted by $\hat{\mathcal{C}}_j^\alpha$ and $\hat{\mathcal{D}}_j^\pm$ which are the modules obtained by acting with all negative current modes $J_{-n}^{3,\pm}$ on the corresponding zero-mode representations. Thus, the full representation space is,

$$\hat{\mathcal{H}} = \mathrm{Span}\{J^{a_1}_{-n_1} J^{a_2}_{-n_2} \cdots |\mathrm{Primary}\rangle\}, \qquad \forall\, |\mathrm{Primary}\rangle \in \mathcal{H} \tag{465}$$

We can show that the quadratic Casimir $C_2 = \kappa_{ab} J_0^a J_0^b$ commutes with all the affine current modes. The affine current algebra satisfies $[J_0^a, J_n^b] = f^{ab}{}_c\, J_n^c$ where $f^{ab}{}_c$ are structure constants.

$$\begin{aligned}
[C_2, J_n^d] &= \kappa_{ab}\big(J_0^a[J_0^b, J_n^d] + [J_0^a, J_n^d]J_0^b\big) = \kappa_{ab}J_0^a f^{bd}{}_c J_n^c + \underbrace{\kappa_{ab} f^{ad}{}_c J_n^c J_0^b}_{a\leftrightarrow b} \\
&= \kappa_{ab} f^{bd}{}_c\big(J_0^a J_n^c + J_n^c J_0^a\big) = \kappa^{bd} f_{abc}\big(J_0^a J_n^c + J_n^c J_0^a\big) = \kappa^{bd} f_{abc}\{J_0^a, J_n^c\} \\
\implies\ & [C_2, J_n^a] = 0
\end{aligned} \tag{466}$$

This was true because f_{abc} is anti-symmetric and $\{J_0^a, J_n^c\}$ is symmetric in 'a' and 'c'. Thus, the descendant states of primary state $|j,m\rangle$ have the same Casimir eigenvalue and conserve the $j-$representation label. The entire affine module built on a primary state forms a representation of the global $\mathrm{SL}(2,\mathbb{R})$ algebra with fixed Casimir eigenvalue $-j(j-1)$. In particular, the action of current modes J^a_{-n} does not change the representation label j.

In general, representations of the affine algebra $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ contain states with negative norm. Such states are problematic for string theory because physical states must have positive norm for the theory to be consistent. For strings propagating on AdS_3, these unwanted negative norm states are removed by imposing the Virasoro constraints on the Hilbert space of a single string state. They are given by,

$$(L_n + \mathcal{L}_n - \delta_{n,0})\,|\text{physical}\rangle = 0, \quad n \geq 0.$$

Here L_n are the Virasoro generators coming from the Sugawara construction of the $\mathrm{SL}(2,\mathbb{R})$ WZW model. $\mathcal{L}_n$ are the Virasoro generators associated with the internal CFT on the manifold $\mathcal{M}$. Then, the $\delta_{n,0}$ appears due to the normal ordering constant in the string mass-shell condition. The Virasoro constraints project the Hilbert space onto a smaller subspace of states that are consistent with string gauge symmetry. These equations enforce worldsheet reparameterization invariance and remove unphysical degrees of freedom. It has been proven that when these Virasoro constraints are imposed, the resulting physical Hilbert space contains no negative norm states (no ghosts), provided $k > 2$, $c_{\text{total}} = 26$ and the representations belong to $\hat{\mathcal{C}}^{\alpha}_{j=1/2+is}$ or $\hat{\mathcal{D}}^{\pm}_j \left(0 < j < \dfrac{k}{2}\right)$. This is the **no-ghost theorem** for strings in AdS_3.

7.2.1 Proof of No-Ghost theorem for unflowed representations

This theorem for the $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ current algebra can be proved when the WZW level $k > 2$ and for $c_{\text{total}} = 26$. First, we can show that the states constructed by acting with the negative modes of the $\mathrm{SL}(2,\mathbb{R})$ Virasoro generators and the J^3 current on the state '$|f\rangle$' which satisfy $L_{n>0}|f\rangle = \mathcal{L}_{n>0}|f\rangle = 0 = J^3_{n>0}|f\rangle$ are linearly independent and their span gives the string's unflowed Hilbert space. And, find the required restrictions for this to be true.

To show the states of type $L_{-n_1}L_{-n_2}\cdots L_{-n_K}\, J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle$ are linearly independent ,

$$\begin{gathered}\text{where } n_1 \geq n_2 \geq \cdots \geq n_K\ ,\ m_1 \geq m_2 \geq \cdots \geq m_M \\ \text{and } L_n|f\rangle = \mathcal{L}_n|f\rangle = 0 = J^3_n|f\rangle\ ,\ \forall\, n \geq 1 \\ \text{To Prove } \mathrm{Span}\big(L_{-n_1}L_{-n_2}\cdots L_{-n_K}\, J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle\big) = \hat{\mathcal{H}}\end{gathered} \tag{467}$$

$|f\rangle$ satisfies $L_{n>0}|f\rangle = \mathcal{L}_{n>0}|f\rangle = J^3_{n>0}|f\rangle = 0$ and belong to the tensor product of an affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ representation, $\hat{\mathcal{D}}^{\pm}_j$ or $\hat{\mathcal{C}}^{\alpha}_j$, and a physical state of the internal $\mathcal{M}$ CFT.

Now, the current modes J^3_n form an affine $\widehat{\mathfrak{u}(1)}_k$ oscillator algebra since their commutation relation is given by $[J^3_m, J^3_n] = -\frac{k}{2}\, m\, \delta_{m+n,0}$. If we rescale these currents as $J^3_n = -i\sqrt{\frac{k}{2}}\, \alpha_n$, then we get $[\alpha_m, \alpha_n] = m\, \delta_{m+n,0}$. This is nothing but the affine extension of the harmonic oscillator algebra, also called the $\widehat{\mathfrak{u}(1)}_k$ current algebra. The states generated by J^3_{-m} correspond to the oscillator states of a free $\widehat{\mathfrak{u}(1)}_k$ current algebra. Different sequences of J^3_{-m} modes therefore produce linearly independent states. The rest is a CFT with a central charge

$$c_{\text{rest}} = 26 - 1 = 25.$$

because the current J_3 behaves as a free boson field $-i\sqrt{\frac{k}{2}}\, \partial\phi(z)$ with a central charge of 1. The Sugawara stress tensor can be split into J^3 current term and $J^{\pm}$ current term. So, we shall define $\hat{L}_n$, as the n^{th} mode of the total $(\text{AdS}_3 + \mathcal{M})$ energy-momentum tensor without the n^{th} free-boson field J^3's energy-momentum tensor.

$$\hat{L}_n = L_n + \mathcal{L}_n - \text{n}^{\text{th}} \text{ mode of } \underbrace{\left(\frac{\kappa_{33}}{k} \; : J^3 J^3 :\right)}_{U(1)\text{ E-M Tensor}} = L_n + \mathcal{L}_n - L_n^{(J_3)} \tag{468}$$

$$\text{Thus, } L_n^{(J^3)} = \frac{\kappa_{33}}{k} \sum_{m\in\mathbb{Z}} : J^3_{n-m} J^3_m : = \frac{\kappa_{33}}{k} \left(: J^3_0 J^3_n : + \sum_{m=1}^{\infty} \left(J^3_{n-m} J^3_m + J^3_{-m} J^3_{n+m} \right) \right) \tag{469}$$

The state $|f\rangle$ satisfies $L_{n>0}|f\rangle = \mathcal{L}_{n>0}|f\rangle = J^3_{n>0}|f\rangle = 0$. Thus, for $n > 0$,

$$L_{n>0}^{(J^3)}|f\rangle = \frac{\kappa_{33}}{k} \left(J^3_0 J^3_n + \sum_{m=1}^{\infty} \left(J^3_{n-m} J^3_m + J^3_{-m} J^3_{n+m} \right) \right) |f\rangle = 0 \tag{470}$$

$$\implies \hat{L}_{n>0}|f\rangle = (L_{n>0} + \mathcal{L}_{n>0} - L_{n>0}^{(J^3)})|f\rangle = 0. \text{ Thus, } \boxed{\hat{L}_n|f\rangle = 0 \quad \forall\ n > 0)}$$

Equality of spans : Since $L_n = \hat{L}_n - \mathcal{L}_n + L_n^{(J^3)}$ and $[\mathcal{L}_n, L_n] = [\mathcal{L}_n, L_n^{(J^3)}] = 0$,

$$\begin{aligned} &\text{Span}\big(\hat{L}_{-n_1} \cdots \hat{L}_{-n_K}\ J^3_{-m_1} \cdots J^3_{-m_M}|f\rangle\big) \\ &\qquad = \text{Span}\Big(\big(L_{-n_1} + L_{-n_1}^{(J^3)}\big) \cdots \big(L_{-n_K} + L_{-n_K}^{(J^3)}\big)\ J^3_{-m_1} \cdots J^3_{-m_M}|f\rangle\Big) \end{aligned}$$

Now, since $L_n^{(J^3)}$ contains only J^3 oscillators, any state generated by L_{-n}, J^3_{-m} can be rewritten as a linear combination of states generated by $\hat{L}_{-n}$, J^3_{-m}. Hence,

$$\text{Span}\big(\hat{L}_{-n_1} \cdots \hat{L}_{-n_K}\ J^3_{-m_1} \cdots J^3_{-m_M}|f\rangle\big) = \text{Span}\big(L_{-n_1} \cdots L_{-n_K}\ J^3_{-m_1} \cdots J^3_{-m_M}|f\rangle\big) \tag{471}$$

Now, we have, $\hat{L}_n = L_n + \mathcal{L}_n - L_n^{(J^3)}$. Then,

$$[\hat{L}_n, J^3_m] = \underbrace{[L_n, J^3_m]}_{-m\, J^3_{n+m}} + \underbrace{[\mathcal{L}_n, J^3_m]}_{0} - [L_n^{(J^3)}, J^3_m] \quad \implies \quad [\hat{L}_n, J^3_m] = -m\, J^3_{n+m} - [L_n^{(J^3)}, J^3_m] \tag{472}$$

To find $[L_n^{(J^3)}, J^3_m]$, expand $L_n^{(J^3)} = -\dfrac{1}{k} \displaystyle\sum_p : J^3_{n-p} J^3_p :$ and use $[AB, C] = A[B, C] + [A, C]B$.

Thus,

$$[L_n^{(J^3)}, J_m^3] = -\frac{1}{k}\sum_p \left(J_{n-p}^3[J_p^3, J_m^3] + [J_{n-p}^3, J_m^3]J_p^3\right) \tag{473}$$

Using the current algebra $[J_p^3, J_m^3] = -\frac{k}{2}p\,\delta_{p+m,0}$ we get the first term as

$$J_{n-p}^3[J_p^3, J_m^3] = -\frac{k}{2}p\,\delta_{p+m,0}\,J_{n-p}^3 = \frac{k}{2}m\,J_{n+m}^3.$$

and the second term becomes $[J_{n-p}^3, J_m^3]J_p^3 = -\frac{k}{2}(n-p)\,\delta_{n-p+m,0}\,J_p^3 = \frac{k}{2}m\,J_{n+m}^3$. Thus,

$$[L_n^{(J^3)}, J_m^3] = -\frac{1}{k}\times k\,m\,J_{n+m}^3 \qquad \Longrightarrow \quad \boxed{[L_n^{(J^3)}, J_m^3] = -m\,J_{n+m}^3} \tag{474}$$

Then,

$$[\hat{L}_n, J_m^3] = -m\,J_{n+m}^3 - [L_n^{(J^3)}, J_m^3] = (-mJ_{n+m}^3) - (-mJ_{n+m}^3) = 0 \tag{475}$$

$$\Longrightarrow \boxed{[\hat{L}_n, J_m^3] = 0}$$

Now, we can also show that the modes $L_n^{(J^3)}$ form a Virasoro algebra with central charge of 1 and the modes $\hat{L}_n$ form a Virasoro algebra with central charge $c_{\mathfrak{sl}(2,\mathbb{R})} - 1$. We compute:

$$[L_n^{(J^3)}, L_m^{(J^3)}] = \frac{1}{k^2}\sum_{p,q}\left[:J_{n-p}^3J_p^3:, :J_{m-q}^3J_q^3:\right] \tag{476}$$

Now, we use the commutator identity $[AB, CD] = A[B,C]D + AC[B,D] + [A,C]DB + C[A,D]B$, and apply it to the normal-ordered operators. First consider the single contractions, use the J^3 current algebra, $[J_m^3, J_n^3] = -\frac{k}{2}m\,\delta_{m+n,0}$ and evaluate each term.

$$\begin{aligned}\text{single contractions} = \frac{1}{k^2}\sum_{p,q}\Big(&J_{n-p}^3[J_p^3, J_{m-q}^3]J_q^3 + J_{n-p}^3J_{m-q}^3[J_p^3, J_q^3] \\ &+ [J_{n-p}^3, J_{m-q}^3]J_q^3J_p^3 + J_{m-q}^3[J_{n-p}^3, J_q^3]J_p^3\Big)\end{aligned} \tag{477}$$

$$J_{n-p}^3[J_p^3, J_{m-q}^3]J_q^3 = -\frac{k}{2}p\,\delta_{p+m-q,0}\,J_{n-p}^3J_q^3.\ \text{Set } q = p+m \implies \text{Term 1} = -\frac{k}{2}p\,J_{n-p}^3J_{p+m}^3$$

$$J_{n-p}^3J_{m-q}^3[J_p^3, J_q^3] = -\frac{k}{2}p\,\delta_{p+q,0}\,J_{n-p}^3J_{m-q}^3.\ \text{Set } q = -p \implies \text{Term 2} = -\frac{k}{2}p\,J_{n-p}^3J_{m+p}^3$$

$$[J_{n-p}^3, J_{m-q}^3]J_q^3J_p^3 = -\frac{k}{2}(n-p)\,\delta_{n-p+m,q}\,J_q^3J_p^3.\ \text{Set } q = n+m-p \implies \text{Term 3} = -\frac{k}{2}(n-p)\,J_{n+m-p}^3J_p^3$$

$$J_{m-q}^3[J_{n-p}^3, J_q^3]J_p^3 = -\frac{k}{2}(n-p)\,\delta_{n-p+q,0}\,J_{m-q}^3J_p^3.\ \text{Set } q = p-n \implies \text{Term 4} = -\frac{k}{2}(n-p)\,J_{m+n-p}^3J_p^3$$

Now, we combine all terms, they are all proportional to $J^3_{n+m-r}J^3_r$. After relabeling indices and summing, we get,

$$[L_n^{(J^3)}, L_m^{(J^3)}] = \frac{1}{k^2}\cdot(-k)\sum_r \Big[(n-m)\Big] J^3_{n+m-r}J^3_r + \text{central term (double contractions)} \quad (478)$$

Thus:

$$[L_n^{(J^3)}, L_m^{(J^3)}] = (n-m)\left(-\frac{1}{k}\sum_r : J^3_{n+m-r}J^3_r :\right) + \text{central term} = (n-m)L^{(J^3)}_{n+m} + \text{central term}$$

Now we find the central term coming from double contractions. The central term comes from contracting twice : $JJ :: JJ :\to \langle JJ\rangle\langle JJ\rangle$ where each contraction gives $\langle J^3_p J^3_q\rangle = -\frac{k}{2}p\,\delta_{p+q,0}$. Performing both contractions yields:

$$\frac{1}{k^2}\sum_p \left(\frac{k}{2}\right)^2 p(n-p)\delta_{n+m,0} = \frac{1}{12}(n^3-n)\delta_{n+m,0}$$

We used $\sum_p p(n-p) = \frac{1}{6}(n^3-n)$ with zeta function regularization. So, the final result is,

$$\boxed{[L_n^{(J^3)}, L_m^{(J^3)}] = (n-m)L^{(J^3)}_{n+m} + \frac{1}{12}n(n^2-1)\delta_{n+m,0}} \quad (479)$$

Thus, the Sugawara stress tensor modes $L_n^{(J^3)}$ of just the $U(1)$ current also form a virasoro algebra with central charge $c_{U(1)} = 1$.

Now, we check the Virasoro algebra of the generators $\hat{L}_n = L_n + \mathcal{L}_n - L_n^{(J^3)}$. So we compute,

$$\begin{aligned}[\hat{L}_n, \hat{L}_m] &= [L_n + \mathcal{L}_n - L_n^{(J^3)}\,,\ L_m + \mathcal{L}_m - L_m^{(J^3)}] \\ &= [L_n, L_m] + [\mathcal{L}_n, \mathcal{L}_m] + [L_n^{(J^3)}, L_m^{(J^3)}] - [L_n, L_m^{(J^3)}] - [L_n^{(J^3)}, L_m]\end{aligned} \quad (480)$$

For the first three terms, we use the respective Virasoro algebra commutation relations.

$$\begin{aligned}[L_n, L_m] &= (n-m)L_{n+m} + \frac{c_{\mathfrak{sl}(2,\mathbb{R})}}{12}(n^3-n)\delta_{n+m,0} \\ [L_n^{(J^3)}, L_m^{(J^3)}] &= (n-m)L^{(J^3)}_{n+m} + \frac{1}{12}(n^3-n)\delta_{n+m,0} \\ [\mathcal{L}_n, \mathcal{L}_m] &= (n-m)\mathcal{L}_{n+m} + \frac{c_{\mathcal{M}}}{12}(n^3-n)\delta_{n+m,0}\end{aligned} \quad (481)$$

Now, we shall compute the commutator

$$[L_n, L_m^{(J^3)}] = -\frac{1}{k}\sum_p \left([L_n, J^3_{m-p}]J^3_p + J^3_{m-p}[L_n, J^3_p]\right) = \frac{1}{k}\sum_p \left((m-p)J^3_{n+m-p}J^3_p + p\,J^3_{m-p}J^3_{n+p}\right)$$

In the second term, set $p \to p-n$. So, $\quad p\,J^3_{m-p}J^3_{n+p} \ \to\ (p-n)J^3_{n+m-p}J^3_p$.

$$\implies [L_n, L_m^{(J^3)}] = \frac{1}{k}\sum_p \left((m-p)+(p-n)\right) J^3_{n+m-p}J^3_p = (n-m)\left(-\frac{1}{k}\sum_p :J^3_{n+m-p}J^3_p:\right) \quad (482)$$

Thus,

$$[L_n, L_m^{(J^3)}] = (n-m)\, L_{n+m}^{(J^3)} \qquad \implies \qquad [L_n^{(J^3)}, L_m] = (n-m)L_{n+m}^{(J^3)} \quad (483)$$

Then, combining all results, we get,

$$\begin{aligned}
[\hat{L}_n, \hat{L}_m] &= (n-m)L_{n+m} + \frac{c_{\mathfrak{sl}(2,\mathbb{R})}}{12}(n^3-n)\delta_{n+m,0} \ + \ (n-m)\mathcal{L}_{n+m} + \frac{c_{\mathcal{M}}}{12}(n^3-n)\delta_{n+m,0} \\
&\quad + (n-m)L_{n+m}^{(J^3)} + \frac{1}{12}(n^3-n)\delta_{n+m,0} - 2(n-m)L_{n+m}^{(J^3)} \\
&= (n-m)(L_{n+m} + \mathcal{L}_{n+m} - L_{n+m}^{(J^3)}) + \frac{c_{\mathfrak{sl}(2,\mathbb{R})} + c_{\mathcal{M}} - 1}{12}(n^3-n)\delta_{n+m,0}
\end{aligned}$$

Thus,

$$[\hat{L}_n, \hat{L}_m] = (n-m)\hat{L}_{n+m} + \frac{c_{\mathfrak{sl}(2,\mathbb{R})} + c_{\mathcal{M}} - 1}{12}(n^3-n)\delta_{n+m,0} \quad (484)$$

Thus, we see that the Virasoro algebra splits into $U(1)$ with $c = 1$, the coset $\mathrm{SL}(2,R)/U(1)$ with $c_{\text{coset}} = c_{\mathfrak{sl}(2,\mathbb{R})} - 1$ and the compact internal $\mathcal{M}$ CFT with central charge $c_{\mathcal{M}}$. Hence,

$$\mathrm{SL}(2,R)\times\mathcal{M} = \Big(\mathrm{SL}(2,R)/U(1)\Big)\times U(1)\times\mathcal{M} \quad (485)$$

For such a CFT factorized into $U(1)$, the coset $\mathrm{SL}(2,R)/U(1)$ and the the internal $\mathcal{M}$ CFT, the '$\hat{L}$' Virasoro descendants of the $|f\rangle$ state contain no null vectors provided the $\mathrm{SL}(2,R)/U(1)\times\mathcal{M}$ conformal weight of $|f\rangle$, i.e., $h_{\textbf{rest}} = h_{\textbf{coset}} + h$ is strictly positive.

Only then the states of the form $\left\{\hat{L}_{-n_1}\cdots\hat{L}_{-n_K}\ J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle\right\}$ be a complete basis of $\hat{\mathcal{H}}$. Now, $\hat{L}_0 = L_0 + \mathcal{L}_0 - L_0^{(J^3)}$. Thus, $h_{\text{rest}} = h_{\text{coset}} + h$, where h_{coset} is the eigenvalue of $L_0 - L_0^{(J^3)}$ and h is the internal CFT's conformal weight (eigenvalue of $\mathcal{L}_0$). Observe that

$$L_0^{(J^3)}|f\rangle = \frac{\kappa_{33}}{k}\left(J^3_0J^3_0 + 2\sum_{k>0}J^3_{-k}J^3_k\right)|f\rangle = -\frac{m^2}{k}|f\rangle \ \text{ where } \ J^3_0|f\rangle = m|f\rangle \quad \implies \quad h_{J^3} = -\frac{m^2}{k}$$

$$\text{and} \ \ L_0|f\rangle = \frac{1}{(k-2)}\left(C_2 + \sum_{m=1}^{\infty}(J^+_{-m}J^-_m + J^-_{-m}J^+_m) - 2\,J^3_{-m}J^3_m\right)|f\rangle = \Big(\frac{C_2}{k-2} + N\Big)|f\rangle$$

where $N \in \{0,1,2,\cdots\}$ is the $\mathrm{SL}(2,\mathbb{R})$ grade of $|f\rangle$. Thus, $h_{\text{coset}} = \Big(\dfrac{C_2}{k-2} + N\Big) - h_{J^3}$.

$$\implies \qquad \boxed{h_{\text{rest}} = \frac{C_2}{k-2} + N + h + \frac{m^2}{k}} \quad (486)$$

If $h_{\text{rest}} > 0$, then $\mathrm{Span}\big(\hat{L}_{-n_1}\hat{L}_{-n_2}\cdots\hat{L}_{-n_K}\ J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle\big) = \hat{\mathcal{H}}$. But note that the internal $\mathcal{M}$ CFT is unitary. Hence, $h \geq 0$. So, our only concern is to have $h_{\text{coset}} > 0$ positive.

For the principal continuous representations, $C_2 = \frac{1}{4} + s^2 > 0$. So, if $k > 2$, then $h_{\text{rest}} > 0$ is satisfied. For Discrete lowest weight representations, D_j^+, we use $C_2 = -j(j-1)$, and rearrange,

$$\frac{-j(j-1)}{k-2} + N + \frac{m^2}{k} = h_{\text{coset}} = \frac{2j(\frac{k}{2}-j)}{k(k-2)} + \frac{2N}{k}\left(\frac{k}{2} - j\right) + \frac{2j}{k}(-j+m+N) + \frac{(j-m)^2}{k} \quad (487)$$

To have $h_{\text{coset}} > 0$, we need each term in the above equation to be positive. Thus, we need,

$$k > 2 \ , \quad 0 < j < \frac{k}{2} \ \text{ and } \ m + N \geq j \quad (488)$$

The last inequality follows from the allowed values of 'm' in the lowest-weight representation.

Then, for the highest-weight representations D_j^-, we have the states of the form, $|j, m\rangle$ where $m = -j, -j-1, -j-2, \ldots$. Now, the conformal weight of the coset sector is

$$h_{\text{coset}} = \frac{-j(j-1)}{k-2} + N + \frac{m^2}{k},$$

To establish positivity, we determine the minimum of h_{coset} over all states in the representation. Since $N \geq 0$, m^2 is minimized at $m = -j$, we obtain the bound

$$h_{\text{coset}} \geq -\frac{j(j-1)}{k-2} + \frac{j^2}{k}.$$

We now analyze the right-hand side:

$$-\frac{j(j-1)}{k-2} + \frac{j^2}{k} = j\left(\frac{j}{k} - \frac{j-1}{k-2}\right) = j\left(\frac{j(k-2) - k(j-1)}{k(k-2)}\right) = j\left(\frac{k-2j}{k(k-2)}\right)$$

Thus,

$$h_{\text{coset}} \geq \frac{j(k-2j)}{k(k-2)}.$$

So, to have $h_{\text{coset}} > 0$, we require $k > 2$, $j > 0$, and $j < \frac{k}{2}$. Therefore, for highest-weight representations with $0 < j < \frac{k}{2}$ and $k > 2$, we get, $h_{\text{coset}} > 0$. This ensures the absence of null states in the coset sector.

Thus, for Discrete representations we need $k > 2$ and $\frac{1}{2} < j < \frac{k}{2}$ and for Continuous representations we just need $k > 2$. Only then, $h_{\text{coset}} > 0 \implies h_{\text{rest}} > 0$.

And then, we will get,

$$\text{Span}\big(L_{-n_1}L_{-n_2}\cdots L_{-n_K}\ J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle\big) = \hat{\mathcal{H}}$$

Now, we can show that imposing the Virasoro constraints on a state in the Hilbert space $\hat{\mathcal{H}}$, when written in the $\Big\{L_{-n_1}L_{-n_2}\cdots L_{-n_K}\ J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle\Big\}$ basis, is equivalent, modulo null states, to those states which are spanned by a basis of the same form but with no $L_{-n_1}\cdots L_{-n_K}$ operators. So that the Virasoro constraints can be imposed on those states written entirely in terms of just the J^3 descendants of $|f\rangle$.

From the previous step, we know that the Hilbert space can be spanned by states of the form $\big\{L_{-n_1}L_{-n_2}\cdots L_{-n_K}\ J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle\big\}$, $n_1\geq n_2\geq\cdots\geq n_K$ and $m_1\geq m_2\geq\cdots\geq m_M$, where $L_{n>0}|f\rangle=\mathcal{L}_{n>0}|f\rangle=J^3_{n>0}|f\rangle=0$. These states form a complete basis of $\hat{\mathcal{H}}$. Consider a state $|\psi\rangle$ in the Hilbert space $\hat{\mathcal{H}}$. Using the basis above, we can decompose it as:

$$|\psi\rangle=|\psi_0\rangle+|\chi\rangle \tag{489}$$

where $|\psi_0\rangle$ contains no L_{-n} operators, and $|\chi\rangle$ contains at least one L_{-n}. Explicitly,

$$|\psi_0\rangle\in\mathrm{Span}\Big(\big\{J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle\big\}\Big) \tag{490}$$

States containing at least one L_{-n} are called spurious states. Since for any physical state $|\phi\rangle$,

$$\langle\phi|L_{-n}=\big(L_n|\phi\rangle\big)^\dagger=0,\quad\forall\ n>0 \tag{491}$$

The spurious states decouple from physical states. When such states also satisfy the mass-shell condition, they are null and are modded out in the physical Hilbert space. Thus, we can write, $|\psi\rangle=|\psi_0\rangle+|\chi\rangle\sim|\psi_0\rangle$. If $|\psi\rangle$ is physical then it satisfies the Virasoro constraints $L_{n>0}|\psi\rangle+\underbrace{\mathcal{L}_{n>0}|\psi\rangle}_{0}=0$ and $(L_0+\mathcal{L}_0-1)|\psi\rangle=0$.

$$\Longrightarrow\ L_{n>0}|\psi_0\rangle+L_{n>0}|\chi\rangle=0\quad\Longrightarrow\quad\sum_{m_1,m_2\cdots m_M}L_{n>0}\big(J^3_{-m_1}\cdots J^3_{-m_K}|f\rangle\big)+L_{n>0}|\chi\rangle=0 \tag{492}$$

Now consider the first term. Using the commutation relation $[L_n,J^3_{-m}]=mJ^3_{n-m}$, we see that acting with L_n $(n>0)$ produces only J^3 modes and does not generate any L_{-n}. Thus $L_n|\psi_0\rangle$ contains no L_{-n} operators. Hence, $L_n|\psi_0\rangle\in\mathrm{Span}\big(\{J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle\}\big)$.

In the second term, when L_n acts on the spurious state, it produces a linear combination of states with all the terms with at least one L_{-n} operator acting on it, provided the total central charge is $c=26$, and the $L_0+\mathcal{L}_0$ eigenvalue of the spurious states is 1. Then, if $|\chi\rangle$ is spurious state, then $L_{n>0}|\chi\rangle$ is a spurious state.

Thus, equation $L_{n>0}|\psi_0\rangle+L_{n>0}|\chi\rangle=0$ contains the sum two types of states. One with no L_{-n} of the form $|\psi_0\rangle$ and the other containing at least one L_{-n} which is spurious.

Since the basis $\{L_{-n_1}L_{-n_2}\cdots L_{-n_K}\ J^3_{-m_1}J^3_{-m_2}\cdots J^3_{-m_M}|f\rangle\}$ is linearly independent, these two sectors cannot cancel each other. Therefore, they must vanish separately:

$$\implies L_n|\psi_0\rangle = 0 \ \text{ and } \ L_n|\chi\rangle = 0 \quad \forall\, n > 0 \tag{493}$$

If the original state is physical, it satisfies $(L_0+\mathcal{L}_0-1)|\psi\rangle = 0 \implies (L_0+\mathcal{L}_0-1)\big(|\psi_0\rangle+|\chi\rangle\big) = 0$. Now, since the two parts are linearly independent, $(L_0+\mathcal{L}_0-1)|\psi_0\rangle = 0$ and $(L_0+\mathcal{L}_0-1)|\chi\rangle = 0$. Thus, $|\psi_0\rangle$ and $|\chi\rangle$ are physical states. So, $|\chi\rangle$ has to be a null state (a spurious physical state). Since the null states are removed from the Hilbert space to get the observable physical states, we can choose the observable physical states $|\psi\rangle$ such that it has no L_{-n} operators.

$$\implies |\psi\rangle \in \text{Span}\Big(\big\{J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle\big\}\Big) \tag{494}$$

Thus, we may choose the physical states to lie in the subclass of states described above. Up to this point, the analysis has established a convenient basis of the Hilbert space in terms of states of the form $\{L_{-n_1}\cdots L_{-n_N}\ J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle\}$. These states form a linearly independent, normalizable, and complete basis provided that $k > 2$, and that $|f\rangle$ satisfies $L_{n>0}|f\rangle = \mathcal{L}_{n>0}|f\rangle = J^3_{n>0}|f\rangle = 0$ which belong to the tensor product the internal $\mathcal{M}$ CFT and an affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ representation, $\hat{\mathcal{D}}^{\pm}_j$ with $\frac{1}{2} < j < \frac{k}{2}$ or $\hat{\mathcal{C}}^{\alpha}_j$. Furthermore, any state $|\psi\rangle \in \hat{\mathcal{H}}$ can be decomposed as $|\psi\rangle = |\psi_0\rangle + |\chi\rangle$, where $|\psi_0\rangle$ contains no L_{-n} excitations, and $|\chi\rangle$ consists of states with at least one L_{-n} excitation and is therefore spurious. Imposing the Virasoro constraints on $|\psi\rangle$ is equivalent to imposing them separately on $|\psi_0\rangle$ and $|\chi\rangle$, provided that the total central charge satisfies $c_{\text{total}} = 26$. If $|\psi\rangle$ satisfies the Virasoro constraints, then $|\chi\rangle$ must also satisfy them. Since $|\chi\rangle$ is spurious and satisfies the physical state conditions, it is a null state. Such spurious (null) states are modded out in the physical Hilbert space. Consequently, imposing the Virasoro constraints on $|\psi\rangle$ is equivalent to imposing them on the reduced part containing only J^3 excitations.

However, the positivity of the norm is not yet fully guaranteed at this stage, since states involving J^3_{-n} oscillators can still contain negative-norm components. The final step of the proof is to show that the Virasoro constraints further restrict the physical states to obey $J^3_{n>0}|\psi\rangle = 0$, thereby eliminating these remaining negative-norm contributions.

Now we shall impose the Virasoro constraints on the Hilbert space (modulo null states) whose states are of the form $|\psi\rangle \in \text{Span}\big(\{J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle\}\big)$. Physical states satisfy $L_{n>0}|\psi\rangle = 0$ and $\mathcal{L}_{n>0}|\psi\rangle = 0$. Recall the decomposition of the Virasoro generators $L_n + \mathcal{L}_n = L^{(J^3)}_n + \hat{L}_n$ where $L^{(J^3)}_n$ is the Sugawara stress tensor of the $U(1)$ current J^3, also we previously saw that $[\hat{L}_n, J^3_m] = 0$. Since the state $|\psi\rangle$ contains only J^3 oscillators, the constraint

$$L_{n>0}|\psi\rangle = 0 \quad\implies\quad \underbrace{\hat{L}_{n>0}|\psi\rangle}_{0} + L^{(J^3)}_{n>0}|\psi\rangle - \underbrace{\mathcal{L}_{n>0}|\psi\rangle}_{0} = 0 \quad\implies\quad L^{(J^3)}_{n>0}|\psi\rangle = 0 \tag{495}$$

This was true because,

$$\hat{L}_{n>0}|\psi\rangle = \hat{L}_{n>0} \sum_{\{m_i\}} c_{\{m_i\}}\, J^3_{-m_1} \cdots J^3_{-m_M}|f\rangle = \sum_{\{m_i\}} c_{\{m_i\}}\, J^3_{-m_1} \cdots J^3_{-m_M} \underbrace{\hat{L}_{n>0}|f\rangle}_{0} = 0 \quad (496)$$

The current J^3 forms a $U(1)$ current algebra $[J^3_m, J^3_n] = -\frac{k}{2}\, m\, \delta_{m+n}$. The corresponding Sugawara stress tensor is,

$$L_n^{(J^3)} = -\frac{1}{k}\left(J_0^3 J_n^3 + \sum_{m=1}^{\infty} J^3_{n-m}J^3_m + J^3_{-m}J^3_{n+m}\right) \quad (497)$$

Now, we shall prove that, if there exists a state $|\psi\rangle$ with J^3 excitations in it, then, the condition that $L^{(J^3)}_{n>0}|\psi\rangle = 0$ will imply that $J^3_{n>0}|\psi\rangle = 0$ provided there are no null states in the $U(1)$ Virasoro descendants. A general state can be written as

$$|\psi\rangle = \sum_{\{m_i\}} c_{\{m_i\}}\, J^3_{-m_1} J^3_{-m_2} \cdots J^3_{-m_M}|f\rangle \quad (498)$$

where $m_i > 0$ and the state $|f\rangle$ satisfies $J^3_{n>0}|f\rangle = 0$. Assume that $|\psi\rangle$ satisfies the Virasoro constraints in the $U(1)$ sector, i.e., $L^{(J^3)}_{n>0}|\psi\rangle = 0$. Also, we previously proved that Virasoro generators built from J^3 satisfy $[L_n^{(J^3)}, J^3_{-\ell}] = \ell\, J^3_{n-\ell}$. Now, separate the highest mode. Assume $|\psi\rangle$ contains at least one J^3 oscillator. Let M be the largest mode number appearing.

$$\Longrightarrow\ |\psi\rangle = c_{\{M\cdots\}}\, J^3_{-M}|\phi\rangle + |\text{lower mode excitations}\rangle \quad (499)$$

where $c_{\{M\cdots\}}$ is the coefficient of the J^3_{-M} term and $|\phi\rangle$ contains no J^3_{-M} excitation in it. Now, act with $L_M^{(J^3)}$. Then,

$$L_M^{(J^3)} J^3_{-M}|\phi\rangle = J^3_{-M}\, L_M^{(J^3)}|\phi\rangle + [L_M^{(J^3)}, J^3_{-M}]|\phi\rangle \quad (500)$$

Use, $[L_M^{(J^3)}, J^3_{-M}] = M\, J^3_0$ and $J^3_0|\phi\rangle = m_f|\phi\rangle$, which is the J^3_0-charge of $|f\rangle$. Since, $[J^3_0, J^3_n] = 0$, $m_\phi = m_\psi = m_f$. Then, we obtain

$$L_M^{(J^3)} J^3_{-M}|\phi\rangle = J^3_{-M}\, L_M^{(J^3)}|\phi\rangle + M m_f|\phi\rangle \quad (501)$$

$$\Longrightarrow\ L_M^{(J^3)}|\psi\rangle = c_{\{M\cdots\}}\, M m_f|\phi\rangle + c_{\{M\cdots\}} \cdot (\text{terms containing } J^3_{-M}) + (\text{lower mode excitations}) \quad (502)$$

In the absence of null states, all descendant states are linearly independent. Therefore, the condition $L_M^{(J^3)}|\psi\rangle = 0$ implies that each independent component must vanish separately. So,

$$c_{\{M\cdots\}}\, M m_f = 0 \quad (503)$$

Now, if $m_f \neq 0$, then this implies that the coefficient of J^3_{-M} which is $c_{\{M\cdots\}}$ must vanish. This contradicts the assumption that such a term was present. Now, repeating the same argument for the next-highest mode, we eliminate all J^3_{-n} excitations. Hence, we can conclude that $|\psi\rangle = |f\rangle$ cannot have any J^3 excitations in it ($|\psi\rangle = |f\rangle$), and thus it can satisfy $J^3_{n>0}|\psi\rangle = 0$. So, in a module without null states and for $m_f \neq 0$, we obtain $L^{(J^3)}_{n>0}|\psi\rangle = 0 \quad \Longrightarrow \quad J^3_{n>0}|\psi\rangle = 0$.

If null states were present, different descendant states could become linearly dependent, and the above argument would fail because one could not isolate the coefficient of J^3_{-N}. This is why the condition of absence of null states is essential.

Also, to make the discussion complete, we should note that a general state can also be a linear combination of J^3 descendants of different primary states $|f_m\rangle$ with different m values. But even then, we can decompose the argument into different J^3_0-eigen-spaces.

Also, since $[L^{(J^3)}_n, J^3_0] = 0$, the operators $L^{(J^3)}_n$ do not mix different m-sectors. Hence, if $L^{(J^3)}_{n>0}|\psi\rangle = 0$, then necessarily $L^{(J^3)}_{n>0}|\psi_m\rangle = 0$ for each $m \neq 0$ where the arguments apply independently. $L^{(J^3)}_{n>0}|\psi_m\rangle = 0 \implies |\psi_m\rangle$ contains no J^3_{-n} excitations. Thus, $|\psi_m\rangle = |f_m\rangle$, where $|f_m\rangle$ is a state with J^3_0-charge of m (whose $m \neq 0$). Then, the full state reduces to

$$|\psi\rangle = \sum_{m\neq 0} |f_m\rangle \; + \; |\psi_{m=0}\rangle \tag{504}$$

where only the $m = 0$ sector may still contain J^3_{-n} excitations. The above arguments fail for $m = 0$ because the crucial term proportional to J^3_0 vanishes. This sector must be analyzed separately using the mass-shell condition, as done below.

For a Virasoro module generated by $L^{(J^3)}_{-n}$ operators, $L^{(J^3)}_{n>0}|\psi\rangle = 0 \implies J^3_{n>0}|\psi\rangle = 0$ only if the module has no null states. Thus, we must analyze whether null states appear in the $U(1)$ Virasoro descendants. The state $|f\rangle$ satisfies $J^3_0|f\rangle = m|f\rangle$. The $U(1)$ conformal weight of $|f\rangle$ is given by $h_{J^3} = -\frac{m^2}{k}$. For generic values of $m \neq 0$, this does not correspond to any degenerate value of h_{J^3}, and hence the $U(1)$ Virasoro module contains no null states. For a free boson CFT, null states occur only when $m = 0$. Thus, if $m \neq 0$, there are no null states and the condition $L^{(J^3)}_{n>0}|\psi\rangle = 0 \implies J^3_{n>0}|\psi\rangle = 0$.

So, we must analyze whether physical states with $m = 0$ exist, and check if they cause any problem to the above conclusion. The mass shell condition is $(L_0 + \mathcal{L}_0 - 1)|\psi\rangle = 0$.

$$\text{As, } L_0 = -\frac{j(j-1)}{k-2} + N \quad \text{and} \quad \mathcal{L}_0 = h \implies -\frac{j(j-1)}{k-2} + N + h - 1 = 0. \tag{505}$$

Thus any physical state should satisfy the above equation where N is the grade of the state in the $\mathrm{SL}(2,\mathbb{R})$ sector and $h > 0$ is the conformal weight of the internal CFT.

When $0 < j < 1$: The discrete representations $\mathcal{D}_j^{\pm}$ have $m = \pm j\,,\ \pm(j+1)\,,\ \pm(j+2), \cdots$. And the affine extension gives states of all possible values of $m = j + n\ , n \in \mathbb{Z}$. So, for $0 < j < 1$, m can never be zero. Therefore, null states cannot occur. For continuous representations, $m = \alpha + n,\ n \in \mathbb{Z}$ and $0 \leq \alpha < 1$ (doesn't depend on j: we will deal with this later)

When $j = 1$: The mass shell condition becomes $N + h - 1 = 0$. There is a possibility of physical states having $m = 0$ in the discrete representations $\hat{\mathcal{D}}_{j=1}^{\pm}$ when we act on the lowest (or highest) state by the affine current modes $J_n^{\pm}$ because it can change the m value towards zero. This can be shown using the commutation relations $[J_0^3, J_{-n}^{\pm}] = \pm J_{-n}^{\pm} \implies J_0^3\big(J_{-n}^{\pm}|j,m\rangle\big) = (m \pm 1)\big(J_{-n}^{\pm}|j,m\rangle\big)$. Thus, even in the zero mode representation, for $j = 1$, if $m = \pm 1\,,\ \pm 2\,,\ \pm 3, \cdots$, we can act the $|j = 1, m = \pm 1\rangle$ with $J_{-n}^{\mp}$ such that,

$$J_0^3\big(J_{-n}^{\mp}|j = 1, m = \pm 1\rangle\big) = (\pm 1 \mp 1)\big(J_{-n}^{\mp}|j = 1, m = \pm 1\rangle\big) = 0\big(J_{-n}^{\mp}|j = 1, m = \pm 1\rangle\big) \quad (506)$$

Thus, the state $J_{-n}^{-}|j = 1, m = 1\rangle$ in the discrete lowest weight representation and the state $J_{-n}^{+}|j = 1, m = -1\rangle$ in the discrete highest weight representation have $m = 0$.

However, these states must also satisfy the physical state (mass-shell) condition. For $j = 1$ the mass shell condition becomes $N + h - 1 = 0$. Since $h \geq 0$, the minimal solution is $N = 1$ and $h = 0$. Therefore, the physical state must contain one current oscillator. Thus, they have to be the states given by,

$$J_{-1}^{-}|j = 1, m = 1\rangle \in \hat{D}_{j=1}^{+} \qquad \text{and} \qquad J_{-1}^{+}|j = 1, m = -1\rangle \in \hat{D}_{j=1}^{-} \quad (507)$$

These are the only physical states that have $m = 0$, with $N = 1$ (one oscillator excitation) and $h = 0$. The norm of these states is

$$\begin{aligned}
&\langle j = 1, m = \pm\, 1|J_1^{\pm}J_{-1}^{\mp}|j = 1, m = \pm 1\rangle \\
&\qquad = \langle j = 1, m = \pm 1|\big([J_1^{\pm}, J_{-1}^{\mp}] + J_{-1}^{\mp}J_1^{\pm}\big)|j = 1, m = \pm 1\rangle \\
&\qquad = \langle j = 1, m = \pm 1|\big(\mp 2J_0^3 + k\big)|j = 1, m = \pm 1\rangle = \mp 2(\pm 1) + k
\end{aligned} \quad (508)$$

$$\implies \langle j = 1, m = \pm\, 1|J_1^{\pm}J_{-1}^{\mp}|j = 1, m = \pm 1\rangle = k - 2$$

Thus, for $k > 2$, these states have positive norm, $m = 0$, and satisfy the Virasoro constraints.

In a unitary CFT, the only state with $h = 0$ is the vacuum of the internal sector, so the full physical state is

$$|\psi_+\rangle = J_{-1}^{-}|j = 1\rangle \otimes |0\rangle_{\text{int}}\ \in \hat{\mathcal{D}}_{j=1}^{+} \otimes \mathcal{M} \qquad \text{and} \qquad |\psi_-\rangle = J_{-1}^{-}|j = 1\rangle \otimes |0\rangle_{\text{int}}\ \in \hat{\mathcal{D}}_{j=1}^{-} \otimes \mathcal{M}$$

Furthermore, the eigenvalue of J_0^3 determines the spacetime energy in AdS_3, and for this state, one finds $m = 0$. Since the spacetime conformal dimension is $\Delta = m + \bar{m}$, this implies $\Delta = 0$, which corresponds to the identity operator of the boundary CFT. Thus, these states represent a normalizable zero mode in the bulk and are associated with the spacetime identity operator.

When $j > 1$: We now explain why a physical state with $m = 0$ is not possible for $j > 1$. The physical states have to satisfy $-\frac{j(j-1)}{k-2} + N + h - 1 = 0$. Consider a lowest-weight representation $\mathcal{D}_j^+$, the primary state is $|j, m = j\rangle$. The action of J_{-n}^- lowers m by one unit, so descendant states have $m = j -$ (number of lowering operators). To reach $m = 0$, we must apply at least j lowering operators, which implies $N \geq j$. Since m changes in integer steps, this is only possible if j is an integer. For $j > 1$, this means $j \geq 2$. Using $h \geq 0$, we obtain that a physical state satisfies,

$$-\frac{j(j-1)}{k-2} + N - 1 = -h \leq 0 \qquad \Longrightarrow \qquad -\frac{j(j-1)}{k-2} + N - 1 \leq 0 \tag{509}$$

However, using $N \geq j$ gives

$$-\frac{j(j-1)}{k-2} + N - 1 \geq -\frac{j(j-1)}{k-2} + j - 1 \tag{510}$$

We now simplify the right-hand side of the above inequality.

$$-\frac{j(j-1)}{k-2} + j - 1 = (j-1)\left(1 - \frac{j}{k-2}\right) = \frac{(j-1)(k-2-j)}{k-2} \tag{511}$$

Thus we obtain

$$-\frac{j(j-1)}{k-2} + N - 1 \geq \frac{(j-1)(k-2-j)}{k-2} \tag{512}$$

In this case, we have $j \geq 2$, and also there are conditions on j as $j < \frac{k}{2} \implies 2j < k$, this gives $j < k - j \implies 2 \leq j < k - j \implies 2 < k - j \implies k - 2 - j > 0$. Hence, we get,

$$\begin{aligned} \frac{(j-1)(k-2-j)}{k-2} > 0 \qquad &\Longrightarrow \qquad -\frac{j(j-1)}{k-2} + N - 1 > 0 \\ &\Longrightarrow \quad -\frac{j(j-1)}{k-2} + N + h - 1 > h > 0 \\ &\Longrightarrow \qquad -\frac{j(j-1)}{k-2} + N + h - 1 > 0 \end{aligned} \tag{513}$$

This contradicts the mass-shell condition for a physical state. Hence, for $j > 1$, a physical state with $m = 0$ cannot exist.

The argument for the discrete highest-weight representation $\hat{\mathcal{D}}_j^-$ proceeds analogously to the lowest-weight case. The primary state satisfies $J_0^3|j,m\rangle = m|j,m\rangle$ with $m = -j, -j-1, \ldots$, and the action of $J_{-n}^{\pm}$ is determined by $[J_0^3, J_{-n}^{\pm}] = \pm J_{-n}^{\pm}$. So, J_{-n}^+ raises m by one unit.

Starting from $m = -j$, to reach $m = 0$, one must apply at least j raising operators, implying $N \geq j$. Since m changes in integer step, this requires $j \in \mathbb{Z}$ with $j \geq 2$ for $j > 1$. The physical state must satisfy the mass-shell condition

$$-\frac{j(j-1)}{k-2} + N + h - 1 = 0,$$

with $h \geq 0$, which implies $-\frac{j(j-1)}{k-2} + N - 1 \leq 0$. However, using $N \geq j$ gives

$$-\frac{j(j-1)}{k-2} + N - 1 \geq -\frac{j(j-1)}{k-2} + j - 1 = \frac{(j-1)(k-2-j)}{k-2}.$$

For the allowed range $2 \leq j \leq \frac{k}{2}$, the right-hand side is strictly positive, leading to a contradiction. Therefore, no physical state with $m = 0$ exists for $j > 1$ in $\hat{\mathcal{D}}_j^-$.

The $m = 0$ Physical State in the Continuous Representation: This representation is $\mathcal{C}_j^\alpha$, $j = \frac{1}{2} + is$, $s \in \mathbb{R}$, with states labeled as $|j, \alpha; m\rangle$, where $m = \alpha + n$, $n \in \mathbb{Z}$, $0 \leq \alpha < 1$ are the eigenvalues of J_0^3. To have a state with $m = 0$, we require $\alpha + n = 0$; this is possible only if $\alpha = 0$. Thus only the representation $\mathcal{C}_j^{\alpha=0}$ contains an $m = 0$ state given by $|j, \alpha = 0, m = 0\rangle$. For it to be a physical state, it should satisfy the mass-shell condition $-\frac{j(j-1)}{k-2} + N + h - 1 = 0$. For the continuous representation, it becomes,

$$\frac{1/4 + s^2}{k-2} + N + h - 1 = 0 \quad \Longrightarrow \quad N + h = 1 - \frac{1/4 + s^2}{k-2} \tag{514}$$

Since $N \geq 0$ and $h \geq 0$ and $\dfrac{1/4 + s^2}{k-2} > 0$, we must have $N + h < 1 \implies N < 1$. Since the allowed values of N are $\{0, 1, 2, 3 \cdots\}$. The only possibility is $N = 0$ (the state has no oscillator excitations) if it should satisfy the physical state condition. Thus, we get,

$$\frac{1/4 + s^2}{k-2} = 1 - h > 0 \implies 0 \leq h \leq 1 \implies 0 \leq \frac{1/4 + s^2}{k-2} \leq 1 \implies -\frac{1}{4} \leq s^2 \leq k - \frac{9}{4} \tag{515}$$

Thus, the only physical state with $m = 0$ in the continuous representation is

$$\left| j = \frac{1}{2} + is, \alpha = 0, m = 0 \right\rangle \in \mathcal{C}^{\alpha=0}_{j=\frac{1}{4}+is} \ , \ s^2 \leq k - \frac{9}{4} \tag{516}$$

The states in this representation are delta-function normalizable. Although this state lies in the $m = 0$ sector, it does not introduce negative-norm states and is harmless.

Conclusion of No-Ghost theorem

Using the Virasoro constraints $(L_n + \mathcal{L}_n - \delta_{n,0})|\psi\rangle = 0$, $\forall\, n \geq 0$ and the fact that $c_{\text{total}} = 26$, any physical state can be chosen, modulo null states, to have no L_{-n} excitations:

$$|\psi\rangle = J^3_{-m_1} \cdots J^3_{-m_M}|f\rangle$$

Then, using the decomposition $L_n + \mathcal{L}_n = L_n^{(J^3)} + \hat{L}_n$, the Virasoro constraint implies $L^{(J^3)}_{n>0}|\psi\rangle = 0$. For the states with J^3_0-charge $m \neq 0$, the $U(1)$ Virasoro module generated by $L^{(J^3)}_{-n}$ has no null states. Therefore, $L^{(J^3)}_{n>0}|\psi\rangle = 0 \implies J^3_{n>0}|\psi\rangle = 0$

But $J^-_{-1}|j, m = 1\rangle \in \hat{\mathcal{D}}^+_{j=1}$, $J^+_{-1}|j, m = -1\rangle \in \hat{\mathcal{D}}^-_{j=1}$, and $\big|j, \alpha = 0, m = 0\big\rangle \in \mathcal{C}^{\alpha=0}_{j=\frac{1}{4}+is}$, $s^2 \leq k - \frac{9}{4}$ are the only states which are physical and have $m = 0$. They all have positive norms and do not introduce ghosts. Thus, modulo null states, all physical states satisfy $J^3_{n>0}|\psi\rangle = 0$ and hence belong to the coset $\mathrm{SL}(2,\mathbb{R})/U(1)$. For this to be true, we require $k > 2$ and $c_{\text{total}} = 26$, and the physical states $|\psi\rangle$ have to be taken from the descendants of discrete series $\mathcal{D}^\pm_j$ with $\frac{1}{2} \leq j \leq \frac{k}{2}$ or from the continuous series $C^\alpha_{j=\frac{1}{2}+is}$, $s \in \mathbb{R}$. The remaining states are built on zero-mode representations $\mathcal{D}^\pm_j$ and $\mathcal{C}^\alpha_j$, which satisfy the usual mass-shell conditions.

Therefore, all physical states satisfying the Virasoro constraints can be expressed, modulo null states, as states in the coset $\mathrm{SL}(2,\mathbb{R})/U(1)$, and all such states have a positive norm.

$$\boxed{\text{The physical Hilbert space contains no negative-norm (ghost) states.}}$$

This establishes the no-ghost theorem for the $\mathrm{SL}(2,\mathbb{R})$ WZW model in the unflowed sector.

7.2.2 Weight diagram in the unflowed representation

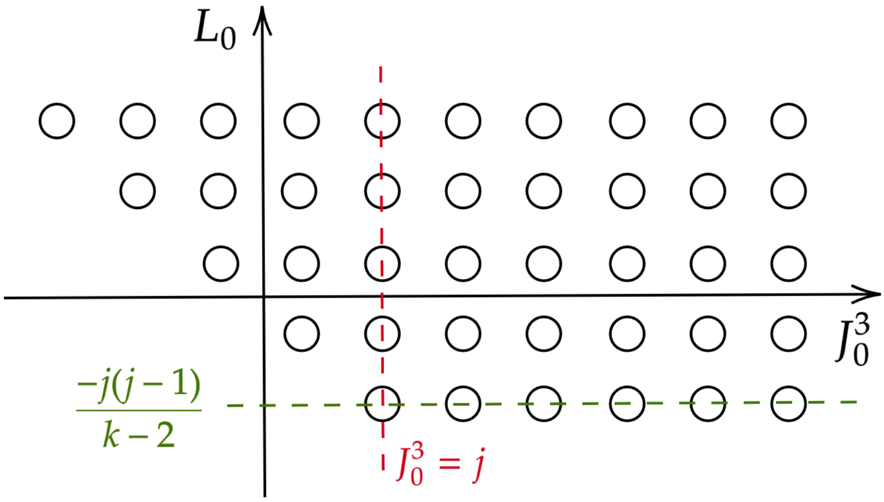

Figure 1: Weight diagram in unflowed lowest-weight representation $\hat{\mathcal{D}}_j^+$

This represents the structure of the lowest-weight representation $\hat{\mathcal{D}}_j^+$ built on a lowest-weight primary state. Each point in the diagram corresponds to a state labeled by the eigenvalues of $J_0^3 \equiv m$ (horizontal axis) and L_0 (vertical axis). Thus, every state is characterized by (J_0^3, L_0).

The bottom-left-most point corresponds to the primary state $|j, m = j\rangle$ which is the lowest-weight state that satisfies $J_0^3|j, m = j\rangle = j|j, m = j\rangle$ and $L_0|j, m = j\rangle = -\frac{j(j-1)}{k-2}|j, m = j\rangle$. This is the lowest-weight state of the discrete representation $\mathcal{D}_j^+$. The green dashed line in the diagram indicate $m = j, j+1, \cdots$ with $L_0 = -\frac{j(j-1)}{k-2}$.

Zero-Mode Descendants: Acting with the zero-mode operator J_0^+ increases m value of a state without changing L_0. $J_0^+|j,m\rangle = A_m|j, m+1\rangle$ where $L_0\big(J_0^+|j,m\rangle\big) = \dfrac{-j(j-1)}{k-2}\big(J_0^+|j,m\rangle\big)$ (since $[L_0, J_0^a] = 0$). Thus, we obtain states $|j,m\rangle$, $m = j, j+1, j+2, \cdots$ all lying on the same horizontal line built towards the right (horizontal line).

Affine Descendants: The affine (current) modes J_{-n}^a with $n > 0$ generate higher-level states. Since $[L_0, J_{-n}^a] = nJ_{-n}^a$, for the states $J_{-n}^a|j,m\rangle$ we have,

$$L_0\big(J_{-n}^a|j,m\rangle\big) = \left(\frac{-j(j-1)}{k-2} + n\right)\big(J_{-n}^a|j,m\rangle\big) \tag{517}$$

Hence, J_{-n}^a takes $L_0 \to L_0 + n$ and increases the oscillator level. This corresponds to moving upward in the diagram. Since we are dealing with the lowest weight representation, all positive current modes acting on the zero-mode representation states give zero. Hence, L_0 value cannot decrease below $L_0^{\min} = \dfrac{-j(j-1)}{k-2}$.

Boundedness of L_0 (Positive Energy Representation): This representation is a *positive energy representation*, meaning that L_0 is bounded from below. The lowest value is $L_0^{\min}$, corresponding to the primary states $|j, m\rangle$ without any oscillator excitations in it where $L_0^{\min} = -\frac{j(j-1)}{k-2}$. All excited states have $L_0 = -\frac{j(j-1)}{k-2} + N$, $N = 0, 1, 2, \cdots$, so that $L_0 \geq L_0^{\min}$. Thus, the spectrum is bounded from below and increases discretely with the level N. A general state with oscillator level N is denoted by $|j, m, N\rangle$ where

$$L_0|j,m,N\rangle = \Big(\frac{-j(j-1)}{k-2} + N\Big)|j,m,N\rangle \quad , \quad N = 0,1,2,3,\cdots \tag{518}$$

Moving upward in the diagram corresponds to increasing the oscillator level N. At each level N, one obtains states of the form

$$J^{a_1}_{-n_1} J^{a_2}_{-n_2} \cdots |j, m\rangle \quad , \quad \sum_i n_i = N \tag{519}$$

Because of the commutator brackets, $[J^3_0, J^3_{-n}] = 0$ and $[J^3_0, J^{\pm}_{-n}] = \pm J^{\pm}_{-n}$, we see that the different current modes act as follows:

$$J^+_{-n} : m \to m+1, \quad J^-_{-n} : m \to m-1, \quad J^3_{-n} : m \to m \tag{520}$$

Thus, at fixed L_0 (at a given row), the states spread horizontally over a range of m values, forming different columns seen in the diagram.

Asymmetry of the Representation: At level $N = 0$, only the zero modes act, so the representation is lowest-weight with $m \geq j$. At higher levels, however, the action of J^-_{-n} allows m to decrease, and the states extend in both directions along the m-axis. However, a decrease in m below j is only possible in the presence of oscillator excitations. Consequently, one cannot arbitrarily lower the value of m without a corresponding increase in the L_0 eigenvalue.

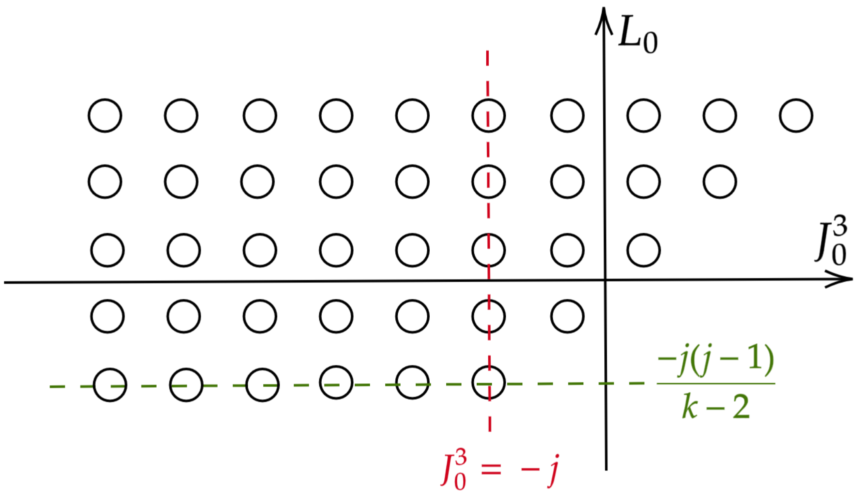

Figure 2: Weight diagram in unflowed highest weight representation $\hat{\mathcal{D}}^-_j$

7.3 Spectral flow and the long strings

The spectral flow is defined as a transformation of the current and Virasoro modes as follows:

$$\begin{aligned}
\tilde{J}^3_n &= J^3_n - \frac{k}{2} w\,\delta_{n,0} \\
\tilde{J}^\pm_n &= J^\pm_{n\pm w} \qquad\qquad\qquad\qquad w \in \mathbb{Z} \\
\tilde{L}_n &= L_n + wJ^3_n - \frac{k}{4} w^2\,\delta_{n,0}
\end{aligned} \tag{521}$$

We have already seen that the commutation relations among the modes $J^{3,\pm}_n$ and L_n are preserved under spectral flow. Hence, it defines an automorphism of the spectrum-generating algebra. Spectral flow maps one representation to another. Since $\mathrm{SL}(2,\mathbb{R})$ is non-compact and the kinetic term of this WZW model is not positive definite, spectral flow may map to representations in which L_0 is not bounded below. The spectrally flowed affine representations built on the zero-mode representations $\mathcal{D}^\pm_j$ and $\mathcal{C}^\alpha_j$ are denoted by $\hat{\mathcal{D}}^{\pm,w}_{\tilde{j}}$ and $\hat{\mathcal{C}}^{\alpha,w}_{\tilde{j}}$, respectively, where $\tilde{j}$ denotes the spin before spectral flow.

Explicitly proving the unboundedness of L_0 and boundedness of $\tilde{L}_0$

Before spectral flow, the commutation relation between the modes is given by,

$$[\tilde{J}^3_0, \tilde{J}^\pm_{-n}] = \pm \tilde{J}^\pm_{-n}\ ,\ [\tilde{J}^3_0, \tilde{J}^3_{-n}] = 0 \quad \text{and} \quad [\tilde{L}_0, \tilde{J}^a_{-n}] = n\tilde{J}^a_{-n}$$

$$\implies \tilde{J}^+_{-n} : \tilde{m} \to \tilde{m} + 1\ , \quad \tilde{J}^-_{-n} : \tilde{m} \to \tilde{m} - 1\ , \quad \tilde{J}^3_{-n} : \tilde{m} \to \tilde{m} \quad \text{and} \quad \tilde{L}_0 \to \tilde{L}_0 + n$$

So, each current raises the level $\tilde{L}_0 \to \tilde{L}_0 + n$. Thus, vertical motion along $\tilde{L}_0$ axis in the diagram corresponds to increasing level N, and motion along $\tilde{J}^3_0$ axis corresponds to changing $\tilde{m}$ via ladder operators.

Due to Spectral Flow on Ladder Structure, the algebra itself is unchanged, so the operators still satisfy $[J^3_0, J^\pm_{-n}] = \pm J^\pm_{-n}$. Thus, the ladder structure in m remains, $J^\pm_{-n} : \quad m \to m \pm 1$. However, the energy now transforms as $L_0 = \tilde{L}_0 - w\tilde{m} - \frac{k}{4}w^2$. Using the commutators,

$$\begin{aligned}
&[J^3_0, \tilde{J}^\pm_{-n}] = [\tilde{J}^3_0, \tilde{J}^\pm_{-n}] = \pm\tilde{J}^\pm_{-n} \quad \text{and} \quad [J^3_0, \tilde{J}^3_{-n}] = [\tilde{J}^3_0, \tilde{J}^3_{-n}] = 0 \\
&[L_0, \tilde{J}^\pm_{-n}] = [L_0, J^\pm_{-n\pm w}] = (n \mp w)J^\pm_{-n\pm w} = (n \mp w)\tilde{J}^\pm_{-n} \quad \text{and} \quad [L_0, \tilde{J}^3_{-n}] = n\tilde{J}^3_{-n}
\end{aligned} \tag{522}$$

When we act with current modes $\tilde{J}^{3,\pm}_{-n}$ on the (m, L_0) states, we see that,

$$\begin{aligned}
&\text{Action of } \tilde{J}^\pm_{-n} :\ m \to m \pm 1, \quad L_0 \to L_0 + n \mp w \\
&\text{Action of } \tilde{J}^3_{-n} :\ m \to m, \quad L_0 \to L_0 + n
\end{aligned} \tag{523}$$

Thus, unlike the unflowed case, the current modes do not uniformly raise L_0. The modes $\tilde{J}^-_{-n}$ and $\tilde{J}^3_{-n}$ always increases L_0, but the mode $\tilde{J}^+_{-n}$ increases L_0 only if $n > w$, decreases it for $n < w$ and for $n = w$ it leaves L_0 unchanged. Thus, the spectrum develops a *tilted structure*, where moving along the ladder in m also shifts L_0. Starting from a state $(\tilde{m}, \tilde{L}_0)$ or (m, L_0)

- Applying $\tilde{J}^-_{-w}$: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m} - 1, \tilde{L}_0 + w)$ and $(m, L_0) \to (m-1,\ L_0 + 2w)$
- Applying $\tilde{J}^+_{-w}$: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m} + 1, \tilde{L}_0 + w)$ and $(m, L_0) \to (m+1,\ L_0)$
- Applying $\tilde{J}^3_{-w}$: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m}, \tilde{L}_0 + w)$ and $(m, L_0) \to (m,\ L_0 + w)$

Thus, upward motion along $\tilde{L}_0$ is generated by all $\tilde{J}^a_{-n}$, diagonal motion arises due to mixing between m and L_0 by the $\tilde{J}^-_{-w}$ mode, and horizontal motion is generated by $\tilde{J}^+_{-w}$ mode which increases $m \to m+1$ but doesn't change L_0.

To explicitly see the unboundedness of L_0, consider,

- Applying $\tilde{J}^-_0$: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m} - 1, \tilde{L}_0)$ and $(m, L_0) \to (m-1,\ L_0 + w)$
- Applying $\tilde{J}^+_0$: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m} + 1, \tilde{L}_0)$ and $(m, L_0) \to (m+1,\ L_0 - w)$

Thus, we see that by repeated action of $\tilde{J}^+_0 = J^+_w$, we can reach arbitrarily large $\tilde{m}$ or m value but keep $\tilde{L}_0$ fixed, whereas L_0 can become arbitrarily negative. Using

$$L_0 = \tilde{L}_0 - \tilde{m}w - \frac{k}{4}w^2,$$

We see that L_0 can become arbitrarily negative as $\tilde{m}$ increases to a large value. Thus, the spectrum is unbounded below in L_0. But we can also show that the spectrum is bounded below in terms of $\tilde{L}_0$ doing a similar analysis.

Now, consider the action of $J^{3,\pm}_{-n}$ on the $(\tilde{m}, \tilde{L}_0)$ states.

$$\begin{aligned}
&[\tilde{J}^3_0, J^\pm_{-n}] = [J^3_0, J^\pm_{-n}] = \pm J^\pm_{-n} \quad \text{and} \quad [\tilde{J}^3_0, J^3_{-n}] = [J^3_0, J^3_{-n}] = 0 \\
&[\tilde{L}_0, J^\pm_{-n}] = [\tilde{L}_0, \tilde{J}^\pm_{-n\mp w}] = (n \pm w)\tilde{J}^\pm_{-n\mp w} = (n \pm w) J^\pm_{-n} \quad \text{and} \quad [\tilde{L}_0, J^3_{-n}] = n J^3_{-n}
\end{aligned} \tag{524}$$

When we act with a current mode on the $(\tilde{m}, \tilde{L}_0)$ states, we see that,

$$\begin{aligned}
&\text{Action of } J^\pm_{-n}: \ \tilde{m} \to \tilde{m} \pm 1, \quad \tilde{L}_0 \to \tilde{L}_0 + n \pm w \\
&\text{Action of } J^3_{-n}: \ \tilde{m} \to \tilde{m}, \quad \tilde{L}_0 \to \tilde{L}_0 + n
\end{aligned} \tag{525}$$

Thus, J^+_{-n} and J^3_{-n} always increases $\tilde{L}_0$, but J^-_{-n} increases $\tilde{L}_0$ if $n > w$, it decreases $\tilde{L}_0$ for $n < w$, and for $n = w$, it leaves $\tilde{L}_0$ unchanged. Starting from a state $(\tilde{m}, \tilde{L}_0)$ or (m, L_0):

- Applying J^-_{-w}: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m}-1, \tilde{L}_0)$ and $(m, L_0) \to (m-1,\ L_0 + w)$
- Applying J^+_{-w}: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m}+1, \tilde{L}_0 + 2w)$ and $(m, L_0) \to (m+1,\ L_0 + w)$
- Applying J^3_{-w}: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m}, \tilde{L}_0 + w)$ and $(m, L_0) \to (m,\ L_0 + w)$

To explicitly see the boundedness of $\tilde{L}_0$, consider,

- Applying J^-_0: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m}-1, \tilde{L}_0 - w)$ and $(m, L_0) \to (m-1,\ L_0)$
- Applying J^+_0: $(\tilde{m}, \tilde{L}_0) \to (\tilde{m}+1, \tilde{L}_0 + w)$ and $(m, L_0) \to (m+1,\ L_0)$

We know that the m or $\tilde{m}$ value is bounded below; we cannot reach a state with an arbitrarily negative m or $\tilde{m}$ value. Thus, even if we repeatedly act by the $J^-_0 = \tilde{J}^-_w$, we will eventually hit a lower bound state, and further action will always lead to zero.

Physical States: Imposing the Virasoro constraint $L_0 + \mathcal{L}_0 = 1 \implies L_0 = 1 - h$ selects a subset of states. On this slice, J^3_0 remains bounded below. We see that this selects a slice of

$$J^3_0 w = \tilde{L}_0 + \frac{k}{4}w^2 - L_0 = \tilde{L}_0 + \frac{k}{4}w^2 - 1 + h \tag{526}$$

Since, $\tilde{L}_0 = -\dfrac{\tilde{j}(\tilde{j}-1)}{k-2} + N$, we get $J^3_0 = -\dfrac{\tilde{j}(\tilde{j}-1)}{k-2} + N + h + \dfrac{k}{4}w^2 - 1$.

7.3.1 Spectrally Flowed Representation ($w = 1$) of $\widehat{D}^+_j$

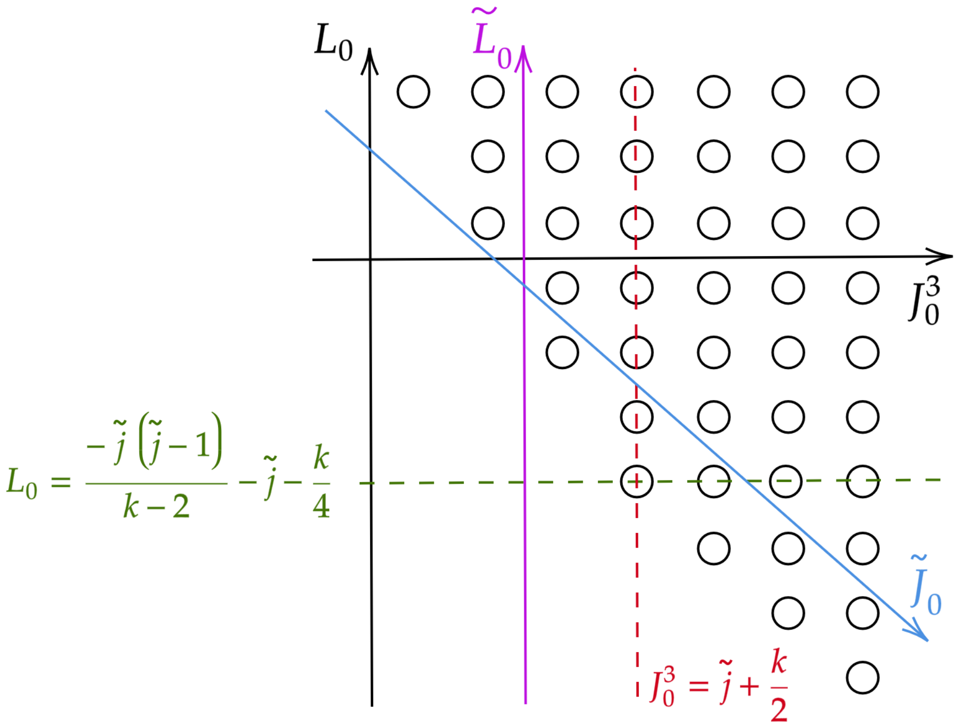

Figure 3: Spectrally flowed $\hat{\mathcal{D}}^{+,w=1}_j$ representation.

The diagram represents the spectral flow ($w = 1$) of the lowest-weight discrete representation $\hat{D}^+_j$. The key feature is that spectral flow mixes the worldsheet energy L_0 with the charge J^3_0, resulting in a tilted structure in the (J^3_0, L_0) plane.

This particular spectrally flowed representation is obtained by a relabeling of states under the transformation $\tilde{J}^3_0 = J^3_0 - \dfrac{k}{2}$ and $\tilde{L}_0 = L_0 + J^3_0 - \dfrac{k}{4}$. Equivalently, we can invert these relations as

$$J^3_0 = \tilde{J}^3_0 + \frac{k}{2} \quad \text{and} \quad L_0 = \tilde{L}_0 - \tilde{J}^3_0 - \frac{k}{4} \tag{527}$$

Thus a state labeled by $(\tilde{m}, \tilde{L}_0)$ before spectral flow is mapped to $m = \tilde{m} + \frac{k}{2}$, $L_0 = \tilde{L}_0 - \tilde{m} - \frac{k}{4}$.

In the unflowed representation, states lie on horizontal lines of fixed $\tilde{L}_0$, with $\tilde{L}_0 \geq -\frac{\tilde{j}(\tilde{j}-1)}{k-2}$. Thus, the spectrum was initially bounded below. But, after spectral flow, the new L_0 depends on both $\tilde{L}_0$ and $\tilde{m}$. Fixing $\tilde{L}_0$, we obtain $L_0 = -\tilde{m} + \text{constant}$. Using $\tilde{m} = J^3_0 - \frac{k}{2}$, this becomes,

$$L_0 = -J^3_0 + \text{constant} \tag{528}$$

Therefore, lines of constant $\tilde{L}_0$ appear as (blue) *diagonal lines with slope* -1 in the (J^3_0, L_0) plane. This explains the tilted structure of the diagram. Although $\tilde{L}_0$ is bounded below, the new L_0 is not. Indeed, $L_0 = \tilde{L}_0 - \tilde{m} - \frac{k}{4}$ and since $\tilde{m}$ can become arbitrarily large, we find $L_0 \to -\infty$. Thus, the spectrally flowed representation is no longer a positive-energy representation with respect to L_0.

However, the J^3_0 spectrum remains bounded below. Originally, $\tilde{m} \geq \tilde{j}$. So, after spectral flow, $J^3_0 = \tilde{m} + \frac{k}{2} \geq \tilde{j} + \frac{k}{2}$. This is represented by the vertical (red) dashed line in the diagram.

Physical states satisfy the Virasoro constraint $L_0 + h = 1$. Substituting this in $L_0 = \tilde{L}_0 - \tilde{m} - \frac{k}{4}$, we obtain $\tilde{m} = \tilde{L}_0 - \frac{k}{4} - 1 + h$. Thus, only specific states in the diagram survive the physical state condition. Since $J^3_0 = \tilde{m} + \frac{k}{2} = \tilde{L}_0 + \frac{k}{4} - 1 + h$, this constraint implies that the physical spectrum has a lower bound on J^3_0 (as there was a lower bound on $\tilde{L}_0$), even though L_0 itself is unbounded below. So, imposing the physical condition $L_0 + h = 1$ selects a slice of states with bounded spacetime energy. Thus, although spectral flow destroys the positivity of L_0, the physical spectrum remains well-behaved due to the Virasoro constraint.

7.3.2 Spectral flow with $w = -1$ on $\hat{\mathcal{D}}^+_{\tilde{j}}$ resulting in $\hat{\mathcal{D}}^{+,w=-1}_{\tilde{j}} = \hat{\mathcal{D}}^-_{j=\frac{k}{2}-\tilde{j}}$

It can be proved that for $w = -1$, the spectral flow of $\hat{\mathcal{D}}^+_{\tilde{j}}$ (a positive-energy representation) where $\tilde{L}_0$ is bounded below will result in $\hat{\mathcal{D}}^{+,w=-1}_{\tilde{j}} = \hat{\mathcal{D}}^-_{j=\frac{k}{2}-\tilde{j}}$ which is also a positive-energy representation with L_0 bounded below. Consider the unflowed highest weight representation where the unflowed generators satisfy $\tilde{J}^3_0|\tilde{j}, \tilde{m}\rangle = \tilde{m}|\tilde{j}, \tilde{m}\rangle$, $\tilde{m} \geq \tilde{j}$. and $\tilde{J}^-_0|\tilde{j}, \tilde{j}\rangle = 0$ and the ladder operator $\tilde{J}^+_0 : \ \tilde{m} \to \tilde{m} + 1$. Thus, $\tilde{m} = \tilde{j}\,, \tilde{j}+1\,, \tilde{j}+2\,, \cdots$

Using the spectral flow transformation with $w = -1$, we get,

$$\tilde{J}^3_n = J^3_n + \frac{k}{2}\,\delta_{n,0} \quad , \quad \tilde{J}^{\pm}_n = J^{\pm}_{n\mp 1} \quad , \quad \tilde{L}_n = L_n - J^3_n - \frac{k}{4}\,\delta_{n,0} \tag{529}$$

$$\implies J^3_0 = \tilde{J}^3_0 - \frac{k}{2} \quad , \quad J^+_0 = \tilde{J}^+_1 \quad , \quad J^-_0 = \tilde{J}^-_{-1} \quad , \quad L_0 = \tilde{L}_0 + \tilde{J}^3_0 - \frac{k}{4} \tag{530}$$

Thus, we obtain the relation between eigenvalues:

$$\begin{aligned} m &= \tilde{m} - \frac{k}{2} \\ L_0 &= \tilde{L}_0 + \tilde{m} - \frac{k}{4} \end{aligned} \tag{531}$$

Consider the state $|\tilde{j}\,,\,\tilde{m} = \tilde{j}\rangle$. Then,

$$J^+_0|\tilde{j},\tilde{j}\rangle = \tilde{J}^+_1|\tilde{j},\tilde{j}\rangle = 0 \quad \text{as } |\tilde{j},\tilde{j}\rangle \text{ is affine primary} \tag{532}$$

Thus, after spectral flow, the state $|\tilde{j},\tilde{m} = \tilde{j}\rangle$ becomes a *highest-weight state* with respect to flowed operators. Now consider the action of J^-_0 on this state and how the J^3_0 value changes.

$$\begin{aligned} & J^-_0|\tilde{j},\tilde{j}\rangle = \tilde{J}^-_{-1}|\tilde{j},\tilde{j}\rangle \neq 0 \quad , \quad J^-_0|\tilde{j},\tilde{j}\rangle \ \text{ is a descendant state} \\ & J^3_0\big(J^-_0|\tilde{j}\,,\,\tilde{j}\rangle\big) = [J^3_0, J^-_0]|\tilde{j}\,,\,\tilde{j}\rangle + J^-_0 J^3_0|\tilde{j}\,,\,\tilde{j}\rangle = [\tilde{J}^3_0, \tilde{J}^-_{-1}]|\tilde{j}\,,\,\tilde{j}\rangle + J^-_0\Big(\tilde{j} - \frac{k}{2}\Big)|\tilde{j}\,,\,\tilde{j}\rangle \\ & \qquad = \bigg(- \tilde{J}^-_{-1} + \Big(\tilde{j} - \frac{k}{2}\Big)\tilde{J}^-_{-1}\bigg)|\tilde{j}\,,\,\tilde{j}\rangle = \Big(\tilde{j} - \frac{k}{2} - 1\Big)\big(J^-_0|\tilde{j}\,,\,\tilde{j}\rangle\big) \\ & \implies J^3_0\big(J^-_0|\tilde{j}\,,\,\tilde{j}\rangle\big) = \Big(\tilde{j} - \frac{k}{2} - 1\Big)\big(J^-_0|\tilde{j}\,,\,\tilde{j}\rangle\big) = (-j-1)\big(J^-_0|\tilde{j}\,,\,\tilde{j}\rangle\big) \quad \text{where } \ j = \frac{k}{2} - \tilde{j} \end{aligned} \tag{533}$$

Thus, we see that the spectrum of $m = \tilde{m} - \frac{k}{2}$ is given by $m = -j, -j-1, -j-2, \cdots$, where $j = \frac{k}{2} - \tilde{j}$. Now, we should check how the eigenvalue of L_0 changes under the spectral flow.

$$L_0|\tilde{j}\,,\,\tilde{m} = \tilde{j}\rangle = \Big(\tilde{L}_0 + \tilde{J}^3_0 - \frac{k}{4}\Big)\,|\tilde{j}\,,\,\tilde{m} = \tilde{j}\rangle = \bigg(\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{j} - \frac{k}{4}\bigg)\,|\tilde{j}\,,\,\tilde{m} = \tilde{j}\rangle \tag{534}$$

Now, substitute $\tilde{j} = \frac{k}{2} - j$ in this eigen value. We see that,

$$\begin{aligned} \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{j} - \frac{k}{4} &= \frac{-1}{k-2}\bigg(\Big(\frac{k}{2} - j\Big)\Big(\frac{k}{2} - j - 1\Big) + \Big(j - \frac{k}{4}\Big)(k-2)\bigg) \\ &= \frac{-1}{k-2}\bigg(j^2 - jk + j + \frac{k^2}{4} - \frac{k}{2} + jk - 2j - \frac{k^2}{4} + \frac{k}{2}\bigg) \\ \implies \frac{-\tilde{j}(\tilde{j}-1)}{k-2} &= \frac{-j\,(j-1)}{k-2} \qquad \text{where } \ \tilde{j} = \frac{k}{2} - j \end{aligned} \tag{535}$$

Thus, we get for the state

$$\left|\tilde{j} = \frac{k}{2} - j\,,\, \tilde{m} = m + \frac{k}{2} = \frac{k}{2} - j\right\rangle = |j, m = -j\rangle$$

$$L_0|j, m = -j\rangle = \frac{-j(j-1)}{k-2}|j, m = -j\rangle$$

Since the commutation relation between the flowed operators is also preserved, we see that $[L_0, J^a_{-n}] = nJ^a_{-n}$. Thus, the spectrum of L_0 is also bounded.

The resulting spectrum after the flow is a positive energy representation with a lower bound for L_0. This representation is equivalent to a highest weight affine representation $\hat{\mathcal{D}}^-_{j=\frac{k}{2}-\tilde{j}}$ whose $j = \frac{k}{2} - \tilde{j}$. Since, $\frac{1}{2} < \tilde{j} < \frac{k}{2}$ (lower bound for normalizibility of states and upper bound is to satisfy no-ghost theorem), we get, $\frac{1}{2} < \frac{k}{2} - j < \frac{k}{2}$ or $\frac{1}{2} - \frac{k}{2} < -j < 0$ or $0 < j < \frac{k-1}{2} < \frac{k}{2}$. Thus, even in this flowed representation, we see that the unitarity bound required for the no-ghost theorem is satisfied. The states here are normalizable provided $\frac{k}{2} - \tilde{j} > \frac{1}{2}$ or $\tilde{j} < \frac{k-1}{2}$.

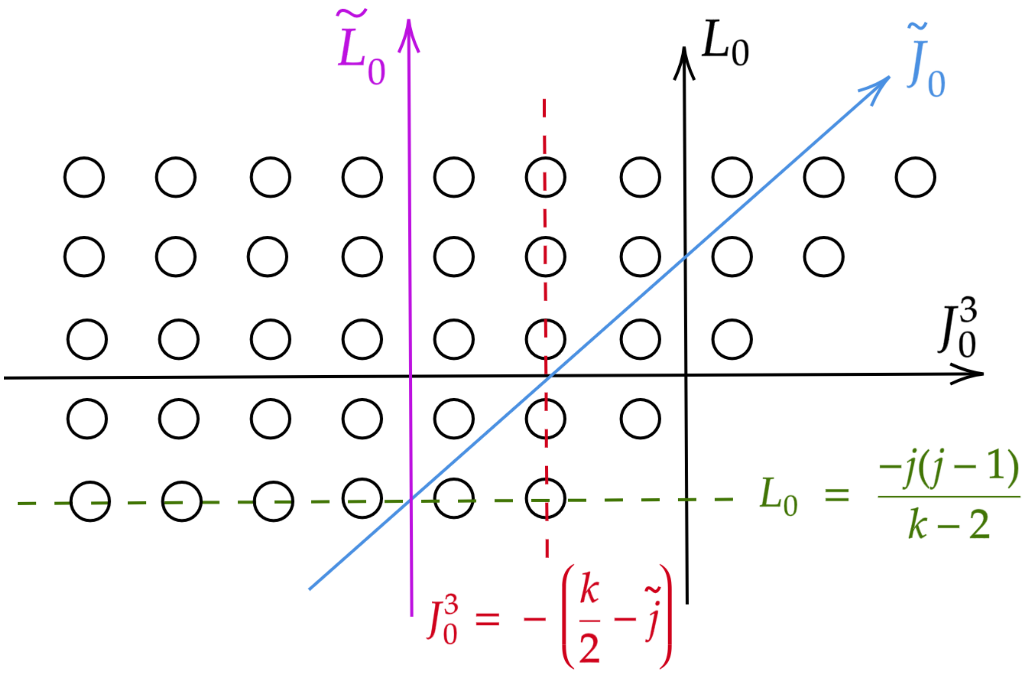

Figure 4: Spectral flow with $w = -1$ on $\hat{\mathcal{D}}^+_{\tilde{j}}$ resulting in $\hat{\mathcal{D}}^{+,w=-1}_{\tilde{j}} = \hat{\mathcal{D}}^-_{j=\frac{k}{2}-\tilde{j}}$

In the unflowed variables $\tilde{m} = \tilde{j},\ \tilde{j}+1,\ \tilde{j}+2, \cdots$. So, the spectrum is bounded from below in $\tilde{J}^3_0 = \tilde{m}$ as shown in the red line. The spectral flow acts as $J^3_0 = \tilde{J}^3_0 - \frac{k}{2}$, $L_0 = \tilde{L}_0 + \tilde{J}^3_0 - \frac{k}{4}$, we see that L_0 depends on $\tilde{J}^3_0$. Therefore, the lines of constant $\tilde{L}_0$ become diagonal in (J^3_0, L_0) coordinates, satisfying the relation $L_0 - J^3_0 =$ constant, such lines will be parallel to the blue line. The original lowest-weight state satisfies $\tilde{m} = \tilde{j}$. After spectral flow $m = \tilde{j} - \frac{k}{2} = -\left(\frac{k}{2} - \tilde{j}\right)$. This corresponds to the red dashed line. Minimal L_0 eigenvalue using $L_0 = -\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{j} - \frac{k}{4}$ and substituting $\tilde{j} = \frac{k}{2} - j$ one obtains $L_0 = -\frac{j(j-1)}{k-2}$. This is shown by the green dashed line.

7.3.3 Spectral flow with $w = 1$ on $\hat{\mathcal{D}}^-_{\tilde{j}}$ resulting in $\hat{\mathcal{D}}^{-,w=+1}_{\tilde{j}} = \hat{\mathcal{D}}^+_{j=\frac{k}{2}-\tilde{j}}$

It can be proved that for $w = +1$, the spectral flow of $\hat{\mathcal{D}}^-_{\tilde{j}}$ (a positive-energy representation) where $\tilde{L}_0$ is bounded below will result in $\hat{\mathcal{D}}^{-,w=+1}_{\tilde{j}} = \hat{\mathcal{D}}^+_{j=\frac{k}{2}-\tilde{j}}$ which is also a positive-energy representation with L_0 bounded below. Consider the unflowed lowest weight representation where the unflowed generators satisfy $\tilde{J}^3_0|\tilde{j},\tilde{m}\rangle = \tilde{m}|\tilde{j},\tilde{m}\rangle$, $\tilde{m} \leq -\tilde{j}$ and $\tilde{J}^+_0|\tilde{j},-\tilde{j}\rangle = 0$ and the ladder operator $\tilde{J}^-_0 :\ \tilde{m} \to \tilde{m} - 1$. Thus, $\tilde{m} = -\tilde{j}\,,\ -\tilde{j}-1\,,\ -\tilde{j}-2\,,\cdots$

Using the spectral flow transformation with $w = +1$, we get,

$$\tilde{J}^3_n = J^3_n - \frac{k}{2}\,\delta_{n,0}\quad,\quad \tilde{J}^\pm_n = J^\pm_{n\pm1}\quad,\quad \tilde{L}_n = L_n + J^3_n - \frac{k}{4}\,\delta_{n,0} \tag{536}$$

$$\implies J^3_0 = \tilde{J}^3_0 + \frac{k}{2}\quad,\quad J^+_0 = \tilde{J}^+_{-1}\quad,\quad J^-_0 = \tilde{J}^-_1\quad,\quad L_n = \tilde{L}_n - \tilde{J}^3_0 - \frac{k}{4} \tag{537}$$

Thus, we obtain the relation between eigenvalues as $m = \tilde{m}+\frac{k}{2}$, and $L_0 = \tilde{L}_0 - \tilde{m} - \frac{k}{4}$. Consider the state $|\tilde{j}\,,\,\tilde{m} = -\tilde{j}\rangle$. Then, $J^-_0|\tilde{j},-\tilde{j}\rangle = \tilde{J}^-_1|\tilde{j},-\tilde{j}\rangle = 0$ as $|\tilde{j},-\tilde{j}\rangle$ is affine primary. Thus, after spectral flow, the state $|\tilde{j},-\tilde{j}\rangle$ becomes a *lowest-weight state* with respect to the flowed operators. Now consider the action of J^+_0 on this state and how the J^3_0 value changes.

$$\begin{gathered} J^+_0|\tilde{j},-\tilde{j}\rangle = \tilde{J}^+_{-1}|\tilde{j},-\tilde{j}\rangle \neq 0\quad,\quad J^+_0|\tilde{j},-\tilde{j}\rangle\ \text{ is a descendant state}\\ J^3_0\big(J^+_0|\tilde{j}\,,-\tilde{j}\rangle\big) = \Big(-\tilde{j}+\frac{k}{2}+1\Big)\big(J^+_0|\tilde{j}\,,-\tilde{j}\rangle\big) = (j+1)\big(J^+_0|\tilde{j}\,,-\tilde{j}\rangle\big)\quad \text{where }\ j = \frac{k}{2}-\tilde{j} \end{gathered} \tag{538}$$

Thus, we see that the spectrum of $m = \tilde{m} + \frac{k}{2}$ is given by $m = j, j+1, j+2, \cdots$, where $j = \frac{k}{2} - \tilde{j}$. Now, we should check how the eigenvalue of L_0 changes under the spectral flow.

$$L_0|\tilde{j}\,,\,\tilde{m} = -\tilde{j}\rangle = \left(\tilde{L}_0 - \tilde{J}^3_0 - \frac{k}{4}\right)|\tilde{j}\,,\,\tilde{m} = -\tilde{j}\rangle = \left(\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{j} - \frac{k}{4}\right)|\tilde{j}\,,\,\tilde{m} = \tilde{j}\rangle \tag{539}$$

Now, substitute $\tilde{j} = \frac{k}{2} - j$ in this eigen value. As done previously, we see that,

$$\frac{-\tilde{j}(\tilde{j}-1)}{k-2} = \frac{-j\,(j-1)}{k-2}\qquad \text{where }\ \tilde{j} = \frac{k}{2} - j \tag{540}$$

Thus, we get for the state $\left|\tilde{j} = \frac{k}{2} - j\,,\,\tilde{m} = m - \frac{k}{2} = -\frac{k}{2} + j\right\rangle = |j, m = j\rangle$,

$$L_0|j, m = j\rangle = \frac{-j(j-1)}{k-2}|j, m = j\rangle \tag{541}$$

Since the commutation relation between the flowed operators is also preserved, we see that $[L_0, J^a_{-n}] = nJ^a_{-n}$. Thus, the spectrum of L_0 is also bounded.

The resulting spectrum after the flow is a positive energy representation with a lower bound for L_0. This representation is equivalent to a lowest weight affine representation $\hat{\mathcal{D}}^+_{j=\frac{k}{2}-\tilde{j}}$ whose $j = \frac{k}{2} - \tilde{j}$. Since, $\frac{1}{2} < \tilde{j} < \frac{k}{2}$ (lower bound for normalizibility of states and upper bound is to satisfy no-ghost theorem), we get, $\frac{1}{2} < \frac{k}{2} - j < \frac{k}{2}$ or $\frac{1}{2} - \frac{k}{2} < -j < 0$ or $0 < j < \frac{k-1}{2} < \frac{k}{2}$. Thus, even in this flowed representation, we see that the unitarity bound required for the no-ghost theorem is satisfied. The states here are normalizable provided $\frac{k}{2} - \tilde{j} > \frac{1}{2}$ or $\tilde{j} < \frac{k-1}{2}$.

7.3.4 Spectral flow with $w = -2$ on $\hat{\mathcal{D}}^+_{\tilde{j}}$

We can see that for $w = -2$, the spectral flow of $\hat{\mathcal{D}}^+_{\tilde{j}}$ (a positive-energy representation) where $\tilde{L}_0$ is bounded below will result in $\hat{\mathcal{D}}^{+,w=-2}_{\tilde{j}}$ which is not a positive-energy representation and here L_0 unbounded below. Consider the unflowed highest weight representation where the unflowed generators satisfy $\tilde{J}^3_0|\tilde{j},\tilde{m}\rangle = \tilde{m}|\tilde{j},\tilde{m}\rangle$, $\tilde{m} \geq \tilde{j}$. and $\tilde{J}^-_0|\tilde{j},\tilde{j}\rangle = 0$ and the ladder operator $\tilde{J}^+_0 : \tilde{m} \to \tilde{m}+1$. Thus, $\tilde{m} = \tilde{j}\,,\, \tilde{j}+1\,,\, \tilde{j}+2\,,\, \cdots$ Using the spectral flow transformation with $w = -2$, we get,

$$\tilde{J}^3_n = J^3_n + k\,\delta_{n,0} \;\;,\quad \tilde{J}^\pm_n = J^\pm_{n\mp2} \;\;,\quad \tilde{L}_n = L_n - 2J^3_n - k\,\delta_{n,0} \tag{542}$$

$$\begin{aligned} \implies J^3_0 = \tilde{J}^3_0 - k \;\;,&\quad L_0 = \tilde{L}_0 + 2\tilde{J}^3_0 - k \\ J^+_0 = \tilde{J}^+_2 \;\;,\quad J^+_{-1} &= \tilde{J}^+_1 \;\;,\quad J^+_{-2} = \tilde{J}^+_0 \\ J^-_0 = \tilde{J}^-_{-2} \;\;,\quad J^-_1 &= \tilde{J}^-_{-1} \;\;,\quad J^-_2 = \tilde{J}^-_0 \end{aligned} \tag{543}$$

Thus, we obtain the relation between eigenvalues $m = \tilde{m} - k$ and $L_0 = \tilde{L}_0 + 2\tilde{m} - k$. Since $\tilde{L}_0 = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + N$, where $N \in \mathbb{Z}^+$ is the grade. We get,

$$L_0 = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + N + 2\tilde{m} - k \tag{544}$$

To make $L_0 \to -\infty$, one would need $N + 2\tilde{m} \to -\infty$. So, we can attempt to achieve this by using the lowering operators. Consider the action of $(\tilde{J}^-_{-1})^p$ on the state $|\tilde{j},\tilde{j}\rangle$ (lowest weight state in unflowed basis), we get,

$$\begin{aligned} L_0 &= \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + N + 2\tilde{m} - k \quad \text{where} \;\; N = p \;\; \text{and} \;\; \tilde{m} = \tilde{j} - p \\ \implies L_0 &= \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + p + 2(\tilde{j} - p) - k = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + 2\tilde{j} - p - k \end{aligned} \tag{545}$$

Since $p = \{0, 1, 2, 3, \cdots \infty\}$, we can achieve $L_0 \to -\infty$.

The same argument wont work for $w = -1$ on $\hat{\mathcal{D}}^+_{\tilde{j}}$. Because here $L_0 = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + N + \tilde{m} - \frac{k}{4}$. Now, $N = p$ and also $\tilde{m} = \tilde{j} - p$ will cancel each other's p dependence. So, here we get $L_0 = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + p + \tilde{j} - p - \frac{k}{4} = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{j} - \frac{k}{4}$ which is bounded below.

Only $\hat{\mathcal{D}}^{\pm,w=\mp1}_{\tilde{j}}$ results in positive energy representation

The spectral flow transformation acts as $L_0 = \tilde{L}_0 - w\,\tilde{J}^3_0 - \dfrac{k}{4}w^2$. And on an eigen state with eigenvalue $\tilde{m}$ as, $L_0 = \tilde{L}_0 - w\,\tilde{m} - \dfrac{k}{4}w^2$.

- Lowest-weight representation $\hat{\mathcal{D}}^+_{\tilde{j}}$, $\tilde{m} = \tilde{j},\ \tilde{j}+1,\ \tilde{j}+2, \ldots$
- Highest-weight representation $\hat{\mathcal{D}}^-_{\tilde{j}}$, $\tilde{m} = -\tilde{j},\ -\tilde{j}-1,\ -\tilde{j}-2, \ldots$

Boundedness of L_0 under spectral flow

For $\hat{\mathcal{D}}^+_{\tilde{j}}$

- When $w = -1$, $L_0 = \tilde{L}_0 + \tilde{m} - \dfrac{k}{4}$. Since $\tilde{m} \geq \tilde{j}$, $L_0 \geq \tilde{L}_0^{\min} + \tilde{j} - \frac{k}{4}$ (bounded below).
- When $w = +1$, $L_0 = \tilde{L}_0 - \tilde{m} - \dfrac{k}{4}$. Since $\tilde{m} \to +\infty$, $L_0 \to -\infty$ (unbounded below).

For $\hat{\mathcal{D}}^-_{\tilde{j}}$

- When $w = +1$, $L_0 = \tilde{L}_0 - \tilde{m} - \dfrac{k}{4}$. Since $\tilde{m} \leq -\tilde{j} \implies -\tilde{m} \geq \tilde{j}$, $L_0 \geq \tilde{L}_0^{\min} + \tilde{j} - \frac{k}{4}$ (bounded below).
- When $w = -1$, $L_0 = \tilde{L}_0 + \tilde{m} - \dfrac{k}{4}$. Since $\tilde{m} \to -\infty$, $L_0 \to -\infty$ (unbounded below).

Thus,

$$\boxed{\hat{\mathcal{D}}^+_{\tilde{j}} \xrightarrow{w=-1} \text{Positive energy representation} \xleftarrow{w=+1} \hat{\mathcal{D}}^-_{\tilde{j}}}$$

Under spectral flow:

$$\boxed{\hat{\mathcal{D}}^+_{\tilde{j}} \xrightarrow{w=-1} \hat{\mathcal{D}}^-_{j}, \qquad j = \frac{k}{2} - \tilde{j}}$$

$$\boxed{\hat{\mathcal{D}}^-_{\tilde{j}} \xrightarrow{w=+1} \hat{\mathcal{D}}^+_{j}, \qquad j = \frac{k}{2} - \tilde{j}}$$

These are positive-energy representations since L_0 remains bounded.

Constraint from harmonic analysis

From $\mathrm{SL}(2,\mathbb{R})$ harmonic analysis we need $\tilde{j} > \frac{1}{2}$, for the wave functions to be normalizable. Using $j = \frac{k}{2} - \tilde{j}$ we get

$$\tilde{j} > \frac{1}{2} \implies j < \frac{k-1}{2}$$

Lower bound on j

We must also ensure that the flowed representation corresponds to a valid $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ representation; we require $j > \frac{1}{2}$. Thus, under inverse spectral flow $\tilde{j} = \frac{k}{2} - j$. The consistency condition from harmonic analysis requires $\tilde{j} < \frac{k-1}{2}$. Thus, if spectral flow is a symmetry, then this forces the truncation and the allowed value of j to be,

$$\frac{1}{2} < \tilde{j} < \frac{k-1}{2} \quad \text{and} \quad \frac{1}{2} < j < \frac{k-1}{2} \tag{546}$$

Thus, the spectral flow of $\hat{\mathcal{D}}^+_{\tilde{j}}$ with $w \geq 1$ or $w \leq -2$ gives a new representation in which L_0 after spectral flow is not bounded below. Similarly, the spectral flow of $\hat{\mathcal{D}}^-_{\tilde{j}}$ with $w \geq 2$ or $w \leq -1$ gives a new representation in which L_0 after spectral flow is not bounded below.

7.3.5 Spectral flow with any $w \neq 0$ on $\hat{\mathcal{C}}^\alpha_{j=\frac{1}{2}+is}$ will be unbounded below in L_0

For continuous representations, $\hat{\mathcal{C}}^\alpha_{j=\frac{1}{2}+is}$, we see that any spectral flow with any $w \neq 0$ will result in a representation in which L_0 is not bounded below. In, $\hat{\mathcal{C}}^\alpha_{j=\frac{1}{2}+is}$, the zero-mode spectrum is $\tilde{m} = \alpha + n$ with $n \in \mathbb{Z}$, so $\tilde{m}$ is unbounded both above and below. Under spectral flow, the energy is $L_0 = \tilde{L}_0 - w\tilde{m} - \frac{k}{4}w^2$, with $\tilde{L}_0 = -\frac{j(j-1)}{k-2} + N$ and $N \geq 0$. Now, for any fixed state one can keep N finite while sending $\tilde{m} \to +\infty$ or $\tilde{m} \to -\infty$; if $w > 0$, taking $\tilde{m} \to +\infty$ gives $L_0 \sim -w\tilde{m} \to -\infty$, while if $w < 0$, taking $\tilde{m} \to -\infty$ gives $L_0 \sim -w\tilde{m} \to -\infty$. Since the linear term in $\tilde{m}$ can always be made arbitrarily negative and cannot be compensated by the non-negative N, it follows that for any $w \neq 0$, L_0 is unbounded below.

7.3.6 No-Ghost Theorem after Spectral Flow

The $\mathfrak{sl}(2,\mathbb{R})$ algebra has an indefinite metric $\kappa^{ab} = \mathrm{diag}(+,+,-)$. Thus, the J^3 direction is time-like, and oscillators built from it can generate negative norm states. In unflowed representations, one can organize the Hilbert space so that physical states have positive norm. However, spectral flow mixes creation and annihilation operators, destroying the simple highest/lowest-weight structure. As a result, Spectrally flowed representations contain negative norm states. To remove the unphysical states, we shall use Virasoro constraints with respect to the spectrally flowed L_n modes. Physical string states must satisfy $L_{n>0}|\mathrm{phys}\rangle = 0$ and $(L_0 + \mathcal{L}_0 - 1)|\mathrm{phys}\rangle = 0$ where $L_0 = \tilde{L}_0 - w\tilde{J}^3_0 - \frac{k}{4}w^2$. This is a generalization of the no-ghost theorem to spectral-flowed sectors. The result holds for $\hat{\mathcal{C}}^{\alpha,w}_{\tilde{j}=\frac{1}{2}+i\tilde{s}}$ and $\hat{\mathcal{D}}^{\pm,w}_{\tilde{j}}$ only if $\tilde{j} < \frac{k}{2}$, $k > 2$.

We extend the proof of the no-ghost theorem from unflowed representations to spectrally flowed representations $\hat{\mathcal{D}}^{\pm,w}_j$ and $\hat{\mathcal{C}}^{\alpha,w}_j$ by mapping the flowed problem to the unflowed one and reusing the previous arguments, with the only modification appearing in the mass-shell condition.

Basis of States for the Hilbert space: Under spectral flow, the Virasoro modes transform as $\tilde{L}_n = L_n + wJ^3_n - \frac{k}{4}w^2\delta_{n,0}$. For $n \neq 0$, this implies $L_{-n} = \tilde{L}_{-n} - w\tilde{J}^3_{-n}$. Thus, the set of states $L_{-n_1}\cdots L_{-n_K}J^3_{-m_1}\cdots J^3_{-m_M}|f\rangle$ can be one-to-one mapped with $\tilde{L}_{-n_1}\cdots\tilde{L}_{-n_K}\,\tilde{J}^3_{-m_1}\cdots\tilde{J}^3_{-m_M}|f\rangle$. As this map is invertible, the basis structure of the Hilbert space is unchanged. Therefore, Step 1 of the unflowed proof carries over directly.

Decoupling of Spurious States: The argument for decoupling of spurious states depends only on the Virasoro algebra and the value of the central charge $c = 26$, which is unchanged under spectral flow. Hence, Step 2 remains valid without modification.

Mass-Shell Condition: The nontrivial part is to show that physical states obey $J^3_{n>0}|\psi\rangle = 0$. For this, we use the modified mass-shell condition. From spectral flow, $L_0 = \tilde{L}_0 - w\tilde{m} - \frac{k}{4}w^2$, where $\tilde{m} = \tilde{J}^3_0$. The physical state condition is $L_0 + \mathcal{L}_0 = 1$. Thus, $\tilde{L}_0 - w\tilde{m} - \frac{k}{4}w^2 + \mathcal{L}_0 = 1$. Using the Sugawara expression, $\tilde{L}_0 = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N}$, $\mathcal{L}_0 = h$ and $\tilde{m} = m - \frac{k}{2}w$ we obtain

$$\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - wm + \frac{kw^2}{4} = 1 \qquad \text{(Flowed Mass-Shell Condition)} \tag{547}$$

In *Continuous representations*, for $\tilde{j} = \frac{1}{2} + is$, and $m = 0$ we have

$$\frac{\frac{1}{4} + s^2}{k-2} + \tilde{N} + h + \frac{kw^2}{4} = 1 \tag{548}$$

Since all terms are non-negative, this can only be satisfied if $N = 0$. Thus, no oscillator excitations are present, and no negative norm states appear.

In *Discrete representations*, without loss of generality, we can assume the lowest-weight representation with $w > 0$ as their complex conjugates cover all the remaining discrete representations. Now, for $0 < \tilde{j} < \frac{k}{2}$, one has $\frac{-\tilde{j}(\tilde{j}-1)}{k-2} > -\frac{k}{4}$. Setting $m = 0$, we get,

$$\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h + \frac{kw^2}{4} > -\frac{k}{4} + \tilde{N} + h + \frac{k}{4}w^2 = \tilde{N} + h + \frac{k}{4}(w^2-1)$$

If $w \geq 2 \implies w^2 - 1 \geq 3$, so $\dfrac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h + \dfrac{kw^2}{4} > \tilde{N} + h + \dfrac{3k}{4} > 1$ and the Virasoro constraint equation cannot be satisfied. Hence, no physical state with $m = 0$ is possible. If $w = 1$, and for $m = 0$, then $\tilde{m} = -\frac{k}{2}$. In a lowest-weight representation, $\tilde{m} \geq \tilde{j}$, so reaching this value requires $\tilde{N} \geq \tilde{j} + \frac{k}{2}$. Substituting into the Virasoro constraint equation, we get,

$$\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h + \frac{k}{4} \geq \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{j} + \frac{k}{2} + \frac{k}{4} > \frac{k}{2} + \tilde{j} > 1$$

Thus, the Virasoro constraint equation cannot be satisfied. In all cases, the condition $m = 0$ is either impossible or leads to trivial states with no oscillators. Therefore, no negative-norm states appear, and all physical states can be mapped to states satisfying $J^3_{n>0}|\psi\rangle = 0$.

7.4 Physical spectra of strings in AdS_3 space-time

7.4.1 Spectra of Discrete and Continuous states

First, let us consider the unflowed basis, whose general eigenstate is $|\tilde{j}, \tilde{m}; \tilde{N}, h\rangle$, where h is the conformal weight of the state in the internal compact CFT ($\mathcal{M}$). Then, the eigen value with respect to $\tilde{L}_0 + \mathcal{L}_0$ (unflowed total conformal weight in $\mathrm{AdS}_3 + \mathcal{M}$) is given by,

$$\tilde{L}_0 + \mathcal{L}_0 = -\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h \tag{549}$$

The physical state condition is $(\tilde{L}_0 + \mathcal{L}_0 - 1)|\psi\rangle = 0$. Thus, we need,

$$-\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - 1 = 0 \tag{550}$$

Rewrite the above equation by substituting $\tilde{j}(\tilde{j}-1) = \left(\tilde{j} - \frac{1}{2}\right)^2 - \frac{1}{4}$.

$$\implies \frac{\left(\tilde{j} - \frac{1}{2}\right)^2}{k-2} = \frac{1}{4(k-2)} + \tilde{N} + h - 1 \tag{551}$$

Now, in the case of discrete representations, if we demand $\frac{1}{2} \leq \tilde{j} \leq \frac{k-1}{2}$, then,

$$\begin{aligned} &\implies 0 \leq \left(\tilde{j} - \frac{1}{2}\right) \leq \frac{k-2}{2} \implies 0 \leq \left(\tilde{j} - \frac{1}{2}\right)^2 \leq \frac{(k-2)^2}{4} \\ &\implies 0 \leq \frac{\left(\tilde{j} - \frac{1}{2}\right)^2}{k-2} \leq \frac{(k-2)}{4} \end{aligned} \tag{552}$$

Thus, we get the range that the quantity $N + h$ has to satisfy for it to be a physical state,

$$0 \leq \frac{1}{4(k-2)} + \tilde{N} + h - 1 \leq \frac{k-2}{4} \implies \boxed{\frac{4k-9}{4(k-2)} \leq \tilde{N} + h \leq \frac{k^2-5}{4(k-2)}} \tag{553}$$

But, if we allow $\tilde{j}$ to go till $\frac{k}{2}$, then,

$$\begin{aligned} &\implies 0 \leq \left(\tilde{j} - \frac{1}{2}\right) \leq \frac{k-1}{2} \implies 0 \leq \left(\tilde{j} - \frac{1}{2}\right)^2 \leq \frac{(k-1)^2}{4} \\ &\implies 0 \leq \frac{\left(\tilde{j} - \frac{1}{2}\right)^2}{k-2} \leq \frac{(k-1)^2}{4(k-2)} \end{aligned} \tag{554}$$

$$\implies 0 \leq \frac{1}{4(k-2)} + \tilde{N} + h - 1 \leq \frac{(k-1)^2}{4(k-2)} \implies \boxed{\frac{4k-9}{4(k-2)} \leq \tilde{N} + h \leq \frac{k+4}{4}} \tag{555}$$

This result shows that the Virasoro constraint restricts physical states to a finite window of $\tilde{N} + h$, ensuring unitarity and enforcing the bound $0 < \tilde{j} < \frac{k}{2}$ or $\frac{k-1}{2}$.

Now, spectral flow acts as, $J_n^{\pm} = \tilde{J}_{n\mp w}^{\pm}$, $J_n^3 = \tilde{J}_n^3 - \dfrac{k}{2} w\, \delta_{n,0}$. The unflowed primary conditions are $\tilde{J}_{n>0}^{3,\pm}|\tilde{j},\tilde{m}\rangle = 0$. Thus $J_{n+w}^{+}|\tilde{j},\tilde{m}\rangle = 0, \quad J_{n-w}^{-}|\tilde{j},\tilde{m}\rangle = 0, \quad J_n^3|\tilde{j},\tilde{m}\rangle = 0, \quad \forall\, n \geq 1$ and for the zero mode $J_0^3 = \tilde{J}_0^3 + \frac{k}{2}w$. Thus,

$$J_0^3|\tilde{j},\tilde{m}\rangle = m|\tilde{j},\tilde{m}\rangle = \left(\tilde{m} + \frac{k}{2}w\right)|\tilde{j},\tilde{m}\rangle \qquad \Longrightarrow \qquad m = \tilde{m} + \frac{k}{2}w \tag{556}$$

Now, the physical states $|\psi\rangle$ have to satisfy the Virasoro constraints with respect to the flowed Virasoro modes $L_n = \tilde{L}_0 - w\tilde{m} - \frac{k}{4}w^2$. Thus, $(L_0 + \mathcal{L}_0 - 1)|\psi\rangle = 0$ and $L_{n>0}|\psi\rangle = 0$. For a state $|\tilde{j},\tilde{m},\tilde{N},h\rangle$, we get, the constraint equation

$$\boxed{-\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - w\tilde{m} - \frac{k}{4}w^2 - 1 = 0} \tag{557}$$

$$\Longrightarrow \ \tilde{m} = -\frac{k}{4}w + \frac{1}{w}\left(-\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - 1\right) \tag{558}$$

Then, we get $J_0^3 \sim m = \tilde{m} + \frac{k}{2}w$ as,

$$J_0^3 = \frac{k}{4}w + \frac{1}{w}\left(-\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - 1\right) \tag{559}$$

This is the quantum version of the classical formula of the J_0^3 value which we found for the spectral flow of the particle like solution moving in AdS_3 with time (or space) like geodesic, with the replacement $\mp\dfrac{k}{4}\alpha^2 \to \dfrac{\tilde{j}(\tilde{j}-1)}{k-2} - \tilde{N} + 1$.

In the case of discrete representations $\hat{\mathcal{D}}_{\tilde{j}}^{\pm,w}$, we need to solve for $\tilde{j}$ using $\tilde{m} = \tilde{j} + q$, $q \in \mathbb{Z}$. But note that all possible integer q values are not allowed because the value of $\tilde{m}$ is bounded. For the lowest weight representation, we can reach the lowest weight state by applying J_{-n}^{-} on a given state, $\tilde{N}$ times. And this decreases the $\tilde{m}$ value to $\tilde{m} - \tilde{N}$. Further decrement of $\tilde{m}$ is not possible. Thus, the minimum allowed value of q is $-\tilde{N}$. Thus for the discrete lowest weight representation, $q \geq -\tilde{N} \implies q = -\tilde{N}, -\tilde{N}+1, -\tilde{N}+2, \cdots$

For the highest weight representation, we can reach the highest weight state by applying J_{-n}^{+} on a given state, $\tilde{N}$ times. And this increases the $\tilde{m}$ value to $\tilde{m} + N$. Further increment of $\tilde{m}$ is not possible. Thus, the maximum allowed value of q is $\tilde{N}$. Thus for the discrete highest weight representation, $q \leq \tilde{N} \implies q = \tilde{N}, \tilde{N}-1, \tilde{N}-2, \cdots$

Substitute $\tilde{m} = \tilde{j} + q$ in the equation of $\tilde{m}$, we get,

$$\tilde{j} + q = -\frac{k}{4}w + \frac{1}{w}\left(-\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - 1\right)$$

$$\Longrightarrow \left(\tilde{j}-\frac{1}{2}\right)^2 + w(k-2)\left(\tilde{j}-\frac{1}{2}\right) - \left(\frac{1}{4} + \left(\tilde{N} - wq + h - 1 - \frac{k}{4}w^2 - \frac{1}{2}w\right)(k-2)\right) = 0$$

$$\Longrightarrow \left(\tilde{j}-\frac{1}{2}\right) = -\frac{w(k-2)}{2} \pm \frac{1}{2}\sqrt{w^2(k-2)^2 + 4\left(\frac{1}{4} + \left(\tilde{N} - wq + h - 1 - \frac{k}{4}w^2 - \frac{1}{2}w\right)(k-2)\right)}$$

Defining $N_w = \tilde{N} - wq$, and the choice of keeping $\tilde{j}$ positive, results in,

$$\boxed{\tilde{j} = \frac{1}{2} - \frac{w}{2}(k-2) + \sqrt{\frac{1}{4} + (k-2)\left(N_w + h - 1 - \frac{1}{2}w(w+1)\right)}} \tag{560}$$

Then, on using $J_0^3 = \tilde{m} + \frac{k}{2}w = \tilde{j} + q + \frac{k}{2}w$, and as it is related to the space-time energy of string as $\frac{E+\ell}{2}$, we get,

$$\frac{E+\ell}{2} = J_0^3 = q + w + \frac{1}{2} + \sqrt{\frac{1}{4} + (k-2)\left(N_w + h - 1 - \frac{1}{2}w(w+1)\right)} \tag{561}$$

Since the value of q is discrete, we get the space-time energy also to be discrete. This condition arises because we are working with the bounded representations (discrete representations), which implies that the string is bounded in the AdS_3 space-time. In this case, the condition $\dfrac{1}{2} \leq \tilde{j} \leq \dfrac{k-1}{2}$ on the $\tilde{j}$ quantum number becomes,

$$\begin{aligned} \frac{1}{2} \leq & \frac{1}{2} - \frac{w}{2}(k-2) + \sqrt{\frac{1}{4} + (k-2)\left(N_w + h - 1 - \frac{1}{2}w(w+1)\right)} \leq \frac{k-1}{2} \\ \Longrightarrow & \boxed{\frac{k}{4}w^2 + \frac{w}{2} \leq \frac{1}{4(k-2)} + N_w + h - 1 \leq \frac{k}{4}(w+1)^2 - \frac{(w+1)}{2}} \end{aligned} \tag{562}$$

This is similar to the condition of the boundedness of the energy. Since, $\dfrac{E+\ell}{2} = \tilde{j} + q + \dfrac{k}{2}w$,

$$\frac{1}{2} + q + \frac{k}{2}w \leq \frac{E+\ell}{2} \leq \frac{k-1}{2} + q + \frac{k}{2}w \tag{563}$$

Now, in continuous representations where $\tilde{j} = \frac{1}{2} + is$, we see that, $-\tilde{j}(\tilde{j}-1) = \frac{1}{4} + s^2$, so the expression for J_0^3 becomes,

$$J_0^3 = \frac{k}{4}w + \frac{1}{w}\left(\frac{\frac{1}{4}+s^2}{k-2} + \tilde{N} + h - 1\right) \tag{564}$$

Comparing with the classical result, we see that $\dfrac{k}{4}\alpha^2 = -\dfrac{\frac{1}{4}+s^2}{k-2} - \tilde{N} + 1$. For large k and s, the quantum formula becomes, $\frac{k}{4}\alpha^2 = \frac{s^2}{k}$. But α controls the radial movement of the string. Thus, we can think that $s \sim \frac{\alpha}{k}$ corresponds to the radial momentum of the string in AdS_3.

7.4.2 Transition between short-strings and long-strings

This can be shown by matching the Discrete and Continuous Representations at the Bounds. Now, the discrete representations require $\frac{1}{2} < \tilde{j} < \frac{k-1}{2}$. So, at the lower bound $\tilde{j} = \frac{1}{2}$, this value of $\tilde{j}$ in discrete representations matches with that of $s = 0$ continuous representations. This also leads to the matching of the space-time energy of both representations. For discrete representations with spectral flow, we have,

$$J_0^3 = q + w + \frac{1}{2} + \sqrt{\frac{1}{4} + (k-2)\left(N_w + h - 1 - \frac{1}{2}w(w+1)\right)}$$

At the lower bound, we get,

$$J_0^3 \quad \left(\text{in } \hat{\mathcal{D}}^{+,w}_{\tilde{j}=\frac{1}{2}}\right) = \frac{1}{2} + q + \frac{k}{2}w \tag{565}$$

Where we have used

$$\frac{1}{4(k-2)} + N_w + h - 1 = \frac{k}{4}w^2 + \frac{w}{2} \tag{566}$$

Take $s = 0$ continuous representation with $\tilde{j} = \frac{1}{2}$ and $\tilde{m} = \alpha + q \ \ , \forall q \in \mathbb{Z}$, for some $0 \leq \alpha < 1$. We get,

$$\begin{aligned} J_0^3 &= \frac{k}{4}w + \frac{1}{w}\left(\frac{1}{4(k-2)} + \tilde{N} + h - 1\right) \\ &= \frac{k}{4}w + \frac{1}{w}\left(\frac{1}{4(k-2)} + N_w + h - 1\right) + q \\ &= \frac{k}{4}w + \frac{1}{w}\left(\frac{k}{4}w^2 + \frac{w}{2}\right) + q \end{aligned} \tag{567}$$

Thus, we get

$$J_0^3 \quad (\text{in } \hat{\mathcal{C}}^{\alpha=\frac{1}{2},w}_{\tilde{j}=\frac{1}{2}}) = \frac{1}{2} + q + \frac{k}{2}w \qquad \Longrightarrow \qquad \alpha = \frac{1}{2} \tag{568}$$

Thus, Discrete and Continuous states have identical J_0^3. At the lower bound of $\tilde{j} = \frac{1}{2}$. Actually, we can check that a state in discrete representation with parameters $(h\,, w\,, q\,, \tilde{N})$ which saturates the lower bound in the inequality with $\frac{1}{4(k-2)} + N_w + h - 1 = \frac{k}{4}w^2 + \frac{w}{2}$, will have the same J_0^3 value with that of the continuous representation for a state with parameters $(h\,, w\,, s = 0\,, \tilde{N})$ whose α parameter to label the representation will be fixed appropriately.

At the upper bound $\tilde{j} = \dfrac{k-1}{2}$ in discrete representation, we have,

$$J_0^3 \quad (\text{in } \hat{\mathcal{D}}^{+,w}_{\tilde{j}=\frac{k-1}{2}}) = \frac{k-1}{2} + q + \frac{k}{2}w \tag{569}$$

where we have used

$$\frac{1}{4(k-2)} + N_w + h - 1 = \frac{k}{4}(w+1)^2 - \frac{(w+1)}{2} \tag{570}$$

Now, take $s=0$ continuous representation with $\tilde{j}=\frac{k-1}{2}$ and $\tilde{m}=\alpha+q$, we get,

$$\begin{aligned} J_0^3 &= \frac{k}{4}w+\frac{1}{w}\left(\frac{k}{4}(w+1)^2-\frac{(w+1)}{2}\right)+q \\ &= \frac{k}{4}w+\left(\frac{k}{4}w+\frac{k}{2}+\frac{k}{4w}-\frac{1}{2}-\frac{1}{2w}\right)+q \\ \implies J_0^3 \quad \left(\text{in } \hat{\mathcal{C}}^{+,w}_{\tilde{j}=\frac{k-1}{2}}\right) &= \frac{k-1}{2}+q+\frac{k}{2}w+\frac{k-2}{4w} \end{aligned} \tag{571}$$

So, the J_0^3 values don't match when the upper bound is saturated. Thus, we are comparing the wrong sectors. So, we shall check if there is equivalence between another spectrally flowed discrete representation and a flowed continuous representation. It can be shown that if we think of the state originally in $\hat{\mathcal{D}}^{+,w}_{\tilde{j}}$ as a state in $\hat{\mathcal{D}}^{-,w+1}_{\hat{j}=\frac{k}{2}-\tilde{j}}$, then this state has $\hat{J}_0^3$ to be equal to that of a state in continuous representation with parameters $(h\,,w+1\,,s=0\,,\tilde{N}+q)$ whose α parameter to label the representation will be fixed appropriately. This can be shown as follows:

Use the identity:

$$\hat{\mathcal{D}}^{+,w}_{\tilde{j}}=\hat{\mathcal{D}}^{-,\,w+1}_{\hat{j}}, \qquad \hat{j}=\frac{k}{2}-\tilde{j} \tag{572}$$

For the upper bound, substitute $\hat{j}=\dfrac{k}{2}-\dfrac{k-1}{2}=\dfrac{1}{2}$. Thus, we can see that,

$$\boxed{\tilde{j}=\frac{k-1}{2} \iff \hat{j}=\frac{1}{2} \text{ in a sector with } w\to w+1}$$

As seen before, this corresponds to the boundary of the continuous representation with $s=0$ with spectral flow of $w\to w+1$. The expression for energy using $\hat{J}_0^3$ in this representation, as seen before, will be,

$$\hat{J}_0^3 \quad \left(\text{in } \hat{\mathcal{D}}^{-,w+1}_{\hat{j}=\frac{1}{2}}\right)=\frac{1}{2}+q+\frac{k}{2}(w+1) \tag{573}$$

Where we have used

$$\frac{1}{4(k-2)}+N_{w+1}+h-1=\frac{k}{4}(w+1)^2+\frac{w+1}{2} \quad \text{(lower bound relation)} \tag{574}$$

Now, in continuous representation with $w\to w+1$, the general expression for J_0^3 is,

$$\begin{aligned} J_0^3 &= \frac{k}{4}(w+1)+\frac{1}{w+1}\left(\frac{1}{4(k-2)}+\tilde{N}+h-1\right) \\ &= \frac{k}{4}(w+1)+\frac{1}{w+1}\left(\frac{1}{4(k-2)}+N_{w+1}+h-1\right)+q \\ &= \frac{k}{4}(w+1)+\frac{1}{w+1}\left(\frac{k}{4}(w+1)^2+\frac{w+1}{2}\right)+q \end{aligned} \tag{575}$$

Thus, we get,

$$J_0^3 \quad \left(\text{in } \hat{\mathcal{C}}_{s=0}^{\alpha=\frac{1}{2},w+1}\right) = \frac{k}{2}(w+1) + \frac{1}{2} + q \tag{576}$$

Thus, we get that,

$$\boxed{J_0^3 \quad \left(\text{in } \hat{\mathcal{D}}_{\tilde{j}=\frac{1}{2}}^{-,w+1}\right) = J_0^3 \quad \left(\text{in } \hat{\mathcal{D}}_{\tilde{j}=\frac{k-1}{2}}^{+,w}\right) = J_0^3 \quad \left(\text{in } \hat{\mathcal{C}}_{s=0}^{\alpha=\frac{1}{2},w+1}\right)}$$

We should note the shift in level for the continuous representation, where the new level is given by $\tilde{N}' = \tilde{N} + q$. This follows from spectral flow $N_w = \tilde{N} - wq$ and consistency under $w \to w+1$. Moreover, as $q \geq -\tilde{N}$ (restriction from the lowest weight state), we see that $\tilde{N}' \geq 0$.

Thus, a discrete state at the upper bound with parameters $(h, w, q, \tilde{N})$ is equivalent to a continuous state with parameters $(h,\ w+1,\ s=0,\ \tilde{N}' = \tilde{N}+q)$. Thus, the upper bound of discrete representations merges into the continuous spectrum.

Thus, the final picture is that, 'Discrete representations exist only within a finite window $\left(\frac{1}{2} \leq \tilde{j} \leq \frac{k-1}{2}\right)$. At $\tilde{j} = \frac{1}{2}$ or $\frac{k-1}{2}$, the states merge with continuous representations. This describes the transition from short strings to long strings.'

7.4.3 Analysis of Lowest energy states: Discrete vs Continuous representations

Here, we are concerned about the lowest energy state and ask to which representation it belongs. For a fixed value of the internal conformal weight h and fixed spectral flow w, we need to find the physical state that has the *smallest* spacetime energy J_0^3.

Discrete representations: We first restrict to the *discrete representations* and impose the allowed range $\frac{k}{4}w^2 + \frac{w}{2} \leq \frac{1}{4(k-2)} + N_w + h - 1 \leq \frac{k}{4}(w+1)^2 - \frac{(w+1)}{2}$. Within this domain, the physical states are parametrized by the integers q and $\tilde{N}$ (with $\tilde{N} \geq 0$), so the energy can be viewed as a function $J_0^3 = J_0^3(h, w, q, \tilde{N})$. To find the lowest energy state, vary q and $\tilde{N}$. Define:

$$X = \frac{1}{4} + (k-2)\left(N_w + h - 1 - \frac{1}{2}w(w+1)\right)$$

Then:

$$J_0^3 = q + w + \frac{1}{2} + \sqrt{X}$$

Monotonicity in q and $\tilde{N}$: One can show that $\dfrac{\partial J_0^3(h, w, q, \tilde{N})}{\partial q} \geq 0$ and $\dfrac{\partial J_0^3(h, w, q, \tilde{N})}{\partial \tilde{N}} > 0$. Use $N_w = \tilde{N} - wq$

$$\implies \frac{\partial X}{\partial q} = -(k-2)w \quad \implies \quad \frac{\partial J_0^3}{\partial q} = 1 + \frac{1}{2\sqrt{X}}\frac{\partial X}{\partial q} = 1 - \frac{(k-2)w}{2\sqrt{X}} \tag{577}$$

Now, from the allowed region, we know:

$$
\begin{aligned}
\frac{k}{4}w^2 + \frac{w}{2} \le \frac{X}{k-2} + \frac{1}{2}w(w+1) \le \frac{k}{4}(w+1)^2 - \frac{(w+1)}{2} \\
\implies \frac{X}{k-2} \ge \frac{k}{4}w^2 + \frac{w}{2} - \frac{1}{2}w(w+1) = \frac{(k-2)}{4}w^2 \\
\implies X \ge \frac{(k-2)^2}{4}w^2 \quad \implies \quad \sqrt{X} \ge \frac{(k-2)}{2}w \quad \implies \quad \frac{(k-2)}{2\sqrt{X}}w \le 1
\end{aligned}
\tag{578}
$$

Hence, $\dfrac{\partial J_0^3}{\partial q} \ge 0$. This means that increasing q never lowers the energy. Physically, q labels how far we move along the J_0^3 tower of the discrete representation, and higher q corresponds to higher energy. So the minimum occurs at the smallest q. Since, for discrete lowest-weight representations $q \ge \tilde{N}$, the lowest energy occurs at $q = -\tilde{N}$. Now, let us find the dependency of J_0^3 on $\tilde{N}$.

$$
\frac{\partial J_0^3}{\partial \tilde{N}} = \frac{1}{2\sqrt{X}}\frac{\partial X}{\partial \tilde{N}} = \frac{k-2}{2\sqrt{X}} \tag{579}
$$

In the allowed region $\sqrt{X} > 0$. Thus, we have, $\dfrac{\partial J_0^3}{\partial \tilde{N}} > 0$. So, excitations always increase energy. So, the lowest energy state is at zero excitations, whose $\tilde{N} = 0$. Hence, within the bounds, the minimum energy discrete state occurs at $q = 0$, $\tilde{N} = 0$. Plug these into the constraint inequality. Thus, we get,

$$
\frac{k}{4}w^2 + \frac{w}{2} \le \frac{1}{4(k-2)} + h - 1 \le \frac{k}{4}(w+1)^2 - \frac{(w+1)}{2} \tag{580}
$$

Thus, only if the conformal weight h satisfies this inequality can the ground state $q = \tilde{N} = 0$ exist. So, the minimum discrete energy is given by,

$$
\begin{aligned}
J_{0_{\min}}^{3(\text{discrete})} = w + \frac{1}{2} + \sqrt{\frac{1}{4} + (k-2)\left(h - 1 - \frac{1}{2}w(w+1)\right)} \\
\implies \frac{1}{(k-2)}\left(J_{0_{\min}}^{3(\text{discrete})} - w - \frac{1}{2}\right)^2 + \frac{1}{2}w(w+1) = \frac{1}{4(k-2)} + h - 1
\end{aligned}
\tag{581}
$$

But, here h follows the bound, $\dfrac{k}{4}w^2 + \dfrac{w}{2} \le h - 1 + \dfrac{1}{4(k-2)} \le \dfrac{k}{4}(w+1)^2 - \dfrac{w+1}{2}$. Thus,

$$
\implies \frac{k}{4}w^2 + \frac{w}{2} \le \frac{1}{(k-2)}\left(J_{0_{\min}}^{3(\text{discrete})} - w - \frac{1}{2}\right)^2 + \frac{1}{2}w(w+1) \le \frac{k}{4}(w+1)^2 - \frac{w+1}{2} \tag{582}
$$

It can be shown that, for h satisfying this bound there is no state in the continuous representations $\hat{\mathcal{C}}_{\tilde{j}}^{\alpha}$ whose spacetime energy J_0^3 is lower than $J_{0_{\min}}^{3(\text{discrete})}$.

For continuous representations, we have,

$$J_0^3 = \frac{k}{4}w + \frac{1}{w}\left(\frac{\frac{1}{4}+s^2}{k-2} + \tilde{N} + h - 1\right) \tag{583}$$

So, the minima occur at $s = \tilde{N} = 0$. Thus,

$$J_{0_{\min}}^{3(\text{continuous})} = \frac{k}{4}w + \frac{1}{w}\left(\frac{1}{4(k-2)} + h - 1\right) \tag{584}$$

Now, if h satisfies the lower bound, then, $\dfrac{1}{4(k-2)} + h - 1 \geq \dfrac{k}{4}w^2 + \dfrac{w}{2}$. Thus,

$$\frac{k}{4}w + \frac{1}{w}\left(\frac{1}{4(k-2)} + h - 1\right) \geq \frac{k}{2}w + \frac{1}{2} \qquad \implies J_{0_{\min}}^{3(\text{continuous})} \geq \frac{k}{2}w + \frac{1}{2} \tag{585}$$

Now, let us compare it with $J_{0_{\min}}^{3(\text{discrete})}$. It satisfies the inequality,

$$\begin{aligned}
&\frac{1}{(k-2)}\left(J_{0_{\min}}^{3(\text{discrete})} - w - \frac{1}{2}\right)^2 + \frac{1}{2}w(w+1) \leq \frac{k}{4}(w+1)^2 - \frac{w+1}{2} \\
\implies &\frac{1}{(k-2)}\left(J_{0_{\min}}^{3(\text{discrete})} - w - \frac{1}{2}\right)^2 \leq \frac{(k-2)}{4}(w+1)^2 \\
\implies &J_{0_{\min}}^{3(\text{discrete})} - w - \frac{1}{2} \leq \frac{(k-2)}{2}(w+1) = \frac{k}{2}w - w + \frac{(k-2)}{2} \\
\implies &J_{0_{\min}}^{3(\text{discrete})} - \frac{(k-2)}{2} \leq \frac{k}{2}w + \frac{1}{2} \leq J_{0_{\min}}^{3(\text{continuous})}
\end{aligned} \tag{586}$$

Thus, within the bounds, we have,

$$J_{0_{\min}}^{3(\text{discrete})} \leq J_{0_{\min}}^{3(\text{continuous})} + \frac{(k-2)}{2} \tag{587}$$

Since, $\frac{(k-2)}{2} > 0$ as $k > 2$, we have $J_{0_{\min}}^{3(\text{discrete})} < J_{0_{\min}}^{3(\text{continuous})}$ (strictly lesser). Thus, for h that satisfies these bounds, the lowest energy state lies in the discrete representation with $\tilde{N} = q = 0$. Now, at the lower bound of $\dfrac{1}{4(k-2)} + h - 1 = \dfrac{k}{4}w^2 + \dfrac{w}{2}$, we have,

$$\begin{aligned}
\frac{k}{4}w^2 + \frac{w}{2} &= \frac{1}{(k-2)}\left(J_{0_{\min}}^{3(\text{discrete})} - w - \frac{1}{2}\right)^2 + \frac{1}{2}w(w+1) \\
\implies J_{0_{\min}}^{3(\text{discrete})} &= \frac{k}{2}w + \frac{w}{2} = J_{0_{\min}}^{3(\text{continuous})}
\end{aligned} \tag{588}$$

Thus, at the lower boundary of h, the discrete and continuous representations have the same energy.

Similarly, at the upper bound of $\frac{1}{4(k-2)}+h-1=\frac{k}{4}(w+1)^2-\frac{w+1}{2}$, we have,

$$\begin{aligned}\frac{1}{(k-2)}\left(J^{3(\text{discrete})}_{0_{\min}}-w-\frac{1}{2}\right)^2+\frac{1}{2}w(w+1)&=\frac{k}{4}(w+1)^2-\frac{w+1}{2}\\ \implies J^{3(\text{discrete})}_{0_{\min}}&=\frac{k}{2}w+\frac{1}{2}+\frac{k-2}{2}=\frac{k}{2}(w+1)-\frac{1}{2}\end{aligned}\tag{589}$$

Now, let us check what happens to $J^{3(\text{continuous})}_{0_{\min}}$ whose $w\to w+1$ at this value of h.

$$\begin{aligned}J^{3(\text{continuous})}_{0_{\min}}&=\frac{k}{4}(w+1)+\frac{1}{w+1}\left(\frac{1}{4(k-2)}+h-1\right)\\ &=\frac{k}{4}(w+1)+\frac{1}{w+1}\left(\frac{k}{4}(w+1)^2-\frac{w+1}{2}\right)\\ &=\frac{k}{4}(w+1)+\frac{k}{4}(w+1)-\frac{1}{2}\\ \implies J^{3(\text{continuous})}_{0_{\min}}\quad(w\to w+1)&=\frac{k}{2}(w+1)-\frac{1}{2}=J^{3(\text{discrete})}_{0_{\min}}\ \text{(upper bound)}\end{aligned}\tag{590}$$

Thus, at the upper-bound of the conformal weight h, we see that the minimum space-time energy of the discrete representation with winding number w matches exactly with that of the continuous representation with winding number $w\to w+1$.

Now, for h outside the allowed range, the condition $q=\tilde{N}=0$ violates the bound $\frac{k}{4}w^2+\frac{w}{2}\leq\frac{1}{4(k-2)}+N_w+h-1\leq\frac{k}{4}(w+1)^2-\frac{(w+1)}{2}$. Thus, no discrete state can realize $q=\tilde{N}=0$ and get the minimum space-time energy condition. So, the discrete spectrum is forced to get higher energy. But, the states in continuous representations do not depend on $\tilde{j}$ bounds. Their minimum remains to be $J^{3(\text{continuous})}_{0_{\min}}$. Since discrete states cannot reach $q=\tilde{N}=0$, we get:

$$\boxed{J^{3(\text{continuous})}_{0_{\min}}<J^{3(\text{discrete})}_{0_{\min}}\quad\text{outside the allowed bound for } h}\tag{591}$$

Quantum vs semi-classical transition. In the semi-classical analysis, the transition between short and long strings occurs sharply at the critical value

$$h=\frac{k}{4}w^2.$$

However, in the full quantum theory, this sharp transition is replaced by a finite *band* of width proportional to w, given by

$$\frac{k}{4}w^2-\frac{w}{2}\ \leq\ h-1+\frac{1}{4(k-2)}\ \leq\ \frac{k}{4}w^2+\frac{w}{2}.\tag{592}$$

Structure of the spectrum. This band organizes the spectrum into three distinct regions:

- **Inside the band:** The lowest-energy states are *short strings.* Their spectrum is discrete, and they belong to the discrete representations $\hat{\mathcal{D}}_{\tilde{j}}^{\pm,w}$
- **Below the band:** The lowest-energy states are *long strings* described by continuous representations $\hat{\mathcal{C}}_{\tilde{j}}^{\alpha,w}$
- **Above the band:** Again the lowest-energy states are *long strings*, but now they appear in spectrally flowed sectors $\hat{\mathcal{C}}_{\tilde{j}}^{\alpha,\,w\pm1}$

Physical interpretation. Thus, the quantum theory replaces the single transition point of the semi-classical picture by a finite interval within which short strings dominate. At the boundaries of this interval, the discrete representations merge smoothly into continuous ones, signaling the transition between short and long strings. Thus, as one moves across the band (long $\to$ short $\to$ long), the effective winding number can shift by one unit. This reflects the fact that spectral flow reorganizes the Hilbert space, and different winding sectors become energetically favorable in different regions of the spectrum.

Left–right spectral flow and worldsheet locality

So far, we considered only one (say, right-moving) sector. In the full string theory, we must combine left- and right-movers. Let the spectral flow act independently as w_L and w_R on the left and right sectors. Under this, a state with (h_L, h_R) and $(\tilde{m}_L, \tilde{m}_R)$ transforms as

$$h_L \;\to\; h_L - w_L\tilde{m}_L - \frac{k}{4}w_L^2, \qquad h_R \;\to\; h_R - w_R\tilde{m}_R - \frac{k}{4}w_R^2.$$

Now, the Worldsheet locality (level matching) gives the quantum condition corresponding to periodicity of the classical solution as $h_L - h_R \in \mathbb{Z}$. If this holds before spectral flow, it must also hold after spectral flow. Subtracting the transformed weights gives the constraint $w_L\tilde{m}_L + \frac{k}{4}w_L^2 = w_R\tilde{m}_R + \frac{k}{4}w_R^2 \pmod{\mathbb{Z}}$. For generic (continuous) values of $(\tilde{m}_L, \tilde{m}_R)$, the above condition can be satisfied only if $w_L = w_R$. Thus, only *left–right symmetric spectral flow* is allowed. Since the target space is the universal cover of $\mathrm{SL}(2,\mathbb{R})$, the charges $(\tilde{m}_L, \tilde{m}_R)$ are continuous. This removes the possibility of compensating different w_L and w_R by integer shifts, forcing $w_L = w_R \equiv w$. Hence, spectral flow must act equally on left and right movers, matching the classical requirement of worldsheet periodicity.

Summary of the spectrum. The spectrum of the $\mathrm{SL}(2,\mathbb{R})$ WZW model (on the universal cover) consists of two classes of representations, both subjected to spectral flow acting *equally* on the left and right sectors ($w_L = w_R = w$).

(1) Continuous representations. These describe long strings and are obtained by spectral flowing the principal continuous series:

$$\boxed{\hat{\mathcal{C}}^{\alpha,w}_{\frac{1}{2}+is,L} \times \hat{\mathcal{C}}^{\alpha,w}_{\frac{1}{2}+is,R}}$$

Here $j = \frac{1}{2} + is$ with $s \in \mathbb{R}$, and α labels the continuous J_0^3 spectrum. These states correspond to strings that can reach the boundary of AdS_3 and hence have a continuous energy spectrum.

(2) Discrete representations. These describe short strings and are given by:

$$\boxed{\hat{\mathcal{D}}^{\pm,w}_{\tilde{j},L} \times \hat{\mathcal{D}}^{\pm,w}_{\tilde{j},R}, \qquad \frac{1}{2} < \tilde{j} < \frac{k-1}{2}.}$$

Both left and right sectors carry the same $\tilde{j}$ and the same spectral flow w. The bound on $\tilde{j}$ ensures unitarity and the absence of negative-norm states.

Coupling to string theory. To obtain the full string spectrum, these representations must be tensored with an internal CFT:

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\mathrm{SL}(2,\mathbb{R})} \otimes \mathcal{H}_{\mathcal{M}}.$$

where the complete $\mathcal{H}_{\mathrm{SL}(2,\mathbb{R})}$ Hilbert space is,

$$\mathcal{H}_{\mathrm{SL}(2,\mathbb{R})} = \bigoplus_{w\in\mathbb{Z}} \left(\bigoplus_{1/2 \leq \tilde{j} \leq (k-1)/2} \left(\hat{\mathcal{D}}^{\pm,w}_{\tilde{j},L} \otimes \hat{\mathcal{D}}^{\pm,w}_{\tilde{j},R} \right) \;\oplus\; \int ds\, \hat{\mathcal{C}}^{\alpha,w}_{\frac{1}{2}+is,L} \otimes \hat{\mathcal{C}}^{\alpha,w}_{\frac{1}{2}+is,R} \right)$$

Virasoro constraints. Physical states must satisfy the string constraints:

$$(L_0 + \mathcal{L}_n - 1)|\text{phys}\rangle = 0, \qquad (\bar{L}_0 + \mathcal{L}_n - 1)|\text{phys}\rangle = 0,$$

along with $L_n = \bar{L}_n = 0$ for $n > 0$.

Final picture.

$$\boxed{\text{Spectrum} = (\text{spectral flow of continuous reps}) \;\oplus\; (\text{spectral flow of discrete reps})}$$

- Continuous reps $\rightarrow$ long strings (continuous spectrum)
- Discrete reps $\rightarrow$ short strings (discrete spectrum)
- Spectral flow w labels different winding sectors
- Left-right symmetry $w_L = w_R$ is required by worldsheet locality

8 Scattering of Long Strings

A long string in AdS_3 is a string configuration that is not confined near the center of the spacetime. Instead, it can extend toward the boundary of AdS_3. Such a string may start near the boundary, move inward toward the center, and then return toward the boundary. Thus, the motion resembles a scattering process, where the incoming and outgoing states are both long strings. During this process, the winding number may change. The reason is geometric. In global AdS_3, the angular circle can shrink to zero size at the center of the space. Therefore, a string wrapping around the angular direction is not protected by topology. As the string passes through the center, it can unwind and rewrap with a different winding number. In the WZW description, this winding information is closely related to the spectral flow number w. Hence, different winding sectors correspond to different spectrally flowed sectors of $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$.

8.1 Wick Rotation of AdS_3 space-time

To study the S-matrix between incoming and outgoing long strings, it is convenient to rotate to Euclidean signature. The S-matrix describes scattering amplitudes between asymptotic incoming and outgoing states. In the Lorentzian signature, one works with real time and physical scattering processes. However, many CFT calculations become easier after performing a Wick rotation to Euclidean signature, where we mainly work with correlation functions. This rotation is performed both on the worldsheet and in the target spacetime. After Wick rotation, the Euclidean continuation of AdS_3 becomes the three-dimensional hyperbolic space H_3 which can be written as $H_3 = \mathrm{SL}(2,\mathbb{C})/SU(2)$. This means that the Euclidean theory is described by a sigma model whose target space is the symmetric space obtained by quotienting $\mathrm{SL}(2,\mathbb{C})$ by $SU(2)$. The Euclidean WZW model based on this target space is commonly called the H_3^+ WZW model. We shall prove all this below.

The Lorentzian AdS_3 can be defined as the hyperboloid $-X_{-1}^2 - X_0^2 + X_1^2 + X_2^2 = -L^2$ embedded inside the flat space $\mathbb{R}^{2,2}$ with metric $ds^2_{\mathbb{R}^{2,2}} = -dX_{-1}^2 - dX_0^2 + dX_1^2 + dX_2^2$. The resulting space has one time direction and two spatial directions, with a constant negative Ricci Scalar. The $\mathrm{SL}(2,\mathbb{R})$ WZW model describes strings propagating on this spacetime. In the global coordinates the Lorentzian AdS_3 is $ds^2_{\mathrm{AdS}_3} = L^2\left(-\cosh^2(\rho)\,dt^2 + d\rho^2 + \sinh^2(\rho)\,d\phi^2\right)$ as seen before. Where t is the time coordinate, ρ is the radial coordinate, and ϕ is the angular coordinate. The metric has Lorentzian signature $(-,+,+)$. If we perform a Wick rotation of the time coordinate $t \to -i\,t_E$. Then,

$$ds_E^2 = L^2\left(\cosh^2(\rho)\,dt_E^2 + d\rho^2 + \sinh^2(\rho)\,d\phi^2\right) \tag{593}$$

This metric is positive definite and therefore has Euclidean signature (+,+,+). We can show that the resulting Euclidean manifold is the three-dimensional hyperbolic space H_3 whose Ricci scalar is also a negative constant, which is the defining property of hyperbolic space.

The three-dimensional hyperbolic space H_3 can be defined as the hyperboloid

$$-X_0^2 + X_1^2 + X_2^2 + X_3^2 = -L^2, \qquad X_0 > 0, \tag{594}$$

embedded in flat Minkowski space $\mathbb{R}^{1,3}$ with metric

$$ds^2_{\mathbb{R}^{1,3}} = -dX_0^2 + dX_1^2 + dX_2^2 + dX_3^2. \tag{595}$$

We shall parameterize the resulting space by the coordinates (T, ρ, ϕ) through the functions,

$$\begin{aligned} X_0 &= L\cosh(\rho)\cosh(T) \\ X_1 &= L\sinh(\rho)\cos(\phi) \\ X_2 &= L\sinh(\rho)\sin(\phi) \\ X_3 &= L\cosh(\rho)\sinh(T) \end{aligned} \tag{596}$$

Now compute the induced metric from the ambient space metric $ds^2_{\mathbb{R}^{1,3}} = -dX_0^2 + dX_1^2 + dX_2^2 + dX_3^2$. Substituting the parametrization and simplifying yields

$$ds^2_{H_3} = L^2\left(\cosh^2(\rho)\, dT^2 + d\rho^2 + \sinh^2(\rho)\, d\phi^2\right) \tag{597}$$

This is exactly the metric obtained from Wick rotating the Lorentzian AdS_3 metric. Therefore the Wick-rotated AdS_3 manifold is precisely the hyperbolic space H_3 and the Ricci scalar is $R = -\frac{6}{L^2}$. Thus, the Euclidean manifold has constant negative curvature.

We now derive explicitly why the Euclidean hyperbolic space H_3 can be represented as the coset $\mathrm{SL}(2,\mathbb{C})/SU(2)$. Note that any 2×2 hermitian matrix can be written as,

$$X = \frac{1}{L}\begin{pmatrix} X_0 + X_3 & X_1 - iX_2 \\ X_1 + iX_2 & X_0 - X_3 \end{pmatrix} = \begin{pmatrix} \cosh(\rho)e^{T} & \sinh(\rho)e^{-i\phi} \\ \sinh(\rho)e^{i\phi} & \cosh(\rho)e^{-T} \end{pmatrix} \tag{598}$$

Then, the determinant is given by,

$$\det(X) = \frac{1}{L^2}(X_0^2 - X_3^2 - X_1^2 - X_2^2) \tag{599}$$

If we want the matrix X to have determinant 1, then this forces $-X_0^2 + X_3^2 + X_1^2 + X_2^2 = -L^2$. This is precisely the equation of three-dimensional hyperbolic space embedded in $\mathbb{R}^{3,1}$. Thus, any 2×2 hermitian matrix with determinant 1 defines a point in H_3 space.

To obtain the upper sheet of the hyperboloid H_3^+, we require the matrix X to be positive definite. That is, all eigenvalues are strictly positive. This implies that $\mathrm{Tr}(X) = 2X_0/L =$ Sum of eigenvalues has to be positive. Thus, we require $X_0 > 0$. Hence, the set of all positive definite, 2×2 hermitian matrices whose determinant is one, forms H_3^+.

Consider any $g \in \mathrm{SL}(2,\mathbb{C}) \implies g^{-1} \in \mathrm{SL}(2,\mathbb{C})$ and $\det(g) = 1$. Now, $(gg^\dagger)^\dagger = (g^\dagger)^\dagger g^\dagger = gg^\dagger$. So, $gg^\dagger$ is Hermitian. Then, consider any non-zero vector $|v\rangle \in \mathbb{C}^2$, $\langle v|(gg^\dagger)|v\rangle = \|g^\dagger|v\rangle\|^2 > 0$ as $g^\dagger|v\rangle$ is a non-zero vector with a positive norm. Hence $gg^\dagger$ is a positive definite matrix. Finally, $\det\big(gg^\dagger\big) = \det(g)\,\det\big(g^\dagger\big) = 1 \cdot (\det(g))^* = 1$. So, we have shown that for any $g \in \mathrm{SL}(2,\mathbb{C})$, $gg^\dagger$ is a positive definite Hermitian 2×2 matrix with determinant 1.

Thus, the factorization of all positive definite, 2×2 hermitian matrices whose determinant is one as $X = gg^\dagger$ (Cholesky factorization) defines a point of H_3^+. Conversely, any $g \in \mathrm{SL}(2,\mathbb{C})$ can be mapped to a point in H_3^+ as $gg^\dagger$. But there is a redundancy under $SU(2)$. Consider multiplying $g \in \mathrm{SL}(2,\mathbb{C})$ on the right by any element $u \in SU(2)$. Then,

$$(gu)(gu)^\dagger = g\,\underbrace{(uu^\dagger)}_{1}\,g^\dagger = gg^\dagger \tag{600}$$

Thus, for any $g \in \mathrm{SL}(2,\mathbb{C})$ and for all $u \in SU(2)$, g and gu determine the same point of H_3^+.

Let G be a group and $H \subset G$ be a subgroup. For any element $g \in G$, the set

$$gH = \{gh \mid h \in H\} \tag{601}$$

is called a left coset of H in G. Two elements $g_1, g_2 \in G$ are said to be equivalent if

$$g_1 = g_2 h, \qquad h \in H. \tag{602}$$

g_1 and g_2h belong to the same coset. Thus, a left coset is an equivalence class under right multiplication by subgroup elements. The set of all such equivalence classes is denoted as G/H.

Now, for any $g \in \mathrm{SL}(2,\mathbb{C})$ and for all $u \in SU(2)$, g and gu determine the same point of H_3^+. Thus, a point of hyperbolic space is not labeled by an individual matrix g, but rather by an entire equivalence class

$$[g] = \{gu \mid u \in SU(2)\}. \tag{603}$$

Hence,

$$H_3 \cong \mathrm{SL}(2,\mathbb{C})/SU(2). \tag{604}$$

Thus, the hyperbolic space is the set of all such equivalence classes, or Cosets.

The hyperbolic space H_3^+ is a non-compact manifold. As one moves farther outward in the radial direction, one approaches the asymptotic boundary of the space. Although this boundary is not part of the manifold itself, it plays a central role in the H_3^+ WZW model.

The center of the space is located at $\rho = 0$, while the asymptotic boundary is reached in the limit $\rho \to \infty$. To understand the geometry near the boundary, consider the large-ρ behavior: $\cosh(\rho) \sim \sinh(\rho) \sim \frac{e^\rho}{2}$. Then, the H_3^+ metric, $ds^2_{H_3} = L^2 \left(\cosh^2(\rho)\, dT^2 + d\rho^2 + \sinh^2(\rho)\, d\phi^2\right)$ becomes

$$ds^2_{H_3}(\rho \to \infty) = L^2 \left(d\rho^2 + \frac{e^{2\rho}}{4}(dT^2 + d\phi^2)\right) \tag{605}$$

Thus the T and ϕ directions grow exponentially as $\rho \to \infty$. Consequently, points at large ρ, are infinitely far from the center of the geometry. This is why $\rho \to \infty$ is interpreted as the boundary of hyperbolic space. Now factor out the divergent conformal factor:

$$ds^2_{H_3}(\rho \to \infty) = \frac{L^2 e^{2\rho}}{4}\left(dT^2 + d\phi^2 + 4e^{-2\rho} d\rho^2\right) \tag{606}$$

As $\rho \to \infty$, $4e^{-2\rho} d\rho^2 \to 0$. Therefore, up to an overall divergent scale factor, the induced metric on the boundary becomes

$$ds^2_{\partial H_3} \propto dT^2 + d\phi^2. \tag{607}$$

Hence, the conformal boundary of H_3^+ is a two-dimensional Euclidean cylinder parameterized by (T, ϕ). The boundary of H_3^+ plays an important role in the Euclidean H_3^+ WZW model.

- Vertex operators are naturally associated with points on the boundary.
- Long strings can propagate toward the asymptotic boundary region.
- Correlation functions are often interpreted as boundary correlators.
- The asymptotic symmetry structure is encoded at the boundary.

Thus, the bulk geometry H_3^+ may be viewed as a curved space whose physical observables are strongly influenced by the behavior near the asymptotic boundary. The scattering states of long strings are characterized by their asymptotic behavior near this boundary region.

8.2 Hermiticity conditions in the Euclidean H_3^+ WZW model

In the Lorentzian worldsheet theory, the worldsheet coordinates are τ, σ, where $\tau \in (-\infty, \infty)$ and $\sigma \in [0, 2\pi)$, these are Lorentzian cylindrical coordinates. But after Wick rotation, even the worldsheet theory becomes Euclidean, where we transform $\tau \to -i\tau_E$. So, (τ_E, σ) are the Euclidean world-sheet coordinates and $\omega = \tau_E + i\sigma$, $\bar{\omega} = \tau_E - i\sigma$ are the complex coordinates on the Lorentzian cylinder. $\implies x^\pm = \tau \pm \sigma = -i\tau_E \pm \sigma = -i(\tau_E \pm i\sigma)$. Thus, $x^+ = -i\omega$ and $x^- = -i\bar{\omega}$. So, the Lorentzian right-moving coordinate x^+ analytically continues into the holomorphic Euclidean complex coordinate, and the left-moving coordinate x^- analytically continues to become the anti-holomorphic Euclidean complex coordinate.

In the Lorentzian signature, the affine currents are functions of the light-cone coordinates $x^{\pm} = \tau \pm \sigma$. The left-moving and right-moving currents satisfy $J_R^a = J^a(x^+)$ and $J_L^a = \bar{J}^a(x^-)$. And the Cartan basis for the lie algebra are $\{J_R^3, J_R^{\pm} = J_R^1 \pm iJ_R^2\}$ and $\{J_L^3, J_L^{\pm} = J_L^1 \mp iJ_L^2\}$. Now, we can define the same currents in terms of the holomorphic and anti-holomorphic coordinates on the cylinder. Thus, the Euclidean theory is regarded as an analytic continuation of the Lorentzian theory rather than an independent Euclidean conformal field theory.

Now, the Lorentzian current modes are defined by,

$$J_R^a(x^+) = \sum_{n\in\mathbb{Z}} J_n^a \, e^{-inx^+} \qquad \text{and} \qquad J_L^a(x^-) = \sum_{n\in\mathbb{Z}} \bar{J}_n^a \, e^{-inx^-} \tag{608}$$

Then, if we take their hermitian conjugate, we get,

$$\begin{aligned} \left[J_R^a(x^+)\right]^\dagger &= \sum_{n\in\mathbb{Z}} (J_n^a)^\dagger \, e^{inx^+} = \sum_{n\in\mathbb{Z}} (J_{-n}^a)^\dagger \, e^{-inx^+} \\ \text{and} \;\; \left[J_L^a(x^-)\right]^\dagger &= \sum_{n\in\mathbb{Z}} (\bar{J}_n^a)^\dagger \, e^{inx^-} = \sum_{n\in\mathbb{Z}} (\bar{J}_{-n}^a)^\dagger \, e^{-inx^-} \end{aligned} \tag{609}$$

We know that for the $\mathrm{SL}(2,\mathbb{R})$ currents, $\left(J_{R,L}^3(x^{\pm})\right)^\dagger = J_{R,L}^3(x^{\pm})$, and $\left(J_{R,L}^{\pm}(x^{\pm})\right)^\dagger = J_{R,L}^{\mp}(x^{\pm})$. Thus, first for the right currents, we get,

$$\begin{aligned} \implies \left[J_R^3(x^+)\right]^\dagger &= \sum_{n\in\mathbb{Z}} (J_{-n}^3)^\dagger \, e^{-inx^+} = \sum_{n\in\mathbb{Z}} J_n^3 \, e^{-inx^+} = J_R^3(x^+) \qquad \implies (J_{-n}^3)^\dagger = J_n^3 \\ \text{and} \;\; \left[J_R^{\pm}(x^+)\right]^\dagger &= \sum_{n\in\mathbb{Z}} (J_{-n}^{\pm})^\dagger \, e^{-inx^+} = \sum_{n\in\mathbb{Z}} J_n^{\mp} \, e^{-inx^+} = J_R^{\mp}(x^+) \qquad \implies (J_{-n}^{\pm})^\dagger = J_n^{\mp} \end{aligned} \tag{610}$$

Then, for the left currents,

$$\begin{aligned} \implies \left[J_L^3(x^-)\right]^\dagger &= \sum_{n\in\mathbb{Z}} (\bar{J}_{-n}^3)^\dagger \, e^{-inx^-} = \sum_{n\in\mathbb{Z}} \bar{J}_n^3 \, e^{-inx^-} = J_L^3(x^-) \qquad \implies (\bar{J}_{-n}^3)^\dagger = \bar{J}_n^3 \\ \text{and} \;\; \left[J_L^{\pm}(x^-)\right]^\dagger &= \sum_{n\in\mathbb{Z}} (\bar{J}_{-n}^{\pm})^\dagger \, e^{-inx^-} = \sum_{n\in\mathbb{Z}} \bar{J}_n^{\mp} \, e^{-inx^-} = J_L^{\mp}(x^-) \qquad \implies (\bar{J}_{-n}^{\pm})^\dagger = \bar{J}_n^{\mp} \end{aligned} \tag{611}$$

Thus, the Hermiticity conditions for the $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ affine currents is,

$$(J_n^3)^\dagger = J_{-n}^3 \;\; , \quad (J_n^{\pm})^\dagger = J_{-n}^{\mp} \quad \text{and} \quad (\bar{J}_n^3)^\dagger = \bar{J}_{-n}^3 \; , \quad (\bar{J}_n^{\pm})^\dagger = \bar{J}_{-n}^{\mp} \tag{612}$$

This is the Hermiticity condition in the Lorentzian theory. Then, after wick rotation, $x^+ = -i\omega$ and $x^- = -i\bar{\omega}$. Thus, the Lorentzian chiral currents analytically continue into holomorphic and anti-holomorphic Euclidean currents:

$$J(x^+) \to J(w), \qquad \bar{J}(x^-) \to \bar{J}(\bar{w}). \tag{613}$$

We can do a conformal mapping of the Euclidean cylinder to the Euclidean complex plane by

$$z = e^{w} = e^{\tau_E + i\sigma} = e^{\tau_E}\, e^{i\sigma}, \qquad \bar{z} = e^{\bar{w}} = e^{\tau_E - i\sigma} = e^{\tau_E}\, e^{-i\sigma}, \tag{614}$$

Thus, the radial distance in the $(z, \bar{z})$ plane relates to the Euclidean time by $r = e^{\tau_E}$. So, $\tau_E \to -\infty$ (Infinite Past) is at origin $(0,0)$ of the plane and $\tau_E = 0$ is the unit circle. Then the spatial coordinate σ gives the phase of the $(z, \bar{z})$ plane. Thus, the time evolution in the world-sheet cylinder corresponds to the radial evolution in the plane.

The conformal mapping of the Euclidean cylinder to the plane will transform the currents as follows. Since the currents behave as tensor operators with conformal weight 1,

$$J_R^{(\text{plane})}(z) = (\partial_w z)^{-1} J_R^{(\text{Cylinder})}(w) \qquad \text{and} \qquad J_L^{(\text{plane})}(z) = (\partial_{\bar{w}} \bar{z})^{-1} J_L^{(\text{Cylinder})}(\bar{w}) \tag{615}$$

$$\begin{aligned} \implies \Big(J_R^a(z)\Big)_{\text{plane}} &= \frac{\Big(J_R^a(w)\Big)_{\text{cylinder}}}{z} = \frac{1}{z}\sum_{n\in\mathbb{Z}} J_n^a e^{-inx^+} = \frac{1}{z}\sum_{n\in\mathbb{Z}} J_n^a e^{-n\omega} = \sum_{n\in\mathbb{Z}} \frac{J_n^a}{z^{n+1}} \\ \text{and} \quad \Big(J_L^a(\bar{z})\Big)_{\text{plane}} &= \frac{\Big(J_L^a(\bar{w})\Big)_{\text{cylinder}}}{z} = \frac{1}{\bar{z}}\sum_{n\in\mathbb{Z}} \bar{J}_n^a e^{-inx^-} = \frac{1}{\bar{z}}\sum_{n\in\mathbb{Z}} \bar{J}_n^a e^{-n\bar{\omega}} = \sum_{n\in\mathbb{Z}} \frac{\bar{J}_n^a}{\bar{z}^{n+1}} \end{aligned} \tag{616}$$

Thus, the affine current modes of the Lorentzian theory now become the coefficients of the Laurent expansion of the holomorphic and the anti-holomorphic currents. Thus, these modes can be extracted using the contour integrals as follows,

$$J_n^a = \frac{1}{2\pi i}\oint_C dz\; z^n\; J^a(z) \qquad \text{and} \qquad \bar{J}_n^a = \frac{1}{2\pi i}\oint_{\bar{C}} d\bar{z}\; \bar{z}^n\; \bar{J}^a(\bar{z}) \tag{617}$$

$C : |z| = \text{const}$ is a contour running anti-clockwise enclosing the origin and $\bar{C} : |\bar{z}| = \text{const}$ is a contour running clockwise enclosing the origin.

Thus, the Laurent modes of the Euclidean affine currents on the complex plane have the same Hermiticity conditions as those of the Fourier modes of the Lorentzian currents on the cylinder.

$$\implies \boxed{(J_n^3)^\dagger = J_{-n}^3 \;,\quad (J_n^\pm)^\dagger = J_{-n}^\mp \quad \text{and} \quad (\bar{J}_n^3)^\dagger = \bar{J}_{-n}^3 \;,\quad (\bar{J}_n^\pm)^\dagger = \bar{J}_{-n}^\mp} \tag{618}$$

Thus, although the target-space geometry is Wick rotated from Lorentzian AdS_3 to Euclidean H_3^+, the Hermiticity structure of the operator algebra is still inherited from the Lorentzian $\text{SL}(2,\mathbb{R})$ theory. Thus, the Euclidean H_3^+ WZW model is not treated as a completely independent Euclidean theory. Rather, it is viewed as the analytic continuation of the Lorentzian AdS_3 string theory, with the same physical Hilbert-space structure where the Euclidean theory still possesses the original $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ affine current algebra.

8.3 Bosonization of the $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ currents

We have previously seen that the Cartan current algebra is isomorphic to the affine $U(1)$ algebra. This can be seen as follows. Consider a time-like real free boson field $\phi(z,\bar{z})$ such that $[\phi(z,\bar{z})]^\dagger = \phi(z,\bar{z})$ on the Euclidean worldsheet whose Polyakov action ($\alpha' = 2$) is,

$$S[\phi] = -\frac{1}{4\pi}\int d^2z \left(\partial\phi(z,\bar{z})\right)\left(\bar{\partial}\phi(z,\bar{z})\right) \tag{619}$$

The equation of motion is given by,

$$\partial\bar{\partial}\phi = \bar{\partial}\partial\phi = 0 \qquad \Longrightarrow \partial\phi \sim \text{Holomorphic} \quad \text{and} \quad \bar{\partial}\phi \sim \text{Anti-Holomorphic} \tag{620}$$

Thys, the field $\phi(z,\bar{z})$ can be decomposed as $\phi(z,\bar{z}) = \phi_R(z) + \phi_L(\bar{z})$. Now the Green's function of the operator $\partial\bar{\partial}$, which is the two-point function $\langle \phi(z,\bar{z})\phi(z',\bar{z}')\rangle$, satisfies

$$\partial\bar{\partial}\left\langle \phi(z,\bar{z})\phi(z',\bar{z}')\right\rangle = 2\pi\delta^{(2)}(z-z',\bar{z}-\bar{z}') \tag{621}$$

In two dimensions, the Green's function of the operator $\partial\bar{\partial}$ is logarithmic.

$$\partial\bar{\partial}\,\ln|z-z'|^2 = 2\pi\,\delta^{(2)}(z-z',\bar{z}-\bar{z}') \tag{622}$$

Thus, on comparing, we get,

$$\langle \phi(z,\bar{z})\phi(z',\bar{z}')\rangle = \log|z-z'|^2 + \text{const}\cdot = \log(z-z') + \log(\bar{z}-\bar{z}') + \text{const}\cdot \tag{623}$$

Then, on differentiating,

$$\begin{aligned} \langle \partial_z\phi(z,\bar{z})\,\phi(z',\bar{z}')\rangle &= \partial_z\left(\langle\phi(z,\bar{z})\phi(z',\bar{z}')\rangle\right) = \frac{1}{z-z'} \\ \Longrightarrow \quad \langle \partial_z\phi(z,\bar{z})\,\partial_{z'}\phi(z',\bar{z}')\rangle &= \partial_z\partial_{z'}\left(\langle\phi(z,\bar{z})\phi(z',\bar{z}')\rangle\right) = \frac{1}{(z-z')^2} \end{aligned} \tag{624}$$

Similarly, we can also get the OPE for the anti-holomorphic 2-point functions. Thus, finally,

$$\langle \partial_z\phi(z,\bar{z})\,\partial_{z'}\phi(z',\bar{z}')\rangle \sim \frac{1}{(z-z')^2} \qquad \text{and} \qquad \langle \partial_{\bar{z}}\phi(z,\bar{z})\,\partial_{\bar{z}'}\phi(z',\bar{z}')\rangle \sim \frac{1}{(\bar{z}-\bar{z}')^2} \tag{625}$$

8.3.1 Bosonization of $J^3_{R,L}$ Cartan currents

We can compare the above OPEs with the $J^3_{R,L}$ Cartan currents' OPE with themselves,

$$\Longrightarrow \langle J^3_R(z)\,J^3_R(z')\rangle \sim -\frac{k}{2}\,\frac{1}{(z-z')^2} \qquad \text{and} \qquad \langle J^3_L(\bar{z})\,J^3_L(\bar{z}')\rangle \sim -\frac{k}{2}\,\frac{1}{(\bar{z}-\bar{z}')^2} \tag{626}$$

Thus, we can identify,

$$J^3_R(z) = -i\sqrt{\frac{k}{2}}\,\partial\phi\,(z) \qquad \text{and} \qquad J^3_L(\bar{z}) = -i\sqrt{\frac{k}{2}}\,\bar{\partial}\phi\,(\bar{z}) \tag{627}$$

The energy-momentum tensor for the free-boson action is given by,

$$T^{(\phi)}(z) = \frac{1}{2} : (\partial\phi)(\partial\phi) : \qquad \text{and} \qquad \bar{T}^{(\phi)}(\bar{z}) = \frac{1}{2} : (\bar{\partial}\phi)(\bar{\partial}\phi) : \tag{628}$$

In terms of the J^3-current, it exactly matches the Sugawara construction as shown below,

$$\begin{aligned}
T^{(\phi)}(z) &= \frac{1}{2} : (\partial\phi)(\partial\phi) : = -\frac{1}{2}\frac{2}{k} : \left(-i\sqrt{\frac{k}{2}}\,\partial\phi\right)\left(-i\sqrt{\frac{k}{2}}\,\partial\phi\right) : \\
\Longrightarrow\ T^{(\phi)}(z) &= -\frac{1}{k} : J^3_R(z)J^3_R(z) : = T^{(J^3)}(z) \\
\text{Similarly,}\quad \bar{T}^{(\phi)}(\bar{z}) &= \frac{1}{2} : (\bar{\partial}\phi)(\bar{\partial}\phi) : = -\frac{1}{2}\frac{2}{k} : \left(-i\sqrt{\frac{k}{2}}\,\bar{\partial}\phi\right)\left(-i\sqrt{\frac{k}{2}}\,\bar{\partial}\phi\right) : \\
\Longrightarrow\ \bar{T}^{(\phi)}(\bar{z}) &= -\frac{1}{k} : J^3_L(\bar{z})J^3_L(\bar{z}) : = \bar{T}^{(J^3)}(\bar{z})
\end{aligned} \tag{629}$$

The OPE of the energy-momentum tensor and the free-boson field is given by,

$$\begin{aligned}
\langle T^{(\phi)}(z)\,\phi(z')\rangle &= \frac{1}{2}\langle : (\partial\phi)(\partial\phi) : \phi(z')\rangle = \big(\partial_z\langle\phi(z)\phi(z')\rangle\big)(\partial\phi(z)) \sim \frac{\partial_z\phi(z)}{z-z'} \sim \frac{\partial_{z'}\phi(z')}{z-z'} \\
\text{Similarly,}\ \langle \bar{T}^{(\phi)}(\bar{z})\,\phi(\bar{z}')\rangle &\sim \frac{\partial_{\bar{z}'}\phi(\bar{z}')}{\bar{z}-\bar{z}'}
\end{aligned} \tag{630}$$

Thus, $\phi(z,\bar{z})$ acts as a primary field of conformal weight $(0,0)$. Now, consider the below OPE,

$$\begin{aligned}
\langle T^{(\phi)}(z)\,\partial_{z'}\phi(z')\rangle &= \frac{1}{2}\langle : (\partial\phi(z))(\partial\phi(z)) : \partial_{z'}\phi(z')\rangle = \big(\partial_z\partial_{z'}\langle\phi(z)\phi(z')\rangle\big)(\partial\phi(z)) \sim \frac{\partial_z\phi(z)}{(z-z')^2} \\
\Longrightarrow\ \langle T^{(\phi)}(z)\,\partial_{z'}\phi(z')\rangle &\sim \frac{\partial_{z'}\phi(z')}{(z-z')^2} + \frac{\partial^2_{z'}\phi(z')}{z-z'} \\
\text{Similarly,}\ \langle \bar{T}^{(\phi)}(\bar{z})\,\partial_{\bar{z}'}\phi(\bar{z}')\rangle &\sim \frac{\partial_{\bar{z}'}\phi(\bar{z}')}{(\bar{z}-\bar{z}')^2} + \frac{\partial^2_{\bar{z}'}\phi(\bar{z}')}{\bar{z}-\bar{z}'}
\end{aligned} \tag{631}$$

Thus, the free-boson fields $\partial\phi$ and $\bar{\partial}\phi$ are primary fields with conformal weight $(1,0)$ and $(0,1)$ respectively. In terms of the J^3 bosonized current, we have $T^{(\phi)}(z) = T^{(J^3)}(z)$ and $\bar{T}^{(\phi)}(\bar{z}) = \bar{T}^{(\bar{J}^3)}(\bar{z})$. Also, since $J^3_R(z) = -i\sqrt{\frac{k}{2}}\,\partial\phi\,(z)$ and $J^3_L(\bar{z}) = -i\sqrt{\frac{k}{2}}\,\bar{\partial}\phi\,(\bar{z})$, we get,

$$\langle T^{(J^3)}(z)\,J^3_R(z')\rangle \sim \frac{J^3_R(z')}{(z-z')^2} + \frac{\partial_{z'}J^3_R(z')}{z-z'} \quad \text{and} \quad \langle \bar{T}^{(\bar{J}^3)}(\bar{z})\,J^3_L(\bar{z}')\rangle \sim \frac{J^3_L(\bar{z}')}{(\bar{z}-\bar{z}')^2} + \frac{\partial_{\bar{z}'}J^3_L(\bar{z}')}{\bar{z}-\bar{z}'} \tag{632}$$

Thus, the Cartan currents $J^3_R(z)$ and $J^3_L(\bar{z})$ have conformal weights $(1,0)$ and $(0,1)$ respectively. Now, we can do a mode expansion of the holomorphic and the anti-holomorphic fields as,

$$\partial\phi\,(z) = \sum_{n\in\mathbb{Z}} \frac{\alpha_n}{z^{n+1}} \qquad \text{and} \qquad \bar{\partial}\phi\,(\bar{z}) = \sum_{n\in\mathbb{Z}} \frac{\bar{\alpha}_n}{\bar{z}^{n+1}} \tag{633}$$

where α_n and $\bar{\alpha}_n$ are the Laurent modes of expansion.

$$\alpha_n = \oint_C \frac{dz}{2\pi i} z^n (\partial\phi) \qquad \text{and} \qquad \bar{\alpha}_n = \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i} \bar{z}^n (\bar{\partial}\phi) \tag{634}$$

We can also find the mode expansion of the boson field $\phi(z,\bar{z})$ as follows. By the chain rule,

$$d\phi(z,\bar{z}) = \partial\phi\,(z)dz + \bar{\partial}\phi\,(\bar{z})d\bar{z} = \sum_{n\in\mathbb{Z}} \left(\frac{\alpha_n}{z^{n+1}} dz + \frac{\bar{\alpha}_n}{\bar{z}^{n+1}} d\bar{z} \right) \tag{635}$$

Integrating this expression gives the mode expansion of the scalar field:

$$\phi(z,\bar{z}) = q + \frac{\alpha_0}{2} \log(z) + \frac{\bar{\alpha}_0}{2} \log(\bar{z}) - \sum_{n\neq 0} \frac{1}{n} \left(\frac{\alpha_n}{z^n} + \frac{\bar{\alpha}_n}{\bar{z}^n} \right), \tag{636}$$

Due to the translation symmetry in the free-boson action, we get $\alpha_0 = \bar{\alpha}_0 = p$, which is the zero-mode momentum. Similarly, q is the zero-mode position. Thus,

$$\phi(z,\bar{z}) = q + p \log|z|^2 - \sum_{n\neq 0} \frac{1}{n} \left(\frac{\alpha_n}{z^n} + \frac{\bar{\alpha}_n}{\bar{z}^n} \right) \tag{637}$$

Now, we can extract the commutation relations of the free boson modes using the OPE and the contour-integration technique. Consider holomorphic objects $A(z)$ and $B(z)$, whose contour integration (enclosing $z = 0$) is Q_1 and Q_2 respectively (which are contour independent). Let $C_1(z_1) > C_2(z_2) > C_3(z_3)$ be three concentric contours with $|z_1| > |z_2| > |z_3|$. Then,

$$Q_1(C_1) = \oint_{C_1} \frac{dz_1}{2\pi i} A(z_1) \quad \text{and} \quad Q_2(C_2) = \oint_{C_2} \frac{dz_z}{2\pi i} B(z_2) \tag{638}$$

$$\begin{aligned}
\implies [Q_1, Q_2] &= Q_1(C_1)Q_2(C_2) - Q_2(C_2)Q_1(C_3) \\
&= \oint_{C_2} \frac{dz_z}{2\pi i} \left(\oint_{C_1} \frac{dz_1}{2\pi i} A(z_1)\, B(z_2) - \oint_{C_3} \frac{dz_3}{2\pi i} B(z_2)\, A(z_3) \right)
\end{aligned} \tag{639}$$

$$\implies [Q_1, Q_2] = \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2} \underbrace{\langle A(z_1) B(z_2) \rangle}_{OPE}$$

$$\begin{aligned}
\implies [\alpha_m, \alpha_n] &= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2} \left(z_1^m\, z_2^n\, \underbrace{\langle \partial_{z_1}\phi(z_1)\, \partial_{z_2}\phi(z_2) \rangle}_{OPE} \right) \\
&= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2} \left(z_2^n \left(z_2^m + m\, z_2^{m-1}(z_1 - z_2) \right) \frac{1}{(z_1 - z_2)^2} \right) \\
&= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2} \left(\frac{z_2^{m+n}}{(z_1 - z_2)^2} + \frac{m\, z_2^{m+n-1}}{z_1 - z_2} \right) = \oint_{C_2} \frac{dz_2}{2\pi i}\, m\, z_2^{m+n-1}
\end{aligned} \tag{640}$$

$$\implies [\alpha_m, \alpha_n] = m\, \delta_{m+n,0}$$

The Cartan-current modes and the free-boson modes are related by,

$$\implies J^3_n = -i\sqrt{\frac{k}{2}}\,\alpha_n \qquad \text{and} \qquad \bar{J}^3_n = -i\sqrt{\frac{k}{2}}\,\bar{\alpha}_n \qquad \implies \qquad \big[J^3_m, J^3_n\big] = -\frac{k}{2}[\alpha_m, \alpha_n] \tag{641}$$

$$\implies \boxed{\big[J^3_m, J^3_n\big] = -\frac{k}{2}\,m\,\delta_{m+n,0}} \qquad \text{Similarly,} \boxed{\big[\bar{J}^3_m, \bar{J}^3_n\big] = -\frac{k}{2}\,m\,\delta_{m+n,0}} \tag{642}$$

Thus, we derived the commutation relation for Cartan current modes using free-boson theory.

8.3.2 Bosonization of $J^{\pm}(z)_{R,L}$ currents

The affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_k$ algebra, $[J^3_m, J^{\pm}_n] = \pm J^{\pm}_{m+n}$ and $[\bar{J}^3_m, \bar{J}^{\pm}_n] = \pm \bar{J}^{\pm}_{m+n}$ implies that the OPE of $J^3_R(z)$ with $J^{\pm}_R(z)$ should be,

$$\langle J^3_R(z) J^{\pm}_R(z')\rangle \sim \frac{\pm J^{\pm}_R(z')}{z - z'} \tag{643}$$

Thus, we need to identify $J^{\pm}_R(z)$ in terms of boson fields whose OPE matches as above. Consider the exponential of the free-boson field $e^{ia\phi(z,\bar{z})}$. We shall find the OPE,

$$\langle \phi(z_1,\bar{z}_1) : e^{ia\phi(z_2,\bar{z}_2)} :\rangle = \Big\langle \phi(z_1,\bar{z}_1) \sum_{n=0}^{\infty} \frac{i^n}{n!}\, a^n : \phi^n(z_2,\bar{z}_2) : \Big\rangle \tag{644}$$

Now use the contraction $\phi(z_1,\bar{z}_1)\phi(z_2,\bar{z}_2) \sim \log|z_1 - z_2|^2$. Hence,

$$\begin{aligned}\langle \phi(z_1,\bar{z}_1) : e^{ia\phi(z_2,\bar{z}_2)} :\rangle \sim \sum_{n=0}^{\infty} \frac{i^n a^n}{n!} \log|z_1 - z_2|^2 \Big(&: \phi^{n-1}(z_2,\bar{z}_2) : \\ &+ : \phi^{n-1}(z_2,\bar{z}_2) : + \cdots + : \phi^{n-1}(z_2,\bar{z}_2) : \Big).\end{aligned} \tag{645}$$

There are n identical contractions, therefore

$$\begin{aligned}\langle \phi(z_1,\bar{z}_1) : e^{ia\phi(z_2,\bar{z}_2)} :\rangle &\sim \log|z_1 - z_2|^2 \sum_{n=0}^{\infty} \frac{i^n a^n}{n!} n : \phi^{n-1}(z_2,\bar{z}_2) : \\ &= \log|z_1 - z_2|^2\, ia \sum_{n=1}^{\infty} \frac{i^{n-1} a^{n-1}}{(n-1)!} : \phi^{n-1}(z_2,\bar{z}_2) : \\ &= \log|z_1 - z_2|^2\, ia \sum_{n=0}^{\infty} \frac{i^n a^n}{(n)!} : \phi^{n}(z_2,\bar{z}_2) :\end{aligned} \tag{646}$$

$$\implies \langle \phi(z_1,\bar{z}_1) : e^{ia\phi(z_2,\bar{z}_2)} :\rangle \sim ia \log|z_1 - z_2|^2 : e^{ia\phi(z_2,\bar{z}_2)} : \tag{647}$$

Now differentiate with respect to z_1:

$$\langle \partial_{z_1}\phi(z_1,\bar{z}_1) : e^{ia\phi(z_2,\bar{z}_2)} :\rangle \sim ia\, \partial_{z_1} \log|z_1 - z_2|^2 : e^{ia\phi(z_2,\bar{z}_2)} : = \frac{ia}{z_1 - z_2} : e^{ia\phi(z_2,\bar{z}_2)} : \tag{648}$$

Hence,

$$
\begin{aligned}
\langle \partial_{z_1}\phi(z_1,\bar z_1) : e^{ia\phi(z_2,\bar z_2)} :\rangle &\sim \frac{ia}{z_1-z_2} : e^{ia\phi(z_2,\bar z_2)} : \\
\implies \langle i\sqrt{\frac{2}{k}} J_R^3(z_1) : e^{ia\phi(z_2,\bar z_2)} :\rangle &\sim \frac{ia}{z_1-z_2} : e^{ia\phi(z_2,\bar z_2)} : \\
\implies \langle J_R^3(z_1) : e^{ia\phi(z_2,\bar z_2)} :\rangle &\sim \frac{\sqrt{\frac{k}{2}}\, a}{z_1-z_2} : e^{ia\phi(z_2,\bar z_2)} :
\end{aligned}
\tag{649}
$$

But note that, $\phi(z,\bar z) = \phi_R(z) + \phi_L(\bar z)$, we denote $\phi_R(z) \equiv \phi(z)$ and $\phi_L(\bar z) \equiv \bar\phi(\bar z)$. Then,

$$
\begin{aligned}
\langle J_R^3(z_1) : e^{ia\phi(z_2,\bar z_2)} :\rangle = \langle J_R^3(z_1) : e^{ia\phi(z)+ia\bar\phi(\bar z_2)} :\rangle \;\approx\; \langle J_R^3(z_1) : e^{ia\phi(z_2)} :\rangle \\
\implies \quad \langle J_R^3(z_1) : e^{ia\phi(z_2)} :\rangle \sim \frac{\sqrt{\frac{k}{2}}\, a}{z_1-z_2} : e^{ia\phi(z_2)} :
\end{aligned}
\tag{650}
$$

This is because $\langle J_R^3(z_1)\, \bar\phi(\bar z_2)\rangle \sim 0$. Now, if we define,

$$
J_R^+(z) = \psi(z) : e^{i\sqrt{\frac{2}{k}}\phi(z)} : \qquad \text{and} \qquad J_R^-(z) = \psi^\dagger(z) : e^{-i\sqrt{\frac{2}{k}}\phi(z)} : \tag{651}
$$

Where the OPE, $\langle J_R^3(z)\, \psi(z')\rangle \sim 0$ and $\langle J_R^3(z)\, \psi^\dagger(z')\rangle \sim 0$, we get,

$$
\begin{aligned}
\langle J_R^3(z_1)\, J_R^+(z_2)\rangle &= \langle J_R^3(z_1)\, \psi(z_2) : e^{i\sqrt{\frac{2}{k}}\phi(z_2)} :\rangle \sim \frac{\psi(z_2) : e^{ia\phi(z_2)} :}{z_1-z_2} = \frac{J_R^+(z_2)}{z_1-z_2} \\
\&\;\; \langle J_R^3(z_1)\, J_R^-(z_2)\rangle &= \langle J_R^3(z_1)\, \psi^\dagger(z_2) : e^{-i\sqrt{\frac{2}{k}}\phi(z_2)} :\rangle \sim \frac{-\,\psi^\dagger(z_2) : e^{-ia\phi(z_2)} :}{z_1-z_2} = \frac{-J_R^-(z_2)}{z_1-z_2}
\end{aligned}
\tag{652}
$$

Then, for the anti-holomorphic or the left-moving current, we can define,

$$
J_L^+(\bar z) = \bar\psi(\bar z) : e^{i\sqrt{\frac{2}{k}}\bar\phi(\bar z)} : \qquad \text{and} \qquad J_L^-(\bar z) = \bar\psi^\dagger(\bar z) : e^{-i\sqrt{\frac{2}{k}}\bar\phi(\bar z)} : \tag{653}
$$

whose OPE with the $J_L^3(\bar z)$ is given by,

$$
\langle J_L^3(\bar z) J_L^\pm(\bar z')\rangle \sim \frac{\pm J_L^\pm(\bar z')}{\bar z - \bar z'} \tag{654}
$$

provided that $\langle J_L^3(\bar z)\, \bar\psi(\bar z')\rangle \sim 0$ and $\langle \bar J_L^3(\bar z)\, \bar\psi^\dagger(\bar z')\rangle \sim 0$.

Thus, we have identified the ladder currents $J_R^\pm(z)$ and $J_L^\pm(\bar z)$ in terms of the boson fields. But we still need to specify more details about the ψ and $\bar\psi$. Until now, we note that these objects are chargeless with respect to the $J_{L,R}^3$ currents. But we still need to set its conformal weight with respect to the $T^{(J^3)}$ energy momentum tensor, such that their conformal weight with respect to the total SL$(2,\mathbb{R})$ energy-momentum tensor is 1. So, for now, let us work assuming the conformal weight of $J_{R,L}^\pm$ is 1. But it will be proved later.

Now, $J_R^{\pm}(z)$ are holomorphic currents of conformal weight $1,0$ and $J_L^{\pm}(\bar{z})$ are anti-holomorphic currents of conformal weight $0,1$. So, they admit the Laurent expansion,

$$J_R^{\pm}(z) = \sum_{n\in\mathbb{Z}} \frac{J_n^{\pm}}{z^{n+1}} \qquad \text{and} \qquad J_L^{\pm}(\bar{z}) = \sum_{n\in\mathbb{Z}} \frac{\bar{J}_n^{\pm}}{\bar{z}^{n+1}} \tag{655}$$

The corresponding modes are extracted using contour integrals:

$$J_n^{\pm} = \oint_C \frac{dz}{2\pi i}\, z^n J_R^{\pm}(z) \qquad \text{and} \qquad \bar{J}_n^{\pm} = \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\, \bar{z}^n J_L^{\pm}(\bar{z}) \tag{656}$$

Now, the OPE-commutator relation gives,

$$\begin{aligned}
[J_m^3, J_n^{\pm}] &= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2}\Big(z_1^m\, z_2^n \, \underbrace{\langle J_R^3(z_1)\, J_R^{\pm}(z_2)\rangle}_{OPE} \Big) = \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2}\Big(z_1^m\, z_2^n \times \frac{\pm J^{\pm}(z_2)}{z_1 - z_2} \Big) \\
&= \pm \oint_{C_2} \frac{dz_2}{2\pi i}\, z_2^{m+n} J^{\pm}(z_2) \qquad \Longrightarrow \; [J_m^3, J_n^{\pm}] = \pm J_{m+n}^{\pm}
\end{aligned} \tag{657}$$

Thus, we need $J_R^{\pm}(z)$ to have conformal weight 1 and to have the appropriate OPE with $J_R^3(z)$. Similarly, we can show that the same relation exists even for the left-moving modes. Thus,

$$\boxed{\left[J_m^3, J_n^{\pm}\right] = \pm J_{m+n}^{\pm}} \qquad \text{and} \qquad \boxed{\left[\bar{J}_m^3, \bar{J}_n^{\pm}\right] = \pm \bar{J}_{m+n}^{\pm}} \tag{658}$$

Then, to derive the commutator between the $J_{R,L}^{\pm}$ modes, we require the OPE as follows,

$$J_R^+(z_1) J_R^-(z_2) \sim \frac{k}{(z_1 - z_2)^2} - \frac{2J_R^3(z_2)}{z_1 - z_2} \quad \text{and} \quad J_L^+(\bar{z}_1) J_L^-(\bar{z}_2) \sim \frac{k}{(\bar{z}_1 - \bar{z}_2)^2} - \frac{2J_L^3(\bar{z}_2)}{\bar{z}_1 - \bar{z}_2} \tag{659}$$

$$\begin{aligned}
\Longrightarrow \; [J_m^+, J_n^-] &= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2}\Big(z_1^m\, z_2^n \, \underbrace{\langle J_R^+(z_1)\, J_R^-(z_2)\rangle}_{OPE} \Big) \\
&= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2}\left[z_1^m\, z_2^n \times \Big(\frac{k}{(z_1 - z_2)^2} - \frac{2J_R^3(z_2)}{z_1 - z_2} \Big) \right] \\
&= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2}\left[\Big(z_2^{m+n} + m\, z_2^{m+n-1}(z_1 - z_2) \Big) \Big(\frac{k}{(z_1 - z_2)^2} - \frac{2J_R^3(z_2)}{z_1 - z_2} \Big) \right] \\
&= \oint_{C_2} \frac{dz_2}{2\pi i} \operatorname*{Res}_{z_1\to z_2}\left[\frac{z_2^{m+n} k}{(z_1 - z_2)^2} + \frac{m\, z_2^{m+n-1} k}{z_1 - z_2} - \frac{2z_2^{m+n} J_R^3(z_2)}{z_1 - z_2} - 2m\, z_2^{m+n-1} J_R^3(z_2) \right] \\
&= \oint_{C_2} \frac{dz_2}{2\pi i} \Big(m\, z_2^{m+n-1} k - 2 z_2^{m+n} J_R^3(z_2) \Big)
\end{aligned}$$

$$\Longrightarrow \; \boxed{[J_m^+, J_n^-] = m\, k\, \delta_{m+n,0} - 2J_{m+n}^3} \quad \text{\& similarly,} \quad \boxed{[\bar{J}_m^+, \bar{J}_n^-] = m\, k\, \delta_{m+n,0} - 2\bar{J}_{m+n}^3} \tag{660}$$

8.4 The Parafermion field

The fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ must satisfy the following properties:

- The fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ should have no singular part in their OPEs with the Cartan currents $J^3_R(z)$ and $J^3_L(\bar{z})$. That is, $J^3_R(z_1)\,\psi(z_2)$ and $J^3_R(z_1)\,\psi^\dagger(z_2)$ are regular and similarly for the antiholomorphic sector, $J^3_L(\bar{z}_1)\,\bar{\psi}(\bar{z}_2)$ and $J^3_L(\bar{z}_1)\,\bar{\psi}^\dagger(\bar{z}_2)$ are regular.
- The conformal weights of these fields, with respect to the total stress tensors $T(z)$ and $\bar{T}(\bar{z})$, should be such that the bosonized ladder currents $J^\pm_R(z) = \psi^\pm(z)\ :e^{\pm i\sqrt{\frac{2}{k}}\phi(z)}:$ and $J^\pm_L(\bar{z}) = \bar{\psi}^\pm(\bar{z})\ :e^{\pm i\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}:$ have conformal weights $(1,0)$ and $(0,1)$ respectively with respect to the full stress tensor.
- The OPEs among these fields must be such that the affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ current algebra is reproduced. In particular, we need,

$$J^+_R(z_1)J^-_R(z_2) \sim \frac{k}{(z_1-z_2)^2} - \frac{2J^3_R(z_2)}{z_1-z_2} \quad \text{and} \quad J^+_L(\bar{z}_1)J^-_L(\bar{z}_2) \sim \frac{k}{(\bar{z}_1-\bar{z}_2)^2} - \frac{2J^3_L(\bar{z}_2)}{\bar{z}_1-\bar{z}_2}$$

8.4.1 Parafermion Energy-momentum tensor

Note that the full Sugawara stress tensor for the affine $\mathrm{SL}(2,\mathbb{R})$ current algebra is

$$\begin{aligned} T(z) &= \frac{1}{k-2}\left[\frac{1}{2}\Big(:J^+_R(z)J^-_R(z): + :J^-_R(z)J^+_R(z):\Big) - :J^3_R(z)J^3_R(z):\right] \\ \bar{T}(\bar{z}) &= \frac{1}{k-2}\left[\frac{1}{2}\Big(:J^+_L(\bar{z})J^-_L(\bar{z}): + :J^-_L(\bar{z})J^+_L(\bar{z}):\Big) - :J^3_L(\bar{z})J^3_L(\bar{z}):\right] \end{aligned} \tag{661}$$

The parafermion stress tensor components $T_{PF}(z)$ and $\bar{T}_{PF}(\bar{z})$ are defined by subtracting the $U(1)$ Cartan-boson energy-momentum tensor contribution from the full stress tensor:

$$T_{\mathrm{PF}}(z) = T(z) - T^{(J^3)}(z) \quad \text{and} \quad \bar{T}_{\mathrm{PF}}(\bar{z}) = \bar{T}(\bar{z}) - \bar{T}^{(J^3)}(\bar{z}) \tag{662}$$

Substituting the explicit expressions gives

$$T_{\mathrm{PF}}(z) = \frac{1}{k-2}\left[\frac{1}{2}\Big(:J^+_R(z)J^-_R(z): + :J^-_R(z)J^+_R(z):\Big) - :J^3_R(z)J^3_R(z):\right] + \frac{1}{k}:J^3_R(z)J^3_R(z): \tag{663}$$

Combining the J^3J^3 terms, we obtain

$$\begin{aligned} T_{\mathrm{PF}}(z) &= \frac{1}{2(k-2)}\Big(:J^+_R(z)J^-_R(z): + :J^-_R(z)J^+_R(z):\Big) - \frac{2}{k(k-2)}:J^3_R(z)J^3_R(z): \\ \text{Similarly,}\ \bar{T}_{\mathrm{PF}}(\bar{z}) &= \frac{1}{2(k-2)}\Big(:J^+_L(\bar{z})J^-_L(\bar{z}): + :J^-_L(\bar{z})J^+_L(\bar{z}):\Big) - \frac{2}{k(k-2)}:J^3_L(\bar{z})J^3_L(\bar{z}): \end{aligned} \tag{664}$$

8.4.2 Conformal weight of $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ fields

The fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ are introduced through the decomposition of the affine $\widehat{\mathrm{SL}(2,\mathbb{R})}_k$ current algebra into a $\widehat{U(1)}$ Cartan sector and a coset sector where the Cartan currents $J^3_R(z)$ and $J^3_L(\bar{z})$ are bosonized in terms of a free scalar fields $\phi(z)$ and $\bar{\phi}(\bar{z})$, while the ladder currents are written as $J^+_R(z) = \psi(z) : e^{i\sqrt{\frac{2}{k}}\phi(z)} :$, $J^-_R(z) = \psi^\dagger(z) : e^{-i\sqrt{\frac{2}{k}}\phi(z)} :$, $J^+_L(\bar{z}) = \bar{\psi}(\bar{z}) : e^{i\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})} :$ and $J^-_L(\bar{z}) = \bar{\psi}^\dagger(\bar{z}) : e^{-i\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})} :$. Here, the exponential factor carries the $U(1)$ charge associated with the Cartan currents $J^3_R(z)$, and $J^3_L(\bar{z})$ whereas the fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ describe the remaining coset degrees of freedom. These fields are not arbitrary auxiliary fields. Their operator product expansions and conformal properties are completely constrained by the requirement that the bosonized currents reproduce the full affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ current algebra. So, we assume that they have the required properties.

First, note that the free-boson field exponential $: e^{\pm ia\phi(z)} :$ is a tensor operator with respect to $T^{(J^3)}(z)$ with the conformal weight $-\dfrac{a^2}{2}$. i.e.,

$$\langle T^{(J^3)}(z_1) :e^{ia\phi(z_2)}:\rangle \sim -\frac{a^2 : e^{ia\,\phi(z_2)} :}{2(z_1 - z_2)^2} + \frac{\partial_{z_2}\big(: e^{ia\phi(z_2)} : \big)}{z_1 - z_2} \tag{665}$$

Also, we need the parafermion energy-momentum tensor to act trivially on $: e^{\pm ia\phi(z)} :$. i.e., $T_{PF}(z_1) : e^{\pm ia\phi(z_2)} : \sim 0$. Thus, even with $T(z) = T^{(J^3)}(z) + T_{PF}(z)$, we get,

$$\begin{aligned} \langle T(z_1) :e^{ia\phi(z_2)}:\rangle &\sim -\frac{a^2 : e^{ia\,\phi(z_2)} :}{2(z_1 - z_2)^2} + \frac{\partial_{z_2}\big(: e^{ia\phi(z_2)} : \big)}{z_1 - z_2} \\ \text{Similarly}\,, \ \langle \bar{T}(\bar{z}_1) :e^{ia\bar{\phi}(\bar{z}_2)}:\rangle &\sim -\frac{a^2 : e^{ia\,\bar{\phi}(\bar{z}_2)} :}{2(\bar{z}_1 - \bar{z}_2)^2} + \frac{\partial_{\bar{z}_2}\big(: e^{ia\bar{\phi}(\bar{z}_2)} : \big)}{\bar{z}_1 - \bar{z}_2} \end{aligned} \tag{666}$$

Thus, the exponential has just the J^3 charges in it. Now, since the fields $J^\pm_R(z) = \psi^\pm(z) : e^{\pm i\sqrt{\frac{2}{k}}\phi(z)} :$ and $J^\pm_L(\bar{z}) = \bar{\psi}^\pm(\bar{z}) : e^{\pm i\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})} :$ should have conformal weights $(1,0)$ and $(0,1)$ respectively with respect to the full stress tensor. We need the total conformal weight of the fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ appropriately. Let,

$$\Big\langle T_{PF}(z_1)\psi(z_2)\Big\rangle \sim \frac{h_\psi \psi(z_2)}{(z_1 - z_2)^2} + \frac{\partial_{z_2}\psi(z_2)}{z_1 - z_2} \tag{667}$$

So, the field $\psi(z_2)$ is a primary field with respect to the parafermion energy-momentum tensor, and hence it is also a primary field with respect to the full energy-momentum tensor. Consider the OPE of $J^+_R(z) = \psi(z) :e^{i\sqrt{\frac{2}{k}}\phi(z)}:$ with total energy-momentum tensor. We can show that,

$$\langle T(z_1) J^+_R(z_2)\rangle = \frac{\left(h_\psi - \frac{1}{k}\right) J^+_R(z_2)}{(z_1 - z_2)^2} + \frac{\partial_{z_2}\big(J^+_R(z_2)\big)}{z_1 - z_2} \tag{668}$$

Proof is given below.

$$
\begin{aligned}
\langle T(z_1)J_R^+(z_2)\rangle &= \Big\langle \big(T_{PF}(z_1)+T^{(J^3)}(z_1)\big)\ \psi(z_2) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:\Big\rangle \\
&= \Big\langle T_{PF}(z_1)\psi(z_2)\Big\rangle :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}: \ + \psi(z_2)\Big\langle T^{(J^3)}(z_1) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:\Big\rangle \\
&= \frac{h_\psi\ \psi(z_2) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:}{(z_1-z_2)^2} + \frac{\big(\partial_{z_2}\psi(z_2)\big) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:}{z_1-z_2} \\
&\qquad\qquad - \frac{\psi(z_2) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:}{k(z_1-z_2)^2} + \frac{\psi(z_2)\big(\partial_{z_2} :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:\big)}{z_1-z_2} \qquad (669) \\
&= \frac{\big(h_\psi-\frac{1}{k}\big)\ \psi(z_2) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:}{(z_1-z_2)^2} + \frac{\partial_{z_2}\big(\psi(z_2) :e^{i\sqrt{\frac{2}{k}}\phi(z_2)}:\big)}{z_1-z_2} \\
\implies \langle T(z_1)J_R^+(z_2)\rangle &= \frac{\big(h_\psi-\frac{1}{k}\big)\ J_R^+(z_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\big(J_R^+(z_2)\big)}{z_1-z_2}
\end{aligned}
$$

Now, to have the total conformal weight of $J_R^+(z_2)$ to be 1, we require, $h_\psi-\frac{1}{k}=1$ or $h_\psi=1+\frac{1}{k}$. In the same way, we can summarise,

$$
\begin{aligned}
\langle T(z_1)J_R^+(z_2)\rangle &= \frac{\big(h_\psi-\frac{1}{k}\big)\ J_R^+(z_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\big(J_R^+(z_2)\big)}{z_1-z_2} &&\implies && h_\psi = 1+\frac{1}{k} \\
\langle T(z_1)J_R^-(z_2)\rangle &= \frac{\big(h_{\psi^\dagger}-\frac{1}{k}\big)\ J_R^-(z_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\big(J_R^-(z_2)\big)}{z_1-z_2} &&\implies && h_{\psi^\dagger} = 1+\frac{1}{k} \\
\langle \bar{T}(\bar{z}_1)J_L^+(\bar{z}_2)\rangle &= \frac{\big(h_{\bar{\psi}}-\frac{1}{k}\big)\ J_L^+(\bar{z}_2)}{(\bar{z}_1-\bar{z}_2)^2} + \frac{\partial_{\bar{z}_2}\big(J_L^+(\bar{z}_2)\big)}{\bar{z}_1-\bar{z}_2} &&\implies && \bar{h}_{\bar{\psi}} = 1+\frac{1}{k} \qquad (670) \\
\langle \bar{T}(\bar{z}_1)J_L^-(\bar{z}_2)\rangle &= \frac{\big(h_{\bar{\psi}^\dagger}-\frac{1}{k}\big)\ J_L^-(\bar{z}_2)}{(\bar{z}_1-\bar{z}_2)^2} + \frac{\partial_{\bar{z}_2}\big(J_L^-(\bar{z}_2)\big)}{\bar{z}_1-\bar{z}_2} &&\implies && \bar{h}_{\bar{\psi}^\dagger} = 1+\frac{1}{k}
\end{aligned}
$$

8.4.3 OPE between $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ fields

Now, we fix the OPE's between the fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ such that they reproduce the correct $\mathrm{SL}(2,\mathbb{R})$ current OPEs. Consider the currents $J_R^+(z)=\psi(z)\ :e^{+i\sqrt{\frac{2}{k}}\phi(z)}:$ and $J_R^-(z)=\psi^\dagger(z)\ :e^{-i\sqrt{\frac{2}{k}}\phi(z)}:$. We require that these ladder currents reproduce the OPE, $J_R^+(z_1)J_R^-(z_2)\sim\frac{k}{(z_1-z_2)^2}-\frac{2J_R^3(z_2)}{z_1-z_2}$. We first compute the contribution from the bosonic exponential factors. For a free-boson exponential, we have the identity,

$$
\begin{aligned}
\Big\langle :e^{ia\phi(z_1)}::e^{ib\phi(z_2)}:\Big\rangle &= (z_1-z_2)^{-ab} :e^{ia\phi(z_1)}e^{ib\phi(z_2)}: \\
\implies \Big\langle :e^{ia\phi(z_1)}::e^{ib\phi(z_2)}:\Big\rangle &= (z_1-z_2)^{-ab} :e^{i(a+b)\phi(z_2)}\Big(1+ia(z_1-z_2)\partial_{z_2}\phi(z_2)+\cdots\Big): \qquad (671) \\
\implies \Big\langle :e^{i\sqrt{\frac{2}{k}}\phi(z_1)}::e^{-i\sqrt{\frac{2}{k}}\phi(z_2)}:\Big\rangle &= (z_1-z_2)^{\frac{2}{k}}\Big(1-\frac{2}{k}(z_1-z_2)\,J_R^3(z_2)+\cdots\Big)
\end{aligned}
$$

Now, we shall derive the OPE between the ψ fields to match the ladder current OPEs.

$$
\begin{aligned}
\left\langle J_R^+(z_1)J_R^-(z_2)\right\rangle &= \left\langle \psi(z_1)\ :e^{+i\sqrt{\frac{2}{k}}\phi(z_1)}:\ \psi^\dagger(z_2)\ :e^{-i\sqrt{\frac{2}{k}}\phi(z_2)}:\right\rangle \\
&\sim \left\langle \psi(z_1)\,\psi^\dagger(z_2)\right\rangle :e^{+i\sqrt{\frac{2}{k}}\phi(z_1)}\ e^{-i\sqrt{\frac{2}{k}}\phi(z_2)}: \\
\implies \left\langle J_R^+(z_1)J_R^-(z_2)\right\rangle &\sim \left\langle \psi(z_1)\,\psi^\dagger(z_2)\right\rangle\ (z_1-z_2)^{\frac{2}{k}}\left(1-\frac{2}{k}(z_1-z_2)\,J_R^3(z_2)+\cdots\right)
\end{aligned} \tag{672}
$$

Now, we can find the leading terms in the ψ OPEs. Assume the OPE to be of the form,

$$
\psi(z_1)\psi^\dagger(z_2) \sim \frac{A}{(z_1-z_2)^{2+2/k}} + \frac{B\,J_R^3(z_2)}{(z_1-z_2)^{1+2/k}} + \cdots . \tag{673}
$$

Substituting into the current product, and neglecting the non-singular terms, we get,

$$
\begin{aligned}
\left\langle J_R^+(z_1)J_R^-(z_2)\right\rangle &\sim :\left(\frac{A}{(z_1-z_2)^{2+2/k}} + \frac{B\,J_R^3(z_2)}{(z_1-z_2)^{1+2/k}} + \cdots\right) \\
&\qquad\times (z_1-z_2)^{2/k}\left(1-\frac{2}{k}(z_1-z_2)J_R^3(z_2)+\cdots\right): \\
\implies \left\langle J_R^+(z_1)J_R^-(z_2)\right\rangle &\sim \frac{A}{(z_1-z_2)^2} - \frac{A}{z_1-z_2}\,\frac{2}{k}J_R^3(z_2) + \frac{B\,J_R^3(z_2)}{(z_1-z_2)} + \cdots
\end{aligned} \tag{674}
$$

Now, for this OPE to match the ladder currents' OPE, we need $A = k$ and $B = 0$. Thus,

$$
\boxed{\left\langle \psi(z_1)\psi^\dagger(z_2)\right\rangle \sim \frac{k}{(z_1-z_2)^{2+\frac{2}{k}}}} \tag{675}
$$

Similarly, for the antiholomorphic or the left-moving sector, we define the currents

$$
J_L^+(\bar z) = \bar\psi(\bar z)\ :e^{+i\sqrt{\frac{2}{k}}\bar\phi(\bar z)}: \quad \text{and} \quad J_L^-(\bar z) = \bar\psi^\dagger(\bar z)\ :e^{-i\sqrt{\frac{2}{k}}\bar\phi(\bar z)}: \tag{676}
$$

We require these currents to reproduce the affine $\mathrm{SL}(2,\mathbb{R})$ current OPE

$$
\left\langle J_L^+(\bar z_1)J_L^-(\bar z_2)\right\rangle \sim \frac{k}{(\bar z_1-\bar z_2)^2} - \frac{2J_L^3(\bar z_2)}{\bar z_1-\bar z_2} \tag{677}
$$

Proceeding exactly as before, we obtain the antiholomorphic $\bar\psi$ OPE as,

$$
\boxed{\left\langle \bar\psi(\bar z_1)\bar\psi^\dagger(\bar z_2)\right\rangle \sim \frac{k}{(\bar z_1-\bar z_2)^{2+\frac{2}{k}}}} \tag{678}
$$

Furthermore, the mixed holomorphic-antiholomorphic OPEs are regular:

$$
\left\langle \psi(z_1)\,\bar\psi(\bar z_2)\right\rangle \sim \text{regular}. \tag{679}
$$

8.4.4 Primary field and its OPE with Virasoro and Affine current modes

A field $\Phi_{jm\bar{m}}(z,\bar{z})$ is called primary of the $\mathrm{SL}(2,\mathbb{R})$ theory if it has the following OPE with the full energy-momentum tensor.

$$\boxed{\begin{aligned}\big\langle T(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &\sim \frac{h_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2}+\frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2}\\ \big\langle \bar{T}(\bar{z}_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &\sim \frac{\bar{h}_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(\bar{z}_1-\bar{z}_2)^2}+\frac{\partial_{\bar{z}_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{\bar{z}_1-\bar{z}_2}\end{aligned}} \tag{680}$$

Where $(h_\Phi,\bar{h}_\Phi)$ is the total conformal weight of the primary field $\Phi_{jm\bar{m}}$. Now, we can see how the Virasoro modes act on these fields. We now expand the holomorphic and antiholomorphic total energy-momentum tensors in Laurent modes:

$$\begin{aligned} T(z) &= \sum_{n\in\mathbb{Z}}\frac{L_n}{z^{n+2}} \quad \text{and} \quad \bar{T}(\bar{z})=\sum_{n\in\mathbb{Z}}\frac{\bar{L}_n}{\bar{z}^{n+2}}\\ \Longrightarrow \quad L_n &= \oint_C\frac{dz}{2\pi i}\,z^{n+1}\,T(z) \quad \text{and} \quad \bar{L}_n=\oint_{\bar{C}}\frac{d\bar{z}}{2\pi i}\,\bar{z}^{n+1}\,\bar{T}(\bar{z})\end{aligned} \tag{681}$$

$$\begin{aligned}\Longrightarrow\ L_n\cdot\Phi_{jm\bar{m}}(z_2,\bar{z}_2) &= \oint_{C:|z_1-z_2|=\epsilon}\frac{dz_1}{2\pi i}\,z_1^{n+1}\,\big\langle T(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle\\ &= \oint_{C:|z_1-z_2|=\epsilon}\frac{dz_1}{2\pi i}\,z_1^{n+1}\left(\frac{h_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2}+\frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2}\right)\\ &= h_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\oint_{C:|z_1-z_2|=\epsilon}\frac{dz_1}{2\pi i}\,\frac{z_1^{n+1}}{(z_1-z_2)^2}\\ &\qquad +\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\oint_{C:|z_1-z_2|=\epsilon}\frac{dz_1}{2\pi i}\,\frac{z_1^{n+1}}{z_1-z_2}\end{aligned} \tag{682}$$

Using $\oint_{C:|z_1-z_2|=\epsilon}\frac{dz_1}{2\pi i}\,\frac{z_1^{n+1}}{(z_1-z_2)^2}=\partial_{z_2}(z_2^{n+1})=(n+1)\,z_2^n$ and $\oint_{C:|z_1-z_2|=\epsilon}\frac{dz_1}{2\pi i}\,\frac{z_1^{n+1}}{z_1-z_2}=z_2^{n+1}$,

$$L_n\cdot\Phi_{jm\bar{m}}(z_2,\bar{z}_2)=h_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)(n+1)\,z_2^n+z_2^{n+1}\,\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2) \tag{683}$$

We can work out similarly for the anti-holomorphic part, whose form will be the same. Thus,

$$\boxed{\begin{aligned}L_n\cdot\Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(z^{n+1}\partial_z+(n+1)\,h_\Phi\,z^n\Big)\Phi_{jm\bar{m}}(z,\bar{z})\\ \bar{L}_n\cdot\Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(\bar{z}^{n+1}\partial_{\bar{z}}+(n+1)h_\Phi\,\bar{z}^n\Big)\Phi_{jm\bar{m}}(z,\bar{z})\end{aligned}} \tag{684}$$

For the Primary field inserted at the origin $(z,\bar{z})\to(0,0)$, we get,

$$L_{n>0}\cdot\Phi_{jm\bar{m}}(0,0)=0 \qquad \text{and} \qquad \bar{L}_{n>0}\cdot\Phi_{jm\bar{m}}(0,0)=0 \tag{685}$$

And, for L_0 and $\bar{L}_0$ it becomes,

$$
\begin{aligned}
L_0 \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(z\,\partial_z + \,h_\Phi\Big)\Phi_{jm\bar{m}}(z,\bar{z}) \\
\bar{L}_0 \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(\bar{z}\,\partial_{\bar{z}} + h_\Phi\Big)\Phi_{jm\bar{m}}(z,\bar{z})
\end{aligned}
\tag{686}
$$

$$
\implies \quad L_0 \cdot \Phi_{jm\bar{m}}(0,0) = h_\Phi \Phi_{jm\bar{m}}(0,0) \qquad \text{and} \qquad \bar{L}_0 \cdot \Phi_{jm\bar{m}}(0,0) = h_\Phi \Phi_{jm\bar{m}}(0,0) \tag{687}
$$

By the state-operator correspondence, we can write $\Phi_{jm\bar{m}}(0,0) \cong |j,m,\bar{m}\rangle$

$$
\begin{aligned}
\implies L_0 \cdot \Phi_{jm\bar{m}}(0,0) &\cong L_0|j,m,\bar{m}\rangle = h_\Phi|j,m,\bar{m}\rangle \\
\text{and} \quad \bar{L}_0 \cdot \Phi_{jm\bar{m}}(0,0) &\cong \bar{L}_0|j,m,\bar{m}\rangle = \bar{h}_\Phi|j,m,\bar{m}\rangle
\end{aligned}
\tag{688}
$$

Thus, the zero-mode Virasoro generators measure the conformal weights of the primary field. We also expect that the relation $J_0^3\,|j,m,\bar{m}\rangle = m\,|j,m,\bar{m}\rangle$ and $\bar{J}_0^3\,|j,m,\bar{m}\rangle = \bar{m}\,|j,m,\bar{m}\rangle$ hold. Thus, we require the OPE's between the $J_{R,L}^3$ currents and the primary field to be,

$$
\boxed{
\begin{aligned}
\big\langle J_R^3(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &\sim \frac{m}{z_1 - z_2}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2) \\
\big\langle J_L^3(\bar{z}_1)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &\sim \frac{\bar{m}}{\bar{z}_1 - \bar{z}_2}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)
\end{aligned}
}
\tag{689}
$$

Now,

$$
\begin{aligned}
J_R^3(z) &= \sum_{n\in\mathbb{Z}} \frac{J_n^3}{z^{n+1}} \qquad \implies J_n^3 = \oint_C \frac{dz}{2\pi i}\,z^n J_R^3(z) \\
\text{and} \qquad J_L^3(\bar{z}) &= \sum_{n\in\mathbb{Z}} \frac{\bar{J}_n^3}{\bar{z}^{n+1}} \qquad \implies \bar{J}_n^3 = \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\,\bar{z}^n J_L^3(\bar{z})
\end{aligned}
\tag{690}
$$

$$
\begin{aligned}
\implies J_n^3 \cdot \Phi_{jm\bar{m}}(z_2,\bar{z}_2) &= \oint_{C:|z_1-z_2|=\epsilon} \frac{dz_1}{2\pi i}\,z_1^n\,\big\langle J_R^3(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle \\
&= \oint_{C:|z_1-z_2|=\epsilon} \frac{dz_1}{2\pi i}\,z_1^n\left(\frac{m}{z_1 - z_2}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right) \\
&= m\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\oint_{C:|z_1-z_2|=\epsilon} \frac{dz_1}{2\pi i}\,\frac{z_1^n}{z_1 - z_2} = m\,z_2^n\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)
\end{aligned}
\tag{691}
$$

Thus, we get, for both the holomorphic and anti-holomorphic sectors,

$$
\boxed{
\begin{aligned}
J_n^3 \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= m\,z^n\,\Phi_{jm\bar{m}}(z,\bar{z}) \qquad \text{and} \qquad \bar{J}_n^3 \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = \bar{m}\,\bar{z}^n\,\Phi_{jm\bar{m}}(z,\bar{z}) \\
\implies J_0^3 \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= m\,\Phi_{jm\bar{m}}(z,\bar{z}) \qquad \text{and} \qquad \bar{J}_0^3 \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = \bar{m}\,\Phi_{jm\bar{m}}(z,\bar{z})
\end{aligned}
}
\tag{692}
$$

When $J_{n>0}^3$ and $\bar{J}_{n>0}^3$ modes act on $\Phi_{jm\bar{m}}(0,0) \cong |j,m,\bar{m}\rangle$, we get,

$$
J_{n>0}^3 \cdot \Phi_{jm\bar{m}}(0,0) = \bar{J}_{n>0}^3 \cdot \Phi_{jm\bar{m}}(0,0) = 0 \implies J_{n>0}^3|j,m,\bar{m}\rangle = \bar{J}_{n>0}^3|j,m,\bar{m}\rangle = 0 \tag{693}
$$

And for the zero mode,

$$\begin{aligned} &J_0^3 \cdot \Phi_{jm\bar{m}}(0,0) = m\,\Phi_{jm\bar{m}}(0,0) &&\implies J_0^3|j,m,\bar{m}\rangle = m|j,m,\bar{m}\rangle \\ \text{and} \quad &\bar{J}_0^3 \cdot \Phi_{jm\bar{m}}(0,0) = \bar{m}\,\Phi_{jm\bar{m}}(0,0) &&\implies \bar{J}_0^3|j,m,\bar{m}\rangle = \bar{m}|j,m,\bar{m}\rangle \end{aligned} \tag{694}$$

Now, we further require that the primary fields exhibit the appropriate ladder structure under the operator product expansions with the ladder currents.

$$\boxed{\begin{aligned} \left\langle J_R^{\pm}(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &\sim \frac{A_{jm}^{\pm}\;\Phi_{j,m\pm1,\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2} \\ \left\langle J_L^{\pm}(\bar{z}_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &\sim \frac{\bar{A}_{j\bar{m}}^{\pm}\;\Phi_{j,m,\bar{m}\pm1}(z_2,\bar{z}_2)}{\bar{z}_1-\bar{z}_2} \end{aligned}} \tag{695}$$

where $A_{jm}^{\pm} = \sqrt{(m \mp j \pm 1)(m \pm j)}$ and $\bar{A}_{j\bar{m}}^{\pm} = \sqrt{(\bar{m} \mp j \pm 1)(\bar{m} \pm j)}$. These OPEs result in,

$$\boxed{\begin{aligned} &J_n^{\pm} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = \sqrt{(m \mp j \pm 1)(m \pm j)}\;\, z^n\,\Phi_{j,m\pm1,\bar{m}}(z,\bar{z}) \\ \text{and} \quad &\bar{J}_n^{\pm} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = \sqrt{(\bar{m} \mp j \pm 1)(\bar{m} \pm j)}\;\, \bar{z}^n\,\Phi_{j,m,\bar{m}\pm1}(z,\bar{z}) \\ \implies \quad &J_0^{\pm} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = \sqrt{(m \mp j \pm 1)(m \pm j)}\;\Phi_{j,m\pm1,\bar{m}}(z,\bar{z}) \\ \text{and} \quad &\bar{J}_0^{\pm} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = \sqrt{(\bar{m} \mp j \pm 1)(\bar{m} \pm j)}\;\Phi_{j,m,\bar{m}\pm1}(z,\bar{z}) \end{aligned}} \tag{696}$$

When $J_{n>0}^{\pm}$ and $\bar{J}_{n>0}^{\pm}$ modes act on $\Phi_{jm\bar{m}}(0,0) \cong |j,m,\bar{m}\rangle$, we get,

$$J_{n>0}^{\pm} \cdot \Phi_{jm\bar{m}}(0,0) = \bar{J}_{n>0}^{\pm} \cdot \Phi_{jm\bar{m}}(0,0) = 0 \implies J_{n>0}^{\pm}|j,m,\bar{m}\rangle = \bar{J}_{n>0}^{\pm}|j,m,\bar{m}\rangle = 0 \tag{697}$$

And for the zero modes,

$$\begin{aligned} &J_0^{\pm} \cdot \Phi_{jm\bar{m}}(0,0) = \sqrt{(m \mp j \pm 1)(m \pm j)}\;\Phi_{j,m\pm1,\bar{m}}(0,0) \\ &\qquad \implies \quad J_0^{\pm}|j,m,\bar{m}\rangle = \sqrt{(m \mp j \pm 1)(m \pm j)}\;|j,m\pm1,\bar{m}\rangle \\ \text{and} \quad &\bar{J}_0^{\pm} \cdot \Phi_{jm\bar{m}}(0,0) = \sqrt{(\bar{m} \mp j \pm 1)(\bar{m} \pm j)}\;\;\Phi_{j,m,\bar{m}\pm1}(0,0) \\ &\qquad \implies \quad \bar{J}_0^{\pm}|j,m,\bar{m}\rangle = \sqrt{(\bar{m} \mp j \pm 1)(\bar{m} \pm j)}\;|j,m,\bar{m}\pm1\rangle \end{aligned} \tag{698}$$

Hence, by imposing the OPEs of the energy-momentum tensors and the $J_{R,L}^{3,\pm}$ currents with the primary fields $\Phi_{jm\bar{m}}(z,\bar{z})$ as discussed previously, we show that these fields satisfy both the Virasoro primary conditions and the affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ primary conditions.

8.4.5 Parafermion and $U(1)$ conformal weight of the Primary field

We know that for a primary state $\Phi_{jm\bar{m}}(0,0) \cong |j,m,\bar{m}\rangle$ in the $\widehat{\mathrm{SL}(2,\mathbb{R})}_L \times \widehat{\mathrm{SL}(2,\mathbb{R})}_R$ algebra, the total conformal weight is given by,

$$h_\Phi = -\frac{j(j-1)}{k-2} \qquad \text{and} \qquad \bar{h}_\Phi = -\frac{j(j-1)}{k-2} \tag{699}$$

This is the eigenvalue of the zeroth Virasoro mode with respect to the primary state. Hence, $L_0 \cdot \Phi_{jm\bar{m}}(0,0) = h_\Phi \Phi_{jm\bar{m}}(0,0)$ and $\bar{L}_0 \cdot \Phi_{jm\bar{m}}(0,0) = \bar{h}_\Phi \Phi_{jm\bar{m}}(0,0)$. Thus,

$$\begin{aligned}
\big\langle T(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &\sim \frac{h_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2} \\
\big\langle \bar{T}(\bar{z}_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &\sim \frac{\bar{h}_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(\bar{z}_1-\bar{z}_2)^2} + \frac{\partial_{\bar{z}_2}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{\bar{z}_1-\bar{z}_2}
\end{aligned}$$

But we can decompose the total $\mathrm{SL}(2,\mathbb{R})$ energy-momentum tensor into the coset $\mathrm{SL}(2,\mathbb{R})/U(1)$ (parafermion) part and the $U(1)$ (cartan) part. Thus,

$$T(z) = T_{PF}(z) + T^{(J^3)}(z) \qquad \text{and} \qquad \bar{T}(\bar{z}) = \bar{T}_{PF}(\bar{z}) + \bar{T}^{(J^3)}(\bar{z}) \tag{700}$$

Then, with $T^{(J^3)}(z) = -\frac{1}{k} : J^3_R(z) J^3_R(z) :$ and $\bar{T}^{(J^3)}(\bar{z}) = -\frac{1}{k} : J^3_L(\bar{z}) J^3_L(\bar{z}) :$, the OPE of the primary field should be,

$$\begin{aligned}
\big\langle T^{(J^3)}(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &= \frac{h_\Phi^{(J^3)}\;\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2} \\
\big\langle \bar{T}^{(J^3)}(\bar{z}_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\big\rangle &= \frac{\bar{h}_\Phi^{(J^3)}\;\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(\bar{z}_1-\bar{z}_2)^2} + \frac{\partial_{\bar{z}_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{\bar{z}_1-\bar{z}_2}
\end{aligned} \tag{701}$$

If we expand $U(1)$ part of the energy-momentum tensor in terms of Virasoro modes, we get,

$$\begin{aligned}
T^{(J^3)}(z) &= \sum_{n\in\mathbb{Z}} \frac{L_n^{(J^3)}}{z^{n+2}} \quad \text{and} \quad \bar{T}^{(J^3)}(\bar{z}) = \sum_{n\in\mathbb{Z}} \frac{\bar{L}_n^{(J^3)}}{\bar{z}^{n+2}} \\
\Longrightarrow \qquad L_n^{(J^3)} &= \oint_C \frac{dz}{2\pi i}\, z^{n+1}\, T^{(J^3)}(z) \quad \text{and} \quad \bar{L}_n^{(J^3)} = \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\, \bar{z}^{n+1}\, \bar{T}^{(J^3)}(\bar{z})
\end{aligned} \tag{702}$$

As seen before, we get,

$$\begin{aligned}
L_n^{(J^3)} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(z^{n+1}\partial_z + (n+1)\,h_\Phi^{(J^3)}\,z^n\Big)\Phi_{jm\bar{m}}(z,\bar{z}) \\
\bar{L}_n^{(J^3)} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(\bar{z}^{n+1}\partial_{\bar{z}} + (n+1)h_\Phi^{(J^3)}\,\bar{z}^n\Big)\Phi_{jm\bar{m}}(z,\bar{z})
\end{aligned} \tag{703}$$

$$\begin{aligned}
\Longrightarrow \qquad L_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(z\,\partial_z + h_\Phi^{(J^3)}\Big)\Phi_{jm\bar{m}}(z,\bar{z}) \\
\bar{L}_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) &= \Big(\bar{z}\,\partial_{\bar{z}} + h_\Phi^{(J^3)}\Big)\Phi_{jm\bar{m}}(z,\bar{z})
\end{aligned} \tag{704}$$

For the Primary field inserted at the origin $(z,\bar{z}) \to (0,0)$, we get,

$$\begin{aligned} L_{n>0}^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) = 0 \qquad &\text{and} \qquad \bar{L}_{n>0}^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) = 0 \\ L_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) = h_\Phi^{(J^3)} \Phi_{jm\bar{m}}(0,0) \qquad &\text{and} \qquad \bar{L}_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) = h_\Phi^{(J^3)} \Phi_{jm\bar{m}}(0,0) \end{aligned} \tag{705}$$

By the state-operator correspondence, we can write $\Phi_{jm\bar{m}}(0,0) \cong |j,m,\bar{m}\rangle$

$$\begin{aligned} \implies L_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) \cong L_0^{(J^3)} |j,m,\bar{m}\rangle = h_\Phi^{(J^3)} |j,m,\bar{m}\rangle \\ \text{and} \quad \bar{L}_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) \cong \bar{L}_0^{(J^3)} |j,m,\bar{m}\rangle = \bar{h}_\Phi^{(J^3)} |j,m,\bar{m}\rangle \end{aligned} \tag{706}$$

$$\begin{aligned} \text{Now,} \qquad L_n^{(J^3)} = \oint_C \frac{dz}{2\pi i}\, z^{n+1}\, T^{(J^3)}(z) \quad &\text{and} \quad \bar{L}_n^{(J^3)} = \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\, \bar{z}^{n+1}\, \bar{T}^{(J^3)}(\bar{z}) \\ L_n^{(J^3)} = -\frac{1}{k} \oint_C \frac{dz}{2\pi i}\, z^{n+1} \, :J_R^3(z) J_R^3(z): \quad &\text{and} \quad \bar{L}_n^{(J^3)} = -\frac{1}{k} \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\, \bar{z}^{n+1} \, :J_L^3(\bar{z}) J_L^3(\bar{z}): \end{aligned} \tag{707}$$

Then, substituting the mode expansion of the $J_{R,L}^3$, we get,

$$\begin{aligned} L_n^{(J^3)} = -\frac{1}{k} \sum_{p,q} \oint_C \frac{dz}{2\pi i}\, z^{n+1} \frac{:J_p^3 J_q^3:}{z^{p+1} z^{q+1}} \quad &\text{and} \quad \bar{L}_n^{(J^3)} = -\frac{1}{k} \sum_{p,q} \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\, \bar{z}^{n+1} \frac{:\bar{J}_p^3 \bar{J}_q^3:}{\bar{z}^{p+1} \bar{z}^{q+1}} \\ \implies L_n^{(J^3)} = -\frac{1}{k} \sum_{p,q} :J_p^3 J_q^3: \ \delta_{p,n-q} \quad &\text{and} \quad \bar{L}_n^{(J^3)} = -\frac{1}{k} \sum_{p,q} :\bar{J}_p^3 \bar{J}_q^3: \ \delta_{p,n-q} \\ \implies L_n^{(J^3)} = -\frac{1}{k} \sum_q :J_{n-q}^3 J_q^3: \quad &\text{and} \quad \bar{L}_n^{(J^3)} = -\frac{1}{k} \sum_q :\bar{J}_{n-q}^3 \bar{J}_q^3: \\ \implies L_0^{(J^3)} = -\frac{1}{k} \sum_q :J_{-q}^3 J_q^3: \quad &\text{and} \quad \bar{L}_0^{(J^3)} = -\frac{1}{k} \sum_q :\bar{J}_{-q}^3 \bar{J}_q^3: \\ \implies L_0^{(J^3)} = -\frac{1}{k} J_0^3 J_0^3 - \frac{2}{k} \sum_{q=1}^{\infty} J_{-q}^3 J_q^3 \quad &\text{and} \quad \bar{L}_0^{(J^3)} = -\frac{1}{k} \bar{J}_0^3 \bar{J}_0^3 - \frac{2}{k} \sum_{q=1}^{\infty} \bar{J}_{-q}^3 \bar{J}_q^3 \end{aligned} \tag{708}$$

Thus, when $L_0^{(J^3)}$ and $\bar{L}_0^{(J^3)}$ acts on $|j,m,\bar{m}\rangle$, we should get,

$$L_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) \cong -\frac{1}{k} J_0^3 J_0^3 |j,m,\bar{m}\rangle - \frac{2}{k} \sum_{q=1}^{\infty} J_{-q}^3 J_q^3 |j,m,\bar{m}\rangle = -\frac{m^2}{k} |j,m,\bar{m}\rangle = h_\Phi^{(J^3)} |j,m,\bar{m}\rangle$$

$$\bar{L}_0^{(J^3)} \cdot \Phi_{jm\bar{m}}(0,0) \cong -\frac{1}{k} \bar{J}_0^3 \bar{J}_0^3 |j,m,\bar{m}\rangle - \frac{2}{k} \sum_{q=1}^{\infty} \bar{J}_{-q}^3 \bar{J}_q^3 |j,m,\bar{m}\rangle = -\frac{\bar{m}^2}{k} |j,m,\bar{m}\rangle = \bar{h}_\Phi^{(J^3)} |j,m,\bar{m}\rangle$$

$$\implies \quad \boxed{h_\Phi^{(J^3)} = -\frac{m^2}{k} \qquad \text{and} \qquad \bar{h}_\Phi^{(J^3)} = -\frac{\bar{m}^2}{k}} \tag{709}$$

Now, we shall derive the Parafermion part of the conformal weight of the primary field $\Phi_{j,m,\bar{m}}(z,\bar{z})$. Split the OPE, of $\langle T(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\rangle$ using $T(z) = T_{PF}(z) + T^{(J^3)}(z)$. Then,

$$\begin{aligned}
\left\langle T(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &= \left\langle T_{PF}(z)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle + \left\langle T^{(J^3)}(z)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \\
\implies \left\langle T_{PF}(z)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &= \left\langle T(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle - \left\langle T^{(J^3)}(z)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \\
&\sim \frac{h_\Phi\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2} \\
&\quad - \left(\frac{h_\Phi^{(J^3)}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2}\right) \\
\implies \left\langle T_{PF}(z)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &\sim \frac{\left(h_\Phi - h_\Phi^{(J^3)}\right)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2}
\end{aligned} \tag{710}$$

Thus, if $\Phi_{jm\bar{m}}(z_2,\bar{z}_2)$ should have the standard OPE with respect to the parafermion energy-momentum tensor, then,

$$\begin{aligned}
\left\langle T_{PF}(z)\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &\sim \frac{h_\Phi^{(PF)}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(z_1-z_2)^2} + \frac{\partial_{z_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{z_1-z_2} \\
\text{Similarly,}\qquad \left\langle \bar{T}_{PF}(\bar{z})\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &\sim \frac{\bar{h}_\Phi^{(PF)}\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{(\bar{z}_1-\bar{z}_2)^2} + \frac{\partial_{\bar{z}_2}\Phi_{jm\bar{m}}(z_2,\bar{z}_2)}{\bar{z}_1-\bar{z}_2}
\end{aligned} \tag{711}$$

$$\implies \boxed{h_\Phi^{(PF)} = h_\Phi - h_\Phi^{(J^3)}} \qquad \text{and} \qquad \boxed{\bar{h}_\Phi^{(PF)} = \bar{h}_\Phi - \bar{h}_\Phi^{(J^3)}} \tag{712}$$

We know that for a primary state $\Phi_{jm\bar{m}}(0,0) \cong |j,m,\bar{m}\rangle$ in the $\widehat{\mathrm{SL}(2,\mathbb{R})}_L \times \widehat{\mathrm{SL}(2,\mathbb{R})}_R$ algebra, the total conformal weight is given by,

$$h_\Phi = -\frac{j(j-1)}{k-2} \qquad \text{and} \qquad \bar{h}_\Phi = -\frac{j(j-1)}{k-2}$$

Thus,

$$\boxed{h_\Phi^{(PF)} = -\frac{j(j-1)}{k-2} + \frac{m^2}{k}} \qquad \text{and} \qquad \boxed{\bar{h}_\Phi^{(PF)} = -\frac{j(j-1)}{k-2} + \frac{\bar{m}^2}{k}} \tag{713}$$

8.4.6 Factorization of Primary Field into Parafermion and $U(1)$-Charged Sectors

We now factorize the primary field into a parafermion component, which carries no $U(1)$ charge, and a pure $U(1)$-charged component.

We previously saw the below OPE,

$$\left\langle J_R^3(z_1) : e^{ia\phi(z_2)} :\right\rangle \sim \frac{\sqrt{\frac{k}{2}}\,a}{z_1-z_2} : e^{ia\phi(z_2)} : \tag{714}$$

If we set $a = m\sqrt{\frac{2}{k}}$, then, $\left\langle J_R^3(z_1) : e^{im\sqrt{\frac{2}{k}}\phi(z_2,\bar{z}_2)} : \right\rangle \sim \dfrac{m}{z_1 - z_2} : e^{im\sqrt{\frac{2}{k}}\phi(z_2)} :$. Thus, the exponential operator $: e^{im\sqrt{\frac{2}{k}}\phi(z_2)} :$ carries the J_0^3 charge m. Similarly, the operator $: e^{i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2)} :$ carries the $\bar{J}_0^3$ charge $\bar{m}$.

$$\left\langle J_L^3(\bar{z}_1) : e^{i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2,\bar{z}_2)} : \right\rangle \sim \frac{\bar{m}}{\bar{z}_1 - \bar{z}_2} : e^{i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2)} : \tag{715}$$

Thus, we can write the primary field in the factorized form

$$\boxed{\Phi_{jm\bar{m}}(z,\bar{z}) = :e^{im\sqrt{\frac{2}{k}}\phi(z) + i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \ \Psi_{jm\bar{m}}(z,\bar{z})} \tag{716}$$

where $\Psi_{jm\bar{m}}(z,\bar{z})$ denotes the parafermion field. By construction, this field has no singular OPE with the Cartan currents $J_R^3(z)$ and $J_L^3(\bar{z})$. Hence, $\Psi_{jm\bar{m}}(z,\bar{z})$ is neutral under the $J_R^3(z)$ and $J_L^3(\bar{z})$ currents and has no any J_0^3 or the $\bar{J}_0^3$ charge. This form of the primary field clearly reproduces all the OPE's with $J_{R,L}^{3,\pm}$ currents and with energy-momentum tensors.

For this to be true, we require,

$$\begin{gathered}
\left\langle J_R^3(z_1)\,\Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \sim 0 \quad , \quad \left\langle J_L^3(\bar{z}_1)\,\Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \sim 0 \\
\left\langle T_{PF}(z_1) :e^{im\sqrt{\frac{2}{k}}\phi(z)+i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \right\rangle \sim 0 \quad , \quad \left\langle \bar{T}_{PF}(\bar{z}_1) :e^{im\sqrt{\frac{2}{k}}\phi(z)+i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \right\rangle \sim 0 \\
\left\langle T^{(J^3)}(z_1)\,\Psi_{jm\bar{m}}(z,\bar{z})\right\rangle \sim 0 \quad , \quad \left\langle \bar{T}^{(J^3)}(\bar{z}_1)\Psi_{jm\bar{m}}(z,\bar{z})\right\rangle \sim 0 \\
h_\Psi = \frac{-j(j-1)}{k-2} + \frac{m^2}{k} \quad , \quad \bar{h}_\Psi = \frac{-j(j-1)}{k-2} + \frac{\bar{m}^2}{k}
\end{gathered} \tag{717}$$

But, to have the correct OPE with the ladder currents $J_R^\pm$ and $J_L^\pm$, we need the parafermion field to have certain OPE with the fields $\psi(z)$, $\psi^\dagger(z)$, $\bar{\psi}(\bar{z})$, and $\bar{\psi}^\dagger(\bar{z})$ which appear in the bosonized expansion of the ladder currents. First, consider the holomorphic ladder current $J_R^\pm(z) = \psi^\pm(z) \ :e^{\pm i\sqrt{\frac{2}{k}}\phi(z)}:$. Then,

$$\begin{aligned}
\left\langle J_R^\pm(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle &= \left\langle \psi^\pm(z_1) \ :e^{\pm i\sqrt{\frac{2}{k}}\phi(z_1)}::e^{im\sqrt{\frac{2}{k}}\phi(z)+i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \ \Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \\
&\sim \left\langle \psi^\pm(z_1)\,\Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \left\langle : e^{\pm i\sqrt{\frac{2}{k}}\phi(z_1)} :: e^{im\sqrt{\frac{2}{k}}\phi(z_2)} : \right\rangle :e^{i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \\
&\sim \left\langle \psi^\pm(z_1)\,\Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle (z_1-z_2)^{\mp\frac{2m}{k}} :e^{i(m\pm1)\sqrt{\frac{2}{k}}\phi(z_2)}::e^{i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \\
&\sim \frac{\left\langle \psi^\pm(z_1)\,\Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle}{\Psi_{j,m\pm1,\bar{m}}(z_2,\bar{z}_2)} (z_1-z_2)^{\mp\frac{2m}{k}} \\
&\qquad \times \left(\Psi_{j,m\pm1,\bar{m}}(z_2,\bar{z}_2) \ :e^{i(m\pm1)\sqrt{\frac{2}{k}}\phi(z_2)+i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}:\right)
\end{aligned} \tag{718}$$

$$\Longrightarrow \left\langle J_R^\pm(z_1)\,\Phi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle \sim \frac{\left\langle \psi^\pm(z_1)\,\Psi_{jm\bar{m}}(z_2,\bar{z}_2)\right\rangle}{\Psi_{j,m\pm1,\bar{m}}(z_2,\bar{z}_2)} (z_1-z_2)^{\mp\frac{2m}{k}}\,\Phi_{j,m\pm1,\bar{m}}(z_2,\bar{z}_2)$$

For the total OPE to reproduce a simple pole, with the factor $A^{\pm}_{jm} = \sqrt{(m \mp j + 1)(m \pm j)}$, we need the parafermion part to behave as,

$$\left\langle \psi^{\pm}(z_1)\, \Psi_{jm\bar{m}}(z_2, \bar{z}_2) \right\rangle \sim \frac{A^{\pm}_{jm}}{(z_1 - z_2)^{1 \mp \frac{2m}{k}}}\, \Psi_{j,m\pm 1,\bar{m}}(z_2, \bar{z}_2) \tag{719}$$

Therefore,

$$\left\langle J^{\pm}_{R}(z_1)\, \Phi_{jm\bar{m}}(z_2, \bar{z}_2) \right\rangle \sim \frac{A^{\pm}_{jm}}{z_1 - z_2} \Phi_{j,m\pm 1,\bar{m}}(z_2, \bar{z}_2) \tag{720}$$

Similarly, for the antiholomorphic ladder currents $J^{\pm}_{L}(\bar{z}) = \bar{\psi}^{\pm}(\bar{z})\; : e^{\pm i\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})} :$. To have required OPE with the primary field $\Phi_{jm\bar{m}}(z, \bar{z})$. We need the OPE

$$\left\langle \psi^{\pm}(\bar{z}_1)\, \Psi_{jm\bar{m}}(z_2, \bar{z}_2) \right\rangle \sim \frac{\bar{A}^{\pm}_{j\bar{m}}}{(\bar{z}_1 - \bar{z}_2)^{1 \mp \frac{2\bar{m}}{k}}}\, \Psi_{j,m,\bar{m}\pm 1}(z_2, \bar{z}_2) \tag{721}$$

where $\bar{A}^{\pm}_{j\bar{m}} = \sqrt{(\bar{m} \mp j + 1)(\bar{m} \pm j)}$. This reproduces,

$$\left\langle J^{\pm}_{L}(\bar{z}_1)\, \Phi_{jm\bar{m}}(z_2, \bar{z}_2) \right\rangle \sim \frac{\bar{A}^{\pm}_{j\bar{m}}}{\bar{z}_1 - \bar{z}_2}\, \Phi_{j,m,\bar{m}\pm 1}(z_2, \bar{z}_2) \tag{722}$$

Identity operator for the Parafermion Theory

The conformal weight of a Parafermion $\Psi_{jm\bar{m}}(z, \bar{z})$ is given by,

$$h_{\Psi} = -\frac{j(j-1)}{k-2} + \frac{m^2}{k} \quad \text{and} \quad \bar{h}_{\Psi} = -\frac{j(j-1)}{k-2} + \frac{\bar{m}^2}{k}$$

Now, for the value of $j = \frac{k}{2}$, and in discrete representations with $m = \bar{m} = \pm\frac{k}{2}$, we see that,

$$h_{\Psi} = \bar{h}_{\Psi} = -\frac{\frac{k}{2}(\frac{k}{2} - 1)}{k-2} + \frac{(\pm\frac{k}{2})^2}{k} = 0 \tag{723}$$

Thus, the parafermion fields corresponding to the states $|j = \frac{k}{2},\, m = \frac{k}{2},\, \bar{m} = \frac{k}{2}\rangle$ in the lowest-weight discrete representations, and $|j = \frac{k}{2},\, m = -\frac{k}{2},\, \bar{m} = -\frac{k}{2}\rangle$ in the highest-weight discrete representations, both have conformal weights $h_{\Psi} = \bar{h}_{\Psi} = 0$. Since the parafermion theory is unitary, fields with vanishing conformal weights are identified, under the state-operator correspondence, with the identity operator.

At these extremal values, the non-Abelian parafermion sector becomes trivial, and the primary field reduces entirely to the $U(1)$-charged bosonic exponential sector. Consequently, these states represent pure $U(1)$ charge excitations and play an important role in the spectral-flow structure and winding sectors of the $\mathrm{SL}(2,\mathbb{R})$ WZW theory.

8.4.7 Spectral Flow of Bosonized Primary Fields

We know that under spectral flow, the eigenvalue of L_0, J_0^3, $\bar{J}_0^3$, changes as,

$$\begin{aligned} m &= \tilde{m} + \frac{k}{2}w \quad , \quad \bar{m} = \tilde{\bar{m}} + \frac{k}{2}w \\ L_0 &= \tilde{L}_0 - w\tilde{m} - \frac{k}{4}w^2 \quad , \quad \bar{L}_0 = \tilde{\bar{L}}_0 - w\tilde{\bar{m}} - \frac{k}{4}w^2 \end{aligned} \tag{724}$$

where '$\tilde{L}_0$, $\tilde{\bar{L}}_0$, $\tilde{m}$, and $\tilde{\bar{m}}$' denotes the eigenvalues before the spectral flow. Now, instead of changing the operators, we shall change the primary field so that, under spectral flow, the eigenvalue changes as shown above.

So, we shall define that under spectral flow, the state changes as

$$\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z,\bar{z}) = :e^{i(\tilde{m}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\phi(z)+i(\tilde{\bar{m}}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}:\ \Psi_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z,\bar{z}) \tag{725}$$

This clearly changes $\tilde{m} \to m = \tilde{m} + \frac{k}{2}w$ and $\tilde{\bar{m}} \to \bar{m} = \tilde{\bar{m}} + \frac{k}{2}w$. Then, the $U(1)$ conformal weight becomes,

$$\begin{aligned} h^{(J^3)}_\Phi &= -\frac{1}{k}\Big(\tilde{m} + \frac{k}{2}w\Big)^2 \qquad \text{and} \qquad \bar{h}^{(J^3)}_\Phi = -\frac{1}{k}\Big(\tilde{\bar{m}} + \frac{k}{2}w\Big)^2 \\ \implies \quad h^{(J^3)}_\Phi &= -\frac{\tilde{m}^2}{k} - w\tilde{m} - \frac{k}{4}w^2 \qquad \text{and} \qquad \bar{h}^{(J^3)}_\Phi = -\frac{\tilde{\bar{m}}^2}{k} - w\tilde{\bar{m}} - \frac{k}{4}w^2 \end{aligned} \tag{726}$$

The conformal weight of the parafermion field is not affected under the spectral flow. So,

$$h^{(PF)}_\Phi = h_\Psi = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \frac{\tilde{m}^2}{k} \qquad \text{and} \qquad \bar{h}^{(PF)}_\Phi = \bar{h}_\Psi = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \frac{\tilde{\bar{m}}^2}{k} \tag{727}$$

Then, the total conformal weight, i.e., the eigenvalue of L_0 and $\bar{L}_0$, changes as follows,

$$\begin{aligned} \tilde{h}_\Phi \to h_\Phi = h_\Psi + h^{(J^3)}_\Phi &= \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \frac{\tilde{m}^2}{k} + \Big(-\frac{\tilde{m}^2}{k} - w\tilde{m} - \frac{k}{4}w^2\Big) \\ \implies & \boxed{h_\Phi = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} - w\tilde{m} - \frac{k}{4}w^2} \end{aligned} \tag{728}$$

Similarly,

$$\begin{aligned} \tilde{\bar{h}}_\Phi \to \bar{h}_\Phi = \bar{h}_\Psi + \bar{h}^{(J^3)}_\Phi &= \frac{-\tilde{j}(\tilde{j}-1)}{k-2} + \frac{\tilde{\bar{m}}^2}{k} + \Big(-\frac{\tilde{\bar{m}}^2}{k} - w\tilde{\bar{m}} - \frac{k}{4}w^2\Big) \\ \implies & \boxed{\bar{h}_\Phi = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} - w\tilde{\bar{m}} - \frac{k}{4}w^2} \end{aligned} \tag{729}$$

Now, we can also see how the ladder operator acts on the spectrally flowed primary field.

$$\left\langle J_R^{\pm}(z_1)\,\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2)\right\rangle = \left\langle \psi^{\pm}(z_1)\ :e^{\pm i\sqrt{\frac{2}{k}}\phi(z_1)}::e^{i(\tilde{m}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\phi(z_2)+i(\tilde{\bar{m}}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2)}:\ \Psi_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2)\right\rangle$$

$$\begin{aligned}
&\sim \left\langle \psi^{\pm}(z_1)\ \Psi_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2)\right\rangle \left\langle\ :e^{\pm i\sqrt{\frac{2}{k}}\phi(z_1)}::e^{i(\tilde{m}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\phi(z_2)+i(\tilde{\bar{m}}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2)}:\right\rangle \\
&\sim \frac{A^{\pm}_{\tilde{j}\tilde{m}}}{(z_1-z_2)^{1\mp\frac{2\tilde{m}}{k}}}\ \Psi_{j,\tilde{m}\pm1,\tilde{\bar{m}}}(z_2,\bar{z}_2)\ \ (z_1-z_2)^{\mp\frac{2}{k}(\tilde{m}+\frac{k}{2}w)} \\
&\qquad\qquad \times :e^{i\sqrt{\frac{2}{k}}\left(\tilde{m}\pm1+\frac{k}{2}w\right)\phi(z_2)+i(\tilde{\bar{m}}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2)}: \\
&\sim \frac{A^{\pm}_{\tilde{j}\tilde{m}}}{(z_1-z_2)^{1\pm w}}\ \Psi_{j,\tilde{m}\pm1,\tilde{\bar{m}}}(z_2,\bar{z}_2)\ :e^{i\sqrt{\frac{2}{k}}\left(\tilde{m}\pm1+\frac{k}{2}w\right)\phi(z_2)+i(\tilde{\bar{m}}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z}_2)}: \\
&\implies \left\langle J_R^{\pm}(z_1)\,\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2)\right\rangle \sim \frac{A^{\pm}_{\tilde{j}\tilde{m}}}{(z_1-z_2)^{1\pm w}}\ \Phi^w_{j,\tilde{m}\pm1,\tilde{\bar{m}}}(z_2,\bar{z}_2)
\end{aligned} \tag{730}$$

Similarly, $$\implies \left\langle J_L^{\pm}(\bar{z}_1)\,\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2)\right\rangle \sim \frac{\bar{A}^{\pm}_{\tilde{j}\tilde{\bar{m}}}}{(\bar{z}_1-\bar{z}_2)^{1\pm w}}\ \Phi^w_{j,\tilde{m},\tilde{\bar{m}}\pm1}(z_2,\bar{z}_2)$$

Now, we can see how the modes of the ladder operators act on the spectrally transformed Primary field $\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z,\bar{z})$ where,

$$\begin{aligned}
&J_R^{\pm}(z) = \sum_{n\in\mathbb{Z}} \frac{J_n^{\pm}}{z^{n+1}} \qquad \implies J_n^{\pm} = \oint_C \frac{dz}{2\pi i}\, z^n J_R^{\pm}(z) \\
\text{and}\qquad &J_L^{\pm}(\bar{z}) = \sum_{n\in\mathbb{Z}} \frac{\bar{J}_n^{\pm}}{\bar{z}^{n+1}} \qquad \implies \bar{J}_n^{\pm} = \oint_{\bar{C}} \frac{d\bar{z}}{2\pi i}\, \bar{z}^n J_L^{\pm}(\bar{z})
\end{aligned} \tag{731}$$

$$\begin{aligned}
&\implies J_n^{\pm}\cdot\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2) = \oint_{C:|z_1-z_2|=\epsilon} \frac{dz_1}{2\pi i}\, z_1^n \left\langle J_R^{\pm}(z_1)\,\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2)\right\rangle \\
&\implies J_n^{\pm}\cdot\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2) = \oint_{C:|z_1-z_2|=\epsilon} \frac{dz_1}{2\pi i}\, z_1^n\, \frac{A^{\pm}_{\tilde{j}\tilde{m}}}{(z_1-z_2)^{1\pm w}}\ \Phi^w_{j,\tilde{m}\pm1,\tilde{\bar{m}}}(z_2,\bar{z}_2)
\end{aligned} \tag{732}$$

Thus, taylor expand z_1 about z_2, $z_1^n = \sum_{k=0}^{\infty} \frac{n!}{(n-k)!\,k!} z_2^{n-k}(z_1-z_2)^k$. Then,

$$J_n^{\pm}\cdot\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z_2,\bar{z}_2) = \sum_{k=0}^{\infty} \frac{n!}{(n-k)!\,k!}\ z_2^{n-k}\ A^{\pm}_{\tilde{j}\tilde{m}}\ \Phi^w_{j,\tilde{m}\pm1,\tilde{\bar{m}}}(z_2,\bar{z}_2)\ \underbrace{\oint_{C:|z_1-z_2|=\epsilon} \frac{dz_1}{2\pi i}\ \frac{1}{(z_1-z_2)^{1-k\pm w}}}_{\delta_{k,\pm w}}$$

$$\implies \boxed{J_n^{\pm}\cdot\Phi^w_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z,\bar{z}) = \frac{n!}{(n\mp w)!\,(\pm w)!}\ z^{n\mp w}\ A^{\pm}_{\tilde{j}\tilde{m}}\ \Phi^w_{j,\tilde{m}\pm1,\tilde{\bar{m}}}(z,\bar{z})} \tag{733}$$

Now, if we change $n \to n \pm w$, then the above relation becomes,

$$J^{\pm}_{n\pm w} \cdot \Phi^{w}_{\tilde{j}\tilde{m}\tilde{\bar{m}}}(z,\bar{z}) = \frac{(n \pm w)!}{n!\,(\pm w)!}\; z^{n}\; A^{\pm}_{\tilde{j}\tilde{m}}\; \Phi^{w}_{j,\tilde{m}\pm 1,\tilde{\bar{m}}}(z,\bar{z}) \tag{734}$$

But, before spectral flow, we had the relation,

$$\tilde{J}^{\pm}_{n} \cdot \Phi_{jm\bar{m}}(z,\bar{z}) = z^{n}\; A^{\pm}_{\tilde{j}\tilde{m}}\; \Phi_{\tilde{j},\tilde{m}\pm 1,\tilde{\bar{m}}}(z,\bar{z}) \tag{735}$$

Thus, apart from the normalization, which appeared just because of the insertion of the field in some non-zero point $(z,\bar{z})$, the mode $J^{\pm}_{n\pm w}$ behaves exactly as it should behave before spectral flow. Thus, we get,

$$\boxed{J^{\pm}_{n\pm w} = \tilde{J}^{\pm}_{n} \qquad \text{or} \qquad J^{+}_{n} = \tilde{J}^{\pm}_{n\mp w}} \tag{736}$$

Thus, in the bosonized formulation, spectral flow reproduces the same shift in the eigenvalues of the affine current zero modes as obtained from the representation-theoretic definition of spectral flow. This arises naturally by redefining the primary fields in the spectrally flowed sector through the shifted $U(1)$-charged bosonic exponential factors. Consequently, the singular structure of the ladder-current OPEs changes, leading to the shifted mode identification

$$\begin{aligned} J^{3}_{n} = \tilde{J}^{3}_{n} + \frac{k}{2} w\,\delta_{n,0} \quad &, \quad \bar{J}^{3}_{n} = \tilde{\bar{J}}^{3}_{n} + \frac{k}{2} w\,\delta_{n,0} \\ J^{+}_{n} = \tilde{J}^{\pm}_{n\mp w} \quad &, \quad \bar{J}^{+}_{n} = \tilde{\bar{J}}^{\pm}_{n\mp w} \\ L_{n} = \tilde{L}_{n} - w\tilde{J}^{3}_{0}\,\delta_{n,0} - \frac{k}{4}w^{2}\,\delta_{n,0} \quad &, \quad \bar{L}_{n} = \tilde{\bar{L}}_{n} - w\tilde{\bar{J}}^{3}_{n}\,\delta_{n,0} - \frac{k}{4}w^{2}\,\delta_{n,0} \end{aligned} \tag{737}$$

This shows that spectral flow reorganizes the affine representation while preserving the underlying $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ current algebra structure.

8.5 Reflection Coefficients

The long-string states of the SL$(2,\mathbb{R})$ WZW model belong to the spectrally flowed continuous representations. Their corresponding vertex operators can be written in the bosonized form as

$$\Phi^{w}_{j\tilde{m}\tilde{\bar{m}}}(z,\bar{z}) =: e^{i\left(\tilde{m}+\frac{k}{2}w\right)\sqrt{\frac{2}{k}}\phi(z) + i\left(\tilde{\bar{m}}+\frac{k}{2}w\right)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})} :\; \Psi_{j\tilde{m}\tilde{\bar{m}}}(z,\bar{z})\, V_{h\bar{h}}(z,\bar{z}) \tag{738}$$

where $j = \frac{1}{2} + is$, $s \in \mathbb{R}$ and $V_{h\bar{h}}(z,\bar{z})$ is the corresponding primary operator in the internal part with conformal weights $(h,\bar{h})$. The spectral flow shifts the J^{3}_{0} eigenvalues according to

$$\begin{aligned} m = \tilde{m} + \frac{k}{2}w \quad &, \quad \bar{m} = \tilde{\bar{m}} + \frac{k}{2}w \\ L_{0} = \tilde{L}_{0} - w\tilde{m} - \frac{k}{4}w^{2} \quad &, \quad \bar{L}_{0} = \tilde{\bar{L}}_{n} - w\tilde{\bar{m}} - \frac{k}{4}w^{2} \end{aligned} \tag{739}$$

The physical energy E and angular momentum ℓ of the corresponding string state are related to these quantum numbers through

$$m = \frac{1}{2}(E + \ell), \qquad \bar{m} = \frac{1}{2}(E - \ell). \tag{740}$$

Also, as seen in the quantum mechanics of a particle in AdS_3, the asymptotic behavior for the radial solution of the particles in the continuous representations is proportional to $e^{\pm is\rho}$, where ρ is the radial direction. Thus, the quantum number 's' plays the role of the radial momentum.

Now, $\tilde{L}_0 = \dfrac{-j(j-1)}{k-2} + \tilde{N}$ and $\tilde{\bar{L}}_0 = \dfrac{-j(j-1)}{k-2} + \tilde{\bar{N}}$, where $\tilde{N}$ and $\tilde{\bar{N}}$ are the oscillator excitations. Using the bosonized form of the spectrally flowed primary field, the $SL(2,\mathbb{R})$ conformal weights are

$$L_0 = -\frac{j(j-1)}{k-2} - w\tilde{m} - \frac{k}{4}w^2 + \tilde{N} \quad \text{and} \quad \bar{L}_0 = -\frac{j(j-1)}{k-2} - w\tilde{\bar{m}} - \frac{k}{4}w^2 + \tilde{\bar{N}} \tag{741}$$

For the continuous representations, $j = \frac{1}{2} + is$, so that $-j(j-1) = \frac{1}{4} + s^2$. Also, the physical-state condition is $L_0 + h = 1$ and $\bar{L}_0 + \bar{h} = 1$. So, for the lowest state in the flowed sector, $\tilde{N} = \tilde{\bar{N}} = 0$, we get,

$$\tilde{m} = -\frac{k}{4}w + \frac{1}{w}\left[\frac{\frac{1}{4} + s^2}{k-2} + h - 1\right] \quad \text{and} \quad \tilde{\bar{m}} = -\frac{k}{4}w + \frac{1}{w}\left[\frac{\frac{1}{4} + s^2}{k-2} + \bar{h} - 1\right] \tag{742}$$

Thus, for a given winding number w, radial momentum s, and internal excitation $(h, \bar{h})$, the values of $\tilde{m}$ and $\tilde{\bar{m}}$ are completely determined by the Virasoro constraints.

The radial motion of a long string in AdS_3 can be viewed as a scattering problem. A long string may propagate from the asymptotic region at large radial distance toward the interior of AdS_3 and subsequently return to infinity. Consequently, the corresponding wavefunction admits two independent asymptotic behaviors, analogous to incoming and outgoing waves in ordinary quantum mechanics. The long-string states belong to the continuous representations of $SL(2,\mathbb{R})$, labelled by $j = \frac{1}{2} + is$, while the representation $1 - j = \frac{1}{2} - is$ describes the same physical state. These two values of j correspond to the two asymptotic radial behaviors of the long-string wavefunction. In the asymptotic region, the radial dependence takes the form $e^{\pm is\rho}$, with $j = \frac{1}{2} + is$ corresponding to the outgoing wave $e^{is\rho}$ and $1 - j = \frac{1}{2} - is$ corresponding to the incoming wave $e^{-is\rho}$. The coefficient relating these two asymptotic solutions is therefore identified with the reflection coefficient. One may thus write

$$\Phi_j = R(j, m, \bar{m})\, \Phi_{1-j} \tag{743}$$

where $R(j, m, \bar{m})$ is known as the reflection coefficient.

The reflection coefficient measures the relative amplitude of the outgoing wave compared to the incoming wave. In this sense, it plays precisely the same role as the reflection amplitude in a one-dimensional quantum-mechanical scattering problem, encoding the exact scattering of long strings in the radial direction of AdS_3.

Now, we can relate the Reflection coefficient to the two-point function $\langle \Phi_j(z,\bar{z})\,\Phi_{j'}(z',\bar{z}')\rangle$. Since the operators Φ_j and Φ_{1-j} describe the same physical state, one may use the reflection relation $\Phi_j = R(j,m,\bar{m})\,\Phi_{1-j}$ to rewrite the correlator as

$$\langle \Phi_j\,\Phi_{j'}\rangle = R(j,m,\bar{m})\ \langle \Phi_{1-j}\,\Phi_{j'}\rangle \tag{744}$$

Therefore, the reflection coefficient necessarily appears as the coefficient multiplying the overlap between a state and its reflected partner. The exact two-point function of the $\mathrm{SL}(2,\mathbb{R})$ WZW model contains two possible overlaps $\langle \Phi_j \Phi_{j'}\rangle \propto \delta(j-j') + R(j,m,\bar{m})\,\delta(j+j'-1)$. The first term corresponds to the overlap of states with identical spin $j'=j$, i.e., $s'=s$, while the second term corresponds to the overlap between a state and its reflected counterpart $j=1-j'$, i.e., $s'=-s$. Thus, $\langle \Phi_{1-j}\Phi_{j'}\rangle \propto \delta(s-s')$. So, the reflected contribution naturally appears with the factor $R(j,m,\bar{m})$. Hence, we can relate the 2-point function to the reflection amplitude as,

$$\langle \Phi_j\,\Phi_{j'}\rangle \propto R(j,m,\bar{m})\ \delta(s-s') \tag{745}$$

Now, we can also fix the coordinate dependence of the 2-point correlation function using the requirement that it has to be invariant under the global conformal transformations. Consider two primary operators $\mathcal{O}_1$ and $\mathcal{O}_2$ with total conformal weights $(h_1^{(\mathrm{tot})}, \bar{h}_1^{(\mathrm{tot})})$ and $(h_2^{(\mathrm{tot})}, \bar{h}_2^{(\mathrm{tot})})$. Translation invariance implies that the two-point function depends only on the coordinate differences $z_1 - z_2$ and $\bar{z}_1 - \bar{z}_2$. Scale invariance then requires

$$\langle \mathcal{O}_1(z_1,\bar{z}_1)\mathcal{O}_2(z_2,\bar{z}_2)\rangle = \frac{C}{(z_1-z_2)^{h_1^{(\mathrm{tot})}+h_2^{(\mathrm{tot})}}\ (\bar{z}_1-\bar{z}_2)^{\bar{h}_1^{(\mathrm{tot})}+\bar{h}_2^{(\mathrm{tot})}}} \tag{746}$$

where C is a constant. Imposing invariance under special conformal transformations further constrains the two operators to have identical conformal weights, $h_1^{(\mathrm{tot})} = h_2^{(\mathrm{tot})} = h^{(\mathrm{tot})}$, and $\bar{h}_1^{(\mathrm{tot})} = \bar{h}_2^{(\mathrm{tot})} = \bar{h}^{(\mathrm{tot})}$. So, the two-point function of a primary field with identical total conformal weights $(h^{(\mathrm{tot})}, \bar{h}^{(\mathrm{tot})})$ takes the universal form

$$\langle \mathcal{O}(z_1,\bar{z}_1)\mathcal{O}(z_2,\bar{z}_2)\rangle = \frac{C}{(z_1-z_2)^{2h^{(\mathrm{tot})}}(\bar{z}_1-\bar{z}_2)^{2\bar{h}^{(\mathrm{tot})}}}. \tag{747}$$

Therefore, the coordinate dependence of the two-point function is completely fixed by conformal symmetry, while all dynamical information is contained in the proportionality coefficient C.

For physical string vertex operators, Φ_j and $\Phi_{j'}$, the Virasoro constraints require the total conformal weights to satisfy $h^{(\text{tot})} = \bar{h}^{(\text{tot})} = 1$. So, substituting these values, we get,

$$\langle \Phi_j \, \Phi_{j'} \rangle \propto R(j, m, \bar{m}) \, \delta(s - s') \, \frac{1}{|z_1 - z_2|^4} \tag{748}$$

A few more delta-function factors appearing in the two-point function are a consequence of the fact that a two-point correlator computes the inner product between physical states. Therefore, the correlator can be non-vanishing only when all conserved quantum numbers of the two states match appropriately. There will be a factor of $\delta(E + E')$ arising from energy conservation. In radial quantization, the operator inserted at infinity corresponds to the conjugate (bra) state, which carries the opposite energy of the state created by the operator inserted at the origin. Consequently, a non-vanishing inner product requires $E + E' = 0$. Similarly, the factor $\delta_{N+N'}$ enforces conservation of the oscillator excitation level. Since affine descendants at different levels are orthogonal, the inner product vanishes unless the excitation levels of the two states coincide (equivalently, $N + N' = 0$ in the convention where one state is represented by its conjugate). Hence, the delta functions encode the conservation of the quantum numbers that characterize the physical string states and ensure that the two-point function is nonzero only for states belonging to the same physical representation.

Although the spectrally flowed vertex operators are labelled by the physical quantum numbers $(m, \bar{m})$, the reflection coefficient is naturally expressed in terms of the unflowed representation labels $(\tilde{m}, \tilde{\bar{m}})$. This is because the reflection relation originates from the underlying unflowed $\mathrm{SL}(2, \mathbb{R})$ representation theory with $\Phi_{j\tilde{m}\tilde{\bar{m}}} = R(j, \tilde{m}, \tilde{\bar{m}}) \, \Phi_{1-j,\tilde{m},\tilde{\bar{m}}}$, which exchanges $j \longleftrightarrow 1 - j$ while leaving $\tilde{m}$ and $\tilde{\bar{m}}$ unchanged. Under spectral flow, the physical J_0^3 eigenvalues become $m = \tilde{m} + \frac{k}{2}w$ and $\bar{m} = \tilde{\bar{m}} + \frac{k}{2}w$. So that m and $\bar{m}$ determine the physical energy and angular momentum of the string state. Nevertheless, the exact reflection coefficient continues to depend on the unflowed labels $(\tilde{m}, \tilde{\bar{m}})$, since it is inherited directly from the unflowed continuous representation. Therefore, the w-dependence of the reflection amplitude is not absent; rather, it is encoded implicitly through $\tilde{m} = m - \frac{k}{2}w$ and $\tilde{\bar{m}} = \bar{m} - \frac{k}{2}w$. Consequently, the left-hand side of the two-point function is labelled by the physical spectrally flowed charges $(m, \bar{m})$, whereas the Gamma-function factor on the right-hand side is naturally written in terms of the unflowed quantum numbers $(\tilde{m}, \tilde{\bar{m}})$ appearing in the reflection relation.

$$\Longrightarrow \; \langle \Phi_{jm\bar{m}}(z, \bar{z}) \, \Phi_{j'm'\bar{m}'}(z', \bar{z}') \rangle = R(j, \tilde{m}, \tilde{\bar{m}}) \, \delta(s - s') \, \delta(E + E') \, \delta_{N+N'} \, \frac{1}{|z - z'|^4} \tag{749}$$

And for the continuous representations with $j = \frac{1}{2} + is$, the reflection coefficient takes the form

$$R(s, \tilde{m}, \tilde{\bar{m}}) = \frac{\Gamma\left(\frac{1}{2} + is - \tilde{m}\right) \Gamma\left(\frac{1}{2} + is + \tilde{\bar{m}}\right) \Gamma(-2is) \Gamma\left(\frac{2is}{k-2}\right)}{\Gamma\left(\frac{1}{2} - is - \tilde{m}\right) \Gamma\left(\frac{1}{2} - is + \tilde{\bar{m}}\right) \Gamma(2is) \Gamma\left(-\frac{2is}{k-2}\right)} \tag{750}$$

8.6 Worldsheet Interpretation of the Reflection Amplitude

The reflection coefficient discussed above arises from a tree-level closed-string scattering amplitude. In perturbative string theory, scattering amplitudes are organized according to the topology of the worldsheet. The Polyakov path integral may be written schematically as

$$\mathcal{A} = \sum_{\text{topologies}} g_s^{-\chi} \int \mathcal{D}X\, \mathcal{D}g\, e^{-S[X,g]}, \tag{751}$$

where $\chi = 2 - 2g$ is the Euler characteristic of the worldsheet and g denotes the genus. For a tree-level closed-string process, $g = 0$. So that $\chi = 2$. The corresponding genus-zero Riemann surface is topologically equivalent to the sphere S^2. Therefore, the long-string reflection amplitude is computed using a two-point correlation function on the sphere. After Wick rotation, the sphere may be identified with the Riemann sphere $S^2 \simeq \mathbb{C} \cup \{\infty\} = \mathbb{CP}^1$.

It is important to note that, although two-dimensional conformal field theories possess an infinite-dimensional local conformal symmetry generated by arbitrary holomorphic transformations $z \to f(z)$, the global conformal transformations on the sphere are much more restricted. Since the sphere is identified with the compact Riemann surface $S^2 \simeq \mathbb{CP}^1$, any globally defined holomorphic map must be well-defined at all points, including infinities. So, such functions are necessarily a rational function of the form, $f(z) = \frac{P(z)}{Q(z)}$, where $P(z)$ and $Q(z)$ are polynomials. Requiring the map to be invertible further restricts it to a rational function of degree one. So, $f(z) = \frac{az+b}{cz+d}$ with $ad - bc \neq 0$, which is a Möbius transformation that forms the $\mathrm{SL}(2, \mathbb{C})$ group. But since multiplying (a, b, c, d) by a common nonzero constant leaves the transformation unchanged, the corresponding symmetry group is actually $P\mathrm{SL}(2, \mathbb{C}) = \mathrm{SL}(2, \mathbb{C})/\mathbb{Z}_2$. But, this group is commonly denoted by $\mathrm{SL}(2, \mathbb{C})$. Therefore, the global conformal group of the sphere is $\mathrm{PSL}(2, \mathbb{C}) \simeq \mathrm{SL}(2, \mathbb{C})$. This is the residual conformal symmetry remaining after fixing the worldsheet metric to conformal gauge.

The Polyakov path integral is invariant under worldsheet diffeomorphisms and Weyl transformations $\mathrm{Diff}(\Sigma) \times \mathrm{Weyl}(\Sigma)$. After imposing conformal gauge, $g_{ab} = e^{2\omega}\delta_{ab}$, most of this gauge freedom is removed. On the sphere, however, a residual symmetry remains, namely the global conformal symmetry, which is the $\mathrm{SL}(2, \mathbb{C})$ symmetry for the Riemann Sphere. These transformations correspond to gauge redundancies rather than physical degrees of freedom. Consequently, the same physical configuration would be counted infinitely many times in the path integral unless one divides by the volume of the residual symmetry group.

Therefore, a string scattering amplitude takes the form

$$\mathcal{A} = \frac{1}{\mathrm{Vol}(\mathrm{SL}(2, \mathbb{C}))} \int \prod_i d^2 z_i \left\langle V_1(z_1) \cdots V_n(z_n) \right\rangle \tag{752}$$

Any Möbius transformation contains three complex degrees of freedom. Consequently, one may use the residual $\mathrm{SL}(2,\mathbb{C})$ invariance to fix the positions of three points on the sphere. For example, for a three-point function, one commonly chooses $z_1 = 0,\ z_2 = 1,\ z_3 = \infty$. Similarly, for a two-point function, one may choose $z = 0$ and $z' = \infty$. This is precisely the gauge choice employed in the computation of the long-string reflection amplitude. But, unlike a three-point function, fixing only two points does not completely remove the $\mathrm{SL}(2,\mathbb{C})$ freedom. A residual subgroup remains, $z \longrightarrow \lambda z,\quad \lambda \in \mathbb{C}-\{0\}$. The volume of this remaining subgroup contributes,

$$\int \frac{d^2\lambda}{|\lambda|^2} \sim \int \frac{d^2 z}{|z|^2} \tag{753}$$

which diverges logarithmically. But the two-point function itself contains $\delta(s-s')\,\delta(E+E')$. Also, for physical scattering processes, whose $E' = -E \implies \delta(E+E') = \delta(0)$. So, the infinite factor $\delta(0)$ arising from energy conservation is precisely cancelled by the divergent volume of the residual $\mathrm{SL}(2,\mathbb{C})$ symmetry. Consequently, the physical amplitude remains finite after gauge fixing and division by the conformal-group volume.

8.7 Poles of the Reflection Amplitude and the Discrete Spectrum

The information about the bound states is contained in the pole structure of the scattering amplitude. Recall that the Gamma function $\Gamma(x)$ possesses simple poles at $x = 0, -1, -2, -3, \cdots$ So, the first Gamma-function in the two-point function, $\Gamma\big(\frac{1}{2} + is - \tilde{m}\big)$, has poles whenever

$$\frac{1}{2} + is - \tilde{m} = -q, \quad q = 0, 1, 2, \cdots \tag{754}$$

or equivalently, whenever $\tilde{m} = \frac{1}{2} + is + q$. Introducing $\tilde{j} = \frac{1}{2} + is$, this may be written as $\tilde{m} = \tilde{j} + q$. Using the physical-state condition derived previously,

$$\tilde{m} = -\frac{k}{4}w + \frac{1}{w}\left[\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + h - 1\right] \tag{755}$$

$$\implies \quad \tilde{j} + q = -\frac{k}{4}w + \frac{1}{w}\left[\frac{-\tilde{j}(\tilde{j}-1)}{k-2} + h - 1\right] \tag{756}$$

Then, solving for $\tilde{j}$, we get,

$$\tilde{j} = \frac{1}{2} - \frac{w}{2}(k-2) + \sqrt{\frac{1}{4} + (k-2)\left(-wq + h - 1 - \frac{1}{2}w(w+1)\right)} \tag{757}$$

Then, after spectral flow, using $J_0^3 = \tilde{m} + \frac{k}{2}w = \tilde{j} + q + \frac{k}{2}w$, we get,

$$J_0^3 = q + w + \frac{1}{2} + \sqrt{\frac{1}{4} + (k-2)\left(-wq + h - 1 - \frac{1}{2}w(w+1)\right)} \tag{758}$$

Remarkably, this is precisely the same condition that appears for the lowest-weight discrete representations $\hat{\mathcal{D}}_j^+$, with $\tilde{N} = 0$. Thus, the poles of the long-string scattering amplitude occur exactly at the locations expected for discrete bound states. This observation is closely related to a general result from scattering theory.

It is worth noting that the pole structure of the reflection coefficient reproduces both the lowest-weight and highest-weight discrete representations. Because the poles of the second gamma function $\Gamma\big(\frac{1}{2} + is + \tilde{\bar{m}}\big)$ occur at $\tilde{\bar{m}} = -\tilde{j} - q, \quad q = 0, 1, 2, \ldots$, which coincides with the highest-weight condition for the discrete representation $\hat{\mathcal{D}}_{\tilde{j}}^-$. Therefore, the pole structure of the reflection amplitude naturally contains both lowest-weight and highest-weight discrete series representations. In the full $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ spectrum, these sectors are related by spectral flow and do not correspond to independent physical families of states.

On the other hand, the poles arising from $\Gamma(-2is)$ are cancelled by corresponding zeros from the denominator factor $\Gamma(2is)$ and therefore do not contribute additional physical poles.

8.8 Emergence of the No-Ghost Bound

At first sight, the pole condition appears to generate an infinite set of candidate bound states. However, the physical discrete representations $\hat{\mathcal{D}}_j^\pm$ satisfy the no-ghost bound $\frac{1}{2} \leq j \leq \frac{k-1}{2}$. But the pole condition itself does not impose this restriction. Consequently, not every pole of the reflection amplitude necessarily corresponds to a physical discrete state. So, one must also examine the residue of the pole. Near a pole located at $E = E_B$, the scattering amplitude behaves as $A(E) \sim \frac{\mathrm{Res}(A)}{E - E_B}$. In general, the residue is proportional to the norm of the intermediate state contributing to the pole. So, a physical state must correspond to a positive residue, whereas a negative residue signals a negative-norm (ghost) state.

Consider the pole-producing factor $\Gamma\big(\frac{1}{2} + is - \tilde{m}\big)$. The poles occur when $\frac{1}{2} + is - \tilde{m} = -q$, $q = 0, 1, 2, \cdots$ or equivalently, when $\tilde{m} = \tilde{j} + q$ where $\tilde{j} = \frac{1}{2} + is$. Near a pole, the Gamma function behaves as

$$\Gamma(-q + \epsilon) = \frac{(-1)^q}{q!}\frac{1}{\epsilon} + \mathcal{O}(1), \qquad \epsilon \to 0. \tag{759}$$

Thus, the residue of an individual Gamma-function pole carries the factor $\frac{(-1)^q}{q!}$ whose sign alternates as q increases. To understand the physical significance of the poles of the reflection amplitude, it is necessary to examine their residues. However, the residue of the full reflection coefficient must be computed by including all Gamma-function factors appearing in $R(s, \tilde{m}, \tilde{\bar{m}})$. Substituting the pole condition $\tilde{m} = \tilde{j} + q$, the denominator contributes

$$\Gamma\left(\frac{1}{2} - is - \tilde{m}\right) = \Gamma(1 - 2\tilde{j} - q). \tag{760}$$

Using the recursion relation

$$\Gamma(z-q) = \frac{\Gamma(z)}{(z-1)(z-2)\cdots(z-q)}, \tag{761}$$

one obtains

$$\Gamma(1-2\tilde{j}-q) = \frac{\Gamma(1-2\tilde{j})}{(-2\tilde{j})(-2\tilde{j}-1)\cdots(-2\tilde{j}-q+1)}. \tag{762}$$

Consequently,

$$\frac{(-1)^q}{\Gamma(1-2\tilde{j}-q)} = \frac{(2\tilde{j})(2\tilde{j}+1)\cdots(2\tilde{j}+q-1)}{\Gamma(1-2\tilde{j})}. \tag{763}$$

The explicit factor $(-1)^q$ arising from the Gamma-function pole is therefore cancelled by the sign generated from the denominator Gamma function. As a result, the sign of the full residue is no longer determined by q. Instead, it is controlled by the remaining $\tilde{j}$-dependent Gamma-function factors, mainly the factor $\Gamma(1-2\tilde{j})$. The Gamma function changes sign whenever its argument crosses one of its poles. Accordingly, as $\tilde{j}$ increases, the residue changes sign when the argument of the relevant Gamma-function factor crosses a pole. In the exact reflection coefficient, this occurs precisely at $1-2\tilde{j} = 2-k$, for some k, or when $\tilde{j} = \frac{k-1}{2}$, which coincides with the upper endpoint of the no-ghost bound.

At $\tilde{j} = \frac{k-1}{2}$, the residue vanishes, while beyond this value its sign reverses. A positive residue corresponds to a positive-norm physical state, while a negative residue signals a negative-norm (ghost) state. Hence, although poles of the reflection amplitude occur whenever the discrete-state mass-shell condition is satisfied, only those poles whose residues remain positive can be identified with genuine physical discrete representations. This provides a scattering-theoretic justification for the no-ghost bound

$$\boxed{\frac{1}{2} \leq \tilde{j} \leq \frac{k-1}{2}.} \tag{764}$$

This does not contradict general scattering theory. The standard one-to-one correspondence between poles and bound states requires the scattering potential to decrease sufficiently rapidly at infinity. The effective potential governing long-string scattering in AdS_3 does not satisfy this condition. As a result, extra poles may appear in the reflection amplitude that do not correspond to normalizable physical states.

Therefore, while the poles of the reflection coefficient reproduce exactly the mass-shell condition of the discrete representations, only those poles whose quantum numbers satisfy $\frac{1}{2} \leq j \leq \frac{k-1}{2}$ and whose residues have the correct sign can be identified with genuine physical bound states.

Conclusion

We studied bosonic string theory in $\mathrm{AdS}_3 \times \mathcal{M}$ using the worldsheet description provided by the $\mathrm{SL}(2,\mathbb{R})$ WZW model. Starting from the realization of AdS_3 as the group manifold of $\mathrm{SL}(2,\mathbb{R})$, we constructed the classical theory, derived its chiral current algebra and Virasoro constraints, and analyzed geodesic solutions. We showed that spectral flow generates new classical solutions corresponding to winding strings, giving rise to short- and long-string configurations.

Upon quantization, the conserved currents generate the affine $\widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2,\mathbb{R})}_{k_R}$ algebra, while the Sugawara construction produces the corresponding Virasoro algebra. We studied the unitary representations of both the global and affine algebras and established the no-ghost theorem, demonstrating that physical states are ghost-free after imposing the Virasoro constraints. Spectral flow was then shown to be an automorphism of the spectrum-generating algebra, making the inclusion of spectrally flowed representations essential for a complete description of the physical Hilbert space, which consists of spectrally flowed discrete representations describing short strings and spectrally flowed continuous representations describing long strings, with a smooth transition between the two sectors at the boundaries of the allowed discrete spectrum.

Finally, we investigated long-string scattering by analytically continuing the theory to the Euclidean H_3^+ WZW model. Using bosonization and parafermionic techniques, the primary fields were constructed, and the reflection amplitude was analyzed. The poles of the reflection coefficient were shown to correspond to normalizable discrete states, while their residues reproduce the unitarity bound required by the no-ghost theorem. Thus, long-string scattering provides an independent and complementary derivation of the physical spectrum, establishing a direct connection between the affine representation theory and the analytic structure of scattering amplitudes in string theory in AdS_3.

References

[1] Maldacena and Ooguri. "Strings in AdS_3 and the SL(2,$\mathbb{R}$) WZW Model". In: (2001).

[2] Lorenz Eberhardt. "Wess–Zumino–Witten Models". In: *YRISW, Lecture notes* (2019).

[3] Philippe Di Francesco and Pierre Mathieu. *Conformal Field Theory.* 1997.

[4] Stephen Hwang. "No-ghost theorem for $SU(1,1)$ string theories". In: *Nuclear Physics B, Particle Physics, Netherlands* 354.1 (1991).

[5] Jonathan M Evans, Matthias R Gaberdiel, and Malcolm J Perry. "The no-ghost theorem for AdS_3 and the stringy exclusion principle". In: *Nuclear Physics B* (1998).

[6] J Teschner. "On structure constants and fusion rules in the H3+-WZNW model". In: *Nuclear Physics B* 546.1-2 (1999), pp. 390–422.

[7] J Teschner. "Operator product expansion and factorization in the H3+-WZNW model". In: *Nuclear Physics B* 571.3 (2000), pp. 555–582.